

A framework for some distinguished series of orthogonal type modular forms

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Diplom-Mathematiker
Martin Woitalla
aus Bremen

Berichter: Univ.-Prof. Dr. Aloys Krieg
Univ.-Prof. Dr. Valery Gritsenko

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Introduction

Modular varieties of orthogonal type have become subject of vast interest since Borcherds published his groundbreaking series of papers in the years 1995-98. In [Bo] and [Bo2] he constructed certain automorphic products whose divisor is completely determined by the Fourier coefficients of some nearly holomorphic $\mathrm{SL}(2, \mathbb{Z})$ modular forms. In [FH], [DK], [K1] and [K2] Borcherds' methods were successfully applied to determine the graded ring of modular forms for some low-dimensional varieties of orthogonal type. All these varieties have interpretations as varieties of unitary or symplectic type due to the exceptional isogenies of signature $(2, 3)$, $(2, 4)$ and $(2, 6)$. In particular the classical results of Igusa and Freitag on the graded rings of Siegel and Hermitian modular forms with respect to the Gaussian number field could be reformulated using Borcherds' automorphic products. In all cases the authors have made large use of the exceptional isogenies to construct new modular forms. The main intent of this text is to present a treatment of orthogonal type modular forms which works regardless of these isogenies. In [G1] the author discovered three towers of reflective modular forms using certain Jacobi forms of singular weight. We develop a framework around these towers which can be used to construct generators for the graded rings of modular forms. Our standard example will be the A_1 -tower of modular forms. We use this tower to illustrate the techniques since in this case there is a particularly easy reduction process which yields the graded rings of modular forms. At the end of the procedure we obtain a very short proof of the classical results of Freitag and Igusa and end up with new interpretations for all generators of the graded ring. Without any further effort we obtain the graded ring for the lattice $4A_1$ with all characters which has not been determined so far, best of the authors knowledge. Moreover we can give a number theoretical application where the formulae of the Fourier coefficients found in [BrK] are involved. By considering Jacobi-Eisenstein series it is possible to treat the other two towers using a similar framework.

We give a short presentation of the contents. Let V be a real quadratic space of signature $(2, n)$. We consider the semisimple and noncompact Lie group $\mathrm{SO}(V)^+$ where the superscript denotes the connected component of zero. In the literature the notation $\mathrm{SO}^0(2, n)$ is widely spread. Its maximal compact subgroup is given by $K = \mathrm{SO}(2) \times \mathrm{SO}(n)$. One can construct a projective model for the Hermitian symmetric space $\mathrm{SO}(V)^+/K$ namely

$$\mathcal{D} = \{[\mathcal{Z}] \in \mathbb{P}(V \otimes \mathbb{C}) \mid (\mathcal{Z}, \mathcal{Z}) = 0, (\mathcal{Z}, \overline{\mathcal{Z}}) > 0\}^+ \xrightarrow{\sim} \mathrm{SO}(V)^+/K.$$

Here the superscript $+$ means that we have chosen one of the two connected components. The associated affine cone is defined as

$$\mathcal{D}^\bullet = \{\mathcal{Z} \in V \otimes \mathbb{C} \mid [\mathcal{Z}] \in \mathcal{D}\}.$$

Let L_2 be a full lattice in V such that L_2 is even and splits two hyperbolic planes. In view of the Gramian of L_2 this means that there exists a basis such that

$$\text{Gram}(L_2) = \begin{pmatrix} 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & -S & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 & 0 \end{pmatrix}$$

where S denotes the Gramian of a positive definite even lattice L . We consider the arithmetic group $\text{O}(L_2)^+$. For any subgroup $\Gamma \leq \text{O}(L_2)^+$ of finite index we consider the modular variety $\Gamma \backslash \mathcal{D}$. This is a noncompact space. Following Baily-Borel and Satake we can add cusps to $\Gamma \backslash \mathcal{D}$ such that

$$\Gamma \backslash \mathcal{D}^* = \Gamma \backslash \mathcal{D} \amalg \coprod_{\Pi} X_{\Pi} \amalg \coprod_l Q_l$$

is compact. Here l runs through the finitely many Γ -orbits of isotropic lines and Π runs through the finitely many Γ -orbits of isotropic planes in $L_2 \otimes \mathbb{Q}$. Each X_{Π} corresponds to a modular curve and is called a one-dimensional cusp of $\Gamma \backslash \mathcal{D}$ and each Q_l corresponds to a point and is called a zero-dimensional cusp. The one-dimensional cusps are represented by rational two-dimensional totally isotropic subspaces of V and the zero-dimensional cusps are represented by primitive isotropic vectors $c \in L_2$. Orthogonal type modular forms are very important tools to investigate the modular variety $\Gamma \backslash \mathcal{D}$ and are the main objects of this text. Here is the definition:

Definition 0.1 Let Γ be a subgroup of $\text{O}(L_2)^+$ of finite index. A *modular form* of weight $k \in \mathbb{Z}$ and character $\chi : \Gamma \rightarrow \mathbb{C}^\times$ with respect to Γ is a holomorphic function $F : \mathcal{D}^\bullet \rightarrow \mathbb{C}$ such that

$$\begin{aligned} F(t\mathcal{Z}) &= t^{-k} F(\mathcal{Z}) && \text{for all } t \in \mathbb{C}^\times, \\ F(g\mathcal{Z}) &= \chi(g) F(\mathcal{Z}) && \text{for all } g \in \Gamma. \end{aligned}$$

A modular form is called a *cuspidal form* if it vanishes at every cusp. The space of modular forms of weight k and character χ for the group Γ will be denoted by $\mathcal{M}_k(\Gamma, \chi)$. For the subspace of cusp forms we will write $\mathcal{S}_k(\Gamma, \chi)$.

Let $F \in \mathcal{M}_k(\Gamma, \chi)$. Any zero-dimensional cusp c yields an affine realization F_c . Since the function F_c has a representation as a Fourier series we will mainly work with the affine realization. The Jacobi group $\Gamma^J(L) \leq \text{O}(L_2)^+$ is a parabolic subgroup of $\text{O}(L_2)^+$ and is of particular importance for our way of constructing modular forms. We denote

by $\mathbb{H}$ the upper half-plane in $\mathbb{C}$. In affine coordinates a Jacobi form of weight k and index t is a function

$$\varphi : \mathbb{H} \times (L \otimes \mathbb{C}) \rightarrow \mathbb{C}$$

such that

$$(\tau, \mathfrak{z}, \omega) \mapsto \varphi(\tau, \mathfrak{z}) \exp(2\pi i t \omega) \quad , \quad \omega \in \mathbb{H}$$

is a modular form with respect to $\Gamma^J(L)$ and satisfies certain regularity conditions at ∞ . This definition is a generalization of the classical Jacobi forms occurring in [EZ]. Besides the examples given there we also consider the Jacobi theta-series

$$\vartheta(\tau, z) = \sum_{n \in \mathbb{Z}, n \equiv 1 \pmod{2}} (-1)^{\frac{n-1}{2}} \exp\left(\frac{\pi i n^2 \tau}{4} + \pi i n z\right) \quad , \quad \tau \in \mathbb{H}, z \in \mathbb{C}.$$

In their paper [GN] Gritsenko and Nikulin reinterpreted this function as a Jacobi form of weight and index $1/2$. We consider the cases where L is one of the root systems NA_1 for $N = 1, \dots, 4$ or D_N for $N = 1, \dots, 8$. In [G1] Gritsenko constructed towers of reflective modular forms for these examples. This construction uses $\vartheta(\tau, z)$ and Dedekind's function $\eta(\tau)$ and requires the choice of a model for the lattice L . We like to illustrate this procedure for the case $L = 4A_1$. We identify this lattice with $\mathbb{Z}^4$ equipped with the bilinear form $2(\cdot, \cdot)_4$ where $(\cdot, \cdot)_4$ denotes the standard scalar product on $\mathbb{Z}^4$. The function

$$\mathbb{H} \times (4A_1 \otimes \mathbb{C}) \rightarrow \mathbb{C} \quad , \quad (\tau, \mathfrak{z}) \mapsto \vartheta(\tau, z_1) \vartheta(\tau, z_2) \vartheta(\tau, z_3) \vartheta(\tau, z_4)$$

yields a Jacobi form of weight 2 and index $1/2$ for the lattice $4A_1$ where we have labelled the coordinates of $\mathfrak{z}$ by $z_1, \dots, z_4$. Another model for this lattice can be given as follows. Let $\varepsilon_1, \dots, \varepsilon_4$ denote the standard basis of $\mathbb{Z}^4$. The pair

$$(\langle \varepsilon_2 + \varepsilon_1, \varepsilon_2 - \varepsilon_1, \varepsilon_4 + \varepsilon_3, \varepsilon_4 - \varepsilon_3 \rangle_{\mathbb{Z}}, (\cdot, \cdot)_4)$$

defines another realization of this lattice. If we label the coordinates of $\mathfrak{z}$ with respect to this basis again by $z_1, \dots, z_4$ we can define another function

$$(\tau, \mathfrak{z}) \mapsto \eta(\tau)^{12} \vartheta(\tau, z_1 + z_2) \vartheta(\tau, z_1 - z_2) \vartheta(\tau, z_3 + z_4) \vartheta(\tau, z_3 - z_4)$$

which yields a Jacobi form of weight 8 and index 1. The Jacobi forms obtained in this way will be called theta-type Jacobi forms. This type of construction provides a very direct method to control the modular behaviour by means of the lattice configuration. In this way we are able to explain the existence of the modular form Φ_{24} for the lattice D_4 which appeared in [FH] and [K2] and establish a connection with the D_N -tower of reflective modular forms in [G1]. Moreover this technique provides a very simple reduction process for the graded rings of modular form for the $4A_1$ -tower.

Theorem 0.2 *There is the following diagram of modular forms for $O(L_2(NA_1))^+$ and trivial character where $N = 1, \dots, 4$. The function at the top of the arrow equals up to multiplication in $\mathbb{C}^*$ the pullback of the lower function. The weights are indicated by the subscript and except for the first column all forms are cuspidal.*

$$\begin{array}{ccccccc}
G_4^{4A_1} & & G_6^{3A_1} & & G_8^{2A_1} & & G_{10}^{A_1} & & \Delta_{12,12} \\
& \nearrow^{z_4=0} & & \nearrow^{z_3=0} & & \nearrow^{z_2=0} & & \nearrow^{z_1=0} & \\
H_6^{4A_1} & & H_8^{3A_1} & & H_{10}^{2A_1} & & H_{12}^{A_1} & & \\
& \nearrow^{z_4=0} & & \nearrow^{z_3=0} & & \nearrow^{z_2=0} & & & \\
H_8^{4A_1} & & H_{10}^{3A_1} & & H_{12}^{2A_1} & & & & \\
& \nearrow^{z_4=0} & & \nearrow^{z_3=0} & & & & & \\
H_{10}^{4A_1} & & H_{12}^{3A_1} & & & & & & \\
& \nearrow^{z_4=0} & & & & & & & \\
H_{12}^{4A_1} & & & & & & & &
\end{array}$$

Here $\Delta_{12}(\tau) = \eta(\tau)^{24}$ denotes the first cusp form for $\mathrm{SL}(2, \mathbb{Z})$ and

$$\Delta_{12,12}(\tau, \omega) := \Delta_{12}(\tau) \Delta_{12}(\omega) \quad , \quad \tau, \omega \in \mathbb{H}.$$

Hence we can consider Δ_{12} as the source of all the modular forms appearing in the diagram. Let $II_{2,26}$ be the unique (up to isomorphism) even unimodular lattice of signature $(2, 26)$. The following Theorem is due to Borcherds [Bo2].

Theorem 0.3 (Borcherds '95) *Suppose that f is a nearly holomorphic modular form of weight $-\frac{s}{2}$ for $\mathrm{SL}(2, \mathbb{Z})$, i.e. a function on the upper half-plane transforming like a $\mathrm{SL}(2, \mathbb{Z})$ modular form with a Fourier expansion of the shape*

$$f(\tau) = \sum_{n \geq -K} c(n) q^n \quad , \quad K \in \mathbb{N}$$

such that $c(n) \in \mathbb{Z}$ and the property $24|c(0)$ if $s = 0$. There is a unique vector $\varrho \in II_{1,1+s}$ such that

$$\Phi(Z) = e^{2\pi i(\varrho, Z)} \prod_{\substack{r \in II_{1,1+s} \\ r > 0}} (1 - e^{2\pi i(r, Z)})^{c((r, r)/2)} \quad , \quad Z \in \mathcal{H}(II_{1,1+s})$$

is a meromorphic automorphic form of weight $c(0)/2$ for the group $\mathrm{O}(II_{2,2+s})^+$. All the zeroes and poles of Φ lie on rational quadratic divisors and the multiplicity of Φ at the rational quadratic divisor of the primitive negative norm vector $r \in II_{2,2+s}$ is

$$\sum_{n > 0} c(n^2(r, r)/2).$$

In particular if this number is always nonnegative then Φ is holomorphic.

If we apply the Theorem to $1/\Delta_{12}$ which is a nearly holomorphic modular form of weight $-12 = -24/2$ for the group $\mathrm{SL}(2, \mathbb{Z})$ we obtain Borchers' famous singular modular form $\Phi_{12} \in \mathcal{M}_{12}(\mathrm{O}(II_{2,26})^+, \det)$. The divisor of this function is exactly the union of all rational quadratic divisors defined by the vectors of norm -2 in $II_{2,26}$. In [Gr] and [GHS2] the authors considered quasi-pullbacks of Φ_{12} to construct modular forms with a good divisor for many lattices. This is the second tool which we will use. Combining these two methods we are able to describe the graded rings of modular forms for the lattices NA_1 , $N = 1, \dots, 4$ where we additionally consider the Jacobi theta series for the unimodular lattice E_8 to produce Eisenstein-like forms of weights 4 and 6. In this way we can reduce all modular forms for the A_1 -tower to the source functions Δ_{12}, e_4, e_6 where e_k denotes the normalized elliptic Eisenstein series of weight k .

The text is organized as follows:

In the first chapter we introduce a projective model for the homogeneous domain of type IV and show how it can be realized as an affine domain by fixing a zero-dimensional cusp. We introduce modular groups of orthogonal type and analyze the action on the homogeneous domain. The corresponding modular varieties are noncompact and can be compactified in various ways. Here we only consider the Baily-Borel compactification. The boundary of the modular variety contains zero- and one-dimensional boundary components represented by totally isotropic subspaces of dimension one and two, respectively. We introduce modular forms for the orthogonal group. If the lattice defining the modular variety is not maximal we have in general different (i.e. nonequivalent) zero-dimensional cusps. Any such cusp admits a Fourier expansion and we investigate how one can switch between the different Fourier expansions of a modular form via application of rational modular substitutions. At the end we collect some results on root systems and their automorphism groups.

In the second chapter we introduce the Jacobi group as a parabolic subgroup of the modular group stabilizing a one-dimensional cusp. The arithmetic lifting introduced in [G] is our basic tool to produce modular forms using Jacobi forms as input functions. We introduce different types of Jacobi forms which appear to be suitable in order to provide natural coordinates for the graded ring of modular forms. We analyze to some extent how the configurations of different root systems can be used to control the modularity of theta-type Jacobi forms.

In the third chapter we have a closer look at the modular group. We formulate two criteria on the positive definite sublattice which guarantee a nice set of generators in the sense of [FH]. Moreover we derive an upper bound for the cardinality of the abelianisation of the modular group. In the last part of the chapter we recall the low-dimensional exceptional isogenies for the orthogonal groups where we are mainly interested in integral realizations.

In the fourth chapter we introduce a generalization of Jacobi-Eisenstein series as they were discussed in [EZ]. We show that they coincide with the vector valued Eisenstein series which appear in [Br] and [BrK]. We use the results of [BrK] to derive an explicit

Fourier expansion for the Jacobi-Eisenstein series at every zero-dimensional cusp. We explain their meaning for the modular variety. Jacobi-Eisenstein series turn out to be another type of natural coordinates for the modular variety.

In chapter five we discuss different applications of Borcherds' automorphic forms. One variant follows the approach in [GN] and [G2] where the authors translated Borcherds' multiplicative lifting using weakly holomorphic Jacobi forms. Following this approach we apply this method to the theta type Jacobi forms defined in chapter 2. This yields several modular forms with a good divisor. Another application of Borcherds' theory appeared in [Gr] where the author used the technique of quasi-pullback to obtain reflective modular forms. The second part of this chapter is a summary of [Gr] considering the cases we are interested in.

In the last chapter we apply the results to the A_1 -tower of modular forms. We obtain new interpretations of classical results. At least in this case our methods seem to be a suitable framework since the proofs turn out to be very simple. We indicate how one can apply the procedure on other series using the formulae for the Jacobi-Eisenstein series.

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Chapter 1

Lattices and orthogonal groups

1.1 Quadratic spaces and lattices

We denote by $\mathbb{R}$ the field of real numbers and by $\mathbb{Z} \subseteq \mathbb{R}$ the set of real integers. Let V be a finite-dimensional vector space over $\mathbb{R}$. A symmetric bilinear form on V is a map $(\cdot, \cdot) : V \times V \rightarrow \mathbb{R}$ such that

$$\begin{aligned} \forall x, y \in V \quad (\cdot, y), (x, \cdot) : V \rightarrow \mathbb{R} \text{ are linear} \\ \forall x, y \in V \quad (x, y) = (y, x) \end{aligned}$$

A quadratic form on V is a map $q : V \rightarrow \mathbb{R}$ such that

$$\begin{aligned} \forall \alpha \in \mathbb{R} \text{ and } x \in V \quad q(\alpha x) &= \alpha^2 q(x) \\ \forall x, y \in V \quad q(x + y) &= q(x) + q(y) + (x, y) \end{aligned}$$

where $(\cdot, \cdot)$ is a symmetric bilinear form on V . The two notions are connected by the polarization identity which holds for every symmetric bilinear form $(\cdot, \cdot)$ on V

$$(x, y) = \frac{(x + y, x + y) - (x, x) - (y, y)}{2}$$

where $x, y \in V$. Hence $q(x) = \frac{1}{2}(x, x)$ according to our definition. A finite-dimensional space V equipped with a quadratic form q is called a (real) *quadratic space*. For every quadratic space we can consider the homomorphism

$$\mathbf{b}_V : V \rightarrow V^*, \mathbf{b}_V(y)(x) = (x, y), x \in V,$$

where V^* denotes the dual vector space.

Definition 1.1.1 Let V be a quadratic space. We call V *non-degenerate* if $\ker(\mathbf{b}_V) = \{0\}$.

For a subset $Y \subseteq V$ we set $Y^\perp = \{x \in V \mid \forall y \in Y (y, x) = 0\}$. The set $Y^\perp$ is a subspace of V and called the *orthogonal complement* of Y in V . The quadratic space is

non-degenerate iff $V^\perp = \{0\}$. Assume that $\{v_1, \dots, v_N\}$ is a basis of V . We define the Gramian of V with respect to this basis as $G := ((v_j, v_k))_{1 \leq j, k \leq N}$. If we change the basis then the determinant of G changes by an element of $(\mathbb{R}^\times)^2$. Consequently we can define the discriminant of V as $\text{disc}(V) = \det(G) \cdot (\mathbb{R}^\times)^2$. For a quadratic space we always have $\text{disc}(V) \in \mathbb{R}/(\mathbb{R}^\times)^2 = \{-1, 0, 1\}$. Obviously V is non-degenerate iff $\text{disc}(V) \neq 0$. We introduce the notation $\det(G) = d(v_1, \dots, v_N)$ where the v_j can be arbitrary.

Two quadratic spaces V, V' are called isomorphic if there exists an isomorphism of vector spaces $\sigma : V \rightarrow V'$ such that for all $x \in V$ we have $q'(\sigma(x)) = q(x)$. Write $N = r + s$ for $r, s \in \mathbb{N}_0$. We define a quadratic form $q_{r,s}$ on the vector space $\mathbb{R}^N = \mathbb{R}^{r+s}$ by

$$q_{r,s}(x) = x_1^2 + \dots + x_r^2 - x_{r+1}^2 - \dots - x_{r+s}^2.$$

By $\mathbb{R}^{r,s}$ we denote the quadratic space $\mathbb{R}^{r+s}$ equipped with the quadratic form $q_{r,s}$. For a proof of the next theorem see e.g. [A], sec. 3.7.

Theorem 1.1.2 (Sylvester's law of inertia) *Assume that V is a non-degenerate real quadratic space. Then V is isomorphic to $\mathbb{R}^{r,s}$ for non-negative integers r, s . The pair (r, s) is uniquely determined by V and is called the signature of V .*

By $\text{GL}(V)$ we denote the group of linear automorphisms of the vector space V . The subgroup which respects the quadratic structure of V is called the orthogonal group of V . It is defined by

$$\text{O}(V) = \{g \in \text{GL}(V) \mid \forall v \in V : q(gv) = q(v)\}.$$

Definition 1.1.3 Let V be a non-degenerate real quadratic space. A subset $L \subseteq V$ is called a *lattice* in V if there exists a basis $b_1, \dots, b_N$ of V such that $L = \mathbb{Z}b_1 + \dots + \mathbb{Z}b_N$. A lattice L in V is called *integral* if for all $h, l \in L$ we have $(h, l) \in \mathbb{Z}$.

Let M be a lattice. A *sublattice* L is a $\mathbb{Z}$ -submodule of M . According to the main theorem on free $\mathbb{Z}$ -modules L is a lattice in some subspace V of W , see below. In particular any lattice is a free $\mathbb{Z}$ -module with $\text{rank}(L) = \dim(L \otimes \mathbb{R})$. Let $\tau : L \rightarrow L$ be a linear map and write $l_j = \tau(b_j)$ for $j = 1, \dots, N$. Then we have

$$d(l_1, \dots, l_N) = \det(\tau)^2 d(b_1, \dots, b_N). \quad (1.1)$$

This formula immediately shows that $d(b_1, \dots, b_N)$ does not depend on the choice of the basis. If $l_1, \dots, l_N$ is another basis of L then $\tau \in \text{GL}(L)$ such that we can define the *discriminant* of the lattice as $d(L) = d(b_1, \dots, b_N)$. If $L \subseteq M$ is a sublattice and $\text{rank}(M) = \text{rank}(L)$ then (1.1) shows that $d(M)$ divides $d(L)$. The next Theorem is well known from linear algebra.

Theorem 1.1.4 *Let L and M be lattices such that $L \leq M$ is a sublattice. Then we have $\text{rank}(L) \leq \text{rank}(M)$. Moreover there exist a basis $b_1, \dots, b_m$ of M and integers $\alpha_1, \dots, \alpha_l$ satisfying $\alpha_j | \alpha_{j+1}$ such that $\alpha_1 b_1, \dots, \alpha_l b_l$ forms a basis of L and $l = \text{rank}(L)$.*

As an immediate consequence we deduce from (1.1) the formula

$$d(L) = [M : L]^2 d(M) \quad (1.2)$$

where $[M : L]$ denotes the index of L in M and $\text{rank}(L) = \text{rank}(M)$.

Remark 1.1.5 If M is a lattice in the quadratic space W we consider a sublattice $L \leq M$ which is a lattice in a subspace $V \leq W$. Consider the map

$$\mathfrak{b}_V : W \rightarrow V^*, \mathfrak{b}_V(w) = \mathfrak{b}_W(w), w \in W.$$

We have $V^\perp = \ker(\mathfrak{b}_V)$ and

$$\dim W = \dim V^\perp + \dim V$$

since $\mathfrak{b}_V$ is surjective because W is non-degenerate. Since $\text{rank } L = \dim V$ and $\text{rank } L_M^\perp = \dim V^\perp$ we obtain the formula

$$\text{rank } M = \text{rank } L + \text{rank } L_M^\perp. \quad (1.3)$$

Moreover the restriction $\mathfrak{b}_V|_V$ is an isomorphism since V is non-degenerate, too. Since $\ker \mathfrak{b}_V|_V = V \cap V_W^\perp$ we see that the sum $V + V^\perp$ is direct. If we summarize our considerations we obtain a decomposition as direct sums

$$L \oplus L^\perp \leq M.$$

Assume that V is non-degenerate. Then we define the set

$$L^\vee := \{x \in V \mid \forall l \in L : (x, l) \in \mathbb{Z}\}.$$

Denote by $b_1^*, \dots, b_N^*$ the dual basis of $b_1, \dots, b_N$. Since $(\cdot, \cdot)$ is non-degenerate we know that for every b_j there exists a $b_j^\vee$ such that $b_j^* = \mathfrak{b}_V(b_j^\vee)$. Then we see that $L^\vee = \mathbb{Z}b_1^\vee + \dots + \mathbb{Z}b_N^\vee$. Thus $L^\vee$ is a lattice again and we have $\text{rank}(L^\vee) = \text{rank}(L)$. We call $L^\vee$ the *dual lattice* of L .

Definition 1.1.6 Let L be a lattice. We call L *unimodular* if $L = L^\vee$.

If L is integral then L is a sublattice of $L^\vee$. Thus there exist real numbers $c_{j,k}$ defined by

$$b_k^\vee = \sum_{l=1}^{\text{rank}(L)} c_{k,l} b_l.$$

The identity

$$(b_j^\vee, b_k^\vee) = b_j^*(b_k^\vee) = c_{k,j}$$

implies

$$\sum_{l=1}^{\text{rank}(L)} (b_j, b_l)(b_l^\vee, b_k^\vee) = \sum_{l=1}^{\text{rank}(L)} (b_j, c_{k,l} b_l) = (b_j, b_k^\vee) = b_k^*(b_j) = \delta_{j,k}.$$

Together with the symmetry of $(\cdot, \cdot)$ this calculation shows that $d(L^\vee) = d(L)^{-1}$ and we obtain the following

Corollary 1.1.7 *Let L be an integral lattice. Then we have*

$$d(L) = [L^\vee : L]. \quad (1.4)$$

Definition 1.1.8 A sublattice $L \leq M$ is called *primitive* in M if there exists a subspace $V \leq W$ such that $L = M \cap V$ where $W = M \otimes_{\mathbb{Z}} \mathbb{R}$.

Let M be a lattice in a quadratic space V . If Y is a subset of M then we immediately see that $Y_M^\perp$ is primitive in M since $Y_M^\perp = M \cap Y_V^\perp$. We give a characterization of primitive sublattices.

Proposition 1.1.9 *Let M be a lattice and L be a sublattice of M . The following statements are equivalent*

- (i) L is primitive in M
- (ii) There exists a sublattice $L' \leq M$ such that $M = L \oplus L'$.
- (iii) M/L is a free $\mathbb{Z}$ -module
- (iv) There is a sublattice $L' \leq M$ such that $L = L_M'^\perp$

Proof

(i) $\Rightarrow$ (ii) : Let $L = M \cap V$ for some subspace V of W . From Theorem 1.1.4 we know that there exists a basis $b_1, \dots, b_m$ of M and integers $a_1, \dots, a_l$ such that $a_1 b_1, \dots, a_l b_l$ forms a basis of L . Thus $V = \langle b_1, \dots, b_l \rangle_{\mathbb{R}}$ and all a_j are units in $\mathbb{Z}$. Hence $L' = \mathbb{Z}b_{l+1} + \dots + \mathbb{Z}b_m$ is a sublattice with the desired properties.

(ii) $\Rightarrow$ (iii) : This is an immediate consequence of (ii).

(iii) $\Rightarrow$ (iv) : Assume that M/L is a free $\mathbb{Z}$ -module. We put $L' = L_M^\perp$. Obviously we have $L \leq L'^\perp$. From (1.3) we deduce $\text{rank } L + \text{rank } L' = \text{rank } L' + \text{rank } L'^\perp$ and obtain $\text{rank}(L) = \text{rank}(L'^\perp)$. Thus $L'^\perp/L$ is a submodule of rank zero of the free submodule M/L . Hence we obtain $L'^\perp = L$.

(iv) $\Rightarrow$ (i) : Any lattice of the shape $L'^\perp$ is primitive according to the above remark. $\square$

For our considerations the notion of an even lattice will play an important role. A lattice L is called *even* if

$$\forall x \in L : (x, x) \in 2\mathbb{Z}.$$

Obviously every even lattice is integral. An integral lattice L is even if there exists a basis $b_1, \dots, b_N$ of L such that $(b_j, b_j) \in 2\mathbb{Z}$ for all $1 \leq j \leq N$.

1.2 Projective and affine model for the homogeneous domain

This section and the following are intended to introduce the basic configuration we will deal with. Let us fix an even lattice L of rank N with a positive definite bilinear form $(\cdot, \cdot)$. By $L(t), t \in \mathbb{R}$ we denote the lattice whose elements lie in L with bilinear form $t(\cdot, \cdot)$. By a *hyperbolic plane* we mean a lattice $\mathbb{Z}e + \mathbb{Z}f$ with $(e, e) = (f, f) = 0$ and $(e, f) = 1$. Let U be a hyperbolic plane. We set

$$L_1 = U \oplus L(-1)$$

and fix the following basis

$$L_1 = \mathbb{Z}e \oplus L(-1) \oplus \mathbb{Z}f \quad , \quad S_1 = \begin{pmatrix} 0 & 0 & 1 \\ 0 & -S & 0 \\ 1 & 0 & 0 \end{pmatrix} \quad (1.5)$$

where S denotes the Gramian of the lattice L . Then L_1 is a lattice of signature $(1, N+1)$. Further on we put L_2 for the corresponding lattice of signature $(2, N+2)$ containing two hyperbolic planes U and $U_1 = \mathbb{Z}e_1 + \mathbb{Z}f_1$, i.e.

$$L_2 = \mathbb{Z}e \oplus \mathbb{Z}e_1 \oplus L(-1) \oplus \mathbb{Z}f_1 \oplus \mathbb{Z}f \quad \text{and} \quad S_2 = \begin{pmatrix} 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & -S & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 & 0 \end{pmatrix} ,$$

for the corresponding Gramian. We denote the bilinear forms of L_1 and L_2 by $(\cdot, \cdot)_{(1)}$ or $(\cdot, \cdot)_{(2)}$, respectively. Therefore for any $v = ne + l + mf \in L_1$ we obtain $(v, v)_{(1)} = 2nm - (l, l)$. We call this quantity the *hyperbolic norm* of $v \in L_1$. The bilinear forms we deal with can be extended to $L_k \otimes_{\mathbb{Z}} \mathbb{R}$ and $L_k \otimes_{\mathbb{Z}} \mathbb{C}$ ($k = 1, 2$) by $\mathbb{R}$ -linearity or $\mathbb{C}$ -linearity, respectively. We use the same notation for them. We will sometimes identify the lattice L_k with $\mathbb{Z}^{N+2k}$ and the bilinear form $(\cdot, \cdot)_{(k)}$ with the matrix S_k .

The *projective space* of $V_{\mathbb{C}}(L_2) = L_2 \otimes_{\mathbb{Z}} \mathbb{C}$ is the set of all complex lines in $L_2 \otimes_{\mathbb{Z}} \mathbb{C}$ namely

$$\mathbb{P}(V_{\mathbb{C}}(L_2)) = \{[\mathcal{Z}]; \mathcal{Z} \in V_{\mathbb{C}}(L_2) \setminus \{0\}\} \quad \text{where} \quad [\mathcal{Z}] := \mathbb{C}\mathcal{Z}.$$

We consider the set

$$\Omega(L_2) = \{[\mathcal{Z}] \in \mathbb{P}(V_{\mathbb{C}}(L_2)) \mid (\mathcal{Z}, \mathcal{Z}) = 0, (\mathcal{Z}, \overline{\mathcal{Z}}) > 0\}$$

which is an open subset of the so called *zero quadric* given by

$$\mathcal{N}^0(L_2) = \{[\mathcal{Z}] \in \mathbb{P}(V_{\mathbb{C}}(L_2)) \mid (\mathcal{Z}, \mathcal{Z}) = 0\}.$$

We write $\mathcal{Z} = \mathcal{X} + i\mathcal{Y}$ where $\mathcal{X}, \mathcal{Y} \in L_2 \otimes_{\mathbb{Z}} \mathbb{R}$. It is easy to see that

$$(\mathcal{Z}, \mathcal{Z}) = 0 \quad , \quad (\mathcal{Z}, \overline{\mathcal{Z}}) > 0$$

is equivalent to the condition

$$(\mathcal{X}, \mathcal{Y})_{(2)} = 0 \quad , \quad (\mathcal{X}, \mathcal{X})_{(2)} = (\mathcal{Y}, \mathcal{Y})_{(2)} \quad \text{and} \quad (\mathcal{Z}, \overline{\mathcal{Z}})_{(2)} = 2(\mathcal{Y}, \mathcal{Y})_{(2)} > 0.$$

In order to give an affine realization of this set we will construct a biholomorphic map. For this purpose consider the (real) quadratic space $V(L_2) = L_2 \otimes_{\mathbb{Z}} \mathbb{R}$ of signature $(2, N+2)$. We consider the cone

$$\mathcal{C} = \{x \in V(L_1) \mid (x, x)_{(1)} > 0\} \subset V(L_1).$$

After a suitable choice of a basis the quadratic form is of the shape $q_1(x) = \frac{1}{2}(x, x) = x_0^2 - x_1^2 - \dots - x_{N+1}^2$ where $x = (x_0, x_1, \dots, x_{N+1})$ and we see that $\mathcal{C}$ possesses two connected components

$$\mathcal{C} = \mathcal{C}^+ \cup \mathcal{C}^-$$

which are defined by the condition $x_0 > 0$ or $x_0 < 0$, respectively. By

$$\tilde{\mathcal{H}}(L_1) := V(L_1) + i\mathcal{C} \subset V_{\mathbb{C}}(L_1)$$

we define an open complex domain. Every $i(a, v, b)^{\text{tr}} \in i\mathcal{C}$ satisfies

$$2ab - (v, v) > 0$$

where we have used the basis fixed in (1.5) and thus $\tilde{\mathcal{H}}(L_1)$ precisely consists of all elements of the shape

$$(\omega, \mathfrak{z}, \tau)^{\text{tr}} \in V_{\mathbb{C}}(L_1) \quad , \quad \text{Im}(\omega) \text{Im}(\tau) > 0 \quad , \quad \mathfrak{z} \in L \otimes_{\mathbb{Z}} \mathbb{C}.$$

Let us define

$$\text{pr} : \tilde{\mathcal{H}}(L_1) \rightarrow L_2 \otimes \mathbb{C} \quad , \quad Z = \begin{pmatrix} \omega \\ \mathfrak{z} \\ \tau \end{pmatrix} \mapsto \begin{pmatrix} -\frac{1}{2}(Z, Z)_1 \\ \omega \\ \mathfrak{z} \\ \tau \\ 1 \end{pmatrix}.$$

Lemma 1.2.1 *We have a biholomorphic map*

$$\tilde{\mathcal{H}}(L_1) \rightarrow \Omega(L_2) \quad , \quad Z \mapsto [\text{pr}(Z)].$$

Proof

We have $[\text{pr}(Z)] = [-1/2(Z, Z)_{(1)} : \omega : \mathfrak{z} : \tau : 1]$. For the well-definedness consider

$$(\mathcal{Z}, \mathcal{Z})_{(2)} = 2 \cdot (-1/2)(Z, Z)_{(1)} + (Z, Z)_{(1)} = 0 \quad \text{and} \quad (\mathcal{Y}, \mathcal{Y})_{(2)} = (\text{Im}(Z), \text{Im}(Z))_{(1)} > 0.$$

Now let $[\mathcal{Z}] = [w' : w : \mathfrak{z} : z : z'] \in \Omega(L_2)$. If one of the four hyperbolic components w', w, z, z' would be zero this would yield two perpendicular vectors of positive norm in a quadratic subspace of signature $(1, N)$ which is a contradiction. So we can write $[\mathcal{Z}] = [w' : \tilde{Z} : 1]$. Now the equation $(\mathcal{Z}, \mathcal{Z})_{(2)} = 0$ implies $w' = -(1/2)(\tilde{Z}, \tilde{Z})_{(1)}$ and $(\mathcal{Z}, \overline{\mathcal{Z}})_{(2)} > 0$ implies $\text{Im}(z) \text{Im}(w) > 0$. This shows the surjectivity. The injectivity is clear. $\square$

This immediately yields the following

Corollary 1.2.2 *The set $\Omega(L_2)$ has two connected components*

$$\Omega(L_2) = \Omega(L_2)^+ \cup \Omega(L_2)^-$$

and we have a bijection which is given by

$$\Omega(L_2)^+ \rightarrow \Omega(L_2)^- \quad , \quad [\mathcal{Z}] \mapsto [\overline{\mathcal{Z}}] .$$

We want to fix $\mathcal{D}(L_2) = \Omega(L_2)^+$ to be the image of

$$\mathcal{H}(L_1) = \{(\omega, \mathfrak{z}, \tau)^{\text{tr}} \in \widetilde{\mathcal{H}}(L_1) \mid \text{Im}(\omega) > 0\}$$

under the bijection defined in Lemma 1.2.1.

The set $\mathcal{D}(L_2)$ is called the *projective homogeneous domain* and $\mathcal{H}(L_1)$ is called its *affine realization* or *tube domain realization*. Later we shall also use the notation $\mathcal{H}_e(L_2) = \mathcal{H}(L_1)$ and the isotropic vector e will be referred to as the *simplest cusp* of the modular variety. Moreover we use the symbol $\mathbb{H}$ for the upper half-plane in $\mathbb{C}$.

1.3 The orthogonal group

Let (V, q) be a non-degenerate quadratic space of signature (r, s) . In suitable coordinates the quadratic form is of the shape

$$q(x) = x_1^2 + \cdots + x_r^2 - x_{r+1}^2 - \cdots - x_{r+s}^2 \quad \text{where} \quad x = (x_1, \dots, x_{r+s}) .$$

The real orthogonal group of V is given by

$$\text{O}_{\mathbb{R}}(V) = \text{O}_{\mathbb{R}}(r, s) = \{g \in \text{GL}(V) \mid \forall v \in V : q(gv) = q(v)\} .$$

Now let L, L_1, L_2 be lattices as in section 1.2. Then $V_2 = V(L_2)$ is a quadratic space of signature $(2, N+2)$ containing two hyperbolic planes. The real orthogonal group $\text{O}_{\mathbb{R}}(V(L_2))$ acts as a linear group on $\Omega(L_2)$:

$$\text{O}_{\mathbb{R}}(V_2) \times \Omega(L_2) \rightarrow \Omega(L_2) \quad , \quad g \cdot [\mathcal{Z}] = [g\mathcal{Z}]$$

and on the set containing the two connected components $\{\Omega(L_2)^+, \Omega(L_2)^-\}$. The subgroup of index 2 which fixes the component $\mathcal{D}(L_2) = \Omega(L_2)^+$ is denoted by $\text{O}_{\mathbb{R}}(V_2)^+$. This group is different from the special orthogonal group $\text{SO}_{\mathbb{R}}(V_2)$ since the element $\text{diag}(1, -1, I_N, -1, 1) \in \text{SO}_{\mathbb{R}}(V_2)$ changes the orientation of $\mathcal{Z} = \mathcal{X} + i\mathcal{Y}$.

If we define $\text{SO}_{\mathbb{R}}(V_2)^+ = \text{SO}_{\mathbb{R}}(V_2) \cap \text{O}_{\mathbb{R}}(V_2)^+ < \text{O}_{\mathbb{R}}(V_2)^+$ we obtain another subgroup of index 2 of $\text{O}_{\mathbb{R}}(V_2)^+$. We can also give an affine realization of the action of $\text{O}_{\mathbb{R}}(V_2)^+$ by virtue of the tube domain realization. We write

$$g = \begin{pmatrix} \alpha & a & \beta \\ b & A & c \\ \gamma & d & \delta \end{pmatrix} \in \text{O}_{\mathbb{R}}(V_2)^+ \quad \text{where} \quad A \in \mathbb{R}^{(N+2) \times (N+2)} .$$

We consider the isomorphism of the domains from the last section. Then we have for $Z = (\tau, \mathfrak{z}, \omega)^{\text{tr}} \in \mathcal{H}(L_1)$

$$[g \operatorname{pr}(Z)] = [\operatorname{pr}(g < Z >)] \quad \text{for some element } g < Z > \in \mathcal{H}(L_1).$$

Thus there exists a $J(g, Z) \in \mathbb{C}^\times$ such that $g \operatorname{pr}(Z) = J(g, Z) \operatorname{pr}(g < Z >)$. The explicit descriptions are given by

$$J(g, Z) = -\frac{\gamma}{2}(Z, Z)_{(1)} + dZ + \delta \quad \text{and} \quad J(g, Z) \cdot g < Z > = -\frac{1}{2}(Z, Z)_{(1)}b + AZ + c \quad (1.6)$$

From the linear action of $\mathrm{O}_{\mathbb{R}}(V_2)^+$ one immediately verifies that $J(g, Z)$ is a holomorphic factor of automorphy on $\mathrm{O}_{\mathbb{R}}(V_2)^+ \times \mathcal{H}(L_1)$ which means that

$$J(gh, Z) = J(h, Z)J(g, h < Z >) \quad \text{for all } g, h \in \mathrm{O}_{\mathbb{R}}(V_2)^+ \text{ and } Z \in \mathcal{H}(L_1)$$

and the assignment $(g, Z) \mapsto g < Z >$ defines a group action on $\mathcal{H}(L_1)$. The kernel of this action is equal to the center of $\mathrm{O}_{\mathbb{R}}(V_2)^+$ which is $\{\pm I\}$. According to [K] we have

$$\operatorname{Bih}(\tilde{\mathcal{H}}(L_1)) \cong \mathrm{O}_{\mathbb{R}}(V_2)/\{\pm I\} \quad \text{and} \quad \operatorname{Bih}(\mathcal{H}(L_1)) \cong \mathrm{O}_{\mathbb{R}}^+(V_2)/\{\pm I\}$$

and in [Ba], Section 6.3, it was shown that this action is transitive. The integral orthogonal group of the lattice L_2 is given by

$$\mathrm{O}(L_2) = \{g \in \mathrm{O}_{\mathbb{R}}(L_2 \otimes \mathbb{R}) \mid g L_2 \subseteq L_2\}.$$

We additionally define

$$\mathrm{O}(L_2)^+ = \mathrm{O}_{\mathbb{R}}(V_2)^+ \cap \mathrm{O}(L_2) \quad \text{and} \quad \mathrm{SO}(L_2)^+ = \mathrm{SO}_{\mathbb{R}}(V_2)^+ \cap \mathrm{O}(L_2)$$

and obtain in this way a chain of subgroups

$$\mathrm{SO}(L_2)^+ < \mathrm{O}(L_2)^+ < \mathrm{O}(L_2)$$

each of index 2 in the bigger group.

The dual lattice of L_2 is given by

$$L_2^\vee = \{x \in L_2 \otimes \mathbb{Q} \mid \forall l \in L_2 : (x, l) \in \mathbb{Z}\}.$$

Since L is an even lattice $L_2 \subseteq L_2^\vee$ is a sublattice in its dual lattice. The discriminant group of the lattice L_2 is defined as

$$D(L_2) = L_2^\vee / L_2 \cong L^\vee / L.$$

The orthogonal group $\mathrm{O}(L_2)$ acts on $D(L_2)$ since for every $l \in L_2$ we have

$$(gl, gx) = (l, x) \in \mathbb{Z} \quad \text{for all } x \in L_2^\vee, g \in \mathrm{O}(L_2).$$

Thus we obtain a homomorphism

$$\mathrm{O}(L_2) \rightarrow \mathrm{O}(D(L_2))$$

into the finite orthogonal group $\mathrm{O}(D(L_2))$. The *stable orthogonal group* is defined as the kernel of this action and denoted by $\tilde{\mathrm{O}}(L_2)$, i.e.

$$\tilde{\mathrm{O}}(L_2) = \{g \in \mathrm{O}(L_2) \mid \forall l \in L_2^\vee : gl - l \in L_2\}.$$

Again we have a chain of subgroups

$$\widetilde{\mathrm{SO}}(L_2)^+ < \tilde{\mathrm{O}}(L_2)^+ < \tilde{\mathrm{O}}(L_2).$$

The next lemma shows that there is an embedding of the stable orthogonal group with respect to a sublattice.

Lemma 1.3.1 *Let M be an integral lattice and $N \subseteq M$ be a sublattice. Then there is an embedding of the groups*

$$\tilde{\mathrm{O}}(N) \hookrightarrow \tilde{\mathrm{O}}(M).$$

Proof

Let $N^\perp \subseteq M$ be the orthogonal complement of N in M . Then we have sublattices

$$N \oplus N^\perp \leq M \leq M^\vee \leq N^\vee \oplus (N^\perp)^\vee.$$

Let $g \in \tilde{\mathrm{O}}(N)$. We extend g to $\tilde{\mathrm{O}}(N \oplus N^\perp)$ by $g = g|_N \oplus \mathrm{id}_{N^\perp}$. Then g is also an element of $\mathrm{O}(N^\vee \oplus (N^\perp)^\vee)$ and for all $n \oplus n' \in N^\vee \oplus (N^\perp)^\vee$ we have

$$g(n \oplus n') - n \oplus n' = g(n) - n \in N \subseteq M$$

which yields $g \in \tilde{\mathrm{O}}(M)$. □

1.4 Discriminant forms and isotropic vectors

We have defined a tube domain realization above. In this realization we have implicitly fixed a cusp. In this section we generalize this procedure and derive the affine realization of the homogenous domain with respect to an arbitrary cusp. Moreover we present a basic calculus with discriminant forms.

Throughout this section We consider an even lattice M of signature $(2, p)$ where $p \geq 3$. A vector $a \in M$ is called *primitive* if

$$\forall k \in \mathbb{Z}, b \in M : (a = kb \implies k = \pm 1).$$

We call $v \in M^\vee$ *isotropic* if $(v, v) = 0$. The *divisor* of $v \in M$ is defined as the positive generator of the ideal $(v, M) = \mathrm{div}(v)\mathbb{Z}$. We state the first elementary Lemma.

Lemma 1.4.1 *Let $a \in M$ be a primitive vector.*

- (i) $a^* := a / \operatorname{div}(a)$ is primitive in $M^\vee$.
- (ii) The order of a^* in $M^\vee / M$ is $\operatorname{div}(a)$.
- (iii) $\operatorname{div}(a) \mid \det(M)$.
- (iv) If M is unimodular we have $\operatorname{div}(a) = 1$.
- (v) There exists an $h \in M^\vee$ such that $(a, h) = 1$.
- (vi) If a is isotropic and h is chosen as in (v) we have

$$a_M^\perp / \mathbb{Z}a \cong M \cap a_M^\perp \cap h_M^\perp.$$

Proof

- (i) Since the lattice M is non-degenerate we have $\operatorname{div}(a) \neq 0$ such that a^* is well-defined. By definition we have $(a^*, M) = \mathbb{Z}$ such that a^* is an element of $M^\vee$. Now let $k \in \mathbb{Z}$ and $b \in M^\vee$ such that $a^* = kb$. Then we have $\mathbb{Z} = (a^*, M) = k(b, M)$ and thus $k = \pm 1$ showing that a^* is primitive.
- (ii) This statement is clear.
- (iii) Since $|\det(M)|$ is the order of the additive group $M^\vee / M$ the assertion follows from (ii).
- (iv) Follows from (iii).
- (v) Since M is non-degenerate there exists a $b \in M$ and $0 \neq d \in \mathbb{Z}$ such that $(a, b) = d$. Now we define $h = \frac{1}{d}b$.
- (vi) Consider the $\mathbb{Z}$ -linear map

$$a_M^\perp \rightarrow M \cap a_M^\perp \cap h_M^\perp, \quad v \mapsto v - (v, h)a$$

whose kernel is exactly $\mathbb{Z}a$. □

The next definition introduces a finite quadratic form on the discriminant group $D(M)$.

Definition 1.4.2 Let M be an even lattice. We define the *discriminant form* on $D(M)$ to be the *finite quadratic form*

$$q : D(M) \rightarrow \mathbb{Q}/2\mathbb{Z}, \quad a + M \mapsto (a, a) + 2\mathbb{Z}.$$

It satisfies the conditions

- (i) $q(rx) = r^2 q(x)$ for all $x \in D(M)$ and $r \in \mathbb{Z}$

(ii) $q(x + y) - q(x) - q(y) \equiv 2(x, y) \pmod{2\mathbb{Z}}$ for all $x, y \in D(M)$

Here $D(M) \times D(M) \rightarrow \mathbb{Q}/\mathbb{Z}$, $(a + M, a' + M) = (a, a') + \mathbb{Z}$ is the associated finite bilinear form.

From the context it will always be clear if q_M is the finite quadratic form associated to $D(M)$ or the real quadratic form associated to $M \otimes \mathbb{R}$. So we use the same notation for both of them. More general a finite quadratic form is a pair (A, q) where A is a finite abelian group and $q : A \rightarrow \mathbb{Q}/2\mathbb{Z}$ is a map satisfying conditions (i) and (ii) above. For a proof of the next statement confer [N], Theorem 1.3.2.

Theorem 1.4.3 *For every finite quadratic form (A, q) there exists a unique even lattice M such that $(A, q) = (D(M), q_M)$. Hence there is a bijective correspondence between finite quadratic forms and discriminant forms.*

The notion of a discriminant form is useful to classify overlattices of a given even lattice. An *overlattice* of M is an embedding $M \hookrightarrow N$ where N is another integral even lattice of the same rank. Let N be an even overlattice of M . The *isotropic group* is the finite abelian group N/M and denoted by $D_N(M)$. The group $D_N(M)$ is called isotropic since $q_M|_{D_N(M)} = 0$. The importance of this group is explained in the following Proposition.

Proposition 1.4.4 *Let M be an even lattice. The map*

$$\{N \mid N \text{ is an overlattice of } M\} \rightarrow \{G \leq D(M) \mid q_M|_G = 0\} \quad , \quad N \mapsto D_N(M)$$

is a one to one correspondence between overlattices of M and isotropic subgroups of $D(M)$. The lattice N is unimodular if and only if $|D_N(M)|^2 = |D(M)|$.

Proof

First it is clear that the map is injective. Let $G \leq D(M)$ be an isotropic subgroup. Thus there exists an abelian group N_G such that $G = D_{N_G}(M)$. The chain of inclusions $M \subseteq N_G \subseteq M^\vee$ shows that N_G is a lattice again. For all $a \in N_G$ we have $q_M(a + M) = 2\mathbb{Z}$ which shows that N_G is even. The second claim follows from the formula

$$|\det(N)|[N : M]^2 = |\det(M)|. \quad \square$$

Remark 1.4.5 If $D(M)$ has no non-trivial isotropic subgroups then the lattice M is a maximal even lattice.

For even lattices M, M' we can define the composite finite quadratic form $q_M \oplus q_{M'}$ as

$$q_M \oplus q_{M'} : D(M) \times D(M') \rightarrow \mathbb{Q}/2\mathbb{Z} \quad , \quad q_M \oplus q_{M'}(m, m') := q_M(m) + q_{M'}(m').$$

Note that we have a chain of embeddings $M \hookrightarrow N \hookrightarrow N^\vee \hookrightarrow M^\vee$ and that $D_N(M) \subseteq N^\vee/M \subseteq D(M)$ and $(N^\vee/M)/D_N(M) \cong D(N)$.

Remark 1.4.6 The notion of orthogonality can also be used for finite quadratic forms in the usual sense. In particular we have $(D_N(M))_{D(M)}^\perp = N^\vee/M$.

For a subgroup $G \leq D_N(M)^\perp$ we define

$$(q_M|_G)/D_N(M) : G/D_N(M) \rightarrow \mathbb{Q}/2\mathbb{Z} \quad , \quad g + D_N(M) \mapsto q_M(g) .$$

This construction is well-defined since for $g + D_N(M) = g' + D_N(M)$ we have

$$q_M(g) - q_M(g') = q_M(g - g') - 2(g, -g') - 2q_M(-g') \equiv 2(g - g', g') \pmod{2\mathbb{Z}}$$

and the latter expression is also zero modulo $2\mathbb{Z}$ since $g' \in N^\vee/M$. Two finite quadratic forms $(A, q), (A', q')$ are called isomorphic if there exists an isomorphism $\sigma : A \rightarrow A'$ such that $q'(\sigma a) = q(a)$ for all $a \in A$. We shortly write $q \cong q'$ in this case. We obtain as a consequence

$$(q_M|_{D_N(M)^\perp})/D_N(M) \cong q_N \tag{1.7}$$

For the next results see [N], Theorem 1.3.1 and Proposition 1.4.2.

Theorem 1.4.7 *Let M, M' be even lattices. We have $q_M \cong q_{M'}$ if and only if there exist even unimodular lattices L, L' such that $L \oplus M \cong L' \oplus M'$.*

Proposition 1.4.8 *Let M be an even lattice and N, N' be two even overlattices. The embeddings are isomorphic if and only if the isotropy groups $D_N(M)$ and $D_{N'}(M)$ are conjugate under some automorphism of M .*

Proof

We start with a comment on the notion of isomorphic embeddings. Let $\sigma : N \rightarrow N'$ be an isomorphism. Then σ extends to a $\mathbb{Z}$ -module isomorphism $N^\vee \rightarrow N'^\vee$. Thus we obtain an isomorphism of the discriminant forms $q_N \cong q_{N'}$. In particular there exists a homomorphism

$$\mathrm{O}(M) \rightarrow \mathrm{O}(q_M) .$$

This makes the statement evident. □

We call two primitive embeddings $M \hookrightarrow N$ and $M \hookrightarrow N'$ isomorphic if there exists an isomorphism $\sigma : N \xrightarrow{\sim} N'$ such that $\sigma(M) = M$. The next statement can be found in [N], Proposition 1.5.1.

Proposition 1.4.9 *A primitive embedding of an even lattice M is determined by a pair (G, γ) where $G \leq D(M)$ is a subgroup and $\gamma : G \rightarrow D(K)$ is group monomorphism satisfying*

$$q_K \circ \gamma = -q_M|_G \quad , \quad (q_M \oplus q_K|_{H_\gamma^\perp})/H_\gamma \cong q_N .$$

Here $K = M^\perp$ is the orthogonal complement of M with respect to the overlattice N and

$$H_\gamma = G \oplus \gamma(G) \subseteq D(M) \oplus D(K) .$$

Two such pairs $(G, \gamma), (G', \gamma')$ determine isomorphic primitive embeddings if and only if there exist isomorphisms $\phi \in \mathrm{O}(M)$ and $\psi : K \rightarrow K'$ such that $\bar{\phi}(G) = G'$ and $\bar{\psi} \circ \gamma = \gamma' \circ \bar{\phi}$. The notation $\bar{\phi}, \bar{\psi}$ stands for the induced morphisms of the discriminant groups and is explained in the proof.

Proof

Let $M \hookrightarrow N$ be a primitive embedding. We set $K = M_N^\perp$ and $\iota : D_N(M \oplus K) = N/(M \oplus K) \hookrightarrow D(M) \oplus D(K)$ as the natural embedding. Moreover we have projections

$$p_M : D_N(M \oplus K) \rightarrow D(M) \quad , \quad p_K : D_N(M \oplus K) \rightarrow D(K).$$

Note that the embedding $K \hookrightarrow N$ is always primitive since $K = N \cap (M \otimes \mathbb{R})_{N \otimes \mathbb{R}}^\perp$. According to Proposition 1.1.9 there exist sublattices $M', K' \leq N$ such that $N = M \oplus M' = K \oplus K'$. This shows that

$$D_N(M \oplus K) \cong M'/K \cong K'/M.$$

Hence the projections p_K, p_M are embeddings if and only if M, K are primitive in N . According to (1.7) we have

$$(q_M \oplus q_K|_{D_N(M \oplus K)^\perp})/D_N(M \oplus K) \cong q_N.$$

Next we define an isomorphism

$$\gamma_{M,K}^N = p_K \circ p_M^{-1} : p_M(D_N(M \oplus K)) \rightarrow p_K(D_N(M \oplus K)). \quad (1.8)$$

From $0 = q_M \oplus q_K(\iota(D_N(M \oplus K))) = q_M(p_M(D_N(M \oplus K))) + q_K(p_K(D_N(M \oplus K)))$ we obtain

$$q_K \circ \gamma_{M,K}^N = -q_M|_{p_M(D_N(M \oplus K))}.$$

If we set for abbreviation $H = \iota(D_N(M \oplus K))$, $G = p_M(D_N(M \oplus K))$ and $\gamma = \gamma_{M,K}^N$ then the pair (G, γ) satisfies

$$q_K \circ \gamma = -q_M|_G \quad , \quad (q_M \oplus q_K|_{H^\perp})/H \cong q_N \quad , \quad H = G \oplus \gamma(G) \quad (1.9)$$

Hence we have constructed a map

$$\begin{aligned} \Psi_M : \quad \{N \mid M \hookrightarrow N \text{ primitive embedding}\} &\longrightarrow \left\{ (G, \gamma) \left| \begin{array}{l} G \leq D(M) \\ \gamma \in \text{Mon}(G, D(K)) \\ (G, \gamma) \text{ satisfies (1.9)} \end{array} \right. \right\} \\ N &\longmapsto (p_M(D_N(M \oplus K)), \gamma_{M,K}^N). \end{aligned}$$

From the considerations above we see that Ψ_M is a bijection. Now assume we have isomorphic primitive embeddings $\sigma : N \rightarrow N'$. Let K' be the corresponding orthogonal complement of M in N' . Now σ extends according to the following diagram

$$\begin{array}{ccc} N/(M \oplus K) & \xleftarrow{\sim} & G \oplus \gamma(G) \\ \sigma \downarrow & & \downarrow \hat{\sigma} \\ N'/(M \oplus K') & \xrightarrow{\sim} & G' \oplus \gamma'(G') \end{array}$$

By assumption we have $\sigma|_M \in \text{O}(M)$ and $\sigma(K) = K'$. Since $K \cong K'$ we can think of $\sigma|_K$ as an element of $\text{O}(K)$. This shows that $\hat{\sigma}$ decomposes as $\hat{\sigma} = \bar{\phi} \oplus \bar{\psi}$ where $\bar{\phi}$ is the image of some $\phi \in \text{O}(M)$ under the natural homomorphism

$$\text{O}(M) \rightarrow \text{O}(D(M))$$

and $\bar{\psi} : D(K) \xrightarrow{\sim} D(K')$ is induced by $\psi : K \xrightarrow{\sim} K'$ and satisfies the condition

$$\bar{\phi} \oplus \bar{\psi}(G \oplus \gamma(G)) = G' \oplus \gamma'(G') \iff \bar{\psi} \circ \gamma = \gamma' \circ \bar{\phi} \text{ and } \bar{\phi}(G) = G'.$$

If on the other hand there exist such morphisms ϕ and ψ and the last condition is satisfied this clearly yields isomorphic embeddings. $\square$

As in [N], Corollary 1.5.2 the condition that decodes isomorphic primitive embeddings can be considered as a criterion for the continuation of lattice morphisms.

Corollary 1.4.10 *Let $M \hookrightarrow N$ and $M' \hookrightarrow N$ be two primitive sublattices and let $\phi : M \rightarrow M'$ be an isomorphism of lattices. Then ϕ extends to an automorphism of N if and only if there exists an isomorphism $\psi : K \rightarrow K'$ where K, K' are the corresponding orthogonal complements such that $\bar{\psi} \circ \gamma_{M,K}^N = \gamma_{M',K'}^N \circ \bar{\phi}$. Here we have used the same notation as in (1.8).*

A particular case is given for unimodular lattices.

Proposition 1.4.11 *A primitive embedding of an even lattice M into an even unimodular lattice is determined by an isomorphism $\gamma : D(M) \rightarrow D(K)$ satisfying*

$$q_K \circ \gamma = -q_M.$$

Two such isomorphisms γ, γ' determine isomorphic primitive embeddings if and only if there exist isomorphisms $\phi \in O(M)$ and $\psi : K \rightarrow K'$ such that $\bar{\psi} \circ \gamma = \gamma' \circ \bar{\phi}$.

Proof

Since the overlattice N is unimodular Proposition 1.4.4 yields $|D_N(M \oplus K)|^2 = |D(M)| \cdot |D(K)|$. Moreover from the injectivity of the projections p_M, p_K we deduce

$$|D_N(M \oplus K)| \leq |D(M)|, \quad |D_N(M \oplus K)| \leq |D(K)|.$$

This shows that the embeddings are in fact isomorphisms between $D_N(M \oplus K)$ and $D(M)$ or $D_N(M \oplus K)$ and $D(K)$, respectively. Now the statement follows from Proposition 1.4.9. $\square$

At this point we generalize our basic configuration from the the last two sections. We define as before

$$\mathcal{D}(M) = \{[\mathcal{Z}] \in \mathbb{P}(V_{\mathbb{C}}(M)) \mid (\mathcal{Z}, \mathcal{Z}) = 0, (\mathcal{Z}, \bar{\mathcal{Z}}) > 0\}^+.$$

Note that we do not require here that M splits two integral hyperbolic planes! The affine cone $\mathcal{D}(M)^{\bullet}$ over $\mathcal{D}(M)$ is given as

$$\mathcal{D}(M)^{\bullet} = \{\mathcal{Z} \in M \otimes \mathbb{C} \mid [\mathcal{Z}] \in \mathcal{D}(M)\}.$$

A *0-dimensional cusp* of $\mathcal{D}(M)$ is determined by a primitive isotropic vector $c \in M$. The vectors c and $-c$ define the same cusp. We define the affine quadric associated to c as

$$\mathcal{D}(M)_c = \{\mathcal{Z} \in \mathcal{D}(M)^{\bullet} \mid (\mathcal{Z}, c) = 1\}.$$

Lemma 1.4.12 *Let $\mathcal{Z} \in \mathcal{D}(M)^\bullet$ and $c \in M$ be a primitive isotropic vector. Then there exists a unique $\alpha \in \mathbb{C}^*$ such that $(\alpha\mathcal{Z}, c) = 1$.*

Proof

We write $\mathcal{Z} = \mathcal{X} + i\mathcal{Y}$. The condition $\mathcal{Z} \in \mathcal{D}(M)^\bullet$ is equivalent to $(\mathcal{X}, \mathcal{Y}) = 0$ and $(\mathcal{X}, \mathcal{X}) = (\mathcal{Y}, \mathcal{Y}) > 0$. Since c is primitive and isotropic the real vector space $V = \langle \mathcal{X}, \mathcal{Y}, c \rangle_{\mathbb{R}}$ has dimension 3. Assume that $\dim c_V^\perp = 3$. Since V is a subspace of a non-degenerate quadratic space of signature $(2, p)$, $p \geq 3$ we can find a vector $w \in M \otimes \mathbb{R}$ such that $H = \langle c, w \rangle$ is a hyperbolic plane and $H \perp \langle \mathcal{X}, \mathcal{Y} \rangle$ which is impossible. Hence the orthogonal complement $c_V^\perp$ has dimension lower than 3 and there are real numbers u, v such that $(u\mathcal{X} + v\mathcal{Y}, c) \neq 0$. This shows that there is a unique $\alpha \in \mathbb{C}^\times$ such that $(\alpha\mathcal{Z}, c) = 1$. $\square$

By virtue of the last lemma we obtain an isomorphism $\mathcal{D}(M)_c \cong \mathcal{D}(M)$. We consider the integral lattice

$$M_c = c_M^\perp / \mathbb{Z}c$$

of signature $(1, p-1)$. According to Lemma 1.4.1 we fix $h \in M^\vee$ to define an embedding of M_c into M by

$$M_c \cong M_{c,h} = M \cap c_M^\perp \cap h_M^\perp.$$

This yields a decomposition

$$M \otimes \mathbb{Q} = M_{c,h} \otimes \mathbb{Q} \oplus (\mathbb{Q}c + \mathbb{Q}h).$$

We define a positive cone for the hyperbolic space $M_c \otimes \mathbb{R}$

$$C(M_c) = \{X \in M_c \otimes \mathbb{R} \mid (X, X) > 0\}.$$

As in section 1.2 we choose one of the two connected components of this cone and denote it by $C^+(M_c)$. We define the tube domain

$$\mathcal{H}_c(M) = M_c \otimes \mathbb{R} + iC^+(M_c).$$

Thus we are able to define the affine realization of $\mathcal{D}(M)$ with respect to the cusp c .

Proposition 1.4.13 *Let $c \in M$ be a 0-dimensional cusp of $\mathcal{D}(M)$. Choose $h \in M^\vee$ such that $(h, c) = 1$. We have an isomorphism of the domains given by*

$$\text{pr}_c : \mathcal{H}_c(M) \rightarrow \mathcal{D}(M)_c \cong \mathcal{D}(M) \quad , \quad Z \mapsto Z \oplus \left[h - \frac{(Z, Z) + (h, h)}{2} c \right].$$

Proof

We obviously have $(\text{pr}_c(Z), c) = 1$. We first show that Z belongs to $\mathcal{D}(M)^\bullet$. Let $Z = X + iY \in \mathcal{H}_c(M)$. Since Z is perpendicular to h and c we obtain

$$(\text{pr}_c(Z), \text{pr}_c(Z)) = (Z, Z) + (Z, 2h - [(Z, Z) + (h, h)]c) + (h, h) - [(Z, Z) + (h, h)](h, c) = 0.$$

On the other hand we have

$$\begin{aligned}
 (\text{pr}_c(Z), \overline{\text{pr}_c(Z)}) &= (Z, \overline{Z}) + (h, h) - \frac{1}{2}[(\overline{Z}, \overline{Z}) + (h, h) + (Z, Z) + (h, h)] \\
 &= (X, X) + (Y, Y) - \frac{1}{2}[(X, X) - (Y, Y) - 2i(X, Y)] \\
 &\quad - \frac{1}{2}[(X, X) - (Y, Y) + 2i(X, Y)] \\
 &= 2(Y, Y) > 0.
 \end{aligned}$$

The injectivity of pr_c follows from Lemma 1.4.12. Now let $\mathcal{Z} \in \mathcal{D}(M)_c$. Consider the decomposition

$$M \otimes \mathbb{C} = M_{c,h} \otimes \mathbb{C} \oplus (\mathbb{C}c + \mathbb{C}h)$$

and write $\mathcal{Z} = Z \oplus \alpha h + \beta c$, $\mathcal{Z} = \mathcal{X} + i\mathcal{Y}$, $Z = X + iY$ where $X, Y \in M_{c,h} \otimes \mathbb{R}$. We have $\alpha = (\mathcal{Z}, c) = 1$ and

$$\begin{aligned}
 0 &= (\mathcal{X}, \mathcal{Y}) = (X + h + \text{Re}(\beta)c, Y + \text{Im}(\beta)c) = (X, Y) + \text{Im}(\beta) \\
 0 &< (\mathcal{Y}, \mathcal{Y}) = (Y + \text{Im}(\beta)c, Y + \text{Im}(\beta)c) = (Y, Y) \\
 (Y, Y) &= (\mathcal{X}, \mathcal{X}) = (X + h + \text{Re}(\beta)c, X + h + \text{Re}(\beta)c) = (X, X) + (h, h) + 2\text{Re}(\beta).
 \end{aligned}$$

Thus we have $Z \in \mathcal{H}_c$ and $\beta = -\frac{1}{2}[(Z, Z) + (h, h)]$. □

1.5 Eichler transvections

In this section let M be an even lattice of signature $(2, p)$ where $p \geq 3$. Throughout this section we assume that M splits two integral hyperbolic planes. We put $V = M \otimes \mathbb{R}$. In order to find generic elements of $\text{O}(M)^+$ we first consider the real orthogonal group. Let $r \in V$ be a non-isotropic vector. The reflection at r is given by

$$\sigma_r : M \otimes_{\mathbb{Z}} \mathbb{R} \rightarrow M \otimes_{\mathbb{Z}} \mathbb{R} \quad , \quad \sigma_r(v) = v - 2 \frac{(r, v)}{(r, r)} r$$

which is an element of $\text{O}_{\mathbb{R}}(M)$ of order two. We state a couple of results on reflections as elements of the orthogonal group. The following result is well-known. For a proof we refer to [Kn], Theorem (4.6).

Theorem 1.5.1 *The orthogonal group $\text{O}_{\mathbb{R}}(M)$ is generated by reflections.*

We now investigate whether a reflection belongs to $\text{O}_{\mathbb{R}}(M)^+$ or not.

Proposition 1.5.2 *Let $r \in M \otimes \mathbb{R}$ satisfying $(r, r) \neq 0$. The reflection σ_r belongs to $\text{O}_{\mathbb{R}}(M)^+$ if and only if $(r, r) < 0$.*

Proof

Let $r \in M \otimes \mathbb{R}$ such that $(r, r) < 0$. Without loss of generality we can assume $(r, r) = -\frac{1}{2}$. Since the signature of the quadratic space is $(2, p)$ there exists an $s \in M \otimes \mathbb{R}$ such that

$(r, s) = 0$ and $(s, s) > 0$. Hence we can assume $(s, s) = \frac{1}{2}$. Consider the hyperbolic plane $H = \langle e_1, e_{p+2} \rangle$ where $e_1 = s + r$ and $e_{p+2} = s - r$. We extend this hyperbolic couple to a basis $e_1, e_2, \dots, e_{p+1}, e_{p+2}$ of V such that e_2, e_{p+1} is another hyperbolic plane and

$$V = \langle e_1, e_{p+2} \rangle \perp \langle e_2, e_{p+1} \rangle \perp \langle e_3, \dots, e_p \rangle.$$

The representation matrix of σ_r with respect to this basis is

$$\begin{pmatrix} 0 & 0 & 1 \\ 0 & I_{N+2} & 0 \\ 1 & 0 & 0 \end{pmatrix}.$$

Using this basis we can consider the special element

$$[\mathcal{Z}] = \begin{bmatrix} \begin{pmatrix} 1 \\ i \\ 0 \\ i \\ 1 \end{pmatrix} \end{bmatrix} \in \mathcal{D}(M)$$

Note that $\begin{bmatrix} 1 & i & 0 & i & 1 \end{bmatrix}$ and $\begin{bmatrix} 1 & -i & 0 & -i & 1 \end{bmatrix}$ belong to different connected components according to Corollary 1.2.2. Then $\sigma_r \cdot [\mathcal{Z}] = [\mathcal{Z}] \in \mathcal{D}(M)$ showing that σ_r belongs to $O_{\mathbb{R}}(M)^+$.

Now assume $(r, r) > 0$. As before we can assume $(r, r) = \frac{1}{2}$ and that there exists an $s \in V$ such that $(r, s) = 0$ and $(s, s) = -\frac{1}{2}$. We set $e_1 = r + s$ and $e_{p+2} = r - s$ and obtain again a hyperbolic couple which can be extended to a basis of the same shape as above. The matrix representation of σ_r then becomes

$$\begin{pmatrix} 0 & 0 & -1 \\ 0 & I_{N+2} & 0 \\ -1 & 0 & 0 \end{pmatrix}$$

which shows $\sigma_r \notin O_{\mathbb{R}}(M)^+$ in this case. □

We want to give another description of the group $O_{\mathbb{R}}(M)^+$. According to Theorem 1.5.1 we can write $g \in O_{\mathbb{R}}(M)$ as a product of reflections, say $g = \sigma_{r_1} \cdots \sigma_{r_m}$. We define the *spinor norm over $\mathbb{R}$* as

$$\text{sn}_{\mathbb{R}}(g) = \left(-\frac{(r_1, r_1)}{2} \right) \cdots \left(-\frac{(r_m, r_m)}{2} \right) (\mathbb{R}^\times)^2$$

According to [Kn] the map $\text{sn}_{\mathbb{R}} : O_{\mathbb{R}}(M) \rightarrow \mathbb{R}^\times / (\mathbb{R}^\times)^2 = \{\pm 1\}$ is a well-defined group homomorphism. From Proposition 1.5.2 we immediately obtain

Corollary 1.5.3 *One has $O_{\mathbb{R}}(M)^+ = \ker(\text{sn}_{\mathbb{R}})$.*

We now return to our standard configuration and assume that

$$M = U \perp U_1 \perp L(-1)$$

where L is an even positive definite lattice of rank N and $U = \mathbb{Z}e + \mathbb{Z}f, U_1$ are hyperbolic planes. Write $L_1 = U_1 \perp L(-1)$ such that $M = U \perp L_1$.

Definition 1.5.4 Let $c \in V$ be an isotropic vector. For any $a \in c_V^\perp$ we define the map

$$t'(c, a) : c_V^\perp \rightarrow c_V^\perp, \quad v \mapsto v - (a, v)c.$$

which belongs to the group $O(c_V^\perp)$.

For a proof of the next Lemma we refer to [GHS].

Lemma 1.5.5 *There is a unique extension of $t'(c, a)$ to an element $t(c, a) \in O_{\mathbb{R}}(M)$ which is defined as*

$$t(c, a) : v \mapsto v - (a, v)c + (c, v)a - \frac{1}{2}(a, a)(c, v)c.$$

This map is called an *Eichler transvection*. We state some elementary properties of Eichler transvections.

Lemma 1.5.6 *Let $c \in V$ be isotropic and $a, b \in c_V^\perp$.*

- (i) $t(c, a)|_{c_V^\perp \cap a_V^\perp} = \text{id}$
- (ii) $t(c, a)t(c, b) = t(c, a + b), t(c, a)^{-1} = t(c, -a)$
- (iii) $\forall g \in O_{\mathbb{R}}(M) : gt(c, a)g^{-1} = t(g(c), g(a))$
- (iv) $\forall \varepsilon \in \mathbb{R}^\times : t(\varepsilon c, a) = t(c, \varepsilon a), t(c, \varepsilon c) = \text{id}$
- (v) $t(c, a) = \sigma_a \sigma_{a + \frac{1}{2}(a, a)c}$ if $(a, a) \neq 0$.
- (vi) *There exists an $n \in \mathbb{N}$ such that $(t(c, a) - \text{id})^n = 0$*
- (vii) *Let $c \in M$ be a 0-dimensional cusp. Then $t(c, a) \in \text{SO}_{\mathbb{R}}(M)^+, t(c, a) \in \widetilde{\text{SO}}(M)^+$ if additionally $a \in M$ such that $(c, a) = 0$*
- (viii) *The map $t(c, \cdot) : c_M^\perp \rightarrow \widetilde{\text{SO}}(M)^+$ is a homomorphism of groups with kernel $\mathbb{Z}c$ where $c \in M$ is a 0-dimensional cusp.*

Proof

The statements (i)-(v) are simple verifications.

- (vi) We choose $f \in M \otimes \mathbb{R}$ such that $H = \langle c, f \rangle_{\mathbb{R}}$ is a hyperbolic plane and $a \in H^\perp$ such that we obtain a decomposition $V = H \oplus H^\perp$. We choose a basis of V of the shape $(c, f, a, \dots)$. Then the representation matrix of $t(c, a) - \text{id}$ is an upper triangular matrix with zeros on the diagonal and hence nilpotent.

- (vii) We use (v) to see that $\det(t(c, a)) = 1$. Since a and $a + \frac{1}{2}(a, a)c$ both have either positive or negative norm the reflections $\sigma_a, \sigma_{a + \frac{1}{2}(a, a)c}$ either both preserve the connected component $\mathcal{D}(M)$ or interchange the connected components. This shows that $t(c, a) \in \mathrm{SO}_{\mathbb{R}}(M)^+$ if $(a, a) \neq 0$. If a is isotropic we can assume $c \notin \langle a \rangle$ by (iv). Now we can find a vector $b \in M \otimes \mathbb{Q}$ such that $H = \langle a, b \rangle$ is a hyperbolic plane and $c \notin H$. Since $M \otimes \mathbb{Q}$ splits another hyperbolic plane we can find an isotropic $a' \in M \otimes \mathbb{Q}$ such that $(a, a') = 1$ and $a' \perp c$. Now define $a_+ = \frac{1}{2}(a + a')$ and $a_- = \frac{1}{2}(a - a')$. Then a_+, a_- are not isotropic and we can decompose $t(c, a) = t(c, a_+)t(c, a_-)$ and apply the previous case. If additionally $a \in M$ then $t(c, a)$ preserves M and for any $m \in M^\vee$ we obtain $t(c, a)(m) - m \in M$ which shows that $t(c, a) \in \widetilde{\mathrm{SO}}(M)^+$.
- (viii) It remains to show the statement on the kernel. Obviously $\mathbb{Z}c \subseteq \ker(t(c, \cdot))$. Now let $a \in \ker(t(c, \cdot))$. This implies $(a, v)c + \frac{1}{2}(a, a)(c, v)c = (c, v)a$ for all $v \in M$. Let $h \in M^\vee$ with $(h, c) = 1$. Since $t(c, \cdot)$ is also the identity on $M \otimes \mathbb{Q}$ we obtain $a \in \mathbb{Z}c$. $\square$

In the situation of part (viii) of the last Lemma the group

$$E_c(M) = \{t(c, a) \mid a \in c_M^\perp\} \cong M_c$$

is called the group of transvections at the cusp c . Moreover we define

$$E(M) = \langle \{t(c, a) \mid c, a \in M, (c, c) = (c, a) = 0, \mathrm{div}(c) = 1\} \rangle \subseteq \widetilde{\mathrm{SO}}(M)^+$$

and

$$E_U(L_1) = \langle \{t(e, a), t(f, a) \mid a \in L_1\} \rangle$$

We continue with a technical Lemma for the special case where $M = U \oplus U_1$ is the direct sum of two integral hyperbolic planes. In this case M is isometric to

$$\mathfrak{M} := \left\{ X = \begin{pmatrix} a & -b \\ c & d \end{pmatrix} \in \mathrm{Mat}(2, 2, \mathbb{Z}) ; a, b, c, d \in \mathbb{Z} \right\}$$

where we equip $\mathfrak{M}$ with the quadratic form $Q(X) := 2 \det(X) = 2(ad + bc)$.

Lemma 1.5.7 *Let $M = U \oplus U_1$ be the sum of two integral hyperbolic planes.*

- (i) *For every $m \in M$ there exists some $g \in E(M)$ such that $g(m) \in U_1$.*
- (ii) *$E(M)$ is isomorphic to $\mathrm{SL}(2, \mathbb{Z}) \times \mathrm{SL}(2, \mathbb{Z}) / \{\pm(I_2, I_2)\}$.*

Proof

We identify M with $\mathfrak{M}$ according to the above remark. We define

$$F : \mathrm{SL}(2, \mathbb{Z}) \times \mathrm{SL}(2, \mathbb{Z}) \rightarrow \widetilde{\mathrm{O}}(M) \quad , \quad F_{A,B} = (X \mapsto AXB^{-1}) .$$

First we note that $F_{A,B}$ defines an orthogonal transformation for all $A, B \in \mathrm{SL}(2, \mathbb{Z})$ since $\det(AXB^{-1}) = \det(X)$ for all $X \in \mathfrak{M}$. Moreover we have

$$F_{AA', BB'} = F_{A,B} F_{A',B'} \text{ for all } A, A', B, B' \in \mathrm{SL}(2, \mathbb{Z}).$$

Thus F defines a homomorphism of groups. We indentify U with $\langle e, f \rangle$ and U_1 with $\langle e_1, f_1 \rangle$ where

$$e = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}, f = \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix}, e_1 = \begin{pmatrix} 0 & -1 \\ 0 & 0 \end{pmatrix}, f_1 = \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix}.$$

We note that $\mathrm{SL}(2, \mathbb{Z})$ is generated by $T = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}$ and $S = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$. One verifies

$$t(e, e_1) = F_{T, I_2}, t(f, f_1) = F_{S, I_2}, t(e_1, f) = F_{I_2, T^{-1}}, t(f_1, e) = F_{I_2, S}$$

where we have identified M with $e\mathbb{Z} \oplus e_1\mathbb{Z} \oplus f_1\mathbb{Z} \oplus f\mathbb{Z}$. Since F is a morphism of groups we see that the image of F is contained in $\mathrm{SO}(M)^+$ by Lemma 1.5.6. Now let $X \in \mathfrak{M}$. By the elementary divisor theorem there exist $A, B \in \mathrm{SL}(2, \mathbb{Z})$ such that

$$AXB^{-1} = \begin{pmatrix} a & 0 \\ 0 & b \end{pmatrix} \text{ where } a, b \in \mathbb{Z}$$

By replacing A by JA where $J = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix} \in \mathrm{SL}(2, \mathbb{Z})$ we obtain $F_{A,B}(X) \in U_1$. Since $F_{A,B}$ is a product of transvections the first part of the assertion is proved. The element (A, B) is in the kernel of F if and only if $AX = XB$ for all $X \in \mathfrak{M}$. If we plug in e or f for X , respectively, we obtain $(A, B) \in \{\pm(I_2, I_2)\}$. $\square$

To apply the statement of the last lemma to the lattice L_2 it is useful to compute the representation matrix of the generators of $\widetilde{\mathrm{SO}}^+(U \oplus U_1)$ with respect to the basis

$$\begin{aligned} & \mathbb{Z}e \oplus \mathbb{Z}e_1 \oplus L(-1) \oplus \mathbb{Z}f_1 \oplus \mathbb{Z}f \\ & t(e, e_1) = \begin{pmatrix} 1 & 0 & 0 & -1 & 0 \\ 0 & 1 & 0 & 0 & 1 \\ 0 & 0 & I_N & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 1 \end{pmatrix}, \quad t(e, f_1) = \begin{pmatrix} 1 & -1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & I_N & 0 & 0 \\ 0 & 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 0 & 1 \end{pmatrix} \\ & t(f, e_1) = \begin{pmatrix} 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 0 & 0 \\ 0 & 0 & I_N & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & -1 & 1 \end{pmatrix}, \quad t(f, f_1) = \begin{pmatrix} 1 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & I_N & 0 & 0 \\ 1 & 0 & 0 & 1 & 0 \\ 0 & -1 & 0 & 0 & 1 \end{pmatrix} \end{aligned} \tag{1.10}$$

We define the group

$$\mathrm{O}(L_1) = \{g \in \mathrm{O}(M) \mid g(e) = e, g(f) = f\}.$$

The next Theorem gives us a possibility to parametrize the orbits of the action of $E(M)$ on M . The first part is known as the *Eichler criterion* and was essentially found by Eichler. The next Proposition can be found in [GHS]. Because of its importance for the sequel we will repeat the proof.

Proposition 1.5.8 *Assume that M is an even integral lattice with a composition $M = U \perp U_1 \perp L(-1)$ as above.*

- (i) *If $r, s \in M$ are primitive, $(s, s) = (r, r)$ and $r^* \equiv s^* \pmod{M}$ then there exists a $\tau \in E_U(L_1)$ such that $\tau(r) = s$.*
- (ii) $E(M) = E_U(L_1)$.
- (iii) $O(M) = \langle E(M), O(L_1) \rangle$.

Proof

- (i) By Lemma 1.4.1 we have $\text{div}(r) = \text{div}(s) =: d$. According to the last Lemma there exist $\tau, \sigma \in E_U(L_1)$ such that $\tau(r) \in L_1$ and $\sigma(s) \in L_1$. Hence we can assume $r, s \in L_1$. Now choose $r', s' \in L_1$ such that $d = (r, r') = (s, s')$. The transvection $t(e, r')$ maps r to $r - de$ and $t(e, s')$ maps s to $s - de$. By assumption the element $w := \frac{r-s}{d}$ is in L_1 . Note that $(w, dw) = (w, r) - (w, s) = 2(w, r)$ since $(w, s - r) = 0$ due to $(s, s) = (r, r)$. Hence we obtain

$$\begin{aligned} t(f, w)(r - de) &= r - de - (w, r - de)f + (f, r - de)w - \frac{1}{2}(w, w)(f, r - de)f \\ &= r - de - (w, r)f - dw + \frac{1}{2}(w, dw)f \\ &= s - de. \end{aligned}$$

Thus the element $t(e, -s')t(f, w)t(e, r') : r \mapsto (r - de) \mapsto (s - de) \mapsto s$ has the desired property.

- (ii) We only have to show that $E(M) \subseteq E_U(L_1)$. Let $t(c, a) \in E(M)$ such that $(c, c) = 0$ and $\text{div}(c) = 1$. From part (i) we obtain a $\tau \in E_U(L_1)$ such that $\tau(c) = e$. By Lemma 1.5.6 (iii) we obtain $\tau^{-1}t(c, a)\tau = t(e, \tau(a))$. Due to Lemma 1.5.7 we can assume that $\tau(a) \in L_1$ which proves the assertion.
- (iii) Let $g \in O(M)$. According to (i) there exists $\tau \in E(M)$ such that $\tau(g(e)) = e$. We have $(e, \tau g(f)) = (\tau g(e), \tau g(f)) = (e, f) = 1$. We write $\tau g(f) = \alpha f + \beta e + l$ where $\alpha, \beta \in \mathbb{Z}$ and $l \in L_1$. The condition $(\tau g(f), \tau g(f)) = 0$ implies $\beta = -\frac{1}{2}(l, l)$ and from $(e, \tau g(f)) = 1$ we obtain $\alpha = 1$. Hence we obtain

$$\tau g(f) = f + l - \frac{1}{2}(l, l)e = t(e, l)(f).$$

If one defines $h = t(e, -l)\tau g$ we obtain

$$h(f) = f, \quad h(e) = t(e, -l)(e) = e$$

which shows that h belongs to $O(L_1)$. □

1.6 Modular forms

In this section we fix the lattice

$$L_2 = \mathbb{Z}e \oplus \mathbb{Z}e_1 \oplus L(-1) \oplus \mathbb{Z}f_1 \oplus \mathbb{Z}f$$

where L is a positive definite even lattice of rank N . We start this section with the definition of modular forms.

Definition 1.6.1 Let Γ be a subgroup of $O(L_2)^+$ of finite index. A *modular form* of weight $k \in \mathbb{Z}$ and character $\chi : \Gamma \rightarrow \mathbb{C}^\times$ with respect to Γ is a holomorphic function $F : \mathcal{D}(L_2)^\bullet \rightarrow \mathbb{C}$ such that

$$\begin{aligned} F(t\mathcal{Z}) &= t^{-k}F(\mathcal{Z}) \quad \text{for all } t \in \mathbb{C}^\times, \\ F(g\mathcal{Z}) &= \chi(g)F(\mathcal{Z}) \quad \text{for all } g \in \Gamma. \end{aligned}$$

A modular form is called a *cusp form* if it vanishes at every cusp. The space of modular forms of weight k and character χ for the group Γ will be denoted by $\mathcal{M}_k(\Gamma, \chi)$. For the subspace of cusp forms we will write $\mathcal{S}_k(\Gamma, \chi)$.

In what follows we shall explain the notions of vanishing and holomorphicity at a cusp. Let $r \in L_2$ be a primitive vector such that $\sigma_r \in O(L_2)$. In particular σ_r fixes L_2 :

$$\forall l \in L_2 : 2 \frac{(r, l)}{(r, r)} r \in L_2.$$

Hence there exists an l satisfying $(l, r) = \text{div}(r)$ and therefore we obtain the identity $r = \frac{(r, r)}{2 \text{div}(r)} s$ for some $s \in L_2$. Since r is primitive and $\text{div}(r) \mid (r, r)$ this implies $\frac{(r, r)}{2 \text{div}(r)} \in \{\pm 1, \pm \frac{1}{2}\}$. We therefore obtain

$$\text{div}(r) \mid (r, r) \mid 2 \text{div}(r) \tag{1.11}$$

We can classify all reflections which act trivially on $D(L_2)$.

Proposition 1.6.2 *Let $r \in L_2$ be primitive. Then $\sigma_r \in \tilde{O}(L_2)$ if and only if $(r, r) = \pm 2$.*

Proof

Assume $(r, r) = \pm 2$. Then we obviously have $\sigma_r \in O(L_2)$ and moreover for every $l \in L_2^\vee$ we have

$$l - \sigma_r(l) = \pm(l, r)r \in L_2$$

showing that σ_r acts trivially on the discriminant group. Now conversely let $\sigma_r \in \tilde{O}(L_2)$. This amounts to the statement

$$\forall l \in L_2^\vee : 2 \frac{(r, l)}{(r, r)} r \in L_2. \tag{1.12}$$

From (1.12) we obtain $2r^* = \frac{2}{\text{div}(r)}r \in L_2$ and this implies $\text{div}(r) = 1$ or $\text{div}(r) = 2$ since r is primitive. Thus $(r, r) = \pm 2$ or $(r, r) = \pm 4$ because L_2 is even. Now suppose r has length ± 4 . By Lemma 1.4.1 there exists an $h \in L_2^\vee$ such that $(h, r) = 1$ and (1.12) yields

$$2 \frac{(r, h)}{(r, r)} r = \pm \frac{r}{2} \in L_2$$

This is impossible because r is primitive. Thus we necessary have $(r, r) = \pm 2$. $\square$

For any $r \in L_2 \otimes \mathbb{Q}$ such that $(r, r) < 0$ we define the *rational quadratic divisor*

$$\mathcal{D}_{[r]} = \mathcal{D}_{[r]}(L_2) = \{[\mathcal{Z}] \in \mathcal{D}(L_2) \mid (\mathcal{Z}, r) = 0\} \cong \mathcal{D}(r_{L_2}^\perp)$$

where $r_M^\perp$ is an even integral lattice of signature $(2, N+1)$. In this case $\sigma_r \in \text{O}_{\mathbb{R}}(L_2)^+$ and thus

$$[\mathcal{Z}] \in \mathcal{D}_{[r]} \iff \sigma_r[\mathcal{Z}] = [\mathcal{Z}].$$

Definition 1.6.3 Let $\Gamma \leq \text{O}(L_2)^+$ be a subgroup of finite index and let $F \in \mathcal{M}_k(\Gamma, \chi)$ be a modular form. F is called *reflective* if

$$\text{div}(F) \subseteq \bigcup_{\substack{\pm r \in L_2 \\ r \text{ is primitive} \\ \sigma_r \in \Gamma \text{ or } -\sigma_r \in \Gamma}} \mathcal{D}_{[r]}(L_2).$$

F is called *strongly reflective* if the multiplicity of every irreducible component of $\text{div}(F)$ is equal to one.

Let $c \in L_2$ be a 0-dimensional cusp and let F be a modular form. Let $h \in L_2^\vee$ such that $(c, h) = 1$. We can identify F with the corresponding modular form on the tube domain $F_{c,h} : \mathcal{H}_c(L_2) \rightarrow \mathbb{C}$ via

$$F_{c,h}(Z) := F(\text{pr}_c(Z)).$$

As already seen in section 1.3 for every $g \in \text{O}(L_2)^+$ and $Z \in \mathcal{H}_c(L_2)$ there exists a $J_c(g, Z) \in \mathbb{C}^*$ and a $g < Z > \in \mathcal{H}_c(L_2)$ such that $g \text{pr}_c(< Z >) = J_c(g, Z) \text{pr}_c(g < Z >)$. It is easy to show that this defines an action on $\mathcal{H}_c(L_2)$.

Lemma 1.6.4 Let $c \in L_2$ be a zero-dimensional cusp and let F be a modular form as in 1.6.1. Then $F_{c,h}$ is holomorphic on $\mathcal{H}_c(L_2)$ and

$$\forall g \in \Gamma : F_{c,h}(g < Z >) = \chi(g) J_c(g, Z)^k F_{c,h}(Z) \quad , \quad Z \in \mathcal{H}_c(L_2). \quad (1.13)$$

If conversely G satisfies (1.13), then there exists a modular form F such that $G(Z) = F(\text{pr}_c(Z))$ for $Z \in \mathcal{H}_c(L_2)$.

Proof

For $Z \in \mathcal{H}_c(L_2)$ and $g \in \Gamma$ the defining properties of a modular form yield

$$\begin{aligned} F_{c,h}(g < Z >) &= F(\text{pr}_c(g < Z >)) = F(J(g, Z)^{-1} g \text{pr}_c(Z)) \\ &= J(g, Z)^k F(g \text{pr}_c(Z)) = \chi(g) J(g, Z)^k F_{c,h}(Z). \end{aligned}$$

Now assume there exists a function G satisfying (1.13). For every $\mathcal{Z} \in \mathcal{D}(L_2)^\bullet$ there exist unique elements $\alpha \in \mathbb{C}^*$ and $Z \in \mathcal{H}_c(L_2)$ such that $\mathcal{Z} = \alpha \cdot \text{pr}_c(Z)$ according to Proposition 1.4.13. So we define

$$F : \mathcal{D}(L_2)^\bullet \rightarrow \mathbb{C} \quad , \quad F(\mathcal{Z}) := \alpha^{-k} G(Z).$$

Then for each $g \in \Gamma$ we have $g\mathcal{Z} = \alpha J(g, Z) \text{pr}_c(g < Z >)$ and therefore

$$F(g\mathcal{Z}) = (\alpha J(g, Z))^{-k} G(g < Z >) = \chi(g) \alpha^{-k} G(Z) = \chi(g) F(\mathcal{Z})$$

and for every $t \in \mathbb{C}^*$ one obtains

$$F(t\mathcal{Z}) = (t\alpha)^{-k} G(Z) = t^{-k} F(\mathcal{Z})$$

which completes the proof. □

For any $Z \in \mathcal{H}_c(L_2)$ and $a \in (L_2)_{c,h}$ we have

$$\begin{aligned} t(c, a)(\mathcal{Z}) &= \mathcal{Z} - \left(a, Z \oplus \left[h - \frac{(Z, Z) + (h, h)}{2} c \right] \right) c + a - \frac{1}{2}(a, a)c \\ &= \mathcal{Z} - (a, Z)c + a - \frac{1}{2}(a, a)c \\ &= \text{pr}_c(Z + a). \end{aligned}$$

Now let $F \in \mathcal{M}_k(\widetilde{\text{SO}}(L_2)^+, 1)$. Then we have $F_{c,h}(Z + a) = F_{c,h}(Z)$ for all $a \in (L_2)_{c,h}$. Hence $F_{c,h}$ possesses a Fourier expansion at the cusp c :

$$F_c(Z) = \sum_{l \in (L_2)_{c,h}^\vee} f(l) \exp(2\pi i(l, Z)).$$

The following well-known result can be found, e.g. in [Ba], p. 209. We present it here for our special configuration. Note that this is true whenever the lattice has signature $(2, p)$ where $p \geq 3$.

Theorem 1.6.5 (Koecher's principle) *Let $F \in \mathcal{M}_k(\Gamma, 1)$ with notation as in Definition 1.6.1. Then F is holomorphic at every cusp. In view of its Fourier expansion with respect to an arbitrary 0-dimensional cusp this means*

$$f(l) \neq 0 \implies l \text{ belongs to the closure of } C^+((L_2)_c).$$

If F is a modular form the Fourier expansion with respect to the simplest cusp is

$$F_e(Z) = \sum_{\substack{m, n \in \mathbb{N}_{\geq 0} \\ l \in L^{\vee-}_0 \\ 2nm - (l, l) \geq 0}} f(m, l, n) \exp(2\pi i[n\tau - (l, \mathfrak{z}) + m\omega]). \quad (1.14)$$

Definition 1.6.6 Let c, c' be two zero-dimensional cusps for $\mathcal{D}(L_2)$ and let $\Gamma \leq \mathrm{O}(L_2)^+$ be a subgroup of finite index containing $\widetilde{\mathrm{SO}}(L_2)^+$. We call c and c' *equivalent with respect to Γ* if there exists a $g \in \Gamma$ such that $g(c) = c'$ or $g(c) = -c'$. Otherwise c and c' are called *inequivalent cusps*. For a cusp c we denote by $[c]_\Gamma$ the class of c under this relation and set

$$\mathcal{D}(L_2)_\Gamma^0 = \{[c]_\Gamma \mid c \text{ is a zero-dimensional cusp for } \mathcal{D}(L_2)\}.$$

The next Lemma tells us how the lattice L parametrizes inequivalent cusps with respect to $\widetilde{\mathrm{SO}}(L_2)^+$.

Lemma 1.6.7 Let $\Gamma = \widetilde{\mathrm{SO}}(L_2)^+$.

- (i) Let $l \in L$ such that $(l^*, l^*) \equiv 0 \pmod{2\mathbb{Z}}$. Then there exists a zero-dimensional cusp $c \in L_2$ such that $c^* \cong l^* \pmod{L_2}$. Moreover any zero-dimensional cusp for $\mathcal{D}(L_2)$ is Γ -equivalent to $\mathrm{div}(c)c_{l^*}$ where

$$c_{l^*} = e_1 + l^* - \frac{(l^*, l^*)}{2} f_1$$

and $l = 0$ if $\mathrm{div}(c) = 1$ or $l \in L \setminus \{0\}$ satisfying $(l^*, l^*) \equiv 0 \pmod{2\mathbb{Z}}$ else.

- (ii) Let

$$\mathfrak{I}(L) = \{l \in D(L) \mid q_L(l) = 0, l = -l \in D(L)\}$$

and

$$\mathfrak{N}(L) = \{l \in D(L) \mid q_L(l) = 0, l \neq -l \in D(L)\}$$

The map

$$\mathfrak{I}(L) \cup \mathfrak{N}(L) \rightarrow L_2, \quad l \mapsto c_l$$

is well-defined and c_l runs through a set of representatives of non-equivalent cusps.

For $l, h \in \mathfrak{N}$ such that $l + h = 0$ the cusps c_l and c_h are Γ -equivalent and

$$\sharp \mathcal{D}(L_2)_\Gamma^0 = \sharp \mathfrak{I}(L) + \sharp \mathfrak{N}(L)/2.$$

Proof

- (i) Let $c = l + u$ be a zero-dimensional cusp where $l \in L$ and $u \in U \perp U_1$. If $\mathrm{div}(c) = 1$ then we can simply choose $l = 0$ and c is equivalent to the simplest cusp by Eichler's criterion. Now assume $\mathrm{div}(c) > 1$. Then we have $l \neq 0$ and $\mathrm{div}(c) = \mathrm{div}(l)$ because $U \perp U_1$ is unimodular. According to Lemma 1.4.1 $c^* = l^* + u/\mathrm{div}(c)$. Note that

$c^* \equiv l^* \pmod{L_2}$. Since c is isotropic c^* is isotropic, too. This implies $(l^*, l^*) \equiv 0 \pmod{2\mathbb{Z}}$. The vector $\text{div}(c)c_{l^*}$ has norm zero and is primitive by construction. According to the Eichler criterion 1.5.8 there exists a $g \in \widetilde{\text{SO}}^+(L_2)$ such that $g(c) = \text{div}(c)c_h$ for some isotropic $h \in D(L)$.

- (ii) The only thing which remains to be shown is the appearance of the factor $1/2$ in the formula. For $l, h \in \mathfrak{N}$ such that $l + h = 0$ we have $\text{ord}(l)c_l \equiv \text{ord}(-h)c_{-h} \equiv -\text{ord}(h)c_h \pmod{L_2}$ and the latter cusp is equivalent to $\text{ord}(h)c_h$ by definition. $\square$

We give a slight modification of Lemma 1.4.1.

Lemma 1.6.8 *Let c be a zero-dimensional cusp. There exists an isotropic $h \in L_2 \otimes \mathbb{Q}$ such that $(c, h) = 1$.*

Proof

Without loss of generality we can assume $c \in \{e_1, \text{div}(c)c_{l^*}\}$ for some $l \in L \setminus \{0\}$ where $(l^*, l^*) \equiv 0 \pmod{2\mathbb{Z}}$. From Lemma 1.4.1 we know there is some $h' \in L^\vee$ such that $(l, h') = 1$. Now we can find $h \in \mathbb{Q}e \oplus h' \oplus \mathbb{Q}f$ such that $(h, c) = 1$ and $(h, h) = 0$. $\square$

We denote an immediate consequence.

Corollary 1.6.9 *Let c be a zero-dimensional cusp and $h \in L_2 \otimes \mathbb{Q}$ as in the last lemma. We have projectors*

$$\begin{aligned} p_{c,h}^\perp : L_2 \otimes \mathbb{C} &\rightarrow (L_2 \cap c^\vee \cap h^\vee) \otimes \mathbb{C} & , & \quad \mathcal{Z} \mapsto \mathcal{Z} - (h, \mathcal{Z})c - (c, \mathcal{Z})h \\ p_{c,h} : L_2 \otimes \mathbb{C} &\rightarrow \mathbb{C}h + \mathbb{C}h & , & \quad \mathcal{Z} \mapsto (h, \mathcal{Z})c + (c, \mathcal{Z})h \end{aligned}$$

We can thus give a construction for the inverse map of pr_c .

Lemma 1.6.10 *Using the same notations as in 1.6.9 we define the map*

$$\text{aff}_c : \mathcal{D}(L_2)_c \rightarrow \mathcal{H}_c(L_2) \quad , \quad \mathcal{Z} \mapsto p_{c,h}^\perp(\mathcal{Z}) .$$

This map is an isomorphism of the domains and is inverse to pr_c from Proposition 1.4.13.

Proof

We first show the well-definedness of this map. Let $\mathcal{Z} \in \mathcal{D}(L_2)_c$. By virtue of Proposition 1.4.13 we have the decomposition

$$\mathcal{Z} = Z \oplus \left[h - \frac{(Z, Z) + (h, h)}{2} c \right]$$

where $Z \in \mathcal{H}_c(L_2)$. Since $\mathcal{H}_c(L_2) \subseteq (L_2 \cap c^\vee \cap h^\vee) \otimes \mathbb{C}$ and $p_{c,h}^\perp$ acts on the latter set as the identity we obtain

$$\text{aff}_c(\mathcal{Z}) = p_{c,h}^\perp(\mathcal{Z}) = Z .$$

This proves the assertion. $\square$

We state a fundamental theorem on extensions of isometries. A proof can be found in [Kn], theorem (3.1).

Theorem 1.6.11 (Witt) *Let V be a quadratic space over a field K where $\text{char}(K) \neq 2$. Let V_1, V_2 be two subspaces and $\sigma : V_1 \rightarrow V_2$ an isomorphism of quadratic spaces. Then there exists a $g \in O(V)$ such that $g|_{V_1} = \sigma$.*

We note an immediate consequence for zero-dimensional cusps.

Corollary 1.6.12 *Let $c, c' \in L_2 \otimes \mathbb{Q}$ be two isotropic vectors. Then there exists some $g \in O(L_2 \otimes \mathbb{Q})$ such that $c' = gc$.*

We can give a refinement of the above corollary using Eichler transvections.

Corollary 1.6.13 *Let L be an even positive definite lattice and $M \supseteq L$ a maximal overlattice of L . Let $c, c' \in L_2$ be two primitive isotropic vectors. Then there exists some $\tau \in \widetilde{SO}^+(M_2)$ and $\alpha \in \mathbb{Q}^\times$ satisfying $\alpha c' = \tau c$. Moreover for $l, h \in L^\vee$ satisfying $(l, l) = (h, h) = 0 \pmod{2\mathbb{Z}}$ and $c^* = c_l, (c')^* = c_h$ we have the following description of τ as a product of Eichler transvections*

$$c^* \xrightarrow{\tau} (c')^* \quad , \quad \tau = t(f - f_1, l - h) t(f - f_1, \varepsilon(l, h) f_1) \quad , \quad \varepsilon(l, h) = \frac{(h, h) - (l, l)}{2}$$

where $\tau \in E(M_2)$. The matrix representation of this τ with respect to the standard basis of M_2 is given by

$$t(f - f_1, \varepsilon(l, h) f_1) = \begin{pmatrix} 1 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & I_N & 0 & 0 \\ \varepsilon & 0 & 0 & 1 & 0 \\ 0 & -\varepsilon & 0 & 0 & 1 \end{pmatrix}$$

$$t(f - f_1, l - h) = \begin{pmatrix} 1 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 \\ l - h & h - l & I_N & 0 & 0 \\ \frac{1}{2}(l - h, l - h) & -\frac{1}{2}(l - h, l - h) & (l - h)^{\text{tr}} S & 1 & 0 \\ -\frac{1}{2}(l - h, l - h) & \frac{1}{2}(l - h, l - h) & -(l - h)^{\text{tr}} S & 0 & 1 \end{pmatrix}.$$

Here S denotes the Gramian of L .

Proof

By lemma 1.6.7 we can assume that the vectors are of the shape βc_l and $\beta' c_h$ for $l, h \in L^\vee$ and $\beta, \beta' \in \mathbb{Q}^\times$ satisfying $(l, l) = (h, h) = 0 \pmod{2\mathbb{Z}}$. Since M is a maximal even lattice of the same rank there are no non-trivial isotropic elements in $D(M_2)$. Hence c_l, c_h are primitive in M_2 and $\text{div}(c_l) = \text{div}(c_h) = 1$ by construction. From the Eichler criterion 1.5.8 we obtain an element $\tau \in E(M_2)$ such that $\tau c_l = c_h$. From the proof of Eichler's criterion we can reconstruct τ . Since

$$(c_l, f_1) = (c_h, f_1) = \text{div}(c_l) = 1$$

the element τ is given by $t(e, -f_1)t(f, c_l - c_h)t(e, f_1)$. Using Lemma 1.5.6 we can rewrite this transvection as

$$\tau = t(f - f_1, l - h + \varepsilon(l, h)f_1) = t(f - f_1, l - h)t(f - f_1, \varepsilon(l, h)f_1)$$

where

$$\varepsilon(l, h) = \frac{(h, h) - (l, l)}{2}.$$

□

Remark 1.6.14 (a) In the situation of the last Corollary the following matrix is contained in $E(L_2)$

$$J := t(e, e_1)t(f, f_1)t(e, f_1)t(f, e_1)t(e, f_1)t(e, e_1), \quad J = \begin{pmatrix} 0 & 0 & 0 & 0 & -1 \\ 0 & 0 & 0 & -1 & 0 \\ 0 & 0 & I_N & 0 & 0 \\ 0 & -1 & 0 & 0 & 0 \\ -1 & 0 & 0 & 0 & 0 \end{pmatrix}$$

If we consider Jc_h, Jc_l instead the transformation $J\tau J \in E(M_2)$ maps Jc_l to Jc_h . The matrix representation of this transformation can be computed as follows

$$Jt(f - f_1, \varepsilon(l, h)f_1)J = \begin{pmatrix} 1 & 0 & 0 & -\varepsilon & 0 \\ 0 & 1 & 0 & 0 & \varepsilon \\ 0 & 0 & I_N & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 1 \end{pmatrix}$$

$$Jt(f - f_1, l - h)J = \begin{pmatrix} 1 & 0 & (l - h)^{\text{tr}}S & \frac{1}{2}(l - h, l - h) & -\frac{1}{2}(l - h, l - h) \\ 0 & 1 & (h - l)^{\text{tr}}S & -\frac{1}{2}(l - h, l - h) & \frac{1}{2}(l - h, l - h) \\ 0 & 0 & I_N & l - h & h - l \\ 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 1 \end{pmatrix}.$$

(b) If $\alpha \in \mathbb{Q}^\times$ and c is a cusp for L_2 then we can construct a model of $\mathcal{D}(L_2)$ with respect to αc . For h chosen as above we consider the module $(L_2)_{\alpha c} \cong (L_2)_{\alpha c, \alpha^{-1}h}$ to obtain $\mathcal{H}_{\alpha c}(L_2) \cong \mathcal{H}_c(L_2)$. Note that the vectors αc and c determine the same rational zero-dimensional boundary component.

In the next Proposition we are going to analyze the dependency of the affine realization of modular forms with respect to the choice of the zero-dimensional part of the boundary.

Proposition 1.6.15 *Let $c, c' \in L_2$ be primitive isotropic vectors and let $F \in \mathcal{M}_k(\Gamma, \chi)$ for some subgroup $\Gamma \leq \text{O}(L_2)^+$ of finite index containing $E(L_2)$. Let $\tau \in \text{SO}(L_2 \otimes \mathbb{Q})^+$ such that $\alpha c' = \tau c$ for some $0 \neq \alpha \in \mathbb{Q}$. By $Z \mapsto \tau Z$ we denote the linear action of $\text{O}(L_2 \otimes \mathbb{R})^+$ on $\mathcal{H}_c(L_2)$ with respect to the natural embedding $L_1 \otimes \mathbb{C} \hookrightarrow L_2 \otimes \mathbb{C}$.*

(i) For all $Z \in \mathcal{H}_c(L_2)$ the identities

$$\begin{aligned} \text{pr}_{\alpha c'}(\tau Z) &= J_c(\tau, Z) \text{pr}_c(\tau < Z >) \\ J_c(\tau, Z) &= (\text{pr}_c(\tau < Z >), \alpha c')^{-1} \end{aligned}$$

are valid.

(ii) There exists an isomorphism of the two affine models for $\mathcal{D}(L_2)$ given by

$$\mathcal{H}_c(L_2) \rightarrow \mathcal{H}_{\alpha c'}(L_2) \quad , \quad Z \mapsto \tau Z .$$

(iii) For $Z \in \mathcal{H}_c(L_2)$ we have the following correspondence between affine realizations of F

$$F_{\alpha c'}(\tau Z) = J_c(\tau, Z)^{-k} F_c(\tau < Z >) .$$

If additionally $c \sim c'$ then τ belongs to $E(L_2)$, $\alpha = 1$ and we obtain

$$F_c(Z) = \chi(\tau)^{-1} F_{c'}(\tau Z) .$$

Proof

(i) We recall the construction from Proposition 1.4.13. The set $\mathcal{H}(L_2)$ was constructed as

$$\mathcal{H}_c(L_2) = (L_2)_c \otimes \mathbb{R} + iC^+ \quad , \quad C = \{X \in (L_2)_c \otimes \mathbb{R} \mid (X, X) > 0\}$$

and the latter set decomposes into two distinct connected components. Here the lattice $(L_2)_c$ was defined by

$$(L_2)_c \cong (L_2)_{c,h} = L_2 \cap c_M^\perp \cap h_M^\perp$$

and h is chosen such that $(c, h) = 1$. We have

$$\tau \text{pr}_c(Z) = \tau Z \oplus \left[\tau h - \frac{(\tau h, \tau h) + (\tau Z, \tau Z)}{2} \alpha c' \right] = \text{pr}_{\alpha c'}(\tau Z) .$$

Since τ preserves the connected component of Z we have $\tau Z \in \mathcal{H}_{\alpha c'}(L_2)$ and $(L_2)_{\alpha c'} \cong L_2 \cap (\alpha c')^\perp \cap (\tau h)^\perp$. Note that $\text{pr}_{\alpha c'}(\tau Z)$ is the unique element $Z \in \mathcal{D}(L_2)^\bullet$ such that $(Z, \alpha c') = 1$. This yields the equation

$$(J_c(\tau, Z) \text{pr}_c(\tau < Z >), \alpha c') = 1 \iff J_c(\tau, Z) = (\text{pr}_c(\tau < Z >), \alpha c')^{-1} .$$

The remaining assertions follow immediately from (i). $\square$

At the end of this section we consider an easy but important consequence of the above considerations. Assume that L is a positive definite even lattice which is not maximal. Let $M \supseteq L$ be a maximal even overlattice. Then there are two zero-dimensional rational cusps c, c' satisfying $c \sim c' \bmod \widehat{\text{SO}}^+(M_2)$ but $c \not\sim c' \bmod \widehat{\text{SO}}^+(L_2)$. Let $F \in \mathcal{M}_k(\Gamma, \chi)$

for some subgroup $\Gamma \subseteq \mathrm{O}(M_2)^+$ containing $\widetilde{\mathrm{SO}}^+(M_2)$. Using the transvection τ from the last Proposition yields

$$F_c(Z) = \chi(\tau)^{-1} F_{c'}(\tau Z) \quad \text{for every } Z \in \mathcal{H}_c(M_2).$$

This shows that up to a non-zero multiple F_c and $F_{c'}$ determine the same function for the two tube domains related to the cusps c, c' . By lemma 1.3.1 F is also a modular form for the group $\widetilde{\mathrm{SO}}^+(L_2)$ and the two functions

$$Z \mapsto F_c(Z) \quad , \quad Z \mapsto F_{c'}(\tau Z) = J_c(\tau, Z)^{-k} F_c(\tau < Z >) \quad \in \quad \mathcal{M}_k(\widetilde{\mathrm{SO}}^+(L_2), \chi)$$

are thus linearly dependent. Now the natural question arises whether there are linearly independent functions $G_1, G_2 \in \mathcal{M}_k(\widetilde{\mathrm{SO}}^+(L_2), \chi)$ such that F is in the span of G_1, G_2 and whether G_1, G_2 are linked by a modular transformation in $E(M_2)$.

1.7 Modular varieties of orthogonal type

In this section we summarize some facts about modular varieties of orthogonal type. The section can be considered as an attempt to investigate the setup we are dealing with from a geometric point of view. Lots of work has been done in this subject. We only collect known results which can be found in the literature. A good survey from the perspective of automorphic forms is given in [Ba]. The standard reference for the theory of compactifications is [BaBor]. For varieties of orthogonal type we follow the descriptions given in [GHS], [GHS2] and [F]. Let V be a real quadratic space of signature $(2, n)$ equipped with a quadratic form q . The group of all isometries of V is the orthogonal group $\mathrm{O}(V)$. This group admits exactly four characters given by $1, \det, \mathrm{sn}_{\mathbb{R}}$ and the product of the last two. Hence $\mathrm{O}(V)$ is not connected and the connected component of zero is denoted by

$$\mathrm{SO}^+(V) := \ker(\mathrm{sn}_{\mathbb{R}}) \cap \ker(\det).$$

In the literature the notation $\mathrm{SO}^0(2, n)$ is widely spread. This group is a semisimple, connected and noncompact Lie group, confer [Kp], Proposition 1.124. Its maximal compact subgroup is given by $K = \mathrm{SO}(2) \times \mathrm{SO}(n)$. The centre of $\mathrm{SO}(m)$ is discrete unless $m = 2$. Hence the centre of K is given by

$$\begin{cases} \mathrm{SO}(2) \times \{\pm I\} & \text{if } n > 2 \\ K & \text{if } n = 2 \end{cases}.$$

An irreducible symmetric space G/K is Hermitian if and only if the centre of the maximal compact subgroup has positive dimension. This is why one is interested in the case of signature $(2, n)$. One can construct a projective model for the Hermitian symmetric space $\mathrm{SO}^+(2, n)/K$ given by

$$\mathcal{D} = \{[Z] \in \mathbb{P}(V \otimes \mathbb{C}) \mid (Z, Z) = 0, (Z, \overline{Z}) > 0\}^+.$$

Here the superscript $+$ means the connected component of $\mathcal{D}$ which contains the element $[1, i, 0, \dots, 0]$ with respect to a basis of the shape

$$\frac{e+f}{2}, \frac{e_1+f_1}{2}, \frac{e-f}{2}, \frac{e_1-f_1}{2}, \dots$$

where we have decomposed V as the orthogonal sum of two hyperbolic planes and a negative definite space V_0 such that

$$V = \langle e, f \rangle \perp \langle e_1, f_1 \rangle \perp V_0$$

and the Gramian of the quadratic form q with respect to this basis is

$$\begin{pmatrix} I_2 & 0 \\ 0 & -I_n \end{pmatrix}.$$

We fix this basis for the rest of this section. The linear action of $\mathrm{SO}^+(2, n)$ on $\mathcal{D}$ is transitive, confer [Ba], section 6.3. For $g \in \mathrm{SO}^+(2, n)$ write

$$g = \begin{pmatrix} g_{11} & g_{12} \\ g_{21} & g_{22} \end{pmatrix} \quad \text{where } g_{11} \in \mathrm{Mat}(2, \mathbb{R}), g_{22} \in \mathrm{Mat}(2, \mathbb{R}).$$

Then we have

$$g(1, i, 0, \dots, 0) = (1, i, 0, \dots, 0) \iff g_{11} \in \{I_2\} \cap \mathrm{SO}(2), g_{12} = g_{21} = 0, g_{22} \in \mathrm{SO}(n).$$

Writing $g_{11} = \begin{pmatrix} \cos(\theta) & \sin(\theta) \\ -\sin(\theta) & \cos(\theta) \end{pmatrix}$ we see that $g_{11}(1, i) = e^{2\pi i \theta}(1, i)$. This shows that the stabilizer of $[1, i, 0, \dots, 0] \in \mathcal{D}$ equals K . Thus we have established a model for the Hermitian domain

$$\mathrm{SO}^+(2, n)/K \xrightarrow{\sim} \mathcal{D}.$$

Any Hermitian domain can be embedded into a bounded symmetric domain, confer [BorJ], Proposition I.5.9.

Proposition 1.7.1 (Harish-Chandra embedding) *Any Hermitian symmetric space $X = G/K$ of noncompact type can be canonically realized as a bounded symmetric domain in the holomorphic tangent space at $x_0 = K$ in X .*

Following [Ba] we can give an explicit realization of this embedding using a variant of the well-known Cayley transform mapping the classical upper half-plane to the unit disc.

Proposition 1.7.2 *Let $n \geq 3$. Define the bounded domain*

$$\mathcal{D}^{IV} := \{\zeta \in \mathbb{C}^n \mid \|\zeta\|^2 < 1 + |\zeta_1 \zeta_2 + \zeta_3^2 + \dots + \zeta_n^2|^2 < 2\}$$

where

$$\|\zeta\|^2 := |\zeta_1|^2 + |\zeta_2|^2 + 2|\zeta_3|^2 + \dots + 2|\zeta_n|^2.$$

We have a biholomorphic isomorphism of the domains given by

$$\mathcal{D} \rightarrow \mathcal{D}^{IV} \quad , \quad [z_1, z_2, z_3, z_4, z_5, \dots, z_{n+2}] \mapsto \left(\frac{z_3 + iz_4}{z_1 - iz_2}, \frac{z_3 - iz_4}{z_1 - iz_2}, \frac{z_5}{z_1 - iz_2}, \dots, \frac{z_{n+2}}{z_1 - iz_2} \right)$$

mapping $[1, i, 0, \dots, 0] \mapsto 0 \in \mathbb{C}^n$. The inverse map is given by

$$\zeta \mapsto \left[\frac{\tau}{\tau - i}, \frac{1}{\tau - i}, \frac{\zeta_1 + \zeta_2}{2}, \frac{\zeta_1 - \zeta_2}{2i}, \zeta_3, \dots, \zeta_n \right]$$

where τ from the lower half-plane in $\mathbb{C}$ is chosen such that

$$\zeta_1 \zeta_2 + \zeta_3^2 + \dots + \zeta_n^2 = \frac{\tau + i}{\tau - i}.$$

Proof

Note that every $[\mathcal{Z}] \in \mathcal{D}$ is a solution of

$$z_1^2 + z_2^2 - z_3^2 - \dots - z_{n+2}^2 = 0 \quad , \quad |z_1|^2 + |z_2|^2 - |z_3|^2 - \dots - |z_{n+2}|^2 > 0. \quad (1.15)$$

Assume there exists a real number α such that $z_2 = \alpha z_1$. Then we would obtain $|z_3|^2 + \dots + z_{n+2}^2 - |z_3|^2 - \dots - |z_{n+2}|^2 > 0$ which is impossible. Thus z_2 is not a real multiple of z_1 and in particular $z_1, z_2 \neq 0$. The choice of the connected component implies $z_1 = \tau z_2$ for some τ in the lower half-plane in $\mathbb{C}$. Hence the denominator $z_1 - iz_2 \neq 0$. If we add the last inequality to (1.15) we obtain an affine characterization of $\mathcal{Z}$ belonging to $\mathcal{D}$. We set

$$\eta_1 := \frac{z_3}{z_1 - iz_2} \quad , \quad \eta_2 := \frac{z_4}{z_1 - iz_2} \quad , \quad \eta_j := \frac{z_{j+2}}{z_1 - iz_2} \quad \text{for } 3 \leq j \leq n.$$

Dividing the equation in (1.15) by $(z_1 - iz_2)^2$ yields

$$\eta_1^2 + \dots + \eta_n^2 = \frac{z_1 + iz_2}{z_1 - iz_2}. \quad (1.16)$$

Clearly $|z_1 + iz_2| < |z_1 - iz_2|$. Dividing the above inequality by $|z_1 - iz_2|^2$ yields

$$\sum_{j=1}^n |\eta_j|^2 < \frac{|z_1|^2 + |z_2|^2}{|z_1 - iz_2|^2} = \frac{1}{2} + \frac{2|z_1|^2 + 2|z_2|^2 - |z_1 - iz_2|^2}{2|z_1 - iz_2|^2} = \frac{1}{2} \left(1 + \left| \frac{z_1 + iz_2}{z_1 - iz_2} \right|^2 \right). \quad (1.17)$$

Let us put

$$\zeta_1 := \eta_1 + i\eta_2 \quad , \quad \zeta_2 := \eta_1 - i\eta_2 \quad , \quad \zeta_j := \eta_j \quad \text{for } j > 2.$$

Then combining (1.16) and (1.17) with the fact that $\zeta_1 \zeta_2 = \eta_1^2 + \eta_2^2$ yields

$$\|\zeta\|^2 < 1 + |\zeta_1 \zeta_2 + \zeta_3^2 + \dots + \zeta_n^2| < 2.$$

Hence the above map is well-defined. On the other hand let $(\zeta_1, \dots, \zeta_n) \in \mathcal{D}^{IV}$. It follows

$$|\zeta_1 \zeta_2 + \zeta_3^2 + \dots + \zeta_n^2| < 1.$$

One obtains from the choice of τ

$$\begin{aligned} \left(\frac{\tau}{\tau-i}\right)^2 + \left(\frac{1}{\tau-i}\right)^2 - \left(\frac{\zeta_1 + \zeta_2}{2}\right)^2 - \left(\frac{\zeta_1 - \zeta_2}{2i}\right)^2 - \zeta_3^2 - \dots - \zeta_n^2 &= 0 \\ \left|\frac{\tau}{\tau-i}\right|^2 + \left|\frac{1}{\tau-i}\right|^2 - \left|\frac{\zeta_1 + \zeta_2}{2}\right|^2 - \left|\frac{\zeta_1 - \zeta_2}{2i}\right|^2 - |\zeta_3|^2 - \dots - |\zeta_n|^2 &> 0 \quad . \end{aligned}$$

The last inequality follows from the boundedness condition defining $\mathcal{D}^{IV}$ combined with the fact

$$|\eta_1|^2 + |\eta_2|^2 = \frac{|\zeta_1|^2 + |\zeta_2|^2}{2}.$$

Moreover by construction the image of ζ belongs to the correct connected component. The fact that the above maps are holomorphic and inverse to each other is obvious. $\square$

Definition 1.7.3 Let X be a Hermitian symmetric space. The *Baily-Borel compactification* X^{BB} is the closure of X in the Harish-Chandra embedding.

We shall study the boundary of $(\mathcal{D}^{IV})^{BB} = (\mathcal{D}_n^{IV})^{BB}$. In our case the boundary can be described using the projective model

$$\mathcal{D}(L_2) \subseteq \mathcal{N}^0(L_2).$$

In this model the boundary is simply the closure of $\mathcal{D}(L_2)$ in the zero-quadric $\mathcal{N}^0(L_2)$ with respect to the topology induced by $\mathbb{P}(V(\mathbb{C}))$. Here the latter space is equipped with the quotient topology. We have already seen that $[\mathcal{Z}] = [\mathcal{X} + i\mathcal{Y}] \in \mathcal{D}(L_2)$ if and only if

$$(\mathcal{X}, \mathcal{Y}) = 0 \quad \text{and} \quad (\mathcal{X}, \mathcal{X}) = (\mathcal{Y}, \mathcal{Y}) > 0.$$

The *Grassmannian* of $V_2 = L_2 \otimes \mathbb{R}$ is the set of all two-dimensional subspaces $W \leq V_2$ such that $(W, (\cdot, \cdot)|_W)$ is positive definite. We denote this set by $\mathcal{G}$. Without proof we note that the map

$$\mathcal{D}(L_2) \rightarrow \mathcal{G} \quad , \quad [\mathcal{Z}] \mapsto \langle \mathcal{X}, \mathcal{Y} \rangle_{\mathbb{R}}$$

is a homeomorphism, confer Lemma 2.17 in [BrvdGHZ].

Zero-dimensional boundary components. Let $[c] \in \mathcal{N}^0(L_2)$ be a point where $c \in L_2 \otimes \mathbb{R}$ is isotropic. Let $d \in L_2 \otimes \mathbb{R}$ be another isotropic vector such that $\langle c, d \rangle$ is a hyperbolic plane. Now by definition $L_2 \otimes \mathbb{R}$ splits another hyperbolic plane $\langle c', d' \rangle$ such that $\langle c', d' \rangle \perp \langle c, d \rangle$. Then the sequence $\mathcal{X}_t := c + \frac{d}{t} + \frac{\sqrt{2}^{-1}}{\sqrt{t}}(c' + d')$ converges to c and we have $(\mathcal{X}_t, \mathcal{X}_t) > 0$ for all $t \in \mathbb{N}$. We define another sequence $\mathcal{Y}_t := c - \frac{\sqrt{2}^{-1}}{\sqrt{t}}(c' + d')$ such that $\mathcal{Y}_t$ converges to c , $(\mathcal{X}_t, \mathcal{Y}_t) = 0$ for all t and $(\mathcal{Y}_t, \mathcal{Y}_t) > 0$. This shows that $[c]$ is contained in the boundary of $\mathcal{D}(L)$.

One-dimensional boundary components. Now let $[\mathcal{X} + i\mathcal{Y}]$ be in the boundary of $\mathcal{D}(L_2)$. Then $(\mathcal{X}, \mathcal{X}) = (\mathcal{Y}, \mathcal{Y}) = 0$ and the space $\langle \mathcal{X}, \mathcal{Y} \rangle_{\mathbb{R}}$ has either dimension two or dimension one because the boundary is contained in $\mathcal{N}^0(L_2)$. If it has dimension

one then $[\mathcal{Z}] = [c]$ where $c \in L_2 \otimes \mathbb{R}$ is isotropic. In this case $[\mathcal{Z}]$ is a zero-dimensional boundary component. If $\langle \mathcal{X}, \mathcal{Y} \rangle_{\mathbb{R}}$ has dimension two then we can find a basis of V_2 of the shape

$$\mathbb{R}c \oplus \mathbb{R}c' \oplus (L \otimes \mathbb{R}) \oplus \mathbb{R}d' \oplus \mathbb{R}d$$

such that $\langle \mathcal{X}, \mathcal{Y} \rangle_{\mathbb{R}} = \langle c, c' \rangle_{\mathbb{R}}$ and $\langle c', d' \rangle, \langle c, d \rangle$ are perpendicular hyperbolic planes. After extending the coefficients to $\mathbb{C}$ we obtain in homogeneous coordinates $[\mathcal{Z}] = [1 : \tau : 0 : 0 : 0]$ with respect to this basis where $\text{Im}(\tau) \neq 0$. Since this point belongs to the boundary there has to be a sequence in $\mathcal{D}(L_2)$ that converges to it. According to Lemma 1.2.1 this sequence has to be of the shape $[\mathcal{Z}]_t = [-\omega\tau t + \frac{1}{2}(\mathfrak{z}, \mathfrak{z}) : t\tau : \mathfrak{z} : \omega : 1]$ where ω, τ belong to the upper half-plane. This sequence tends to $[1 : -\omega^{-1} : 0 : 0 : 0]$ showing that $[\mathcal{Z}]$ belongs to the boundary if and only if $[\mathcal{Z}] = [1 : \tau : 0 : 0 : 0]$ where $\tau \in \mathbb{H}$.

We summarize the description of the boundary components in the following Proposition.

Proposition 1.7.4 (a) *There is a bijective correspondence between the boundary components of $\mathcal{D}(L_2)$ and non-zero isotropic subspaces of V_2 .*

(b) *There is a holomorphic embedding of the upper half-plane $\mathbb{H}$ into the zero-quadric $\mathcal{N}^0(L_2)$ such that the image is exactly a one-dimensional boundary component. Moreover the embedding extends to a topological map of $\mathbb{H} \cup \mathbb{R} \cup \{\infty\}$ such that the image consists of all boundary points which belong to a distinguished isotropic subspace F .*

Proof

The only thing which remains to be shown is the existence of the forementioned extension in part (b). Suppose we have constructed an embedding of $\mathbb{H}$ with respect to the basis fixed above

$$\mathbb{H} \rightarrow F \otimes \mathbb{C} \quad , \quad \tau \mapsto [1 : \tau : 0 : 0 : 0]$$

where $F = \langle c, c' \rangle_{\mathbb{R}}$ as before. This map can be extended via

$$t \mapsto [1 : t : 0 : 0 : 0], \quad \text{if } t \in \mathbb{R} \quad , \quad \infty \mapsto [0 : 1 : 0 : 0 : 0]. \quad \square$$

Let $F, F' \leq V_2$ be two isotropic subspaces such that $F = \langle c, c' \rangle_{\mathbb{R}}, F' = \langle d, d' \rangle_{\mathbb{R}}$ and $\langle c, d \rangle, \langle c', d' \rangle$ are perpendicular hyperbolic planes. With a similar identification as in Lemma 1.5.7 we obtain an epimorphism

$$\Psi : \text{SL}(2, \mathbb{R}) \times \text{SL}(2, \mathbb{R}) \rightarrow \text{SO}(F \oplus F')^+ \quad , \quad (A, B) \mapsto (X \mapsto AXB^{\text{tr}})$$

with kernel $(\pm I_2, \pm I_2)$. We define

$$\mathbb{P}^1(\mathbb{C}) := \mathbb{C} \cup \{\infty\}.$$

For the special case $V_2 = F \oplus F'$ we can construct the following biholomorphic map

$$\begin{aligned} \mathbb{P}^1(\mathbb{C}) \times \mathbb{P}^1(\mathbb{C}) &\xrightarrow{\sim} \mathcal{N}^0(F \oplus F') \quad , \quad (z, w) \mapsto \begin{bmatrix} zw & z \\ w & 1 \end{bmatrix} \quad , \quad (\infty, \infty) \mapsto \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix} , \\ (z, \infty) &\mapsto \begin{bmatrix} z & 0 \\ 1 & 0 \end{bmatrix} \quad , \quad (\infty, w) \mapsto \begin{bmatrix} w & 1 \\ 0 & 0 \end{bmatrix} \end{aligned}$$

with the forementioned identification and obtain a commuting diagram

$$\begin{array}{ccc} \mathbb{P}^1(\mathbb{C}) \times \mathbb{P}^1(\mathbb{C}) & \xrightarrow{(A,B)} & \mathbb{P}^1(\mathbb{C}) \times \mathbb{P}^1(\mathbb{C}) \\ \downarrow \wr & & \downarrow \wr \\ \mathcal{N}^0(F \oplus F') & \xrightarrow{\Psi(A,B)} & \mathcal{N}^0(F \oplus F') \end{array} \quad \begin{array}{ccc} (z, w) & \longrightarrow & (A.z, A.w) \\ \downarrow & & \downarrow \\ [\mathcal{Z}] & \longrightarrow & \Psi(A, B).[\mathcal{Z}] \end{array}$$

where $A.z = \frac{az+b}{cz+d}$ is the usual fractional linear action of $\mathrm{SL}(2, \mathbb{R})$ on $\mathbb{P}^1(\mathbb{C})$. Note that in this case the zero-dimensional boundary components correspond to points in $(\mathbb{R} \cup \{\infty\}) \times (\mathbb{R} \cup \{\infty\})$ and the one-dimensional boundary components are of the shape $\{a\} \times \mathbb{H}$ or $\mathbb{H} \times \{a\}$ where $a \in \mathbb{R} \cup \{\infty\}$. Let $G_{\mathbb{R}} = \mathrm{SO}(V_2)^+$. Consider the normalizer of F which is defined as

$$N_{G_{\mathbb{R}}}(F) := \{g \in G_{\mathbb{R}} \mid g(F) = F\}.$$

Then $N_{G_{\mathbb{R}}}(F)$ acts on the boundary component F . Moreover $g(F \oplus F') = F \oplus F'$. Note that by Witt's Theorem 1.6.11 every element of $\mathrm{SO}(F \oplus F')^+$ extends to an element of $\mathrm{O}(V_2)$. One observes that in this case we even obtain an embedding $\mathrm{SO}(F \oplus F')^+ \hookrightarrow N_{G_{\mathbb{R}}}(F)$. For a one-dimensional boundary component the preimage of the stabilizer of F in $\mathrm{SO}(F \oplus F')^+$ under the above correspondence is the stabilizer of $\mathbb{H}$ under $\mathrm{SL}(2, \mathbb{R})/\{\pm I_2\}$ times the stabilizer of a point $a \in \mathbb{R} \cup \{\infty\}$ under $\mathrm{SL}(2, \mathbb{R})/\{\pm I_2\}$. By considering the projection onto the first component we obtain the following

Proposition 1.7.5 *Let $F \leq V_2$ be a two-dimensional isotropic subspace and choose F' as before. Then we have an epimorphism*

$$N_{G_{\mathbb{R}}}(F) \rightarrow \mathrm{SL}(2, \mathbb{R})/\{\pm I_2\}$$

such that the action of $N_{G_{\mathbb{R}}}(F)$ on the boundary component is compatible with the action of $\mathrm{SL}(2, \mathbb{R})$ on $\mathbb{H}$. The kernel of this action is

$$Z_{G_{\mathbb{R}}}(F) := \{g \in N_{G_{\mathbb{R}}}(F) \mid \forall x \in F : g(x) = x\}.$$

Let $F \leq V_2$ be an isotropic subspace representing a boundary component. The boundary component is called *rational* if

$$F = (F \cap (L_2 \otimes \mathbb{Q})) \otimes \mathbb{R}.$$

The rational boundary components are in one to one correspondence to the isotropic subspaces of $V_2(\mathbb{Q}) = L_2 \otimes \mathbb{Q}$. Let $\Gamma \leq \mathrm{O}(L_2 \otimes \mathbb{Q}) \cap \mathrm{O}(V_2)^+$ be an arithmetic subgroup. Then Γ acts on

$$\mathcal{D}(L_2)^* = \mathcal{D}(L_2) \amalg \coprod_{F \leq V_2(\mathbb{Q})} \mathcal{D}_F$$

where $\mathcal{D}_F$ denotes the rational boundary component corresponding to F . Hence we can consider the quotients

$$\Gamma \backslash \mathcal{D}(L_2) \quad , \quad \Gamma \backslash \mathcal{D}(L_2)^* .$$

The space $\Gamma \backslash \mathcal{D}(L_2)$ is called *modular variety of orthogonal type*. This is a noncompact space. The compactification is described in the next Theorem. For details we refer to [BorJ].

Theorem 1.7.6 *There exists a unique topology on $\mathcal{D}(L_2)^*$ such that the space*

$$\Gamma \backslash \mathcal{D}(L_2)^*$$

equipped with the quotient topology is compact. This topology is called the Satake topology. There is the following decomposition into boundary components

$$\Gamma \backslash \mathcal{D}(L_2)^* = \Gamma \backslash \mathcal{D}(L_2) \amalg \coprod_{\Pi} X_{\Pi} \amalg \coprod_l Q_l$$

where l runs through the finitely many Γ -orbits of isotropic lines and Π runs through the finitely many Γ -orbits of isotropic planes in $L_2 \otimes \mathbb{Q}$. Each X_{Π} is a modular curve and each Q_l is a point.

In this context the boundary components are also called cusps. At this point we can return to the terminology from the last section. We have already discussed zero-dimensional cusps which correspond exactly to zero-dimensional boundary components of $\mathcal{D}(L_2)$. Moreover Proposition 1.7.4 is also true for rational boundary components with obvious modifications. We fix a zero-dimensional rational boundary component which is represented by a primitive isotropic vector $c \in L_2$. Let $G \in \mathcal{M}_k(\Gamma, 1)$ be a modular form for a subgroup $\Gamma \leq \mathrm{O}(L_2)^+$ of finite index containing the group $\widetilde{\mathrm{SO}}(L_2)^+$. Let $Z \in \mathcal{H}_c(L_2)$ be arbitrary. The boundary component belonging to c is obtained as the limit of

$$\mathrm{pr}_c(tZ) = \left[-\frac{(tZ, tZ) + (h, h)}{2} : tZ : 1 \right] \xrightarrow{t \rightarrow \infty} [1 : 0 : 0]$$

in the Satake topology. There is some rational hyperbolic plane $U' = \langle d, d' \rangle \leq L_2 \otimes \mathbb{Q}$ and a negative definite subspace $K \leq L_2 \otimes \mathbb{Q}$ such that $Z \in \mathcal{H}_c(L_2)$ admits a decomposition

$$Z = \tau d + \mathfrak{z}_K + \omega d' .$$

The subspace $\langle c, d \rangle$ determines a one-dimensional rational boundary component which can be obtained as the Satake-limit of

$$\mathrm{pr}_c(t\tau d + \mathfrak{z}_K + \omega d') = \left[\frac{-2t\tau\omega + (\mathfrak{z}, \mathfrak{z}) - (h, h)}{2} : t\tau : \mathfrak{z}_K : \omega : 1 \right] \rightarrow [1 : -\omega^{-1} : 0 : 0 : 0] .$$

Definition 1.7.7 Let $F \leq L_2 \otimes \mathbb{Q}$ be a two-dimensional isotropic subspace representing a one-dimensional cusp. Let $G \in \mathcal{M}_k(\Gamma, 1)$ with the Fourier expansion at the 0-dimensional cusp $c \in F$

$$G_c(Z) = \sum_{\substack{m, n \in \mathbb{N}_{\geq 0} \\ l \in K^\vee \\ 2nm - (l, l) \geq 0}} g(m, l, n) \exp(2\pi i[n\tau - (l, \mathfrak{z}) + m\omega]) \quad (1.18)$$

for some negative definite even lattice $K \leq L_2$. The linear operator

$$\Phi_F : \mathcal{M}_k(\Gamma, 1) \rightarrow \mathcal{M}_k(\mathrm{SL}(2, \mathbb{Z})) \quad , \quad \Phi_F(G)(\omega) := \sum_{m=0}^{\infty} g(m, 0, 0) e^{2\pi i m \omega}$$

maps modular forms to elliptic modular forms for $\mathrm{SL}(2, \mathbb{Z})$ and is called *Siegel's phi-operator* with respect to the cusp F .

Remark 1.7.8 The modularity of $\Phi_F(G)$ is explained in Section 2.1.

Note that

$$\Phi_F(G)(\omega) = \lim_{t \rightarrow \infty} G(t\tau d + \mathfrak{z}_K + \omega d').$$

This leads us to the following characterization.

Theorem 1.7.9 *Under the same assumptions as before let $G \in \mathcal{M}_k(\Gamma, 1)$. We assume additionally that the lattice L_2 is maximal. For each 0-dimensional cusp c we define*

$$G_c(Z) = \sum_{l \in (L_2)_{c,h}^\vee} g_c(l) \exp(2\pi i(l, Z)) \quad , \quad Z \in \mathcal{H}_c(L_2).$$

The following statements are equivalent:

- (i) G is a cusp form.
- (ii) For every one-dimensional cusp represented by a two-dimensional isotropic subspace $F \leq L_2 \otimes \mathbb{Q}$ we have

$$\Phi_F(G) \equiv 0.$$

- (iii) For all 0-dimensional cusps c we have

$$g_c(l) \neq 0 \implies l \in C^+((L_2)_c).$$

Proof

We first like to mention that Siegel's operator is the correct tool to formulate the fact of being a cusp form. Thus (ii) is the formal definition for G being a cusp form. Hence it remains to show that (ii) is equivalent to (iii). We first show that (iii) implies (ii). Let $F \leq L_2 \otimes \mathbb{Q}$ be a one-dimensional cusp and let $c \in F$. By considering the coordinates introduced in (1.18) we see that (iii) implies (ii). Now assume that (ii) is satisfied. Let c

be an arbitrary 0-dimensional cusp. Since the lattice is maximal there are no non-trivial isotropic vectors in $D(L_2)$. Hence by Corollary 1.6.13 c is equivalent to the simplest cusp with respect to Γ and by virtue of Proposition 1.6.15 we can assume that $c = e$ and $(L_2)_{c,h}^\vee = L_1^\vee$. Now let $h = ne_1 + l + mf_1 \in L_1^\vee \cap \partial C^+((L_2)_e)$ such that $2nm - (l, l) = 0$. Since the lattice is maximal this implies $l \in L$ where $L \leq L_2$ is the defining positive definite sublattice. Again by using Corollary 1.6.13 we can find a $g \in E(L_2)$ such that $g(h) = \kappa e_1$ for some $\kappa \in \mathbb{N}$. The construction of g in the forementioned corollary implies $g(e) = e$ and $g(f) = w \in \langle f, e_1, L \rangle$. As in the proof of part (iii) in Proposition 1.5.8 we define $g' = t(e, -w)g \in E(L_2) \cap O(L_1)$. Then $g'(h) = t(e, -w)(e_1) = e_1$. Note that g' acts on $\mathcal{H}(L_1)$ as $g' \cdot Z = g' \cdot Z$. Since G is modular (ii) yields

$$g_c(m, l, n) = g_c(\kappa, 0, 0) = 0$$

which remained to be shown. $\square$

This considerations show that the zeroth coefficient $g_c(0, 0, 0)$ is connected to the boundary component defined by c and the one-dimensional boundary component F containing c is connected to the value Φ_F which is a modular form for the group $\mathrm{SL}(2, \mathbb{Z})$.

Remark 1.7.10 The characterization given in the last Theorem can be extended to lattices having the property that every isotropic subgroup in $D(L_2)$ is cyclic. A proof can be found in [GHS1], Theorem 4.2.

1.8 Positive definite lattices and root systems

In this section we recall some facts on positive definite lattices. In large parts of our exposition we follow [E]. For any natural number $N \in \mathbb{N}$ we consider the vector space $\mathbb{R}^N$. The standard basis of $\mathbb{R}^N$ is denoted by $\varepsilon_1, \dots, \varepsilon_N$. We equip $\mathbb{R}^N$ with the Euclidean scalar product $(\cdot, \cdot) = (\cdot, \cdot)_N$ which is defined by

$$(\varepsilon_j, \varepsilon_k) = \delta_{j,k}$$

where the right hand side denotes Kronecker's delta. The Euclidean scalar product defines a non-degenerate, positive definite bilinear form. For the rest of this section we will identify $\mathbb{R}^N$ with the positive definite quadratic space equipped with the Euclidean scalar product.

Let L be an even lattice in $\mathbb{R}^N$. An element $r \in L$ is called a *root* if $(r, r) = 2$. The set of all roots is given by

$$R_L = \{l \in L \mid (l, l) = 2\}.$$

We call L a *root lattice* if L is generated by R_L . For a proof of the next theorem we refer to [E], section 1.4.

Theorem 1.8.1 *Let L be a root lattice in $\mathbb{R}^N$. Then there exists a basis $b_1, \dots, b_N$ such that*

$$\begin{aligned} (b_j, b_j) &= 2, & 1 \leq j \leq N \\ (b_j, b_k) &\in \{0, -1\}, & 1 \leq j, k \leq N, j \neq k. \end{aligned}$$

The existence of such a basis is guaranteed if we can construct a so called fundamental system of roots. A subset $S \subseteq R_L$ is called a *fundamental system of roots* if

- (i) S is a basis of L
- (ii) Each $r \in R_L$ can be written as a linear combination $r = \sum_{s \in S} \alpha_s s$ with integral coefficients α_s , which are all non-positive or all non-negative

Proposition 1.8.2 *For every root lattice L there exists a fundamental system of roots S_L .*

A lattice L is called reducible if it can be written as the orthogonal direct sum $L = L_1 \perp L_2$ of two lattices L_1, L_2 . Otherwise L is called irreducible. According to theorem 1.8.1 every root lattice contains a basis of roots whose elements have scalar products with values in $\{0, 2, -1\}$. If such a basis, say $b_1, \dots, b_N$, is given we associate a graph to it. We represent each b_j by a vertex. Two vertices b_j and b_k are connected by an edge iff $(b_j, b_k) = -1$. This graph is called the *Coxeter diagram* corresponding to the basis $b_1, \dots, b_N$. We will construct several examples of irreducible root lattices which will turn out to be the fundamental components of every root lattice.

The root system A_N : We consider the standard basis of $\mathbb{R}^{N+1}$ and define the vector $e := \varepsilon_1 + \dots + \varepsilon_{N+1} \in \mathbb{Z}^{N+1}$. The set $e_{\mathbb{Z}^{N+1}}^\perp$ is a (non-degenerate) lattice in a subspace of $\mathbb{R}^{N+1}$ isomorphic to $\mathbb{R}^N$. We have

$$e_{\mathbb{Z}^{N+1}}^\perp = \{x \in \mathbb{Z}^{N+1} \mid x_1 + \dots + x_N = 0\}$$

and we see that the vectors $\varepsilon_j - \varepsilon_k$ where $j \neq k$ belong to $e_{\mathbb{Z}^{N+1}}^\perp$. On the other hand any $x \in e_{\mathbb{Z}^{N+1}}^\perp$ can be written as $x = x_1(\varepsilon_1 - \varepsilon_{N+1}) + \dots + x_N(\varepsilon_N - \varepsilon_{N+1})$ and we immediately see that $\varepsilon_2 - \varepsilon_1, \dots, \varepsilon_{N+1} - \varepsilon_N$ form a basis of this lattice which one calls A_N . The Coxeter diagram with respect to this basis is given below. Using induction one computes $d(A_N) = N + 1$. The set of roots is

$$R_{A_N} = \{\varepsilon_j - \varepsilon_k \mid 1 \leq j, k \leq N + 1, j \neq k\}$$

and is of order $N(N + 1)$. We want to mention that A_1 is isomorphic to the lattice $\mathbb{Z}$ equipped with the bilinear form $(x, x) = 2x^2$. For this reason we will often write $< 2 > = < 2 >_{\mathbb{Z}}$ for this lattice and more general we will write $< t >$ for the lattice $\mathbb{Z}$ equipped with the quadratic form $(x, x) = tx^2$, $t \in \mathbb{Z}$.



Figure 1.1: Coxeter diagram of A_N

The root system D_N : We consider a subset of $\mathbb{Z}^N$ given by

$$D_N = \{(x_1, \dots, x_N) \in \mathbb{Z}^N \mid x_1 + \dots + x_N \equiv 0 \pmod{2}\}$$

where $N \in \mathbb{N}$. Then D_N is a sublattice of $\mathbb{Z}^N$ of index 2. Write $x = \sum_j x_j \varepsilon_j$. For $N = 1$ we have $D_1 \cong \langle 4 \rangle$ and one easily sees that $D_2 \cong A_1 \perp A_1 =: 2A_1$. For $N \geq 3$ we note that x can be written in the form

$$\begin{aligned} \frac{x_1 + \dots + x_N}{2}(\varepsilon_2 + \varepsilon_1) &+ \frac{-x_1 + \dots + x_N}{2}(\varepsilon_2 - \varepsilon_1) \\ &+ (x_3 + \dots + x_N)(\varepsilon_3 - \varepsilon_2) + \dots + x_N(\varepsilon_N - \varepsilon_{N-1}). \end{aligned}$$

This shows that $\varepsilon_2 + \varepsilon_1, \varepsilon_2 - \varepsilon_1, \varepsilon_3 - \varepsilon_2, \dots, \varepsilon_N - \varepsilon_{N-1}$ forms a basis of D_N . The set of roots is given by

$$R_{D_N} = \{\pm \varepsilon_j \pm \varepsilon_k, \pm \varepsilon_j \mp \varepsilon_k \mid 1 \leq j, k \leq N, j \neq k\}$$

which is of order $2N(N-1)$. Since D_N is of index 2 in $\mathbb{Z}^N$ formula (1.2) gives $d(D_N) = 4$. The corresponding Coxeter diagram is given below.

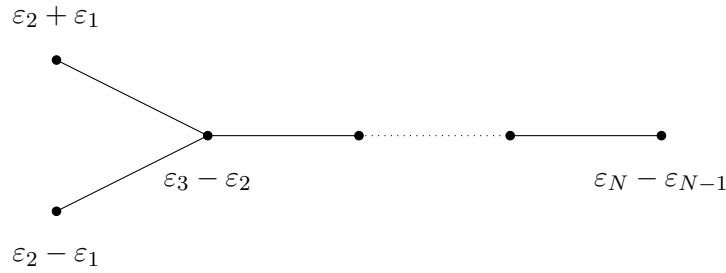


Figure 1.2: Coxeter diagram of D_N

The root system E_8 : There are various ways to define the root lattice E_8 . We give the following description. Let us define

$$E_8 = \{(x_1, \dots, x_8) \in \mathbb{Z}^8/2 \mid x_1 = \dots = x_8 \pmod{1}, x_1 + \dots + x_8 = 2x_8 \pmod{2}\}$$

Then E_8 is a lattice. The condition $x_1 = \dots = x_8$ implies that E_8 is an integral lattice. Since E_8 contains D_8 as a sublattice we have $\text{rank}(E_8) = 8$. Thus it contains the 7 roots $\varepsilon_2 - \varepsilon_1, \dots, \varepsilon_8 - \varepsilon_7$. We add the vector $v := \frac{1}{2}(\varepsilon_1 + \varepsilon_2 + \varepsilon_3 + \varepsilon_4 + \varepsilon_5 - \varepsilon_6 - \varepsilon_7 - \varepsilon_8) \in E_8$ to this sequence which is again a root. One immediately checks that these 8 vectors are a root system and we can draw its Coxeter diagram.

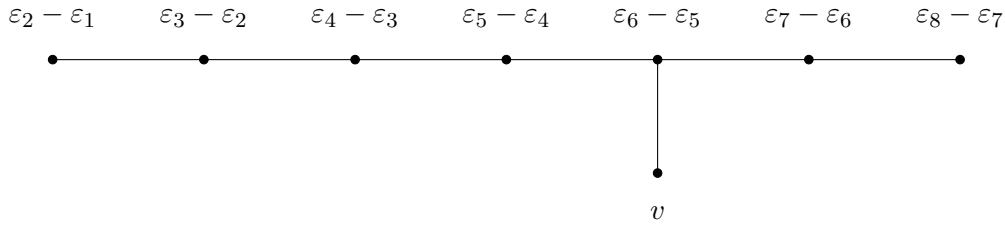


Figure 1.3: Coxeter diagram of E_8

Thus the 8 roots span a sublattice L of E_8 of rank 8. One verifies that $d(L) = 1$. From formula (1.2) we conclude that E_8 is unimodular itself and coincides with this sublattice which is a root lattice. Moreover we have $E_8 = D_8 \cup (v + D_8)$ since $4 = d(D_8) = [E_8 : D_8]^2$. The lattice E_8 contains exactly 240 roots. For a more careful treatment of this important lattice we refer to [CS].

The root system E_7 : For convenience we rename the above basis of E_8 by $e_j = \varepsilon_{j+1} - \varepsilon_j$ for $j = 1, \dots, 7$ and we put $e_8 = v$. Let us define $w := 2e_1 + 3e_2 + 4e_3 + 5e_4 + 6e_5 + 4e_6 + 2e_7 + 3e_8$ and define

$$E_7 = w^\perp_{E_8}.$$

Then $e_2, \dots, e_8$ are perpendicular to w and hence contained in E_7 . Since $(w, e_1) = 1$ we see that they already form a basis of E_7 . An immediate calculation shows $d(E_7) = 2$.

The root system E_6 , According to the last example we define w as above and put $w' = e_1$. We define

$$E_6 = \langle w, w' \rangle^\perp_{E_8}.$$

We immediately see that $e_3, \dots, e_8$ is contained in E_6 and using the fact

$$(e_1, w) = 1, (e_1, w') = 2, (e_2, w') = -1$$

we conclude that $e_3, \dots, e_8$ is in fact a basis of E_6 . One has $d(E_6) = 3$.

A proof of the next Theorem can be found in [E], Theorem 1.2.

Theorem 1.8.3 *Every root lattice has a representation as an orthogonal direct sum of the lattices A_N, D_N, E_6, E_7, E_8 .*

The next theorem can again be found in [E].

Theorem 1.8.4 *Let L be an irreducible root lattice of rank N . The following statements are equivalent:*

(i) *L contains N pairwise orthogonal roots*

(ii) *$NA_1 \subseteq L$*

(iii) *$2L^\vee \subseteq L$*

(iv) *L is of the type A_1, D_N where $N \geq 2$, even, E_7 or E_8 .*

Now let L be an irreducible root lattice. If we fix a fundamental system of roots for L we can define a partial ordering on R_L . Relative to this ordering there exists a unique maximal root. For details we refer to [E]. We call this root β_L . At the moment we only need the following property: Let $e_1, \dots, e_N$ be a fundamental system of roots of L and put $e_{N+1} := -\beta_L$. Then we have

$$(e_{N+1}, e_j) \in \{0, -1\} \text{ for all } 1 \leq j \leq N$$

except for the root systems A_1, D_1, D_2 . We can draw a graph of the so extended basis $e_1, \dots, e_N, e_{N+1}$ where the e_j form the edges. As before two distinct edges are connected by a vertex iff their scalar product equals -1. This diagram is called *extended Coxeter diagram*. In the figure below we give the extended Coxeter diagrams for the irreducible root lattices with respect to the standard root systems defined above.

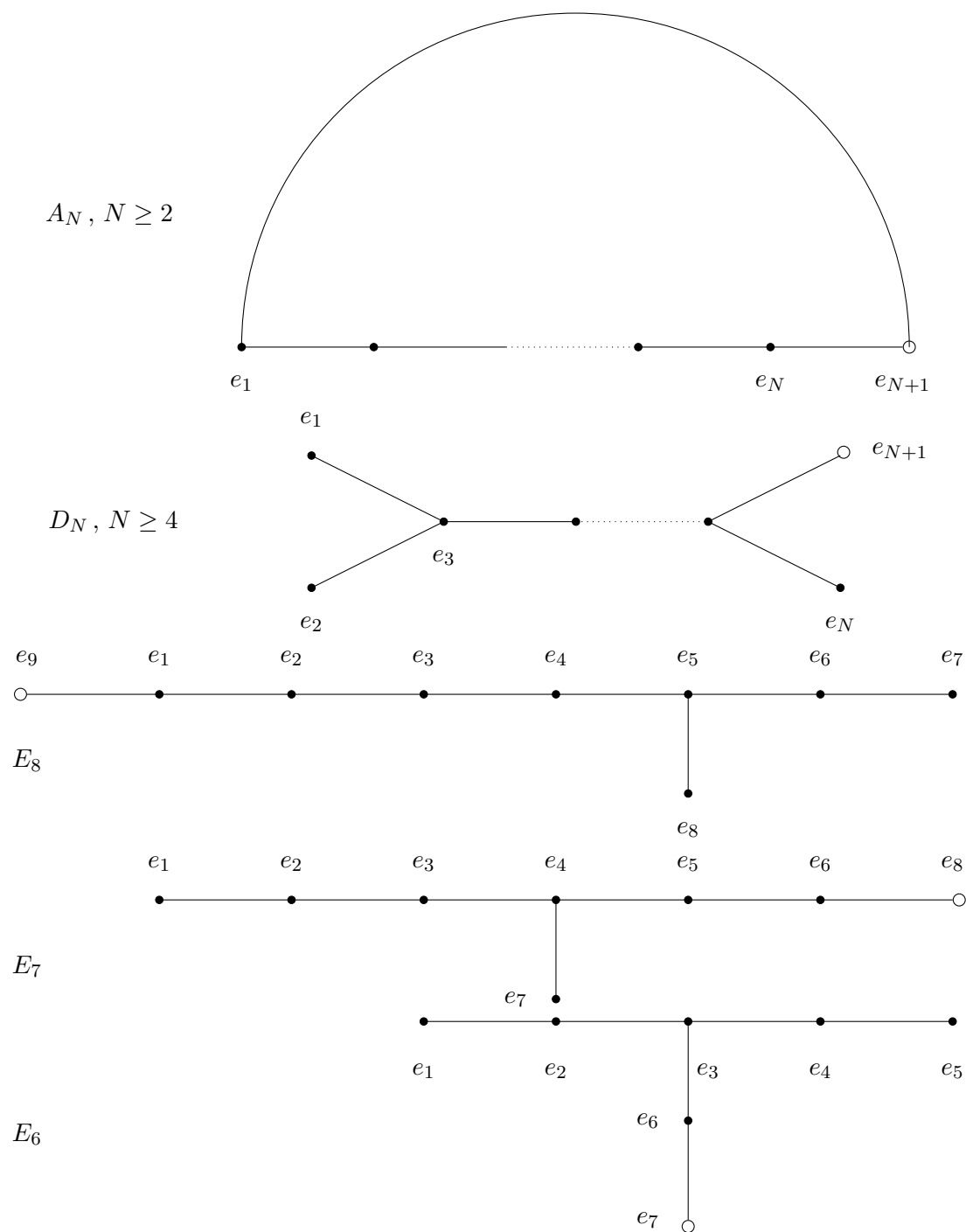


Figure 1.4: Extended Coxeter diagrams of irreducible root lattices

1.9 Automorphism groups of root lattices

Now we consider the orthogonal group $O(L)$ of a positive definite even lattice. We will mainly concentrate on root lattices. It is well known that the orthogonal group of a positive definite lattice is finite. In particular, if L is a root lattice any $g \in O(L)$ is uniquely determined by the images of a basis of roots. Since the number of roots is finite we have only finitely many possibilities for the choice of the images. Another important subgroup is the *Weyl group* of a root lattice L which is defined as

$$W(L) = \{\sigma_r \mid r \in R_L\}$$

where σ_r denotes the reflection at the hyperplane $(r)^\perp$. For a proof of the next Proposition we refer to [E], section 1.

Proposition 1.9.1 *Let L be an irreducible root lattice.*

- (i) *If $U \subseteq \mathbb{R}^N$ is a subspace which is invariant under $W(L)$ then either $U = \{0\}$ or $U = \mathbb{R}^N$.*
- (ii) *The Weyl group $W(L)$ acts transitively on R_L .*

We remind to the definition of the stable orthogonal group which was defined as

$$\tilde{O}(L) = \ker(O(L) \rightarrow O(D(L))).$$

By definition we have $W(L) \leq \tilde{O}(L)$.

The root lattice D_N , $N \geq 2$ The dual lattice of D_N is given by

$$D_N^\vee = \{x \in D_N \otimes \mathbb{Q} \mid \forall l \in D_N : (x, l) \in \mathbb{Z}\}.$$

We know $d(D_N) = 4$. We consider $v_N := \frac{1}{2}(\varepsilon_1 + \cdots + \varepsilon_N) \in D_N^\vee$. We have $2v_N \in D_N \iff N \in 2\mathbb{Z}$. Hence $D_N^\vee$ is generated by $v_N + D_N$ if and only if N is odd. In this case $D(D_N) \cong \mathbb{Z}/4\mathbb{Z}$.

Now let N be even. We consider $\varepsilon_1 \in D_N^\vee \setminus D_N$. Both $v_N + D_N$ and $\varepsilon_1 + D_N$ have order 2 in $D(D_N)$. Moreover $\varepsilon_1 - v_N \notin D_N$. Thus we have shown

$$D(D_N) = \begin{cases} \langle v_N + D_N, \varepsilon_1 + D_N \rangle \cong \mathcal{C}_2 \times \mathcal{C}_2 & \text{if } N \in 2\mathbb{Z} \\ \langle v_N + D_N \rangle \cong \mathcal{C}_4 & \text{if } N \notin 2\mathbb{Z} \end{cases}$$

and

$$D_N^\vee = D_N \cup (\varepsilon_1 + D_N) \cup (v_N + D_N) \cup (\varepsilon_1 + v_N + D_N).$$

We want to investigate the orthogonal group $O(D(D_N))$. Let $N \in 2\mathbb{Z}$. In this case $D(D_N) = \langle \mathfrak{e}, \mathfrak{v} \rangle := \langle \varepsilon_1 + D_N, v_N + D_N \rangle$. We have

$$\text{Aut}(D(D_N)) = \left\langle \begin{array}{cc} \mathfrak{e} \mapsto \mathfrak{v} & \mathfrak{e} \mapsto \mathfrak{v} \\ \mathfrak{v} \mapsto \mathfrak{e} & \mathfrak{v} \mapsto \mathfrak{e} + \mathfrak{v} \end{array} \right\rangle \cong \mathcal{S}_3.$$

We compute the images of the representatives of $D_N^\vee/D_N$ under the discriminant form $q = q_{D(D_N)}$

$$q(0) = 0 + 2\mathbb{Z} \quad , \quad q(\mathfrak{e}) = 1 + 2\mathbb{Z} \quad , \quad q(\mathfrak{v}) = \frac{N}{4} + 2\mathbb{Z} \quad , \quad q(\mathfrak{e} + \mathfrak{v}) = \frac{N}{4} + 2\mathbb{Z}.$$

If $N \equiv 4 \pmod{8}$ we immediately see that $\text{Aut}(D(D_N)) = \text{O}(D(D_N))$. If $N \not\equiv 4 \pmod{8}$ then every element of $\text{O}(D(D_N))$ fixes $\mathfrak{e}$. Hence

$$\text{O}(D(D_N)) = \left\langle \begin{array}{cc} \mathfrak{e} + \mathfrak{v} & \mapsto \mathfrak{v} \\ \mathfrak{v} & \mapsto \mathfrak{e} + \mathfrak{v} \end{array} \right\rangle \cong \mathcal{C}_2 \quad , \quad \text{if } N \not\equiv 4 \pmod{8}, N \in 2\mathbb{Z}.$$

Now let N be odd. Then $D(D_N) = \langle \mathfrak{v} \rangle$ and $\text{Aut}(D(D_N)) = \langle \mathfrak{v} \mapsto -\mathfrak{v} \rangle \cong \mathcal{C}_2$. Thus $\text{Aut}(D(D_N))$ acts on $D_N^\vee$ by permutation of the classes $\mathfrak{v}$ and $\mathfrak{e} + \mathfrak{v}$ and we obtain

$$\text{O}(D(D_N)) = \langle \mathfrak{v} \mapsto -\mathfrak{v} \rangle \cong \mathcal{C}_2 \quad , \quad \text{if } N \equiv 1 \pmod{2}.$$

Summarizing this considerations we obtain

$$\text{O}(D(D_N)) \cong \begin{cases} \mathcal{S}_3 & \text{if } N \equiv 4 \pmod{8} \\ \mathcal{C}_2 & \text{if } N \not\equiv 4 \pmod{8} \end{cases} \quad (1.19)$$

What about the surjectivity of the projection morphism $\text{O}(L) \rightarrow \text{O}(D(D_N))$? The permutations $\mathfrak{v} \mapsto -\mathfrak{v}$ or $\mathfrak{v} \mapsto \mathfrak{e} + \mathfrak{v}$, if N is even, can be realized by the reflection σ_{ε_1} since

$$\sigma_{\varepsilon_1}(\varepsilon_1) = -\varepsilon_1 \in \mathfrak{e} \quad \text{and} \quad \sigma_{\varepsilon_1}(v_N) = v_N - \frac{2(\varepsilon_1, v_N)}{(\varepsilon_1, \varepsilon_1)} \varepsilon_1 = v_N - \varepsilon_1 \in \mathfrak{v} + \mathfrak{e}.$$

In the case where $N = 4$ we meet a particular constellation because $(\varepsilon_1, \varepsilon_1) = (v_4, v_4) = (v_4 - \varepsilon_1, v_4 - \varepsilon_1) = 1$. Besides σ_{ε_1} we have the reflection $\sigma_{v_4} \in \text{O}(D_4)$ with the properties

$$\sigma_{v_4}(\varepsilon_1) = \varepsilon_1 - (\varepsilon_1, 2v_4)v_4 \in \mathfrak{e} + \mathfrak{v} \quad \text{and} \quad \sigma_{v_4}(v_4) = -v_4 \in \mathfrak{v}$$

and $\sigma_{\varepsilon_1}, \sigma_{v_4}$ generate $\text{O}(D(D_4)) \cong \mathcal{S}_3$. Now let $N \equiv 4 \pmod{8}$ and $N \neq 4$. Assume there exists a $g \in \text{O}(D_N)$ such that $g(\varepsilon_1) \in \mathfrak{v}$. Then there exists an $h = \sum_{j=1}^N h_j \varepsilon_j \in D_N$ such that

$$1 = (g(\varepsilon_1), \varepsilon_1) = (v_N + h, v_N + h) = \frac{N}{4} + \sum_j h_j(h_j + 1)$$

Since $h_j \in \mathbb{Z}$ the last sum is always greater or equal to zero which yields a contradiction because $N \neq 4$. Hence such a g cannot exist. The same argument yields that there is no orthogonal transformation in D_N realizing the permutation of $\mathfrak{e}$ and $\mathfrak{v} + \mathfrak{e}$. Hence we obtain the surjectivity of $\text{O}(D_N) \rightarrow \text{O}(D(D_N))$ if and only if $N \not\equiv 4 \pmod{8}$ or $N = 4$.

The root lattice NA_1 , $N \in \mathbb{N}$ The lattice NA_1 can be identified with $L = \mathbb{Z}^N(2)$. The dual lattice of L is given by

$$L^\vee := \{x \in L \otimes \mathbb{Q} : 2x \in L\}.$$

Therefore we have $d(NA_1) = 2^N$ and the discriminant group is given by

$$D(L) \cong \{(a_1, \dots, a_N) \in \mathcal{C}_2^N\}$$

Thus we can regard $D(L)$ as the vector space $\mathbb{F}_2^N$. We want to determine the finite orthogonal group $O(D(L))$. We set $q = q_{D(L)}$. Let $\alpha_1, \dots, \alpha_N$ be a basis of $D(L)$ considered as a vector space over $\mathbb{F}_2$ such that $q(\alpha_j) = \frac{1}{2} + 2\mathbb{Z}$. Let $g \in \text{Aut}(L)$. Then $g \in O(D(L))$ if and only if $q(g(\alpha_j)) = q(\alpha_j)$ for $j = 1, \dots, N$. This immediately yields $O(D(L)) \cong \mathcal{S}_N$ and $O(D(L))$ acts by permutation of the generators α_j .

The root lattice E_7 In the last section we constructed this lattice as the orthogonal complement in E_8 of a certain vector w_{E_8} and a basis was given by $e_2, \dots, e_8$. We put $v_{E_7} := \frac{1}{4}(3\varepsilon_1 + \varepsilon_2 + \varepsilon_3 + \varepsilon_4 + \varepsilon_5 + \varepsilon_6 + \varepsilon_7 - 3\varepsilon_1)$. Then $v_{E_7} \in E_7^\vee$, $2v_{E_7} \in E_8$ and $v_{E_7} \perp w$. Hence $E_7^\vee = E_7 \cup v_{E_7} + E_7$ and $q_{D(E_7)}(v_{E_7}) = \frac{3}{2} + 2\mathbb{Z}$. This shows that $O(D(E_7)) = \{\text{id}\}$. In particular the canonical homomorphism $O(E_7) \rightarrow O(D(E_7))$ is surjective.

The root lattice E_6 By definition $E_6 = \langle w, w' \rangle_{E_8}^\perp$. We put $v_{E_6} = \frac{1}{3}(-2\varepsilon_3 - 2\varepsilon_4 + \varepsilon_5 + \varepsilon_6 + \varepsilon_7 + \varepsilon_8)$. One immediately verifies that $v_{E_6} \in E_6^\vee$ and the order in $D(E_6)$ is three. Moreover $-v_{E_6} = 2v_{E_6} + 2(\varepsilon_3 + \varepsilon_4 + \varepsilon_5 + \varepsilon_6 + \varepsilon_7 + \varepsilon_8)$ and the latter element belongs to E_6 . Hence we obtain

$$E_6^\vee = E_6 \cup v_{E_6} + E_6 \cup -v_{E_6} + E_6.$$

The two non-trivial classes both take the value $\frac{4}{3} + 2\mathbb{Z}$ under the discriminant form $q_{D(E_6)}$. Hence the orthogonal group of the discriminant group is isomorphic to $\mathcal{C}_2$ and consists of the permutation interchanging v_{E_6} and $-v_{E_6}$. This transformation is induced by the image of $-\text{id} \in O(E_6)$.

The lattice $D_N(2)$ The rescaled lattice $D_N(2)$ is the lattice with the underlying set of D_N and bilinear form $(\cdot, \cdot)_{D_N(2)} = 2(\cdot, \cdot)_{D_N}$. We obviously have $d(D_N(2)) = 2^N d(D_N) = 2^{N+2}$. For abbreviation we set

$$b_1 := \varepsilon_2 + \varepsilon_1, b_2 := \varepsilon_2 - \varepsilon_1, b_3 := \varepsilon_3 - \varepsilon_2, \dots, b_N := \varepsilon_N - \varepsilon_{N-1}.$$

From the identity $D_N(2)^\vee = \frac{1}{2}D_N^\vee$ we obtain the disjoint union

$$D_N(2)^\vee = \frac{1}{2}D_N(2) \cup \frac{\varepsilon_1}{2} + \frac{1}{2}D_N(2) \cup \frac{v_N}{2} + \frac{1}{2}D_N(2) \cup \frac{\varepsilon_1 + v_N}{2} + \frac{1}{2}D_N(2).$$

Moreover one has

$$\frac{1}{2}D_N(2) = \bigcup_{\substack{j=1 \\ t_j \in \{1,2\}}}^N \sum \left(\frac{t_j}{2}b_j + D_N(2) \right)$$

which gives us a decomposition of $D_N(2)^\vee$ into $d(D_N(2))$ classes. The elementary divisors of the Gramian matrix of $D_N(2)$ are

$$\begin{cases} [2(N-2\text{-times}), 4, 4] & , \text{ if } N \equiv 0 \pmod{2} \\ [2(N-1\text{-times}), 8] & , \text{ if } N \equiv 1 \pmod{2}. \end{cases}$$

Let $N \geq 2$ be even. The class $\varepsilon_1 + D_N(2)$ contains all of the standard basis vectors ε_j of $\mathbb{Z}^N$. Moreover the elements $\frac{\varepsilon_j}{2} + D_N(2)$ are contained in the subgroup generated by the classes $\frac{b_2}{2} + D_N(2), \dots, \frac{b_{N-1}}{2} + D_N(2), \frac{\varepsilon_1}{2} + D_N(2), v_N + D_N(2)$. This shows that the discriminant group $D(D_N(2))$ is generated by the N classes

$$\frac{b_2}{2} + D_N(2), \dots, \frac{b_{N-1}}{2} + D_N(2), \frac{\varepsilon_1}{2} + D_N(2), \frac{v_N}{2} + D_N(2)$$

and is isomorphic to $\mathcal{C}_2^{N-2} \times \mathcal{C}_4^2$. Now let $N \geq 3$ be odd. The class $\frac{v_N}{2} + D_N(2)$ is of order 8 in the discriminant group. The vectors ε_j are contained in the class $2v_N + D_N(2)$ and therefore the vectors $\frac{\varepsilon_2 + \varepsilon_3}{2}, \dots, \frac{\varepsilon_{N-1} + \varepsilon_N}{2}$ are contained in the subgroup generated by

$$v + D_N(2), \frac{b_2}{2} + D_N(2), \dots, \frac{b_N}{2} + D_N(2)$$

and consequently the vector $\frac{\varepsilon_1}{2}$, too. This shows that

$$\frac{v_N}{2} + D_N(2), \frac{b_2}{2} + D_N(2), \dots, \frac{b_N}{2} + D_N(2)$$

already generate $D(D_N(2))$ and this group is isomorphic to $\mathcal{C}_2^{N-1} \times \mathcal{C}_8$.

We consider again the lattice D_4 . It is known that $\sharp R_{D_4} = 24$ and $R_{D_4}^4 = 24$ where for a positive integer n we define

$$R_L^{2n} := \{r \in L \mid (r, r) = 2n\}.$$

The sequence $(\sharp R_L^{2n})_{n \in \mathbb{N}_0}$ contains exactly the Fourier coefficients of the theta series associated to the positive definite even lattice L .

Lemma 1.9.2 *Let $r \in D_4$. Then $r \in R_{D_4}^4$ if and only if $r/2 \in D_4^\vee \setminus D_4$.*

Proof

If $r/2 \in D_4^\vee \setminus D_4$ then $r \in R_{D_4}^4$ because the order of any element in $D(D_4)$ is at most two. If conversely $r \in R_{D_4}^4$ write $r = al$ for some $l \in D_4$. Then we obtain

$$4 = (r, r) \in 2a^2\mathbb{Z}$$

which implies $a = \pm 1$. Hence r is primitive in D_4 . Now we can apply (1.11) and obtain $2 \mid \text{div}(r)$. This yields $r/2 \in D_4^\vee \setminus D_4$. $\square$

We define the generalized root system of D_4 as

$$F_4 := R_{D_4}^2 \cup R_{D_4}^4 \quad (1.20)$$

The next Corollary shows that $O(D_4)$ is generated by F_4 -reflections.

Corollary 1.9.3 *We have $O(D_4) = \langle \sigma_r \mid r \in F_4 \rangle$ and the Weyl group $W(D_4)$ coincides with the stable orthogonal group $\tilde{O}(D_4)$.*

Proof

One can compute the cardinality of the finite group as $\#O(D_4) = 1152$ as well as $\#W(D_4) = 192$. We have shown above that $O(D(D_4)) \cong \mathcal{S}_3$ and proved the surjectivity of $O(D_4) \rightarrow O(D(D_4))$. Moreover the image of this morphism is generated by σ_{ε_1} and σ_{v_4} where $\varepsilon_1, v_4 \in D_4^\vee$. If $r \in R_{D_4}^4$ Lemma 1.9.2 yields for $l \in D_4$

$$\sigma_r(l) = l - \frac{2(r, l)}{(r, r)}r = l - (r/2, l)r \in D_4.$$

Now the claim follows from $\sigma_{ar} = \sigma_r$ for $a \in \mathbb{Z}$. $\square$

Chapter 2

Jacobi forms

2.1 Jacobi group and Jacobi forms

In this section we introduce Jacobi forms. The presentation of the theory is influenced by the classical book on Jacobi forms in one variable of Eichler and Zagier [EZ]. For the description of Jacobi forms in several variables we follow [G], [CG]. Let L be a positive definite lattice of rank N with bilinear form $(\cdot, \cdot)_L = (\cdot, \cdot)$. By an *integral hyperbolic plane* we mean a lattice U in a quadratic space H of signature $(1, 1)$ which is generated by two isotropic elements e, f such that

$$(e, e)_H = (f, f)_H = 0 \text{ and } (e, f)_H = 1.$$

Let $U_1 = \mathbb{Z}e_1 + \mathbb{Z}f_1$ be another hyperbolic plane. As before we define the lattice L_2 as the orthogonal direct sum of the hyperbolic planes U, U_1 and the negative definite lattice $L(-1)$

$$L_2 = U \perp U_1 \perp L(-1)$$

Then L_2 is an even lattice in $\mathbb{R}^{2, N+2}$. We use the basis (e, e_1, L, f_1, f) such that the Gramian of L_2 reads

$$\begin{pmatrix} 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & -S & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 & 0 \end{pmatrix}$$

where S is the Gramian of the positive definite lattice L . Let F be the totally isotropic plane spanned by e and e_1 . We define the parabolic subgroup of $\mathrm{SO}(L_2)^+$ fixing F as

$$N_G(F) = \{g \in \mathrm{SO}(L_2)^+ \mid gF = F\} \quad , \quad G = \mathrm{SO}(L_2)^+.$$

We can give a matrix realization of $N_G(F)$ with respect to our standard basis of L_2 . A direct calculation leads us to the following Lemma, compare [G].

Lemma 2.1.1 *The group $N_G(F)$ consists of all elements of the shape*

$$\begin{pmatrix} A^* & X_1 & T \\ 0 & U & X \\ 0 & 0 & A \end{pmatrix}, \quad A = \begin{pmatrix} \alpha & \beta \\ \gamma & \delta \end{pmatrix} \in \mathrm{SL}(2, \mathbb{Z})$$

where X_1, T, U, X denote matrices with integral entries satisfying

$$(i) \quad A^* = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} (A^{-1})^{\mathrm{tr}} \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} = \begin{pmatrix} \alpha & -\beta \\ -\gamma & \delta \end{pmatrix},$$

$$(ii) \quad U^{\mathrm{tr}} S U = S, \text{ this means } U \in \mathrm{O}(L),$$

$$(iii) \quad X_1 = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} (A^{-1})^{\mathrm{tr}} X^{\mathrm{tr}} S U,$$

$$(iv) \quad X^{\mathrm{tr}} S X = A^{\mathrm{tr}} \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} T + T^{\mathrm{tr}} \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} A.$$

The *integral Heisenberg group* $H(L)$ of the lattice L is generated by the elements

$$[x, y : r] := \begin{pmatrix} 1 & 0 & -y^{\mathrm{tr}} S & -\frac{1}{2}(x, y) - r & -\frac{1}{2}(y, y) \\ 0 & 1 & -x^{\mathrm{tr}} S & -\frac{1}{2}(x, x) & -\frac{1}{2}(x, y) + r \\ 0 & 0 & I_N & x & y \\ 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 1 \end{pmatrix}$$

where $x, y \in L$ and $r \in \mathbb{Z}/2$ such that $r + \frac{1}{2}(x, y) \in \mathbb{Z}$. The group law of $H(L)$ is

$$[x, y : r] \cdot [x', y' : r'] = [x + x', y + y', r + r' + \frac{1}{2}[(x, y') - (x', y)]] .$$

We define an embedding of $\mathrm{SL}(2, \mathbb{Z})$ into $\mathrm{SO}(L_2)^+$ as

$$\{A\} := \begin{pmatrix} A^* & 0 & 0 \\ 0 & I_N & 0 \\ 0 & 0 & A \end{pmatrix} \quad \text{where } A \in \mathrm{SL}(2, \mathbb{Z}), \quad A^* = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} (A^{-1})^{\mathrm{tr}} \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$$

The group $\mathrm{SL}(2, \mathbb{Z})$ acts on the Heisenberg group by conjugation

$$A.[x, y : r] = \{A\}[x, y : r]\{A\}^{-1} = [dx - cy, ay - bx : r] .$$

Note that $\frac{1}{2}(dx - cy, ay - bx) = \frac{1}{2}(x, y) \bmod \mathbb{Z}$ since $A \in \mathrm{SL}(2, \mathbb{Z})$. The *integral Jacobi group* $\Gamma^J(L)$ is defined as the subgroup of $N_G(F)$ which acts trivially on the sublattice L . In the following Lemma we give a realization of the integral Jacobi group.

Lemma 2.1.2 *The group $\Gamma^J(L)$ is isomorphic to the semi-direct product*

$$\mathrm{SL}(2, \mathbb{Z}) \ltimes H(L) .$$

Proof

Let M be the matrix representation of an element of $\Gamma^J(L)$. From Lemma 2.1.1 we deduce that

$$M = \begin{pmatrix} A^* & X_1 & T \\ 0 & I_N & X \\ 0 & 0 & A \end{pmatrix}.$$

In a first step we show that the normal subgroup which is generated by all elements of the shape

$$h = \begin{pmatrix} I_2 & X_1 & T \\ 0 & I_N & X \\ 0 & 0 & I_2 \end{pmatrix}$$

is isomorphic to $H(L)$. We write $X = (x, y)$ with $x, y \in L$. Then Lemma 2.1.1 yields

$$X_1 = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \begin{pmatrix} x^{\text{tr}} \\ y^{\text{tr}} \end{pmatrix} S = \begin{pmatrix} y^{\text{tr}} S \\ x^{\text{tr}} S \end{pmatrix}.$$

The matrix T is determined by the equation

$$\begin{pmatrix} x^{\text{tr}} S x & x^{\text{tr}} S y \\ y^{\text{tr}} S x & y^{\text{tr}} S y \end{pmatrix} = X^{\text{tr}} S X = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} T + T^{\text{tr}} \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}.$$

This yields

$$T = \begin{pmatrix} \alpha & \frac{1}{2}(y, y) \\ \frac{1}{2}(x, x) & \beta \end{pmatrix}$$

where $\alpha, \beta \in \mathbb{Z}$ such that $\alpha + \beta = (x, y)$. By choosing $r \in \mathbb{Z}/2$ such that $\beta = r + \frac{1}{2}(x, y) \in \mathbb{Z}$ we fix a particular realization of h . Thus we obtain

$$h = \begin{pmatrix} 1 & 0 & y^{\text{tr}} S & \frac{1}{2}(x, y) - r & \frac{1}{2}(y, y) \\ 0 & 1 & x^{\text{tr}} S & \frac{1}{2}(x, x) & \frac{1}{2}(x, y) + r \\ 0 & 0 & I_N & x & y \\ 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 1 \end{pmatrix}.$$

Then one easily sees that the assignment $h \mapsto [x, y : r]$ defines a group isomorphism with $H(L)$. We have already seen that the group $\text{SL}(2, \mathbb{Z})$ acts on $H(L)$ by conjugation. This proves the assertion. $\square$

We define two special elements of the orthogonal group

$$J_N = \begin{pmatrix} 0 & 0 & 0 & 0 & -1 \\ 0 & 0 & 0 & -1 & 0 \\ 0 & 0 & I_N & 0 & 0 \\ 0 & -1 & 0 & 0 & 0 \\ -1 & 0 & 0 & 0 & 0 \end{pmatrix} \in O^+(L_2) \quad ,$$

$$R_U = \begin{pmatrix} 1 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & U & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 1 \end{pmatrix} \in O^+(L_2) \quad , \quad U \in O(L) \quad .$$

Obviously R_U and J_N commute. The latter element defines an embedding of $O(L)$ into $O(L_2)^+$. In the next step we consider finite characters of the Jacobi group.

Lemma 2.1.3 *Let χ be a finite character of the Jacobi group $\Gamma^J(L)$. Then we have*

$$\chi = v_{\mathrm{SL}(2, \mathbb{Z})} \cdot \nu_{H(L)}$$

where $v_{\mathrm{SL}(2, \mathbb{Z})}$ is a finite character of $\mathrm{SL}(2, \mathbb{Z})$ and $\nu_{H(L)}$ is a finite character of the group $H(L)$. The character $\nu_{H(L)}$ satisfies the equation

$$\forall A \in \mathrm{SL}(2, \mathbb{Z}) \forall x, y \in L, r \in \mathbb{Z}/2, r + \frac{1}{2}(x, y) \in \mathbb{Z} : \nu_{H(L)}(A.[x, y : r]) = \nu_{H(L)}([x, y : r]) \quad .$$

Proof

Since $\Gamma^J(L) \cong \mathrm{SL}(2, \mathbb{Z}) \ltimes H(L)$ it is clear that $\chi|_{H(L)} = \nu_{H(L)}$ and $\chi|_{\mathrm{SL}(2, \mathbb{Z})} = v_{\mathrm{SL}(2, \mathbb{Z})}$ for characters of $\mathrm{SL}(2, \mathbb{Z})$ or $H(L)$, respectively. Since the group is a semi-direct product we know that each character is of the above shape. For $A \in \mathrm{SL}(2, \mathbb{Z})$ and $[x, y : r] \in H(L)$ we have

$$\chi((A, [x, y : r])) = \chi((I_2, [x, y : r])) \cdot \chi((A, [0, 0 : 0])) = \chi((A, A.[x, y : r]))$$

which implies

$$\nu_{H(L)}(A.[x, y : r]) = \nu_{H(L)}([x, y : r]) \quad . \quad \square$$

The next two Lemmas describe all finite characters of the groups $\mathrm{SL}(2, \mathbb{Z})$ and the Heisenberg group. We remind to the definition of *Dedekind's eta function* $\eta : \mathbb{H} \rightarrow \mathbb{C}$ where $\mathbb{H}$ denotes the upper half plane in $\mathbb{C}$. This function is defined by the product expansion

$$\eta(\tau) = q^{1/24} \prod_{n=1}^{\infty} (1 - q^n) \quad , \quad q = e^{2\pi i \tau} \quad .$$

The transformation law for the generators of $\mathrm{SL}(2, \mathbb{Z})$ is

$$\begin{aligned} \eta(\tau + 1) &= e^{\pi i/12} \eta(\tau) \\ \eta(-1/\tau) &= (-i\tau)^{1/2} \eta(\tau) \end{aligned}$$

where we have chosen the branch of the square root $z^{1/2}$ which is positive if $z > 0$. Thus there exists a multiplier system $v_\eta : \mathbb{H} \rightarrow \mathbb{C}^*$ of order 24 whose square is a finite character such that

$$\eta\left(\frac{a\tau + b}{c\tau + d}\right) = v_\eta(A) \eta(\tau) \quad \text{for } A = \begin{pmatrix} a & b \\ c & d \end{pmatrix} \in \text{SL}(2, \mathbb{Z}).$$

For more details on the function $\eta(\tau)$ we refer to [Ap], section 3.

Lemma 2.1.4 *The group of finite characters of $\text{SL}(2, \mathbb{Z})$ is a cyclic group of order 12 and is generated by v_η^2 .*

Proof

From [R] we know that the index of $[\text{SL}(2, \mathbb{Z}), \text{SL}(2, \mathbb{Z})]$ in $\text{SL}(2, \mathbb{Z})$ is 12. This proves the claim. $\square$

The following Proposition describes all finite characters of the Heisenberg group. The centre of the Heisenberg group is the group $\{[0, 0 : r] | r \in \mathbb{Z}\}$ which is isomorphic to $\mathbb{Z}$. Let us introduce the subgroup

$$H_s(L) = \langle [x, 0 : 0], [0, y : 0] | x, y \in L \rangle.$$

We define the quantities $s(L), n(L) \in \mathbb{N}$ as the generators of the ideal generated by (x, y) or (x, x) , respectively, where $x, y \in L$. These quantities are called *scale* and *norm* of the lattice L . Obviously $s(L)|n(L)$. The centre of $H_s(L)$ is a subgroup of $\{[0, 0 : r] | r \in \mathbb{Z}\}$ and is generated by $[0, 0 : s(L)]$. A proof of the next lemma can be found in [CG].

Lemma 2.1.5 *Let L be a positive definite even lattice.*

(a) *We have*

$$[H(L), H(L)] = [H_s(L), H_s(L)] = \langle [0, 0 : s(L)] \rangle.$$

(b) *Let $\nu : H(L) \rightarrow \mathbb{C}^*$ be a character of finite order such that $\nu(A.[x, y : r]) = \nu([x, y : r])$ for all $[x, y : r] \in H(L)$ and $A \in \text{SL}(2, \mathbb{Z})$. Then ν is of the shape*

$$\nu([x, y : r]) = e^{\pi i t((x,x)+(y,y)-(x,y)+2r)}$$

where $t \in \mathbb{Q}$ such that $t \cdot s(L) \in \mathbb{Z}$. The restriction $\nu|_{H_s(L)}$ is a binary character which is trivial if $t \cdot n(L) \in 2\mathbb{Z}$.

From Section 1.3 we know that there exists a holomorphic factor of automorphy for the orthogonal group $J(g, Z)$ where $h \in \text{O}(L_2)^+$ and $Z \in \mathcal{H}(L_1)$. In the affine model of the homogeneous domain with respect to the simplest cusp a holomorphic function $\psi : \mathcal{H}(L_1) \rightarrow \mathbb{C}$ is a modular form of weight k for a subgroup $\Gamma \leq \text{O}(L_2)^+$ if and only if

$$\forall g \in \Gamma \quad \forall Z \in \mathcal{H}(L_1) : \psi(g < Z >) = J(g, Z)^k \psi(Z).$$

We want to consider modular forms for the special subgroup $\Gamma^J(L) \leq O(L_2)^+$. For this purpose we compute $g < Z >$ for the generators of the Jacobi group. Let $Z = (\omega, \mathfrak{z}, \tau)^{\text{tr}} \in \mathcal{H}(L_1)$. Let $A \in \text{SL}(2, \mathbb{Z})$. We have $J(\{A\}, Z) = c\tau + d$ and thus we obtain

$$\{A\} < Z > = \left(\frac{d\omega}{c\tau + d}, \frac{\mathfrak{z}}{c\tau + d}, \frac{a\tau + b}{c\tau + d} \right)^{\text{tr}} = \left(\omega - \frac{c(\mathfrak{z}, \mathfrak{z})}{c\tau + d}, \frac{\mathfrak{z}}{c\tau + d}, \frac{a\tau + b}{c\tau + d} \right)^{\text{tr}} \quad (2.1)$$

from (1.6). For $[x, y : r] \in H(L)$ we have $J([x, y : r], Z) = 1$ and

$$[x, y : r] < Z > = \left(\omega + (x, \mathfrak{z}) + \frac{1}{2}(x, x)\tau + \frac{1}{2}(x, y) + r, \mathfrak{z} + x\tau + y, \tau \right)^{\text{tr}} \quad (2.2)$$

Now let $\tilde{\varphi}(Z) := \varphi(\tau, \mathfrak{z}) \cdot e^{2\pi i t \omega}$ be modular of weight k with respect to the group $\Gamma^J(L)$ and character χ . Here $\varphi : \mathbb{H} \times (L \otimes \mathbb{C}) \rightarrow \mathbb{C}$ denotes a holomorphic function. According to Lemma 2.1.3, Lemma 2.1.4 and Lemma 2.1.5 there exists an even integer D and a rational number $t \in \mathbb{Q}$ satisfying $ts(L) \in \mathbb{Z}$ such that

$$\chi = v_\eta^D \cdot \nu \quad \text{where} \quad \nu([x, y : r]) = e^{\pi i t((x, x) + (y, y) - (x, y) + 2r)}.$$

The Jacobi group acts on the domain $\mathbb{H} \times (L \otimes \mathbb{C})$ in the following way. For $(\tau, \mathfrak{z})^{\text{tr}} \in \mathbb{H} \times (L \otimes \mathbb{C})$ there exists $\omega \in \mathbb{H}$ such that $Z = (\omega, \mathfrak{z}, \tau)^{\text{tr}} \in \mathcal{H}(L_1)$. For $g \in \Gamma^J(L)$ we define

$$g < \tau, \mathfrak{z} > := (\tau(g < Z >), \mathfrak{z}(g < Z >))^{\text{tr}}$$

where we have put

$$g < Z > = (\omega(g < Z >), \mathfrak{z}(g < Z >), \tau(g < Z >))^{\text{tr}} \in \mathcal{H}(L_1).$$

Lemma 2.1.6 *Let $k \in \mathbb{Z}$ and $t \in \mathbb{Q}$. The map*

$$J_{k,t} : \Gamma^J(L) \times (\mathbb{H} \times (L \otimes \mathbb{C})) \rightarrow \mathbb{C}^*, \quad J_{k,t}(g; \tau, \mathfrak{z}) = J(g, Z)^k \cdot e^{-2\pi i t \omega\{g < Z >\}} e^{2\pi i t \omega}$$

defines a factor of automorphy for $\Gamma^J(L)$. The parameters k and t are called weight and index. Moreover we have

$$\begin{aligned} J_{k,t}(\{A\}; \tau, \mathfrak{z}) &= (c\tau + d)^k e^{\pi i t \frac{c(\mathfrak{z}, \mathfrak{z})}{c\tau + d}}, & A \in \text{SL}(2, \mathbb{Z}) \\ J_{k,t}([x, y : r]; \tau, \mathfrak{z}) &= e^{-2\pi i t (\frac{1}{2}(x, x)\tau + (x, \mathfrak{z}) + \frac{1}{2}(x, y) + r)}, & [x, y : r] \in H(L). \end{aligned}$$

Proof

Let $g, h \in \Gamma^J(L)$. From (1.6) we deduce

$$\begin{aligned} J_{k,t}(gh; \tau, \mathfrak{z}) &= J(gh, Z)^k e^{-2\pi i t \omega\{gh < Z >\}} e^{2\pi i t \omega} \\ &= J(h, Z)^k e^{-2\pi i t \omega\{h < Z >\}} e^{2\pi i t \omega} J(g, h < Z >)^k \\ &\quad \times e^{-2\pi i t \omega\{gh < Z >\}} e^{2\pi i t \omega\{h < Z >\}} \\ &= J_{k,t}(g; h < \tau, \mathfrak{z} >) J_{k,t}(h; \tau, \mathfrak{z}). \end{aligned}$$

From (2.1) and (2.2) we obtain the formulae for the generators of $\Gamma^J(L)$. □

We define an action of the Jacobi group on the space of holomorphic functions on $\mathbb{H} \times (L \otimes \mathbb{C})$:

$$(\varphi|_{k,t} g)(\tau, \mathfrak{z}) := J_{k,t}^{-1}(g; \tau, \mathfrak{z}) \varphi(g < \tau, \mathfrak{z} >).$$

Then for $g \in \Gamma^J(L)$ and $Z \in \mathcal{H}(L_1)$ we obviously have

$$\tilde{\varphi}(g < Z >) = \chi(g) J(g, Z)^k \tilde{\varphi}(Z) \iff (\varphi|_{k,t} g)(\tau, \mathfrak{z}) = \chi(g) \varphi(\tau, \mathfrak{z}) \quad (2.3)$$

where $\tilde{\varphi}(Z) := \varphi(\tau, \mathfrak{z}) \cdot e^{2\pi i t \omega}$. Now let φ be a function on $\mathbb{H} \times (L \otimes \mathbb{C})$ satisfying (2.3) for all $g \in \Gamma^J(L)$. Considering the generators of $\Gamma^J(L)$ this is equivalent to the statements

$$\varphi\left(\frac{a\tau + b}{c\tau + d}, \frac{\mathfrak{z}}{c\tau + d}\right) = v_\eta^D(A) (c\tau + d)^k e^{\pi i t \frac{c(\mathfrak{z}, \mathfrak{z})}{c\tau + d}} \varphi(\tau, \mathfrak{z})$$

for all $A = \begin{pmatrix} a & b \\ c & d \end{pmatrix} \in \mathrm{SL}(2, \mathbb{Z})$ and

$$\begin{aligned} \varphi(\tau, \mathfrak{z} + x\tau + y) &= \nu([x, y : r]) \cdot e^{-2\pi i t (\frac{1}{2}(x, x)\tau + (x, \mathfrak{z}) + \frac{1}{2}(x, y) + r)} \varphi(\tau, \mathfrak{z}) \\ &= \nu([x, y : -(x, y)/2]) \cdot e^{-2\pi i t (\frac{1}{2}(x, x)\tau + (x, \mathfrak{z}))} \varphi(\tau, \mathfrak{z}) \end{aligned}$$

for all $[x, y : r] \in H(L)$. Then φ is periodic in τ and $\mathfrak{z}$ since

$$\varphi(\tau + 1, \mathfrak{z}) = e^{2\pi i \frac{D}{24}} \varphi(\tau, \mathfrak{z}) \quad \text{and} \quad \varphi(\tau, \mathfrak{z} + 2y) = \nu(0, 2y : 0) \varphi(\tau, \mathfrak{z}) = \varphi(\tau, \mathfrak{z}), \quad y \in L.$$

Thus the function has a Fourier expansion of the shape

$$\varphi(\tau, \mathfrak{z}) = \sum_{\substack{n \in \mathbb{Q}, n \equiv \frac{D}{24} \pmod{\mathbb{Z}} \\ l \in \frac{1}{2}L^\vee}} f(n, l) e^{2\pi i (n\tau + (l, \mathfrak{z}))}. \quad (2.4)$$

We now introduce the notion of a Jacobi form.

Definition 2.1.7 Let $k \in \frac{1}{2}\mathbb{Z}$ and $t \in \mathbb{Q}_{\geq 0}$. A holomorphic function $\varphi : \mathbb{H} \times (L \otimes \mathbb{C}) \rightarrow \mathbb{C}$ is called a *weak Jacobi form* of weight k and index t with character χ if it satisfies

$$\forall g \in \Gamma^J(L) : \varphi|_{k,t} g = \chi(g) \varphi$$

and has a Fourier expansion as in (2.4) where additionally

$$f(n, l) \neq 0 \implies n \geq 0.$$

We call φ a *holomorphic Jacobi form* if the Fourier expansion ranges over all n, l with $2nt - (l, l) \geq 0$ and φ is called a *Jacobi cusp form* if it ranges over all n, l satisfying $2nt - (l, l) > 0$.

Suppose φ is a non-vanishing Jacobi form of index t and weight k with character $\chi = v_\eta^D \cdot \nu$. Consider the extended Jacobi form $\tilde{\varphi}(\tau, \mathfrak{z}) = \varphi(\tau, \mathfrak{z})e^{2\pi i t \omega}$. If we substitute $g = [0, 0 : 1]$ in formula (2.3) we obtain $e^{2\pi i t} = \nu([0, 0 : 1])$ which implies $\nu([x, y : r]) = e^{\pi i t((x,x)+(y,y)-(x,y)+2r)}$ since φ does not vanish identically. We denote by $J_{k,L;t}^{(weak)}(\chi)$ the vector space of weak Jacobi forms and the corresponding spaces of holomorphic forms and cusp forms are denoted by $J_{k,L;t}(\chi)$ and $J_{k,L;t}^{(cusp)}(\chi)$, respectively. If the character is trivial we also write $J_{k,L;t} = J_{k,L;t}(1)$ for short. We will also use the notation $J_{k,L;t}(\chi)$ if $\chi|_{H(L)}$ is a character of the Heisenberg group and $\chi|_{\text{SL}(2,\mathbb{Z})} = v_\eta^m$ for some $m \in \mathbb{N}$.

Remark 2.1.8 The Jacobi forms for a fixed lattice L and with trivial character form a bigraded ring. The transformation behaviour of elements belonging to $J_{k_1,L;t_1} J_{k_2,L;t_2}$ is that of a Jacobi form of weight $k_1 + k_2$ and index $t_1 + t_2$. If both factors are holomorphic then the product is also a holomorphic Jacobi form by virtue of the identity

$$k_1 + k_2 - \frac{(x+y, x+y)}{2(t_1 + t_2)} = \left(k_1 - \frac{(x, x)}{2t_1}\right) + \left(k_2 - \frac{(y, y)}{2t_2}\right) + \frac{(t_1 y - t_2 x, t_1 y - t_2 x)}{2t_1 t_2 (t_1 + t_2)}$$

which holds for all $k_1, k_2 \in \mathbb{Q}_{\geq 0}$, $t_1, t_2 > 0$ and $x, y \in L \otimes \mathbb{R}$.

In view of section 1.4 the function φ is a holomorphic Jacobi form if $\tilde{\varphi}$ is holomorphic at the zero-dimensional cusp e .

Lemma 2.1.9 *Let φ be a weak Jacobi form with trivial character and positive integral index $t \in \mathbb{N}_{>0}$. Then the coefficients $f(n, l)$ in (2.4) only depend on the class of l in $L^\vee/tL$ and $2nt - (l, l)$.*

Proof

Let $\varphi \in J_{k,L;t}(1)$ with a Fourier expansion as in (2.4). The invariance under generators of the Heisenberg group applied to the Fourier expansion yields

$$\begin{aligned} \varphi(\tau, \mathfrak{z}) &= \sum_{\substack{n \in \mathbb{N} \\ l \in L^\vee}} f(n, l) e^{2\pi i (\tau(n + \frac{t}{2}(x, x)) + (l, \mathfrak{z} + x\tau + y) + t(x, \mathfrak{z}))} \\ &= \sum_{\substack{n \in \mathbb{N} \\ l \in L^\vee}} f(n, l) e^{2\pi i (\tau(n + \frac{t}{2}(x, x) + (l, x)) + (l + tx, \mathfrak{z}))}. \end{aligned}$$

Therefore we have $f(n + \frac{t}{2}(x, x) + (l, x), l + tx) = f(n, l)$. We have

$$2nt - (l, l) = 2t(n + (t/2)(x, x) + (l, x)) - (l + tx, l + tx)$$

If there are two pairs (n, l) and $(n', l + tx)$ such that $2nt - (l, l) = 2n't - (l + tx, l + tx)$ we conclude $n' = n + (t/2)(x, x) + (l, x)$ which proves the assertion. $\square$

We introduce theta-functions for the positive definite lattice L given by

$$\theta_{\mu,t}^L = \sum_{l \in L} \exp(\pi i t(l + \mu/t, l + \mu/t)\tau + 2\pi i t(l + \mu/t, \mathfrak{z})) \quad , \text{ where } \mu \in L^\vee/tL.$$

These series converge locally uniformly according to [K1] and are invariant with respect to the action of the Heisenberg group.

Proposition 2.1.10 *Let $\varphi \in J_{k,L;t}^{(weak)}(1)$ where $k, t \in \mathbb{Z}$. Then we have*

$$\varphi(\tau, \mathfrak{z}) = \sum_{\mu \in L^\vee / tL} \phi_\mu(\tau) \theta_{\mu,t}^L(\tau, \mathfrak{z})$$

where

$$\phi_\mu(\tau) = \sum_{\substack{r \in \mathbb{Z} \\ r \equiv -\frac{q}{2}(\mu, \mu) \pmod{qt}}} f\left(\frac{2r + q(\mu, \mu)}{2qt}, \mu\right) \exp\left(\frac{2\pi i r \tau}{qt}\right)$$

Proof

According to Lemma 2.1.9 the Fourier coefficients of φ depend only on the class of l in $L^\vee / tL$ and the hyperbolic norm $2nt - (l, l)$. Let q be the level of L , i.e. $q(l, l) \in 2\mathbb{Z}$ for all $l \in L^\vee$. We have $2nt - (l, l) \in \mathbb{Z}/q$. If we parametrize this quantity by $r = qnt - \frac{q}{2}(l, l)$ we can rewrite the Fourier expansion as

$$\begin{aligned} \varphi(\tau, \mathfrak{z}) &= \sum_{\mu \in L^\vee / tL} \sum_{x \in L} \sum_{\substack{r \in \mathbb{Z} \\ r \equiv -\frac{q}{2}(\mu, \mu) \pmod{qt}}} f\left(\frac{2r + q(\mu + tx, \mu + tx)}{2qt}, \mu + tx\right) \\ &\quad \times \exp(2\pi i(\mu + tx, \mathfrak{z})) \exp\left(2\pi i \frac{2r\tau + q(\mu + tx, \mu + tx)\tau}{2qt}\right) \\ &= \sum_{\mu \in L^\vee / tL} \sum_{x \in L} \sum_{\substack{r \in \mathbb{Z} \\ r \equiv -\frac{q}{2}(\mu, \mu) \pmod{qt}}} f\left(\frac{2r + q(\mu + tx, \mu + tx)}{2qt}, \mu + tx\right) \\ &\quad \times \exp\left(\frac{2\pi i r \tau}{qt}\right) \exp(\pi i t(2(x + \mu/t, \mathfrak{z}) + (x + \mu/t, x + \mu/t)\tau)) \end{aligned}$$

By Lemma 2.1.9 we know $f\left(\frac{2r + q(\mu + tx, \mu + tx)}{2qt}, \mu + tx\right) = f\left(\frac{2r + q(\mu, \mu)}{2qt}, \mu\right)$. This completes the proof. $\square$

The set of all holomorphic functions

$$\mathcal{O}_{\mathbb{H} \times (L \otimes \mathbb{C})} = \{\varphi : \mathbb{H} \times (L \otimes \mathbb{C}) \rightarrow \mathbb{C} : \varphi \text{ is holomorphic}\}$$

is a module over the ring of holomorphic functions on the upper half plane $\mathbb{H}$. Let $\mathcal{O}(\Theta_t^L)$ be the submodule generated by the theta-series

$$\mathcal{O}(\Theta_t^L) = \langle \theta_{\mu,t}^L ; \mu \in L^\vee / tL \rangle_{\mathcal{O}(\mathbb{H})} \subseteq \mathcal{O}_{\mathbb{H} \times (L \otimes \mathbb{C})}.$$

Lemma 2.1.11 *Let $t \in \mathbb{Z}$. The module $\mathcal{O}(\Theta_t^L)$ is free and a basis is given by*

$$\{\theta_{\mu,t}^L ; \mu \in L^\vee / tL\}.$$

Proof

Let $f_\mu \in \mathcal{O}(\mathbb{H})$ such that $\sum_{\mu \in L^\vee/tL} f_\mu(\tau) \theta_{\mu,t}^L(\tau, \mathfrak{z}) = 0$ for all $\tau \in \mathbb{H}$ and $\mathfrak{z} \in L \otimes \mathbb{C}$. It is well-known that $\{\theta_{\mu,t}^L(\tau, \cdot); \mu \in L^\vee/tL\}$ is linearly independent for fixed τ . Thus we obtain $f_\mu(\tau) = 0$ for all $\mu \in L^\vee/tL$. Since τ was arbitrary we deduce that f_μ vanishes identically for every μ . $\square$

Corollary 2.1.12 *Let $\varphi \in J_{k,L;t}^{(weak)}(1)$ with $k \in \mathbb{Z}$, $t \in \mathbb{N}$ and a Fourier expansion as in (2.4). Then we have*

$$f(n, l) \neq 0 \implies 2nt - (l, l) \geq - \min_{v \in l+tL} (v, v).$$

Proof

From the previous Proposition we know that each summand is of the shape

$$f(n, l) \exp \left(\pi i t \left[\frac{2r}{qt^2} + (x + \mu/t, x + \mu/t) \right] \tau + 2\pi i t (x + \mu, \mathfrak{z}) \right)$$

where $n \in \mathbb{N}$, $l \in L^\vee$ such that $n = \frac{r}{qt} + (1/2t)(tx + \mu, tx + \mu)$ and $l = tx + \mu$. Here $\mu, r = r(\mu), q$ and x are chosen as in the proof of the Proposition above. This implies

$$\forall x \in L : r \geq -(q/2)(tx + \mu, tx + \mu)$$

and hence $r \geq -\frac{q}{2} \min_{x \in L} (tx + \mu, tx + \mu)$. Finally we obtain $2nt - (tx + \mu, tx + \mu) = \frac{2r}{q} \geq -\min_{x \in L} (tx + \mu, tx + \mu)$ which completes the proof. $\square$

We can rewrite the statement of the last lemma. For this purpose we define the column vectors

$$\Phi(\tau) = (\phi_\mu(\tau))_{\mu \in L^\vee/tL} \quad , \quad \Theta_t^L(\tau, \mathfrak{z}) = (\theta_{\mu,t}^L(\tau, \mathfrak{z}))_{\mu \in L^\vee/tL}.$$

Then in accordance with Proposition 2.1.10 we have $\varphi(\tau, \mathfrak{z}) = \Phi(\tau)^{\text{tr}} \cdot \Theta_t^L(\tau, \mathfrak{z})$. For a proof of the next Lemma we refer to [G1] and [Od].

Lemma 2.1.13 *Let $A = \begin{pmatrix} a & b \\ c & d \end{pmatrix} \in \text{SL}(2, \mathbb{Z})$. Then we have*

$$\Theta_t^L \left(\frac{a\tau + b}{c\tau + d}, \frac{\mathfrak{z}}{c\tau + d} \right) = (c\tau + d)^{N/2} U(A) \exp \left(\frac{\pi i t c(\mathfrak{z}, \mathfrak{z})}{c\tau + d} \right) \Theta_t^L(\tau, \mathfrak{z})$$

where $U(A)$ is a unitary matrix defined by

$$U(M)_{\mu, \nu} = \begin{cases} \varepsilon(c) \cdot d(L(t))^{-\frac{1}{2}} |c|^{-\frac{N}{2}} \lambda_{\mu, \nu} & \text{if } c \neq 0 \\ \sqrt{i}^{N(1-\text{sgn}(d))} \cdot \exp \left(\frac{\pi i a b(\mu, \mu)}{t} \right) \delta_{\mu, a\nu} & \text{if } c = 0 \end{cases}$$

where $\delta_{\alpha, \beta}$ denotes Kronecker's delta and

$$\lambda_{\mu, \nu} = \sum_{l \in L/cL} \exp \left(\frac{\pi i}{c} (at(l + \mu/t, l + \mu/t) - 2(\nu, l + \mu/t) + (d/t)(\nu, \nu)) \right)$$

and

$$\varepsilon(c) = \begin{cases} \exp\left(\frac{\pi i N}{4}\right) & \text{if } c > 0 \\ \exp\left(-\frac{\pi i N}{4}\right) & \text{if } c < 0 \end{cases}$$

Here we have chosen this branch of the root which is positive for positive arguments.

Using this formula we can compute the $U(J)$ and $U(T)$ for $T = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}$, $J = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}$.

Corollary 2.1.14 *Let $\varphi \in J_{k,L;t}^{(weak)}(1)$ as in Proposition 2.1.10.*

(i) *We have*

$$U(J) = d(L)^{-\frac{1}{2}}(i/t)^{\frac{N}{2}} \left(\exp\left(-2\pi i \frac{(\mu, \nu)}{t}\right) \right)_{\mu, \nu \in L^\vee / tL}$$

$$U(T) = \left(\delta_{\mu, \nu} \cdot \exp\left(\frac{\pi i (\mu, \nu)}{t}\right) \right)_{\mu, \nu \in L^\vee / tL}$$

(ii) *We have for $\mu \in L^\vee / tL$*

$$\phi_\mu(\tau + 1) = \exp\left(\frac{-\pi i (\mu, \nu)}{t}\right) \phi_\mu(\tau)$$

and

$$\Phi(-1/\tau) = \tau^{k-N/2} \overline{U(J)} \Phi(\tau)$$

where $\overline{U(J)}$ is the complex conjugated matrix of $U(J)$.

Proof

The first part follows by a direct computation. For the second part we use the transformation behaviour of Jacobi forms under $\mathrm{SL}(2, \mathbb{Z})$. We have $\varphi(\tau, \mathfrak{z}) = \Phi(\tau)^{\mathrm{tr}} \cdot \Theta_t^L(\tau, \mathfrak{z})$. From $\varphi(\tau + 1, \mathfrak{z}) = \varphi|_{k,t} T(\tau, \mathfrak{z}) = \varphi(\tau, \mathfrak{z})$ we deduce

$$\Phi(\tau)^{\mathrm{tr}} \cdot \Theta_t^L(\tau, \mathfrak{z}) = \Phi(\tau + 1)^{\mathrm{tr}} U(T) \Theta_t^L(\tau, \mathfrak{z}).$$

Moreover $\tau^{-k} e^{-\pi i t \frac{(\mathfrak{z}, \mathfrak{z})}{\tau}} \varphi(-1/\tau, \mathfrak{z}/\tau) = \varphi|_{k,t} J(\tau, \mathfrak{z}) = \varphi(\tau, \mathfrak{z})$ yields

$$\tau^k e^{\pi i t \frac{(\mathfrak{z}, \mathfrak{z})}{\tau}} \Phi(\tau)^{\mathrm{tr}} \cdot \Theta_t^L(\tau, \mathfrak{z}) = \tau^{N/2} e^{\pi i t \frac{(\mathfrak{z}, \mathfrak{z})}{\tau}} \Phi(-1/\tau)^{\mathrm{tr}} U(J) \Theta_t^L(\tau, \mathfrak{z}).$$

The assertion now follows from Lemma 2.1.11. □

If we assume additionally that φ is a holomorphic Jacobi form then for each $\mu \in L^\vee / tL$ the function ϕ_μ is a holomorphic function on the upper half plane which is holomorphic at infinity. The transformation behaviour is clarified by the next Lemma which is a direct consequence of Lemma 2.1.13, compare for example [G].

Corollary 2.1.15 *Let q be the level of L and let $\varphi \in J_{k,L;t}(1)$ be a Jacobi form with decomposition $\varphi(\tau, \mathfrak{z}) = \Phi(\tau)^{\text{tr}} \cdot \Theta_t^L(\tau, \mathfrak{z})$. We define the congruence subgroup*

$$\Gamma_0(tq) = \left\{ A = \begin{pmatrix} a & b \\ c & d \end{pmatrix} \in \text{SL}(2, \mathbb{Z}) ; c \equiv 0 \pmod{tq} \right\}$$

of level tq . Then for every $\mu \in L^\vee/tL$ the function ϕ_μ is a classical modular form of weight $k - \frac{N}{2}$ for the main congruence subgroup $\Gamma(tq)$ and ϕ_0 is a modular form for $\Gamma_0(tq)$.

This yields an upper bound for the dimension of vector spaces of holomorphic Jacobi forms and index $t > 0$, namely

$$\dim J_{k,L;t}(1) \leq \dim \mathcal{M}_{k-\frac{N}{2}}(\Gamma_0(tq)) + (|L^\vee/tL| - 1) \dim \mathcal{M}_{k-\frac{N}{2}}(\Gamma(tq)) \quad (2.5)$$

where $\mathcal{M}_k(\Gamma(n))$ denotes the space of classical modular forms of weight k for the principal congruence subgroup of level n . This shows that the space of holomorphic Jacobi forms is always finite-dimensional. In particular there are no holomorphic Jacobi forms of negative weight and the minimal possible positive weight $N/2$ is called *singular weight*. The next lemma is a direct consequence of the definition of Jacobi forms.

Lemma 2.1.16 *Let L be a positive definite even lattice of rank one and let $\varphi \in J_{k,L;t}(\chi)$ be a Jacobi form where $k \in \frac{1}{2}\mathbb{Z}$. Then the function $\left. \frac{\partial \varphi}{\partial \mathfrak{z}} \right|_{\mathfrak{z}=0}$ is a modular form for the group $\text{SL}(2, \mathbb{Z})$ of weight $k+1$ and character $\chi|_{\text{SL}(2, \mathbb{Z})}$.*

We consider two standard examples of Jacobi forms of half-integral weight.

Example 2.1.17 (a) Dedekind's eta function is a Jacobi form of weight $1/2$ and index 0 for every positive definite even lattice L , thus $\eta \in J_{1/2,L;0}^{(\text{cusp})}(v_\eta)$ where v_η is not a character but a multiplier system of $\text{SL}(2, \mathbb{Z})$. We want to mention the following infinite sum-expansion for $\eta^3(\tau)$

$$\eta^3(\tau) = \frac{1}{2} \sum_{n \in \mathbb{Z}, n \equiv 1 \pmod{2}} \left(\frac{-4}{n} \right) n q^{\frac{n^2}{8}} \quad (2.6)$$

which follows from the Jacobi triple identity.

(b) The Jacobi theta-series of characteristic $(\frac{1}{2}, \frac{1}{2})$ is given by

$$\vartheta(\tau, z) = \sum_{n \in \mathbb{Z}} \left(\frac{-4}{n} \right) q^{\frac{n^2}{8}} r^{\frac{n}{2}} = \sum_{n \in \mathbb{Z}, n \equiv 1 \pmod{2}} (-1)^{\frac{n-1}{2}} \exp\left(\frac{\pi i n^2 \tau}{4} + \pi i n z\right)$$

where $q = e^{2\pi i \tau}$, $\tau \in \mathbb{H}$ and $r = e^{2\pi i z}$, $z \in \mathbb{C}$ and for $m \in \mathbb{Z}$ one defines

$$\left(\frac{-4}{m} \right) := \begin{cases} 1 & , m \equiv 1 \pmod{4} \\ -1 & , m \equiv -1 \pmod{4} \\ 0 & , m \equiv 0 \pmod{2} \end{cases}.$$

This function was originally discovered by Carl Gustav Jacob Jacobi. In [GN] the authors reinterpreted this function as a modular form of half-integral weight and index. Apart from the classical examples given in [EZ] nowadays this function can be considered as an additional classical function. The same proof as in [KK], section I.6 can be used to show that ϑ defines a holomorphic function on $\mathbb{H} \times \mathbb{C}$. The transformation behaviour under the action of $H(A_1)$ is

$$\vartheta(\tau, z + x\tau + y) = (-1)^{x+y} \exp(-\pi i(x^2\tau + 2xz))\vartheta(\tau, z).$$

Moreover we have

$$\vartheta(\tau + 1, z) = \exp(\pi i/4)\vartheta(\tau, z)$$

and

$$\vartheta(-1/\tau, z/\tau) = \exp(-3\pi i/4)\tau^{1/2} \exp(\pi i z^2/\tau)\vartheta(\tau, z)$$

by the theta transformation formula. This shows that ϑ transforms like a Jacobi form of weight $1/2$ and index $1/2$. If we consider the extended Jacobi form $\tilde{\vartheta}(\tau, z, \omega) = \vartheta(\tau, z)e^{\pi i\omega}$ we obtain

$$\begin{aligned} \tilde{\vartheta}(\tau, z, \omega) \Big|_{\frac{1}{2}} [x, y : r] &= e^{\pi i(x+y)} \exp(-\pi i(x^2\tau + 2xz))\vartheta(\tau, z) \\ &\quad \times \exp(\pi i(\omega + 2xz + x^2\tau + xy + r)) \\ &= e^{\pi i(x+y+xy+r)} \tilde{\vartheta}(\tau, z, \omega) \\ &=: \nu_{H(A_1)}([x, y : r]) \tilde{\vartheta}(\tau, z, \omega) \end{aligned}$$

From the infinite sum expansion we obtain

$$\frac{\partial \vartheta(\tau, z)}{\partial z} \Big|_{z=0} = \pi i \sum_{n \in \mathbb{Z}, n \equiv 1 \pmod{2}} \left(\frac{-4}{n} \right) n q^{\frac{n^2}{8}} = 2\pi i \eta(\tau)^3$$

by (2.6). This shows that ϑ transforms like a Jacobi form with multiplier system $v_\eta^3 \times \nu_{H(A_1)}$. Moreover Jacobi's triple identity yields

$$\vartheta(\tau, z) = -q^{1/8}r^{-1/2} \prod_{n \geq 1} (1 - q^{n-1}r)(1 - q^n r^{-1})(1 - q^n).$$

Since each summand in the infinite sum defining ϑ is of the shape $e^{2\pi i \tau n^2/8} e^{2\pi i \cdot 2(z, m/4)}$ we see that ϑ is holomorphic Jacobi form. Here $(\cdot, \cdot)$ denotes the standard scalar product on $\mathbb{Z}$. The function has the properties

$$\vartheta(\tau, -z) = -\vartheta(\tau, z), \quad \vartheta(\tau, z + x\tau + y) = (-1)^{x+y} \exp(-\pi i(x^2\tau + 2xz))\vartheta(\tau, z)$$

for all $x, y \in \mathbb{Z}$. Hence the divisor of ϑ contains $\{z = x\tau + y \mid x, y \in \mathbb{Z}\}$. Finally a classical argument from the theory of complex functions yields $\text{div}(\vartheta) = \{x\tau + y \mid x, y \in \mathbb{Z}\}$, compare for example, [FB] section 5.6.

We present a type of construction which will be fundamental in the following sections. We assume $L \subseteq \mathbb{R}^N$ and consider $\mathfrak{z}_L \in L \otimes \mathbb{C}$. Let $\varepsilon_1, \dots, \varepsilon_N$ be the standard basis of $\mathbb{C}^N$ where $N = \text{rank}(L)$. We define the function

$$(\tau, \mathfrak{z}_L) \mapsto \vartheta(\tau, (\mathfrak{z}_L, \varepsilon_1)) \cdots \vartheta(\tau, (\mathfrak{z}_L, \varepsilon_N)).$$

This defines a function on $\mathbb{H} \times (L \otimes \mathbb{C})$. We like to investigate the modular behaviour of this function under the action of $\Gamma^J(L)$. In the next Lemma we consider the example $L = D_N$.

Lemma 2.1.18 *Let $N \in \mathbb{N}$. Then there is a Jacobi form of singular weight for D_N given by*

$$\vartheta_{D_N}(\tau, \mathfrak{z}_N) := \vartheta(\tau, z_1) \cdots \vartheta(\tau, z_N) \quad \text{where} \quad \mathfrak{z}_N = \sum_{j=1}^N z_j \varepsilon_j$$

is given in the standard basis of $\mathbb{C}^N$. More precisely $\vartheta_{D_N} \in J_{\frac{N}{2}, D_N; 1}(v_\eta^{3N})$.

Proof

We do not present all details of the proof. This will be done in the following sections after fixing a suitable model for D_N . We first show that ϑ_{D_N} is holomorphic at infinity. The lattice D_N consists of all $l \in \mathbb{Z}^N$ such that $l_1 + \cdots + l_N$ is even. Let us consider the case $N = 1$. From Example 2.1.17 we know the Fourier expansion of $\vartheta(\tau, z)$. The expansion ranges over all terms of the shape $q^n r^l$ where $n \equiv 1 \pmod{8}$ and $l \in \frac{1}{2}\mathbb{Z}$ such that $2n - l^2 = 0$. This means that ϑ is a holomorphic Jacobi form of index 1 and weight $1/2$ for the lattice $D_1 \cong A_1(2)$. If $N > 1$ the terms appearing in the Fourier expansion are of the shape

$$q^{n_1} r_1^{l_1} \cdots q^{n_N} r_N^{l_N} \quad \text{where} \quad n_j \equiv \frac{1}{8} \pmod{\mathbb{Z}}, l_j \in \mathbb{Z}/2.$$

Since $2l = 2(l_1, \dots, l_N) \in \mathbb{Z}^N(2)^\vee$ and $2(n_1 + \cdots + n_N) - (l, l) \geq 0$ by assumption we obtain that ϑ_{D_N} is also holomorphic at infinity in the general case. By definition ϑ_{D_N} transforms like a Jacobi form of index 1 and weight $N/2$ with character v_η^{3N} under elements of $\text{SL}(2, \mathbb{Z})$. Now let $x, y \in D_N$. Then the sum of the coordinates of x and y is even and we obtain

$$\begin{aligned} \vartheta_{D_N}(\tau, \mathfrak{z}_N + x\tau + y) &= (-1)^{\sum_j x_j + y_j} \cdot e^{-\pi i (\sum_j (x_j^2 \tau + 2x_j z_j))} \vartheta_{D_N}(\tau, \mathfrak{z}_N) \\ &= e^{-\pi i ((x, x)\tau + 2(x, \mathfrak{z}))} \vartheta_{D_N}(\tau, \mathfrak{z}_N). \end{aligned} \quad \square$$

In the next Lemma we consider the special case where $N = 4$. The construction was mentioned first in [CG].

Lemma 2.1.19 *We have two additional singular Jacobi forms $\vartheta_{D_4}^{(2)}(\tau, \mathfrak{z}_4), \vartheta_{D_4}^{(3)}(\tau, \mathfrak{z}_4) \in J_{2, D_4; 1}(v_\eta^{12})$ where*

$$\begin{aligned}\vartheta_{D_4}^{(2)}(\tau, \mathfrak{z}_4) &= \vartheta\left(\tau, \frac{-z_1 + z_2 + z_3 + z_4}{2}\right) \vartheta\left(\tau, \frac{z_1 - z_2 + z_3 + z_4}{2}\right) \\ &\quad \vartheta\left(\tau, \frac{z_1 + z_2 - z_3 + z_4}{2}\right) \vartheta\left(\tau, \frac{z_1 + z_2 + z_3 - z_4}{2}\right), \\ \vartheta_{D_4}^{(3)}(\tau, \mathfrak{z}_4) &= \vartheta\left(\tau, \frac{z_1 + z_2 + z_3 + z_4}{2}\right) \vartheta\left(\tau, \frac{z_1 + z_2 - z_3 - z_4}{2}\right) \\ &\quad \vartheta\left(\tau, \frac{z_1 - z_2 - z_3 + z_4}{2}\right) \vartheta\left(\tau, \frac{z_1 - z_2 + z_3 - z_4}{2}\right).\end{aligned}$$

These functions are linearly independent and we have the relation $\vartheta_{D_4} = \vartheta_{D_4}^{(2)} + \vartheta_{D_4}^{(3)}$.

Proof

A similiar calculation as in the last lemma shows that both functions are holomorphic at infinity and have the right transformation behaviour under the group $\mathrm{SL}(2, \mathbb{Z})$. Let $x, y \in D_4$. For simplicity we write

$$\vartheta(\tau, w_1 + x^{(1)}\tau + y^{(1)}) \cdots \vartheta(\tau, w_4 + x^{(4)}\tau + y^{(4)}) := \vartheta_{D_4}^{(2)}(\tau, \mathfrak{z}_4 + x\tau + y)$$

Then we have

$$\begin{aligned}\sum_{j=1}^4 x^{(j)} + \sum_{j=1}^4 y^{(j)} &= \sum_{j=1}^4 (x_j + y_j) \quad , \quad \sum_{j=1}^4 (x^{(j)})^2 = \sum_{j=1}^4 x_j^2 \\ &\quad , \quad \sum_{j=1}^4 2x^{(j)}w_j = \sum_{j=1}^4 2x_j z_j\end{aligned}$$

which yields the Heisenberg invariance of this function and the same calculation can be performed for $\vartheta_{D_4}^{(3)}$. The function ϑ_{D_4} vanishes along the set

$$\mathrm{div}(\vartheta_{D_4}) = \{(z_1, \dots, z_4) \in D_4 \otimes \mathbb{C} \mid z_j \in \mathbb{Z}\tau + \mathbb{Z} \text{ for some } j\}.$$

Let $\mathfrak{z} = (z_1, \dots, z_4) \in \mathrm{div}(\vartheta_{D_4})$. Without loss of generality we can assume $\mathfrak{z} = (x\tau + y, z_2, z_3, z_4)$ for $x, y \in \mathbb{Z}$. Since

$$\vartheta(\tau, -z) = -\vartheta(\tau, z)$$

and

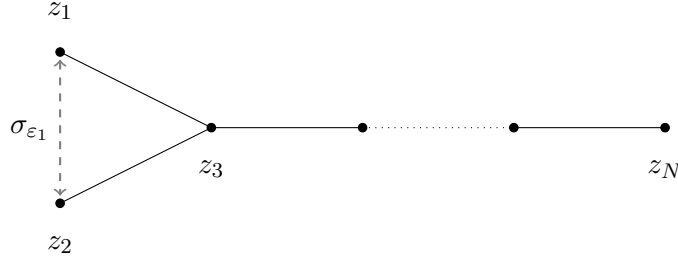
$$\vartheta(\tau, z + x\tau + y) = (-1)^{x+y} \exp(-\pi i(x^2\tau + 2xz))\vartheta(\tau, \mathfrak{z})$$

we obtain

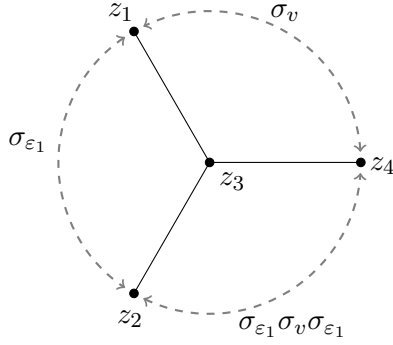
$$\begin{aligned}
(\vartheta_{D_4}^{(2)} + \vartheta_{D_4}^{(3)})(\tau, \mathfrak{z}) &= \vartheta\left(\tau, \frac{z_2 + z_3 + z_4 - (x\tau + y)}{2}\right) \vartheta\left(\tau, \frac{x\tau + y - z_2 + z_3 + z_4}{2}\right) \\
&\quad \vartheta\left(\tau, \frac{x\tau + y + z_2 - z_3 + z_4}{2}\right) \vartheta\left(\tau, \frac{x\tau + y + z_2 + z_3 - z_4}{2}\right) \\
&+ \vartheta\left(\tau, \frac{x\tau + y + z_2 + z_3 + z_4}{2}\right) \vartheta\left(\tau, \frac{x\tau + y + z_2 - z_3 - z_4}{2}\right) \\
&\quad \vartheta\left(\tau, \frac{x\tau + y - z_2 - z_3 + z_4}{2}\right) \vartheta\left(\tau, \frac{x\tau + y - z_2 + z_3 - z_4}{2}\right) \\
&= -\exp(\pi i(2x^2\tau + 2xz_2)) \times \\
&\quad \vartheta\left(\tau, \frac{x\tau + y + z_2 + z_3 + z_4}{2}\right) \vartheta\left(\tau, \frac{(x\tau + y) + z_2 - z_3 - z_4}{2}\right) \\
&\quad \vartheta\left(\tau, \frac{x\tau + y + z_2 - z_3 + z_4}{2}\right) \vartheta\left(\tau, \frac{x\tau + y + z_2 + z_3 - z_4}{2}\right) \\
&+ \exp(\pi i(2x^2\tau + 2xz_2)) \times \\
&\quad \vartheta\left(\tau, \frac{x\tau + y + z_2 + z_3 + z_4}{2}\right) \vartheta\left(\tau, \frac{x\tau + y + z_2 - z_3 - z_4}{2}\right) \\
&\quad \vartheta\left(\tau, \frac{x\tau + y + z_2 + z_3 - z_4}{2}\right) \vartheta\left(\tau, \frac{x\tau + y + z_2 - z_3 + z_4}{2}\right) \\
&= 0.
\end{aligned}$$

This shows $\text{div}(\vartheta_{D_4}) \subseteq \text{div}(\vartheta_{D_4}^{(2)} + \vartheta_{D_4}^{(3)})$ and the function $\frac{\vartheta_{D_4}^{(2)} + \vartheta_{D_4}^{(3)}}{\vartheta_{D_4}}$ is a holomorphic Jacobi form of weight and index zero. Thus it must be constant. Finally the Fourier coefficient of $q^{1/2}r_1^{-1/2}r_2^{-1/2}r_3^{-1/2}r_4^{-1/2}$ for each of the functions $\vartheta_{D_4}^{(2)} + \vartheta_{D_4}^{(3)}$ and ϑ_{D_4} , where $r_j = \exp(2\pi iz_j)$, equals -1. The linear independence of $\vartheta_{D_4}^{(2)}$ and $\vartheta_{D_4}^{(3)}$ follows since the Fourier expansions are mutually disjoint. This completes the proof. $\square$

The existence of the additional Jacobi forms $\vartheta_{D_4}^{(2)}$ and $\vartheta_{D_4}^{(3)}$ is due to a extended symmetry of the Coxeter diagram of D_4 . Let us consider the Coxeter diagram of D_N which was described in Section 1.8. If we choose the basis given there and label the coordinates of $\mathfrak{z}_{D_N} \in D_N \otimes \mathbb{C}$ accordingly we see that in general the only automorphism of the diagram is given by the permutation of z_1 and z_2 :



If $N = 4$ the diagram has additional symmetries:



$$v = \frac{\varepsilon_1 + \varepsilon_2 + \varepsilon_3 - \varepsilon_4}{2} \in D_4^\vee.$$

If we denote the coordinates with respect to this basis by $\mathfrak{z}_{D_4}$ we obtain:

$$\begin{aligned} \vartheta_{D_4}(\tau, \mathfrak{z}_{D_4}) &= \vartheta(\tau, z_1 - z_2) \vartheta(\tau, z_1 + z_2 - z_3) \vartheta(\tau, z_3 - z_4) \vartheta(\tau, z_4) \\ \vartheta_{D_4}^{(2)}(\tau, \mathfrak{z}_{D_4}) &= \vartheta(\tau, z_2) \vartheta(\tau, z_3 - z_2) \vartheta(\tau, z_1 - z_3 + z_4) \vartheta(\tau, z_1 - z_4) \\ \vartheta_{D_4}^{(3)}(\tau, \mathfrak{z}_{D_4}) &= \vartheta(\tau, z_1) \vartheta(\tau, z_1 - z_3) \vartheta(\tau, z_4 - z_2) \vartheta(\tau, -z_2 + z_3 - z_4) \end{aligned}$$

and hence

$$\vartheta_{D_4}|_{2,1} \sigma_v = \vartheta_{D_4}^{(3)} \quad , \quad \vartheta_{D_4}|_{2,1} \sigma_{\varepsilon_1} \sigma_v \sigma_{\varepsilon_1} = \vartheta_{D_4}^{(2)}.$$

2.2 Pullbacks of Jacobi forms

In this section we assume that L is a positive definite even lattice.

Lemma 2.2.1 *Let $M \leq L$ be a sublattice of L such that $\text{rank } M < \text{rank } L$ and $\varphi \in J_{k,L,t}(\chi)$ be a Jacobi form of weight k and index t for the character χ . Consider the decomposition $\mathfrak{z}_L = \mathfrak{z}_M \oplus \mathfrak{z}_{M^\perp} \in L \otimes \mathbb{C} = (M \oplus (M)_L^\perp) \otimes \mathbb{C}$. We define the pullback of φ to M as the function $\varphi|_M$ on $\mathbb{H} \times (M \otimes \mathbb{C})$ defined by*

$$\varphi|_M(\tau, \mathfrak{z}_M) := \varphi(\tau, \mathfrak{z}_M \oplus 0).$$

Then $\varphi|_M \in J_{k,M,t}(\chi|_M)$ and the pullback maps cusp forms to cusp forms.

Proof

Since $(\mathfrak{z}_M \oplus 0_{M^\perp}, \mathfrak{z}_M \oplus 0_{M^\perp})_L = (\mathfrak{z}_M, \mathfrak{z}_M)_M$ we immediately see that $\varphi|_M(\tau, \mathfrak{z}_M)$ transforms like a Jacobi form of weight k and index t for the lattice M . To prove the second assertion we have to consider the Fourier expansion. For this purpose we consider the direct sum of lattices $M \oplus M_L^\perp$ which has rank L due to (1.3). Hence we obtain a chain of sublattices each of finite index

$$M \oplus M_L^\perp \leq L \leq L^\vee \leq M^\vee \oplus (M_L^\perp)^\vee.$$

According to (2.4) the Fourier expansion is of the shape

$$\varphi(\tau, \mathfrak{z}) = \sum_{\substack{n \in \mathbb{Q}_{\geq 0}, n \equiv \frac{D}{24} \pmod{\mathbb{Z}} \\ l \in L^\vee}} f(n, l) e^{2\pi i(n\tau + (l, \mathfrak{z}))}$$

if the binary character $\chi|_{H_s(L)}$ is trivial. In this case any $l \in L^\vee$ can be written as $l = l_M \oplus l_{M^\perp}$ and the Fourier expansion of the pullback is

$$\begin{aligned} \varphi|_M(\tau, \mathfrak{z}_M) &= \sum_{\substack{n \in \mathbb{Q}_{\geq 0}, n \equiv \frac{D}{24} \pmod{\mathbb{Z}} \\ l \in L^\vee}} f(n, l) e^{2\pi i(n\tau + (l_M, \mathfrak{z}_M))} \\ &= \sum_{\substack{n \in \mathbb{Q}_{\geq 0}, n \equiv \frac{D}{24} \pmod{\mathbb{Z}} \\ l_M \in M^\vee}} \left(\sum_{\substack{l_{M^\perp} \in (M_L^\perp)^\vee \\ l_M + l_{M^\perp} \in L^\vee}} f(n, l_M \oplus l_{M^\perp}) \right) e^{2\pi i(n\tau + (l_M, \mathfrak{z}_M))} \end{aligned}$$

and since φ is a Jacobi form $f(n, l) \neq 0$ implies $2nt - (l, l) \geq 0$ and thus $f(n, l_M \oplus l_{M^\perp}) \neq 0$ implies $2nt - (l_M, l_M) \geq 2nt - (l, l) \geq 0$ for every $l_{M^\perp} \in (M_L^\perp)^\vee$. This shows that the pullback is indeed a Jacobi form. If the character $\chi|_{H_s(L)}$ is not trivial the summation ranges over $l \in \frac{1}{2}L^\vee = L(2)$. We obtain the assertion in the general case if we replace L, M in the considerations above by the rescaled lattices $L(2)$ and $M(2)$. $\square$

Up to now we have not been concerned much with the concrete realization of the lattice L . We have already used the isometry of the root lattices D_2 and $2A_1$. In Section 1.8 D_2 was realized as the $\mathbb{Z}$ -module generated by $\varepsilon_2 + \varepsilon_1$ and $\varepsilon_2 - \varepsilon_1$ equipped with the standard scalar product and $2A_1$ was considered as $\mathbb{Z}^2$ equipped with the quadratic form $(x, x) = 2x_1^2 + 2x_2^2$. Another possible realization of this lattice is $(\sqrt{2}\mathbb{Z}^2, (\cdot, \cdot))$. Because of the importance of this concept for later purposes we introduce some notion to distinguish between the isometry class and the model for a lattice.

Definition 2.2.2 Let L be a positive definite integral lattice equipped with a bilinear form $(\cdot, \cdot)_L$. Let $\text{Cl}(L)$ be the class of isometries of L , i.e.

$$\text{Cl}(L) := \{M \mid M \text{ a lattice in } L \otimes \mathbb{R} \text{ and there ex. a bijective isometry } \sigma_M : M \rightarrow L\}.$$

A *model* of L is a pair (M, b) where $M \in \text{Cl}(L)$, $M \subseteq \mathbb{R}^{\text{rank}(L)}$ and $b : M \rightarrow \mathbb{Z}$ is a bilinear form such that

$$\forall x, y \in M : (\sigma_M(x), \sigma_M(y))_L = b(x, y).$$

For convenience we will often identify root lattices with their classes so we may write $2A_1$ for the root lattice and likewise for the class $\text{Cl}(2A_1) = \text{Cl}(D_2)$. We again denote by $\varepsilon_1, \dots, \varepsilon_N$ the standard basis of $\mathbb{R}^N$ and we fix the notation $(\cdot, \cdot)_N$ for the standard scalar product on $\mathbb{R}^N$. Let us fix our standard models for the lattices D_N and NA_1 for $N \geq 1$:

$(\langle \varepsilon_1, \dots, \varepsilon_N \rangle_{\mathbb{Z}}, 2(\cdot, \cdot)_N)$	Basic model for NA_1
$(\langle \varepsilon_2 + \varepsilon_1, \varepsilon_2 - \varepsilon_1, \varepsilon_3 - \varepsilon_2, \dots, \varepsilon_N - \varepsilon_{N-1} \rangle_{\mathbb{Z}}, (\cdot, \cdot)_N)$	Basic model for $D_N, N \geq 3$

Figure 2.1: Basic models for D_N and NA_1

We want to mention the importance of this notion for the construction of Jacobi forms.

Remark 2.2.3 We consider again the two different models for A_1 given by $\mathcal{M}(A_1) = (\mathbb{Z}\varepsilon_1, 2x^2)$ and $\mathcal{M}'(A_1) = (\mathbb{Z}(\sqrt{2}\varepsilon_1), x^2)$. We consider the Jacobi form

$$\phi_{\mathcal{M}(A_1)}(\tau, z\varepsilon_1) := \vartheta(\tau, z) \quad , \quad \phi_{\mathcal{M}(A_1)} \in J_{\frac{1}{2}, A_1; \frac{1}{2}}(\chi)$$

as already seen in Example 2.1.17 where χ is some character for the Jacobi group. The function $\phi_{\mathcal{M}(A_1)}(\tau, \cdot)$ satisfies

$$\begin{aligned} & \phi_{\mathcal{M}(A_1)}(\tau, z\varepsilon_1 + x\varepsilon_1 + y\varepsilon_1) \\ &= (-1)^{x+y} \exp(-\pi i((x\varepsilon_1, x\varepsilon_1)_{\mathcal{M}(A_1)}\tau + 2(x, z\varepsilon_1)_{\mathcal{M}(A_1)})) \phi_{\mathcal{M}(A_1)}(\tau, z\varepsilon_1) \end{aligned}$$

If one would like to construct the same function with respect to the coordinate system of the model $\mathcal{M}'(A_1)$ by using ϑ one certainly obtains as a necessary condition

$$\phi_{\mathcal{M}'(A_1)}(\tau, \sqrt{2}z\varepsilon_1) = \vartheta(\tau, \sqrt{2}z)$$

from $\phi_{\mathcal{M}(A_1)}(\tau, z\varepsilon_1) = \phi_{\mathcal{M}'(A_1)}(\tau, (z/\sqrt{2})\sqrt{2}\varepsilon_1)$. Now we meet with the obstacle to determine

$$\phi_{\mathcal{M}(A_1)}(\tau, z(\sqrt{2}\varepsilon_1) + x\tau(\sqrt{2}\varepsilon_1) + y(\sqrt{2}\varepsilon_1)) = \vartheta(\tau, z + \sqrt{2}x\tau + \sqrt{2}y)$$

from the definition of $\vartheta(\tau, z)$. So we see that the model $\mathcal{M}'(A_1)$ is not compatible with the transformation properties of ϑ .

Let $\sigma : L \rightarrow \sigma L \in \text{Cl}(L)$ be an isometry. Then σ extends to an isometry $L_2(L) \rightarrow L_2(\sigma L)$ and to an isomorphism of the corresponding quadratic spaces $L_2(L) \otimes \mathbb{C} \rightarrow L_2(\sigma L) \otimes \mathbb{C}$. Moreover for $Z = (\omega, \mathfrak{z}, \tau) \in \mathcal{H}(L_1)$ we obtain an isomorphism of the half-spaces by

$$\mathcal{H}(L_1(L)) \rightarrow \mathcal{H}(L_1(\sigma L)) \quad , \quad Z \mapsto \sigma Z = (\omega, \sigma \mathfrak{z}, \tau) .$$

If $\Gamma \leq \text{O}(L_2(L))^+$ is a subgroup then σ induces an isomorphism of groups

$$\Gamma \rightarrow \Gamma^\sigma \leq \text{O}(L_2(\sigma L))^+ \quad , \quad g \mapsto \sigma \circ g \circ \sigma^{-1}$$

and a commuting diagram

$$\begin{array}{ccc} \mathcal{D}(L_2(L))^\bullet & \longrightarrow & \mathcal{D}(L_2(L))^\bullet \\ \downarrow & & \downarrow \\ \mathcal{D}(L_2(\sigma L))^\bullet & \longrightarrow & \mathcal{D}(L_2(\sigma L))^\bullet \end{array} \qquad \begin{array}{ccc} \text{pr}(Z) & \xrightarrow{g} & g \text{pr}(Z) \\ \sigma \downarrow & & \downarrow \sigma \\ \text{pr}(\sigma Z) & \xrightarrow{g^\sigma} & g^\sigma \text{pr}(\sigma Z) \end{array}$$

since $\text{pr}(\sigma Z) = \sigma \text{pr}(Z)$ and $g^\sigma \text{pr}(\sigma Z) = (\sigma \circ g) \text{pr}(Z)$. Here we denoted the extended isomorphisms by σ as well. Here the action of σ on $\text{pr}(Z)$ is again defined as the extension of the action of σ on $L \otimes \mathbb{C}$ by acting trivially on the four hyperbolic components. Let χ be a character for the subgroup Γ . Then χ induces a character χ^σ for Γ^σ defined by $\chi^\sigma(g) = \chi(\sigma^{-1} \circ g \circ \sigma)$. Two models for a lattice L induce an isomorphism of the corresponding spaces of Jacobi forms and modular forms.

Lemma 2.2.4 *Let L be a positive definite even lattice and according to Definition 2.2.2 let*

$$\sigma : L \rightarrow \sigma L \in \text{Cl}(L)$$

be an isometry. Then we have the following isomorphisms

$$\begin{aligned} J_{k,L;t}(\lambda) &\xrightarrow{\sim} J_{k,\sigma L;t}(\lambda^\sigma) \quad , \quad \varphi \mapsto ((\tau, \mathfrak{z}) \mapsto \varphi(\tau, \sigma^{-1}\mathfrak{z})) \\ \mathcal{M}_k(\Gamma, \chi) &\xrightarrow{\sim} \mathcal{M}_k(\Gamma^\sigma, \chi^\sigma) \quad , \quad F \mapsto (Z \mapsto F(\sigma^{-1}Z)) \end{aligned}$$

and for the factor of automorphy the identity $J(g, Z) = J(g^\sigma, \sigma Z)$ is valid for $g \in \Gamma$ and $Z \in \mathcal{H}(L_1)$.

Proof

The transformation property of $\varphi(\tau, \sigma^{-1}\mathfrak{z})$ under elements of $\mathrm{SL}(2, \mathbb{Z})$ is clear. Let $x, y \in \sigma L$. From $\sigma^{-1} \circ [x, y; \frac{1}{2}(x, y)] \circ \sigma = [\sigma^{-1}x, \sigma^{-1}y; \frac{1}{2}(x, y)]$ we obtain

$$\lambda^\sigma|_{H_s(\sigma L)}([x, y; \frac{1}{2}(x, y)]) = \lambda|_{H_s(L)}([\sigma^{-1}x, \sigma^{-1}y; \frac{1}{2}(\sigma^{-1}x, \sigma^{-1}y)])$$

which shows that $\varphi(\tau, \sigma^{-1}\mathfrak{z})$ also transforms correctly under $H(L)$. Now we want to investigate the transformation properties for $F(\sigma^{-1}Z)$. Let $g \in \Gamma$ and $Z \in \mathcal{H}(L_1)$. From the above constructions we deduce

$$J(g^\sigma, \sigma Z) \operatorname{pr}(g^\sigma < \sigma Z >) = g^\sigma \operatorname{pr}(\sigma Z) = \sigma(g \operatorname{pr}(Z)) = J(g, Z) \sigma(\operatorname{pr}(g < Z >))$$

A comparison of the last components on both sides of the equation yields $J(g^\sigma, \sigma Z) = J(g, Z)$ and consequently $\sigma(g < Z >) = g^\sigma < \sigma Z >$. Hence we obtain

$$F(\sigma^{-1}(g^\sigma < \sigma Z >)) = \chi(g)J(g, Z)^k F(Z) = \chi^g(g^\sigma)J(g^\sigma, \sigma Z)F(\sigma^{-1}(\sigma Z)).$$

According to Lemma 1.6.4 the assertion is proved. $\square$

2.3 Theta type Jacobi forms for the lattice $4A_1$

We now construct a series of pullbacks for the lattices NA_1 where $N = 1, \dots, 4$. In [G1] Gritsenko constructed a series of reflective modular forms by means of Jacobi forms. At the moment we are only interested in Jacobi forms. The following lemma repeats Gritsenko's construction. Throughout this section we assume that NA_1 is represented by the standard model from the last section.

Lemma 2.3.1 *We consider the standard model of $4A_1$ from figure 2.1 and write*

$$\mathfrak{z}_{NA_1} = \sum_{j=1}^N z_j \varepsilon_j \in NA_1 \otimes \mathbb{C} \quad , \quad N = 1, \dots, 4.$$

We define

$$\vartheta_{NA_1} : \mathbb{H} \times (NA_1 \otimes \mathbb{C}) \rightarrow \mathbb{C} \quad , \quad \vartheta_{NA_1}(\tau, \mathfrak{z}_{NA_1}) := \vartheta(\tau, z_1) \cdots \vartheta(\tau, z_N).$$

Then we have $\vartheta_{NA_1} \in J_{N/2, NA_1; \frac{1}{2}}(v_\eta^{3N} \times \nu_{H(NA_1)})$ where

$$\nu_{H(NA_1)}([x, y : r]) = (-1)^{r + \sum_{j=1}^N x_j + y_j + x_j y_j} \quad \text{for all } [x, y : r] \in H(NA_1).$$

Proof

We first show that ϑ_{NA_1} transforms like a Jacobi form with the given character.

For $A = \begin{pmatrix} a & b \\ c & d \end{pmatrix} \in \mathrm{SL}(2, \mathbb{Z})$ we have

$$\begin{aligned} \vartheta_{NA_1} \left(\frac{a\tau + b}{c\tau + d}, \frac{\mathfrak{z}_{NA_1}}{c\tau + d} \right) &= \vartheta \left(\frac{a\tau + b}{c\tau + d}, \frac{z_1}{c\tau + d} \right) \cdots \vartheta \left(\frac{a\tau + b}{c\tau + d}, \frac{z_N}{c\tau + d} \right) \\ &= v_\eta^3(A)(c\tau + d)^{\frac{1}{2}} \vartheta(\tau, z_1) \cdots v_\eta^3(A)(c\tau + d)^{\frac{1}{2}} \vartheta(\tau, z_N) \\ &\times \exp \left(\pi i \frac{cz_1^2}{c\tau + d} \right) \cdots \exp \left(\pi i \frac{cz_N^2}{c\tau + d} \right) \\ &= v_\eta(A)^{3N} (c\tau + d)^{N/2} \exp \left(\frac{\pi i}{2} \frac{c2(\mathfrak{z}, \mathfrak{z})_N}{c\tau + d} \right) \vartheta_{NA_1}(\tau, \mathfrak{z}_{NA_1}) \end{aligned}$$

and for $x, y \in NA_1$ we obtain

$$\begin{aligned} &\vartheta_{NA_1}(\tau, \mathfrak{z}_{NA_1} + x\tau + y) \\ &= \vartheta(\tau, z_1 + x_1\tau + y_1) \cdots \vartheta(\tau, z_N + x_N\tau + y_N) \\ &= (-1)^{x_1+y_1} \exp(-\pi i(x_1^2\tau + 2x_1z_1)) \vartheta(\tau, z_1) \cdots \\ &\times (-1)^{x_N+y_N} \exp(-\pi i(x_N^2\tau + 2x_Nz_N)) \vartheta(\tau, z_N) \\ &= (-1)^{\sum_{j=1}^N x_j+y_j} \exp(-\pi i(1/2)[2(x, x)_N\tau + 2 \cdot 2(x, \mathfrak{z}_{NA_1})_N]) \vartheta_{NA_1}(\tau, \mathfrak{z}_{NA_1}) \end{aligned}$$

and $\nu_{H(NA_1)}([x, y : -\frac{1}{2}(x, y)]) = (-1)^{\sum_{j=1}^N x_j+y_j}$ according to Lemma 2.1.5 and Example 2.1.17. From the same example we know that $\vartheta(\tau, \mathfrak{z}_{A_1}) := \vartheta(\tau, z)$ is a holomorphic Jacobi form of weight $1/2$ and index $1/2$ for the lattice A_1 and the Fourier expansion given there yields

$$\vartheta_{NA_1}(\tau, \mathfrak{z}_{NA_1}) = \sum_{\substack{n \in \mathbb{Q}_{\geq 0}, n \equiv \frac{3N}{24} \pmod{\mathbb{Z}} \\ l \in (\frac{\mathbb{Z}}{4})^N \\ n - 2(l, l)_N \geq 0}} f(n, l) q^n \exp(2\pi i \cdot 2(l, \mathfrak{z}_{NA_1})_N)$$

which completes the proof. $\square$

We hence obtain four Jacobi forms of singular weight transforming like a Jacobi form with some power of the multiplier system v_η . For the construction of further Jacobi forms we consider the following functions. In [EZ] the authors constructed the Jacobi-Eisenstein series $e_{k,1} \in J_{k,A_1;1}$ with Fourier expansion

$$e_{k,1}(\tau, \mathfrak{z}_{A_1}) = \sum_{l \in A_1} q^{2(l, l)_1} e^{2\pi i \cdot 2(l, z)_1} + \sum_{\substack{n \in \mathbb{N}_0, l \in A_1^\vee \\ 2n - 2(l, l)_1 > 0}} \frac{L(2 - k, 4(l, l)_1 - 4n)}{\zeta(3 - 2k)} q^n e^{2\pi i \cdot 2(l, z)_1}$$

where L is a generalized Dirichlet L -series. One defines a Jacobi cusp form $\varphi_{12,A_1} \in J_{12,A_1;1}$ as

$$\varphi_{12,A_1}(\tau, \mathfrak{z}_{A_1}) := \frac{1}{144} (e_4^2(\tau) e_{4,1}(\tau, \mathfrak{z}_{A_1}) - e_6(\tau) e_{6,1}(\tau, \mathfrak{z}_{A_1}))$$

and

$$\phi_{12,1}(\tau, z) := \varphi_{12,A_1}(\tau, z\varepsilon_1)$$

where

$$e_k(\tau) = 1 - \frac{2k}{B_k} \cdot \sum_{n=1}^{\infty} \sigma_{k-1}(n) q^n$$

denotes the classical Eisenstein series of weight k for the group $\mathrm{SL}(2, \mathbb{Z})$, confer e.g. [KK], p. 161. Using this functions we can define

$$\varphi_{0,1}(\tau, \mathfrak{z}_{A_1}) := \frac{\varphi_{12,A_1}(\tau, \mathfrak{z}_{A_1})}{\eta^{24}(\tau)} = \frac{e_4^2(\tau) e_{4,1}(\tau, \mathfrak{z}_{A_1}) - e_6(\tau) e_{6,1}(\tau, \mathfrak{z}_{A_1})}{\frac{1}{12} (e_4^3(\tau) - e_6^2(\tau))}$$

and

$$\phi_{0,1}(\tau, z) := \varphi_{0,1}(\tau, z\varepsilon_1),$$

respectively. In accordance with the classical theory of elliptic modular forms we write

$$\Delta_{12}(\tau) := \eta^{24}(\tau)$$

for the first cusp form for $\mathrm{SL}(2, \mathbb{Z})$.

Proposition 2.3.2 (a) $\varphi_{0,1} \in J_{0,A_1;1}^{(weak)}(1)$ and one has

$$\phi_{0,1}(\tau, z) = \sum_{\substack{n,s \in \mathbb{Z}, n \geq 0 \\ 4n-s^2 \geq -1}} q^n r^s = (r + 10 + r^{-1}) + q(\dots) \quad , \quad r = \exp(2\pi i z).$$

(b) The function defined by $(\tau, \mathfrak{z}_{A_1}) \mapsto \eta^6(\tau) \varphi_{0,1}(\tau, \mathfrak{z}_{A_1})$ is a holomorphic Jacobi form of weight 3 and index 1 for the lattice A_1 with Fourier expansion

$$\eta^6(\tau) \phi_{0,1}(\tau, z) = q^{\frac{1}{4}}(r + 10 + r^{-1}) + q^{\frac{5}{4}}(\dots).$$

Proof

(a) From the definition it is clear that the function transforms like an element of $J_{0,A_1;1}^{(weak)}(1)$. From the Fourier expansions of $e_{k,1}, e_k$ one obtains (comp. [EZ], p.39) $\Delta(\tau) = q + q^2(\dots)$ and $\phi_{12,1}(\tau, z) = (r + 10 + r^{-1})q + q^2(\dots)$ which shows that $\varphi_{0,1}$ is a weak Jacobi form with the above expansion.

(b) From Example 2.1.17 we know the Fourier expansion of $\eta(\tau)^3$ which implies

$$\eta(\tau)^6 = \sum_{n \in \mathbb{Q}_{\geq 0}, n \equiv \frac{1}{4} \pmod{\mathbb{Z}}} \alpha(n) q^n$$

where $\alpha(1/4) = 1$. For $m \in \mathbb{Q}_{\geq 0}$ such that $\alpha(m) \neq 0$ and any term

$$q^n \exp(2\pi i 2(l, \mathfrak{z}_{A_1})_1)$$

in the Fourier expansion of $\varphi_{0,1}(\tau, z)$ with non-vanishing coefficient we obtain

$$2(m+n) - (l, l) = 2m + (2n - (l, l)) \geq 2\frac{1}{4} - \frac{1}{2} = 0.$$

Hence

$$\eta(\tau)^6 \varphi_{0,1}(\tau, \mathfrak{z}_{A_1}) = \sum_{\substack{n \in \mathbb{Q}_{\geq 0}, n \equiv \frac{1}{4} \pmod{\mathbb{Z}} \\ l \in A_1^\vee \\ 2n - 2(l, l)_1 \geq 0}} f(n, l) q^n \exp(2\pi i \cdot 2(l, \mathfrak{z}_{A_1})_1)$$

is a holomorphic Jacobi form. $\square$

Assume the lattice L is the orthogonal sum $L = M \oplus M'$. Then every element $l \in L$ has a unique representation as $l = l_M \oplus l_{M'}$ and l_M is the image under the projection $L \rightarrow M$.

Lemma 2.3.3 *Assume we have a decomposition $L = M \oplus M'$ as orthogonal sublattices. Then for each $t \in \mathbb{Z}_{\geq 0}$ we have an operator*

$$J_{k,M;t} \times J_{k',M';t} \rightarrow J_{k+k',L;t} \quad , \quad \varphi \otimes \varphi'(\tau, \mathfrak{z}_L) = \varphi(\tau, \mathfrak{z}_M) \varphi'(\tau, \mathfrak{z}_{M'}).$$

Proof

For all $A = \begin{pmatrix} a & b \\ c & d \end{pmatrix} \in \mathrm{SL}(2, \mathbb{Z})$ we have

$$\varphi \otimes \varphi' \left(\frac{a\tau + b}{c\tau + d}, \frac{\mathfrak{z}_L}{c\tau + d} \right) = (c\tau + d)^{k+k'} e^{\pi i \left(t \frac{c(\mathfrak{z}_M, \mathfrak{z}_M)}{c\tau + d} + t \frac{c(\mathfrak{z}_{M'}, \mathfrak{z}_{M'})}{c\tau + d} \right)} \varphi(\tau, \mathfrak{z}_M) \varphi'(\tau, \mathfrak{z}_{M'})$$

and for $x = x_M \oplus x_{M'}, y = y_M \oplus y_{M'} \in L$

$$\begin{aligned} \varphi \otimes \varphi'(\tau, \mathfrak{z}_L + x\tau + y) &= e^{-2\pi i t (\frac{1}{2}(x_M, x_M)\tau + (x_M, \mathfrak{z}_{M'}))} e^{-2\pi i t (\frac{1}{2}(x_{M'}, x_{M'})\tau + (x_{M'}, \mathfrak{z}_{M'}))} \\ &\times \varphi(\tau, \mathfrak{z}_M) \varphi'(\tau, \mathfrak{z}_{M'}) \\ &= e^{-2\pi i t (\frac{1}{2}(x, x)\tau + (x, \mathfrak{z}))} \cdot \varphi \otimes \varphi'(\tau, \mathfrak{z}_L). \end{aligned}$$

By assumption the functions φ, φ' have the following kind of Fourier expansion

$$\begin{aligned} \varphi(\tau, \mathfrak{z}_M) &= \sum_{\substack{n \in \mathbb{N} \\ l \in M^\vee \\ 2nt - (l, l) \geq 0}} f(n, l) q^n \exp(2\pi i(l, \mathfrak{z}_M)) \\ \varphi(\tau, \mathfrak{z}_{M'}) &= \sum_{\substack{n' \in \mathbb{N} \\ l' \in (M')^\vee \\ 2n't - (l', l') \geq 0}} g(n', l') q^{n'} \exp(2\pi i(l', \mathfrak{z}_{M'})). \end{aligned}$$

Therefore only terms of the shape $q^{n+n'} \exp(2\pi i(l, \mathfrak{z}_M)) \exp(2\pi i(l', \mathfrak{z}_{M'}))$ appear in the expansion of $\varphi \otimes \varphi'$ where $2(n+n')t - (l+l', l+l') \geq 0$. $\square$

Note that the same type of construction works for holomorphic Jacobi forms of half-integral index and a binary character with obvious modifications. In the next step we apply this construction to Jacobi forms for the lattice NA_1 .

Lemma 2.3.4 *Let $\varphi \in J_{k,NA_1;1}^{(weak)}(1)$ where $N \geq 1$. Then the function*

$$\varphi \otimes \phi_{0,1} : \mathbb{H} \times ((N+1)A_1 \otimes \mathbb{C}) \rightarrow \mathbb{C} \quad , \quad \varphi \otimes \phi_{0,1}(\tau, \mathfrak{z}_{(N+1)A_1}) := \varphi(\tau, \mathfrak{z}_{NA_1})\phi_{0,1}(\tau, z_{N+1})$$

is an element of $J_{k,(N+1)A_1;1}^{(weak)}(1)$. Moreover we have

$$\varphi \otimes \phi_{0,1} \downarrow_{NA_1} \in \mathbb{C}^\times \varphi.$$

Proof

We only have to investigate the transformation behaviour under $H((N+1)A_1)$. From Proposition 2.3.2 we obtain for all $x, y \in (N+1)A_1$

$$\begin{aligned} & \varphi \otimes \phi_{0,1}(\tau, \mathfrak{z}_{(N+1)A_1} + x\tau + y) \\ &= \varphi(\tau, \mathfrak{z}_{NA_1} + x_{NA_1}\tau + y_{NA_1})\phi_{0,1}(\tau, z_{N+1} + x_{N+1}\tau + y_{N+1}) \\ &= \exp(-\pi i((x_{NA_1}, x_{NA_1})\tau + 2(x_{NA_1}, \mathfrak{z}_{NA_1}))) \exp(-2\pi i(x_{N+1}^2 + 2x_{N+1}z_{N+1})) \\ &\quad \times \varphi(\tau, \mathfrak{z}_{NA_1})\phi_{0,1}(\tau, z_{N+1}) \\ &= \exp(-\pi i((x, x)\tau + 2(x, \mathfrak{z}_{(N+1)A_1}))) \cdot \varphi \otimes \phi_{0,1}(\tau, \mathfrak{z}_{(N+1)A_1}) \end{aligned}$$

by using our standard model for NA_1 in figure 2.1. The statement on the pullbacks follows from $\phi_{0,1}(\tau, 0) = 12$. $\square$

The last lemma shows that there exists an operator

$$\otimes_{NA_1} \phi_{0,1} : J_{k,NA_1;1}^{(weak)}(1) \rightarrow J_{k,(N+1)A_1;1}^{(weak)}(1) \quad , \quad \varphi \mapsto \varphi \otimes \phi_{0,1}. \quad (2.7)$$

which extends to Jacobi forms of index zero being isomorphic to $\mathcal{M}_k(\mathrm{SL}(2, \mathbb{Z}))$ by the same construction

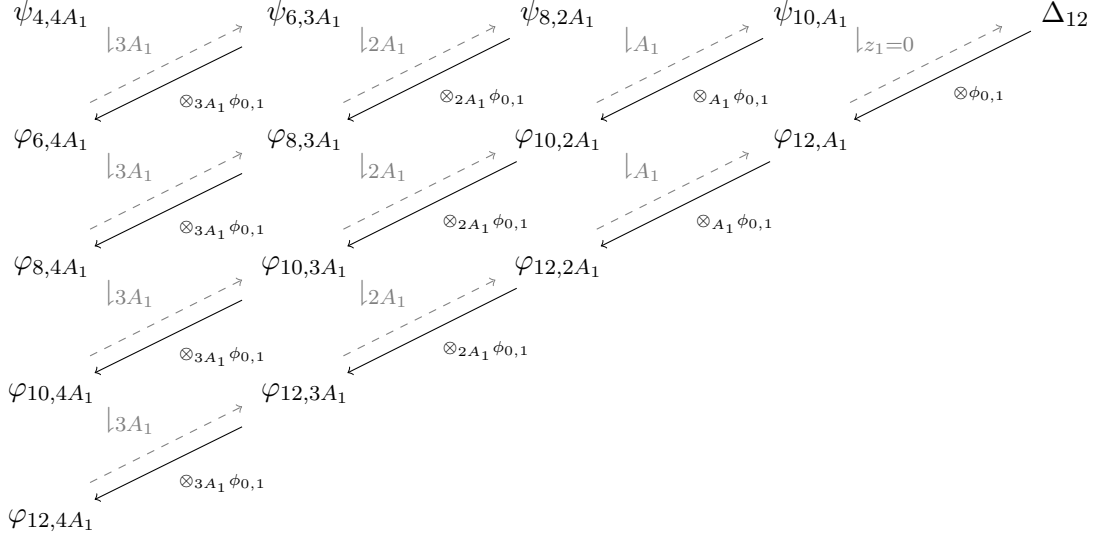
$$\otimes \phi_{0,1} : J_{k,L;0}^{(weak)}(1) \rightarrow J_{k,A_1;1}^{(weak)}(1).$$

Following [G1] we define the following functions for $N = 1, \dots, 4$

$$\begin{aligned} \varrho_{6-N,NA_1} : \mathbb{H} \times (NA_1 \otimes \mathbb{C}) &\rightarrow \mathbb{C} \quad , \quad \varrho_{6-N,NA_1}(\tau, \mathfrak{z}_{NA_1}) := \eta(\tau)^{12-3N} \vartheta_{NA_1}(\tau, \mathfrak{z}_{NA_1}) \\ \psi_{12-2N,NA_1} &:= \varrho_{6-N,NA_1}^2. \end{aligned}$$

Then obviously $\varrho_{6-N,NA_1} \in J_{6-N,NA_1;\frac{1}{2}}(v_\eta^{12} \times \nu_{H(NA_1)})$ and $\psi_{12-2N,NA_1} \in J_{12-2N,NA_1;1}$.

Proposition 2.3.5 *Let $\varepsilon_1, \dots, \varepsilon_4$ be the standard basis of $\mathbb{R}^4$. We identify NA_1 with $(\langle \varepsilon_1, \dots, \varepsilon_N \rangle_{\mathbb{Z}}, 2(\cdot, \cdot)_4)$ for $N = 1, \dots, 4$. This yields a chain of sublattices $A_1 \leq 2A_1 \leq 3A_1 \leq 4A_1$. We have the following diagram of holomorphic Jacobi forms*



where $\varphi_{k,NA_1} \in J_{k,NA_1;1}(1)$. If $N < 4$ then φ_{k,NA_1} is a cusp form.

Proof

The first assertion follows from Proposition 2.3.2, Lemma 2.3.3 and (2.7). The assertion on cusp forms follows from the fact that $\eta(\tau)$ is a cusp form such that φ_{k,NA_1} is a product of a holomorphic Jacobi form and a Jacobi cusp form whenever $N < 4$. $\square$

We have already seen in Section 1.9 that all permutations of the components $z_1, \dots, z_N$ of $\mathfrak{z}_{NA_1}$ are induced by elements of $O(L_2(NA_1))^+$. The symmetric group on N letters acts on $J_{k,NA_1;t}$ from the left by permutations of the abelian variables, i.e.

$$\pi.\varphi(\tau, \mathfrak{z}_{NA_1}) := \varphi(\tau, z_{\pi^{-1}(1)}\varepsilon_1 + \dots + z_{\pi^{-1}(N)}\varepsilon_N) \quad , \pi \in \mathcal{S}_N.$$

If $g \in O(NA_1)$ induces π by virtue of the natural epimorphism

$$O(NA_1) \twoheadrightarrow O(D(NA_1)) \cong \mathcal{S}_N$$

then we have

$$\pi.\varphi = \varphi|_{k,t}g.$$

Definition 2.3.6 Let $\varphi \in J_{k,NA_1;t}(1)$. We call φ *symmetric* if φ is fixed under the action of $\mathcal{S}_N$, i.e.

$$\pi.\varphi = \varphi \quad \text{for all } \pi \in \mathcal{S}_N.$$

The function φ is called *anti-symmetric* if

$$\pi.\varphi = \text{sign}(\pi)\varphi \quad \text{for all } \pi \in \mathcal{S}_N.$$

The space of symmetric Jacobi forms is denoted by $J_{k,NA_1;t}(1)^+$ and the space of anti-symmetric Jacobi forms is denoted by $J_{k,NA_1;t}(1)^-$.

In the previous definition we have used our standard model for NA_1 . By virtue of Lemma 2.2.4 the notion of a symmetric respectively anti-symmetric Jacobi form does in fact not depend on the choice of the basis of NA_1 . We have two projectors

$$J_{k,NA_1;t}(1) \rightarrow J_{k,NA_1;t}(1)^+, \quad \varphi^+ := \frac{1}{|S_N|} \sum_{\pi \in S_N} \pi \cdot \varphi$$

$$J_{k,NA_1;t}(1) \rightarrow J_{k,NA_1;t}(1)^-, \quad \varphi^- := \frac{1}{|S_N|} \sum_{\pi \in S_N} \text{sign}(\pi) \pi \cdot \varphi$$

We now consider a different model for NA_1 which will give us more Jacobi forms. Let $N \in \mathbb{N}$ be an even positive integer. Then we consider the model

$$(\varepsilon_2 - \varepsilon_1, \varepsilon_2 + \varepsilon_1, \dots, \varepsilon_N - \varepsilon_{N-1}, \varepsilon_N + \varepsilon_{N-1}, (\cdot, \cdot)_N).$$

For convenience we will denote this basis by $\mathcal{B}_{NA_1}$. The Jacobi forms $\vartheta_{D_4}, \vartheta_{D_4}^{(2)}, \vartheta_{D_4}^{(3)}$ induce three Jacobi forms for the lattice $4A_1 \leq D_4$.

Proposition 2.3.7 *The following functions define holomorphic Jacobi forms:*

(a) *Define*

$$\begin{aligned} \vartheta_{4A_1}^{(1)}(\tau, \mathfrak{z}_{\mathcal{B}_{4A_1}}) &= \vartheta(\tau, z_1 - z_2) \vartheta(\tau, z_1 + z_2) \vartheta(\tau, z_3 - z_4) \vartheta(\tau, z_3 + z_4) \\ \vartheta_{4A_1}^{(2)}(\tau, \mathfrak{z}_{\mathcal{B}_{4A_1}}) &= \vartheta(\tau, z_3 - z_2) \vartheta(\tau, z_3 + z_2) \vartheta(\tau, z_1 - z_4) \vartheta(\tau, z_1 + z_4) \\ \vartheta_{4A_1}^{(3)}(\tau, \mathfrak{z}_{\mathcal{B}_{4A_1}}) &= \vartheta(\tau, z_1 - z_3) \vartheta(\tau, z_1 + z_3) \vartheta(\tau, z_2 - z_4) \vartheta(\tau, z_2 + z_4). \end{aligned}$$

Then we have $\vartheta_{4A_1}^{(1)}, \vartheta_{4A_1}^{(2)}, \vartheta_{4A_1}^{(3)} \in J_{2,4A_1;1}(v_\eta^{12})$.

(b) *Define*

$$\vartheta_{2A_1}^{(1)}(\tau, \mathfrak{z}_{\mathcal{B}_{2A_1}}) = \vartheta(\tau, z_2 - z_1) \vartheta(\tau, z_2 + z_1).$$

Then we have $\vartheta_{2A_1}^{(1)} \in J_{1,2A_1;1}(v_\eta^6)$.

Proof

For all $A = \begin{pmatrix} a & b \\ c & d \end{pmatrix} \in \text{SL}(2, \mathbb{Z})$ we have

$$\begin{aligned} \vartheta_{4A_1}^{(1)}\left(\frac{a\tau + b}{c\tau + d}, \frac{\mathfrak{z}_{\mathcal{B}_{4A_1}}}{c\tau + d}\right) &= \exp\left(\pi i \frac{c(z_2 - z_1)^2}{c\tau + d}\right) \cdots \exp\left(\pi i \frac{c(z_4 + z_3)^2}{c\tau + d}\right) \\ &\times v_\eta^{12}(A)(c\tau + d)^2 \vartheta(\tau, z_2 - z_1) \cdots \vartheta(\tau, z_4 + z_3) \end{aligned}$$

and

$$(\mathfrak{z}_{\mathcal{B}_{4A_1}}, \mathfrak{z}_{\mathcal{B}_{4A_1}})_4 = 2(z_1^2 + \cdots + z_4^2) = (z_2 - z_1)^2 + (z_2 + z_1)^2 + (z_4 - z_3)^2 + (z_4 + z_3)^2$$

proving that $\vartheta_{4A_1}^{(1)}$ has the right transformation behaviour under $\text{SL}(2, \mathbb{Z})$. Using the last equation we also see that the transformation behaviour under the Heisenberg group is the claimed one. Analogue calculations are used to verify the transformation behaviour of the other forms. Since $\mathcal{B}_{4A_1}$ is an orthogonal basis the claim follows from Lemma 2.3.3. $\square$

In the next lemma we construct three non-trivial anti-symmetric Jacobi forms using the functions from the last Proposition.

Proposition 2.3.8 (a) We have a function $\varrho_{10,2A_1} \in J_{10,2A_1;1}(1)^-$ defined by

$$\varrho_{10,2A_1}(\tau, \mathfrak{z}_{\mathcal{B}_{2A_1}}) = \eta(\tau)^{18} \vartheta_{2A_1}^{(1)}(\tau, \mathfrak{z}_{\mathcal{B}_{2A_1}}).$$

Moreover, $-144\varrho_{10,2A_1} = 2(e_{4,1} \otimes e_{6,1})^-$ where the tensor product is taken with respect to the basis $\mathcal{B}_{2A_1}$.

(b) There exists an anti-symmetric Jacobi form $\varrho_{6,4A_1(3)} \in J_{6,4A_1(3),1}(v_\eta^{12})^-$

$$\varrho_{6,4A_1(3)} = \vartheta_{4A_1}^{(1)} \vartheta_{4A_1}^{(2)} \vartheta_{4A_1}^{(3)}.$$

(c) There exists a non-vanishing Jacobi form $\varrho_{6,3A_1(3)} \in J_{6,3A_1;3}(v_\eta^{12})^-$ given by

$$\varrho_{6,3A_1(3)} = \varrho_{6,4A_1(3)} \downarrow_{3A_1}.$$

Proof

(a) We have seen in Example 2.1.17 that $\vartheta(\tau, -z) = -\vartheta(\tau, z)$. This shows that $\varrho_{10,2A_1}$ is an anti-symmetric Jacobi form which vanishes exactly on the set

$$\{\mathfrak{z}_{\mathcal{B}_{2A_1}}; z_2 - z_1 \in \mathbb{Z}\tau + \mathbb{Z} \text{ or } z_2 + z_1 \in \mathbb{Z}\tau + \mathbb{Z}\}.$$

By construction we have

$$2(e_{4,1} \otimes e_{6,1})^-(\tau, \mathfrak{z}_{\mathcal{B}_{2A_1}}) = e_{4,1}(\tau, z_1)e_{6,1}(\tau, z_2) - e_{4,1}(\tau, z_2)e_{6,1}(\tau, z_1)$$

and the latter function vanishes on the same set. Hence we can consider the quotient $2(e_{4,1} \otimes e_{6,1})^-/\varrho_{10,2A_1}$ which transforms like a Jacobi form of weight 0 and index 0. Since the Fourier expansion of both functions is of order q the quotient has to be constant. From the Fourier expansion of $e_{k,1}$ we obtain the above identity.

(b) Each permutation in $\mathcal{S}_4$ acts on $\varrho_{6,4A_1(3)}$ by changing the sign of exactly one of the factors $\vartheta_{4A_1}^{(1)}, \vartheta_{4A_1}^{(2)}, \vartheta_{4A_1}^{(3)}$ and by interchanging the remaining factors with the same sign. By construction this function has the indicated weight and is of index 3.

(c) This statement follows immediately from (b) where the pullback is performed with respect to the basis $\varepsilon_2 - \varepsilon_1, \varepsilon_2 + \varepsilon_1, \varepsilon_4 - \varepsilon_3$ which is a root system of type $3A_1$. $\square$

2.4 Theta type Jacobi forms for several root lattices

In this section we consider pullbacks for several root lattices. We start with another series of Jacobi forms for the lattice D_8 which was already considered in [G1]. For abbreviation we define

$$b_1 = \varepsilon_2 - \varepsilon_1, b_2 = \varepsilon_2 + \varepsilon_1, b_3 = \varepsilon_3 - \varepsilon_2, \dots, b_N = \varepsilon_N - \varepsilon_{N-1}$$

where $N \geq 3$ and $\varepsilon_1, \dots, \varepsilon_N$ denotes the first N standard basis vectors of $\mathbb{R}^k$ for some $k \geq N$. We restate Lemma 2.1.18 by using the new model for D_N .

Lemma 2.4.1 *We consider the standard model of D_8 from figure 2.1 and write*

$$\mathfrak{z}_{D_N} = \sum_{j=1}^N z_j b_j \in D_N \otimes \mathbb{C} \quad , \quad N = 1, \dots, 8.$$

We define

$$\begin{aligned} \vartheta_{D_1}(\tau, \mathfrak{z}_{D_1}) &= \vartheta(\tau, 2z_1) \quad , \quad \vartheta_{D_2}(\tau, \mathfrak{z}_{D_2}) = \vartheta(\tau, z_2 - z_1)\vartheta(\tau, z_2 + z_1) , \\ \vartheta_{D_3}(\tau, \mathfrak{z}_{D_3}) &= \vartheta(\tau, z_2 - z_1)\vartheta(\tau, z_2 + z_1 - z_3)\vartheta(\tau, z_3) \end{aligned}$$

and for $N \geq 4$

$$\vartheta_{D_N}(\tau, \mathfrak{z}_{D_N}) := \vartheta(\tau, z_2 - z_1)\vartheta(\tau, z_2 + z_1 - z_3) \cdots \vartheta(\tau, z_{N-1} - z_N)\vartheta(\tau, z_N) .$$

Then we have $\vartheta_{D_N} \in J_{N/2, D_N, 1}(v_\eta^{3N})$. In particular each ϑ_{D_N} is singular.

Proof

We first show that ϑ_{NA_1} transforms like a Jacobi form with the given character.

Let $N \geq 4$. For $A = \begin{pmatrix} a & b \\ c & d \end{pmatrix} \in \mathrm{SL}(2, \mathbb{Z})$ we have

$$\begin{aligned} & \vartheta_{D_N} \left(\frac{a\tau + b}{c\tau + d}, \frac{\mathfrak{z}_{D_N}}{c\tau + d} \right) \\ &= \vartheta \left(\frac{a\tau + b}{c\tau + d}, \frac{z_2 - z_1}{c\tau + d} \right) \vartheta \left(\frac{a\tau + b}{c\tau + d}, \frac{z_2 + z_1 - z_3}{c\tau + d} \right) \cdots \\ &\times \vartheta \left(\frac{a\tau + b}{c\tau + d}, \frac{z_{N-1} - z_N}{c\tau + d} \right) \vartheta \left(\frac{a\tau + b}{c\tau + d}, \frac{z_N}{c\tau + d} \right) \\ &= v_\eta^{3N}(A)(c\tau + d)^{\frac{N}{2}} \vartheta(\tau, z_2 - z_1) \vartheta(\tau, z_2 + z_1 - z_3) \cdots \vartheta(\tau, z_N - z_{N-1}) \vartheta(\tau, z_N) \\ &\times \exp \left(\pi i \frac{c(z_2 - z_1)^2}{c\tau + d} \right) \exp \left(\pi i \frac{c(z_2 + z_1 - z_3)^2}{c\tau + d} \right) \cdots \\ &\times \exp \left(\pi i \frac{c(z_{N-1} - z_N)^2}{c\tau + d} \right) \exp \left(\pi i \frac{cz_N^2}{c\tau + d} \right) \\ &= v_\eta(A)^{3N} (c\tau + d)^{N/2} \exp \left(\pi i \frac{c(\mathfrak{z}_{D_N}, \mathfrak{z}_{D_N})N}{c\tau + d} \right) \vartheta_{D_N}(\tau, \mathfrak{z}_{D_N}) \end{aligned}$$

and for $x, y \in D_N$ we obtain

$$\begin{aligned} & \vartheta_{D_N}(\tau, \mathfrak{z}_{D_N} + x\tau + y) \\ &= \vartheta(\tau, z_2 + x_2\tau + y_2 - z_1 - x_1\tau - y_1) \\ &\times \vartheta(\tau, z_2 + x_2\tau + y_2 + z_1 + x_1\tau + y_1 - z_3 - x_3\tau - y_3) \cdots \\ &= (-1)^{2x_2+2y_2} \exp(-\pi i((\mathfrak{z}_{D_N}, \mathfrak{z}_{D_N})_N \tau + 2(x, \mathfrak{z}_{D_N})_N)) \vartheta_{D_N}(\tau, \mathfrak{z}_{D_N}) \end{aligned}$$

where we have used the identity

$$\begin{aligned} (\mathfrak{z}_{D_N}, \mathfrak{w}_{D_N})_N &= (z_2 - z_1)(w_2 - w_1) + (z_2 + z_1 - z_3)(w_2 + w_1 - w_3) + \cdots \\ &+ (z_{N-1} - z_N)(w_{N-1} - w_N) + z_N w_N . \end{aligned}$$

The transformation properties for $N \leq 3$ are verified with similiar calculations. It remains to show that ϑ_{D_N} is a holomorphic Jacobi form. To this end we note that all terms arising in the Fourier expansion are of the shape

$$q^{\frac{n_1^2 + \dots + n_N^2}{8}} \exp(\pi i(n_1(z_2 - z_1) + \dots)) \quad \text{where } n_1, \dots, n_N \in \mathbb{Z}.$$

The lattice $D_N^\vee$ is contained in $(\frac{1}{2}\mathbb{Z}^N, (\cdot, \cdot))$ where $\mathbb{Z}^N$ is equipped with the basis

$$\varepsilon_1, \dots, \varepsilon_N.$$

Hence the terms appearing in the Fourier expansion of ϑ_{D_N} can be rewritten as

$$q^{\frac{n_1^2 + \dots + n_N^2}{8}} \exp(2\pi i(v, \mathfrak{z}_{D_N})) \quad \text{where } v := \frac{1}{2}(n_1\varepsilon_1 + \dots + n_N\varepsilon_N).$$

Finally the hyperbolic norm of the last term with respect to D_N equals

$$2\frac{n_1^2 + \dots + n_N^2}{8} - (v, v) = 0$$

showing that ϑ_{D_N} is a holomorphic Jacobi form of singular weight and index 1. $\square$

In view of pullback constructions we analyze orthogonal complements for the D_N -series.

Lemma 2.4.2 *Let $N \geq 2$. The orthogonal complement $(D_{N-1})_{D_N}^\perp$ is isomorphic to $A_1(2)$. More precisely, if we consider our standard model for D_N we have*

$$(D_{N-1})_{D_N}^\perp = \langle 2\varepsilon_N \rangle_{\mathbb{Z}}.$$

Proof

Obviously we have $D_1^\perp = \langle 2\varepsilon_2 \rangle_{\mathbb{Z}}$ so we can assume $N \geq 3$. Let us denote our standard basis for D_N by $b_1, \dots, b_N$ and let $l = \sum_{j=1}^N \alpha_j b_j \in D_N$. We have $l \in (D_{N-1})_{D_N}^\perp$ if and only if

$$(l, b_j) = 0 \quad \text{for all } j = 0, \dots, N-1.$$

This leads us to the linear system

$$\left\{ \begin{array}{cccccccc} 2\alpha_1 & & & - & \alpha_3 & & & = & 0 \\ & 2\alpha_2 & & - & \alpha_3 & & & = & 0 \\ -\alpha_1 & & \alpha_2 & + & 2\alpha_3 & - & \alpha_4 & = & 0 \\ & & & - & \alpha_3 & + & 2\alpha_4 & - & \alpha_5 & = & 0 \\ & & & & & & \ddots & & & \vdots \\ & & & & & - & \alpha_{N-2} & + & 2\alpha_{N-1} & - & \alpha_N & = & 0 \end{array} \right.$$

whose solutions exactly correspond to the coordinates of elements belonging to the orthogonal complement for D_N where $N \geq 5$. The cases D_3 and D_4 correspond to solutions

of the subsystem given by the first two and three rows, respectively. One immediately computes the module of solutions as

$$(\alpha_1, \alpha_2, \alpha_3, \dots, \alpha_N) = t(1, 1, 2, \dots, 2) \quad , \quad t \in \mathbb{Z}.$$

This yields $(D_{N-1})_{D_N}^\perp = \langle 2\varepsilon_N \rangle_{\mathbb{Z}}$ which is isomorphic to $A_1(2)$. $\square$

Following Gritsenko [G1] we apply a quasi-pullback procedure to ϑ_{D_8} to obtain a chain of Jacobi forms with trivial character.

Theorem 2.4.3 *For $N \in \{1, \dots, 8\}$ we have Jacobi forms $\psi_{12-N, D_N} \in J_{12-N, D_N; 1}(1)$ given by*

$$\psi_{12-N, D_N}(\tau, \mathfrak{z}_{D_N}) = \eta(\tau)^{24-3N} \vartheta_{D_N}(\tau, \mathfrak{z}_{D_N}).$$

For $N < 8$ these functions define Jacobi cusp forms. These forms are obtained by iterative quasi-pullbacks of ϑ_{D_8} , namely

$$\left. \frac{\partial \psi_{12-N, D_N}}{\partial z_N} \right|_{z_N=0} = 2\pi i \psi_{12-N+1, D_{N-1}}$$

where the operator is of course taken with respect to the standard coordinate system of D_N .

Proof

From Lemma 2.4.1 and the fact that $\eta(\tau)$ is a Jacobi cusp form of weight $1/2$ and index 0 for any lattice it is clear that ψ_{12-N, D_N} is a holomorphic Jacobi form with trivial character and the cusp form property is evident as well. In Example 2.1.17 we have already analyzed the function $\vartheta(\tau, z)$. Now the statement about the quasi pullbacks is a direct consequence of the structure of the set of zeroes of ϑ and the fact that

$$\left. \frac{\partial \vartheta(\tau, z)}{\partial z} \right|_{z=0} = 2\pi i \eta(\tau)^3. \quad \square$$

We consider again the classical weak Jacobi form $\phi_{0,1}(\tau, z)$ from [EZ]. Similiar to the construction given in Proposition 2.3.5 we construct a twist of Jacobi forms.

Lemma 2.4.4 *There is the following sequence*

$$J_{k, D_1(2); 1}^{(weak)} \xrightarrow{D_1(2) \Pi^{D_2(2)}(\cdot, \phi_{0,1})} J_{k, D_2(2); 1}^\Pi \xrightarrow{D_2(2) \Pi^{D_3(2)}(\cdot, \phi_{0,1})} J_{k, D_3(2); 1}^\Pi \cdots$$

where the spaces $J_{k, D_N(2); 1}^\Pi$ are defined recursively as the images of the operator

$$D_N(2) \prod^{D_{N+1}(2)}(\cdot, \phi_{0,1}) : J_{k, D_N(2); 1}^\Pi \rightarrow J_{k, D_{N+1}(2); 1}^\Pi$$

where

$$D_N(2) \prod^{D_{N+1}(2)}(\varphi, \phi_{0,1})(\tau, \mathfrak{z}) := \varphi(\tau, \mathfrak{z} - (\mathfrak{z}, \varepsilon_{N+1})\varepsilon_{N+1})\phi_{0,1}(\tau, (\mathfrak{z}, \varepsilon_{N+1}))$$

for $N \in \mathbb{N}_{\geq 1}$ and $J_{k, D_1(2); 1}^\Pi := J_{k, D_1(2); 1}^{(weak)}$.

Proof

Considering our standard model for D_N the proof follows from the identities

$$\begin{aligned} 2(z_2 - z_1)^2 + 2(z_2 + z_1)^2 &= 2(\mathfrak{z}_{D_2}, \mathfrak{z}_{D_2}) \\ 2(z_2 - z_1)^2 + 2(z_2 + z_1 - z_3)^2 + 2z_3^2 &= 2(\mathfrak{z}_{D_3}, \mathfrak{z}_{D_3}) \\ 2(\mathfrak{z}_{D_N}, \mathfrak{z}_{D_N}) - 2z_N^2 + 2(z_N - z_{N+1})^2 + z_{N+1}^2 &= 2(\mathfrak{z}_{D_{N+1}}, \mathfrak{z}_{D_{N+1}}). \end{aligned}$$

where $N \geq 3$. The facts that both $\vartheta(\tau, z)^2$ and $\phi_{0,1}$ have the same transformation behaviour under the Heisenberg group make this functions to be modular under the Jacobi group Γ^J and the weight of the input function is kept. Hence we obtain weak Jacobi forms. A more direct way to verify the assertion is to consider the model

$$D_N = \{x \in \mathbb{Z}^N; x_1 + \cdots + x_N \equiv 0 \pmod{2}\}$$

and proceed as in the proof of Lemma 2.1.18. □

The above construction is motivated by the identity

$$\vartheta_{D_{N+1}} = \prod_{D_N}^{D_{N+1}} (\vartheta_{D_N}, \vartheta).$$

Before we state our next theorem we like to write down some examples to see how this construction looks like in our coordinates:

$$\begin{aligned} \phi_{22, D_2(2)}(\tau, \mathfrak{z}_{D_2}) &:= \prod_{D_1(2)}^{D_2(2)} (\psi_{11, D_1}^2, \phi_{0,1})(\tau, \mathfrak{z}_{D_2}) \\ &= \psi_{11, D_1}^2(\tau, z_2 - z_1) \phi_{0,1}(\tau, z_2 + z_1), \end{aligned}$$

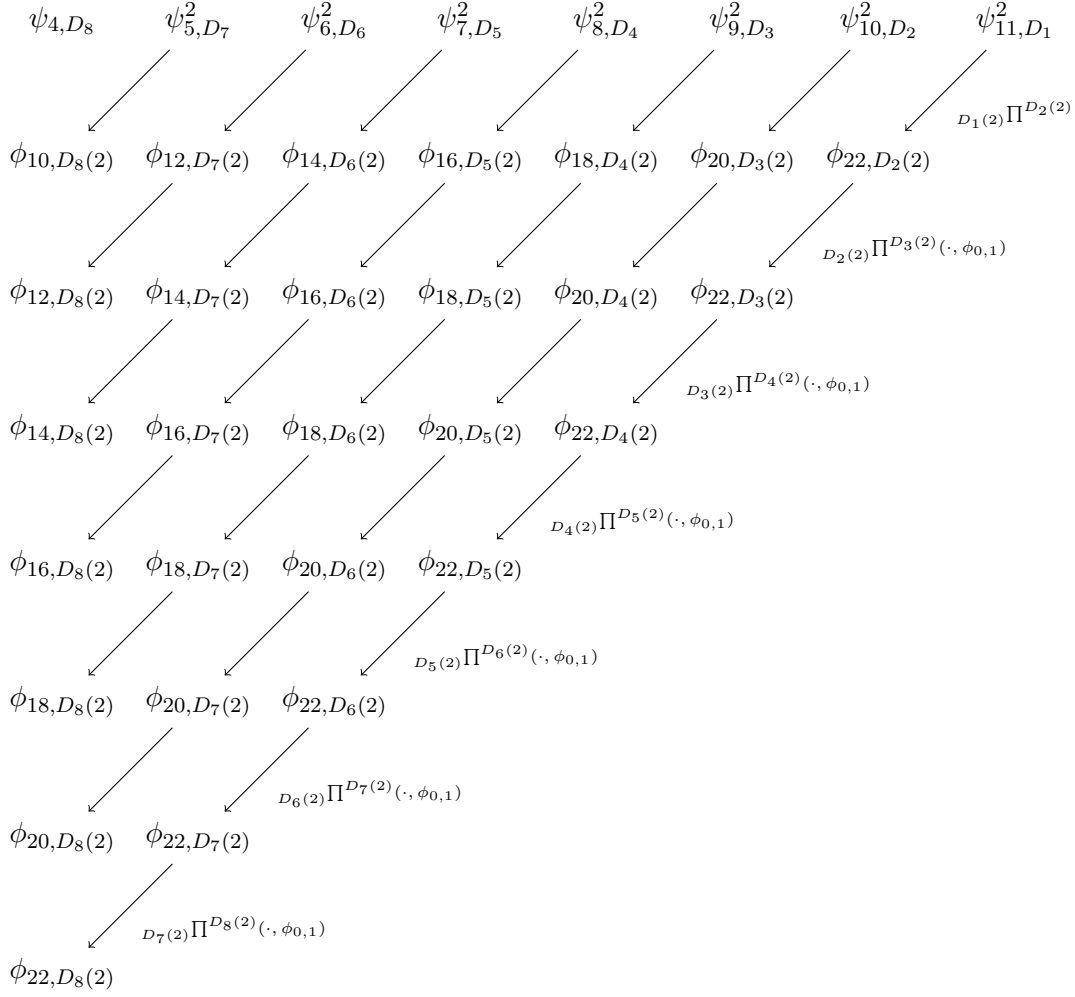
$$\begin{aligned} \phi_{20, D_3(2)}(\tau, \mathfrak{z}_{D_3}) &:= \prod_{D_2(2)}^{D_3(2)} (\psi_{10, D_2}^2, \phi_{0,1})(\tau, \mathfrak{z}_{D_3}) \\ &= \eta(\tau)^{36} \vartheta(\tau, z_2 - z_1)^2 \vartheta(\tau, z_2 + z_1 - z_3)^2 \phi_{0,1}(\tau, z_3), \end{aligned}$$

$$\begin{aligned} \phi_{22, D_3(2)}(\tau, \mathfrak{z}_{D_3}) &:= \prod_{D_2(2)}^{D_3(2)} (\phi_{22, D_2(2)}, \phi_{0,1})(\tau, \mathfrak{z}_{D_3}) \\ &= \psi_{11, D_1}^2(\tau, z_2 - z_1) \phi_{0,1}(\tau, z_2 + z_1 - z_3) \phi_{0,1}(\tau, z_3), \end{aligned}$$

$$\begin{aligned} \phi_{18, D_4(2)}(\tau, \mathfrak{z}_{D_4}) &:= \prod_{D_3(2)}^{D_4(2)} (\psi_{9, D_3}^2, \phi_{0,1})(\tau, \mathfrak{z}_{D_4}) \\ &= \eta(\tau)^{30} \vartheta(\tau, z_2 - z_1)^2 \vartheta(\tau, z_2 + z_1 - z_3)^2 \vartheta(\tau, z_3 - z_4)^2 \phi_{0,1}(\tau, z_4). \end{aligned}$$

This yields another tower of holomorphic Jacobi forms for $D_8(2)$.

Theorem 2.4.5 *We have the following diagram of holomorphic Jacobi forms where all forms except for the first column are cups forms of the weight indicated by the index*



More precisely we have

$$\phi_{k,D_N(2)} \in J_{k,D_N(2),1}(1)$$

and $\phi_{k,D_N(2)}$ is a cusp form if and only if $N < 8$. All the functions except for the first line have the property

$$D_{N-1}(2) \prod^{D_N(2)}(\phi_{k,D_{N-1}(2)}, \phi_{0,1}) \downarrow_{D_{N-1}(2)} \in \mathbb{C}^\times \phi_{k,D_{N-1}(2)} \text{ where } N \geq 2$$

regarding the convention $\phi_{26-2N,D_{N-1}(2)} := \psi_{13-N,D_{N-1}}^2$

Proof

The transformation behaviour of the functions was already discussed in Lemma 2.4.4. The character is trivial because only factors either of the shape ϑ^2 or $\phi_{0,1}$ appear. The behaviour under pullbacks is granted by $\phi_{0,1}(\tau, 0) = 12$. We need to observe the holomorphicity at infinity since according to Proposition 2.3.2 $\phi_{0,1}$ is only a weak Jacobi

form. Assume we have a holomorphic Jacobi form φ for the lattice $D_N(2)$ which is of the above shape. We consider the Fourier expansion of the twisted form:

$$\begin{aligned} {}_{D_N(2)}\prod^{D_{N+1}(2)}(\varphi, \phi_{0,1})(\tau, \mathfrak{z}_{D_{N+1}}) &= \sum_{\substack{n \geq 0 \\ l \in D_N(2)^\vee \\ 2n - (l, l) \geq M}} g(n, l) q^n \exp(2\pi i(l, \hat{\mathfrak{z}}_{D_{N+1}})) \\ &\times \sum_{\substack{m \geq 0 \\ h \in A_1^\vee \\ 2m - (h, h) \geq -\frac{1}{2}}} f(m, h) q^n \exp(2\pi i(h, z_{N+1} \varepsilon_{N+1})_{A_1}). \end{aligned}$$

In this context the lattice A_1 is realized as a sublattice in $\mathbb{R}^{N+1}$ by $(\langle \varepsilon_{N+1} \rangle_{\mathbb{Z}}, 2(\cdot, \cdot)_{N+1})$ and $\hat{\mathfrak{z}}_{D_{N+1}}$ is the coordinate transformation such that

$${}_{D_N(2)}\prod^{D_{N+1}(2)}(\varphi, \phi_{0,1})(\tau, \mathfrak{z}_{D_{N+1}}) = \varphi(\tau, \hat{\mathfrak{z}}_{D_{N+1}}) \cdot \phi_{0,1}(\tau, z_{N+1}).$$

Observing that for any positive definite lattice L and $t \in \mathbb{N}$ the relation $L(t)^\vee = \frac{1}{t}L^\vee$ is valid we can rewrite the second Fourier series as

$$\sum_{\substack{m \geq 0 \\ h \in A_1(4)^\vee \\ 2m - (h, h)_{A_1(4)} \geq -1/2}} \tilde{f}(m, h) q^n \exp(2\pi i(h, z_{N+1} 2\varepsilon_{N+1})_{A_1(4)}).$$

Here the realization of $A_1(4)$ as a sublattice of $D_{N+1}(2)$ is given by $(\langle 2\varepsilon_{N+1} \rangle_{\mathbb{Z}}, 2(\cdot, \cdot)_{N+1})$. By Lemma 2.4.2 we have $D_N^\perp = A_1(2)$ and thus $D_N(2)^\perp = A_1(4)$. Hence the Fourier expansion of ${}_{D_N(2)}\prod^{D_{N+1}(2)}(\varphi, \phi_{0,1})$ ranges over all $n \geq 0$ and $l \oplus h \in D_{N+1}(2)^\vee \subseteq D_N(2)^\vee \oplus A_1(4)^\vee$ satisfying

$$2n - (l \oplus h, l \oplus h)_{D_{N+1}(2)} \geq M - 1/2$$

where M is some positive integer. Hence application of the operator

$${}_{D_N(2)}\prod^{D_{N+1}(2)}(\cdot, \phi_{0,1})$$

has the effect to change the lower bound for the hyperbolic norm of all non-vanishing Fourier coefficients by adding $-1/2$ to the lower bound of the input function φ . Conversely, multiplication with $\eta(\tau)^6$ has the effect to change the hyperbolic norm of the function by adding $1/2$ as we have observed in Proposition 2.3.2. Since ϑ_{D_N} is a Jacobi form of singular weight the lower bound for the hyperbolic norm of $\phi_{k, D_N(2)}$ is given by $\frac{8-N}{2}$ according to the preceding investigation, which completes the proof (including the statement on cusp forms). $\square$

There are pullbacks for other root lattices, too. We start again with the lattice A_1 which corresponds to the case of classical Jacobi forms as treated in [EZ]. The root lattice A_2 can be found as a sublattice of the lattice D_3 which in our model is realized as $\langle b_1, b_2, b_3 \rangle_{\mathbb{Z}}$ equipped with the standard scalar product. Then we identify A_2 with the sublattice $\langle b_1, b_3 \rangle_{\mathbb{Z}}$. Note that this realization differs from that in Section 1.8.

Proposition 2.4.6 (a) *The function*

$$\vartheta_{1,A_2}(\tau, \mathfrak{z}_{A_2}) := \eta(\tau)^{-1} \vartheta(\tau, z_1 - z_2) \vartheta(\tau, z_1) \vartheta(\tau, z_2)$$

is an element of $J_{1,A_2;1}(v_\eta^8)$ and $\psi_{9,A_2}(\tau, \mathfrak{z}_{A_2}) := \eta(\tau)^{16} \vartheta_{1,A_2}(\tau, \mathfrak{z}_{A_2})$ is a holomorphic Jacobi form of index 1 and weight 9 with trivial character for A_2 .

(b) *There is an operator*

$$_{A_1} \prod^{A_2(2)}(\cdot, \phi_{0,1}) : J_{k,A_1;1}(1) \rightarrow J_{k,A_2(2);1}^{(weak)}(1)$$

given by

$$_{A_1} \prod^{A_2(2)}(\varphi, \phi_{0,1})(\tau, \mathfrak{z}_{A_2}) := \varphi(\tau, z_1) \phi_{0,1}(\tau, z_1 - z_2) \phi_{0,1}(\tau, z_2).$$

The functions $_{A_1} \prod^{A_2(2)}(\varphi_{12,A_1}, \phi_{0,1})$ and $_{A_1} \prod^{A_2(2)}(\psi_{10,A_1}, \phi_{0,1})$ are holomorphic Jacobi forms and cusp forms for the lattice $A_2(2)$.

(c) *The functions*

$$\begin{aligned} \varphi_{10,D_3(2)}(\tau, \mathfrak{z}_{D_3}) &:= \psi_{10,A_1}(\tau, z_1 - z_2) \phi_{0,1}(z_1 + z_2 - z_3) \phi_{0,1}(z_3), \\ \varphi_{12,D_3(2)}(\tau, \mathfrak{z}_{D_3}) &:= \varphi_{12,A_1}(\tau, z_1 - z_2) \phi_{0,1}(z_1 + z_2 - z_3) \phi_{0,1}(z_3) \end{aligned}$$

are holomorphic cusp forms with trivial character for the lattice $D_3(2)$ of weight 10 and 12, respectively. Moreover we have a cusp form $\psi_{8,D_3} \in J_{8,D_3;1}^{(cusp)}(1)$ defined by the formula

$$\psi_{8,D_3}(\tau, \mathfrak{z}_{D_3}) = \eta(\tau)^{12} \vartheta(\tau, z_1) \vartheta(\tau, z_2) \vartheta(\tau, z_1 - z_3) \vartheta(\tau, z_2 - z_3).$$

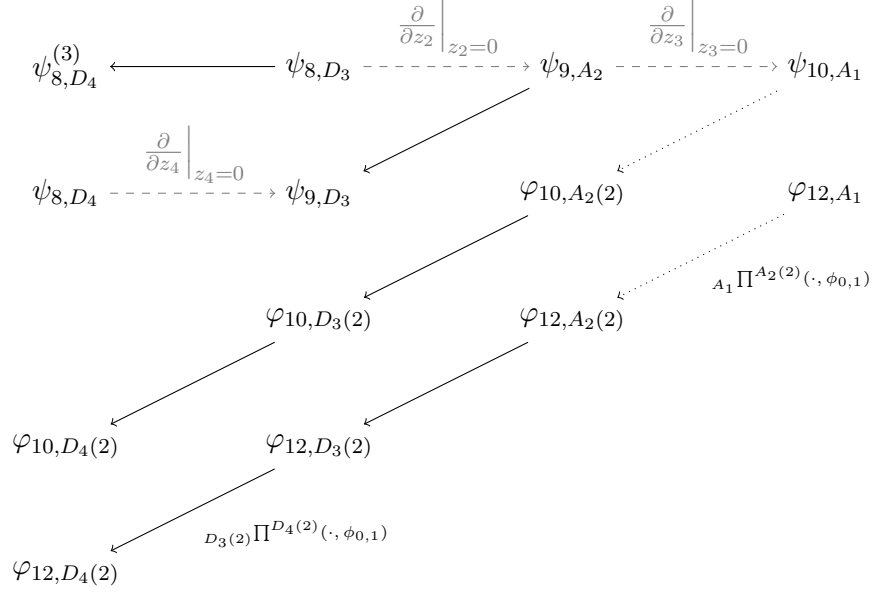
(d) *We have another singular Jacobi form for D_4 given by*

$$\vartheta_{2,D_4}^{(3)}(\tau, \mathfrak{z}_{D_4}) := \vartheta(\tau, z_1) \vartheta(\tau, z_2 - z_4) \vartheta(\tau, z_1 - z_3) \vartheta(\tau, z_2 - z_3 + z_4)$$

and this definition coincides with Lemma 2.1.19 where we find this function for the basis $\varepsilon_1, \dots, \varepsilon_4$. Moreover we have the function

$$\psi_{8,D_4}^{(3)}(\tau, \mathfrak{z}_{D_4}) := \eta(\tau)^{12} \vartheta_{2,D_4}^{(3)}(\tau, \mathfrak{z}_{D_4}) \in J_{8,D_4;1}(1).$$

The following diagram gives an overview for the functions from above



All functions appearing in the diagram are holomorphic Jacobi forms and cusp forms except for $\varphi_{k,D_4(2)}$, $k = 10, 12$. The continuous arrows indicate a pullback structure whereas the dashed arrows indicate a quasi-pullback structure. The dotted arrows mean that the pullback of the function listed below is not a $\mathbb{C}^\times$ -multiple of the original function.

Proof

(a) We recall the Fourier expansion of $\vartheta(\tau, z)$ from Example 2.1.17

$$\vartheta(\tau, z) = \sum_{n \in \mathbb{Z}, n \equiv 1 \pmod{2}} (-1)^{\frac{n-1}{2}} \exp\left(\frac{\pi i n^2 \tau}{4} + \pi i n z\right).$$

The transformation behaviour of ϑ_{1,A_2} under the generators of the Jacobi group is that of an element of $J_{1,A_2;1}^{(weak)}(1)$ by virtue of the identity

$$z_1^2 + z_2^2 + (z_1 - z_2)^2 = (\mathfrak{z}_{A_2}, \mathfrak{z}_{A_2}) \quad \text{where} \quad \mathfrak{z}_{A_2} = z_1 b_1 + z_2 b_3.$$

Again we have to analyze the Fourier expansion of this function to check holomorphicity. We start with the expansion of the function $\eta(\tau) \vartheta_{1,A_2;1}(\tau, \mathfrak{z}_{A_2})$. Each summand is of the shape

$$q^{\frac{n_1^2 + n_2^2 + n_3^2}{8}} \exp(\pi i (n_1 z_1 + n_2 z_2 + n_3 (z_1 - z_2)))$$

where $n_1 \equiv n_2 \equiv n_3 \equiv 1 \pmod{2}$. Since $A_2^\vee \subseteq \frac{1}{6} A_2$ the last term can be rewritten as

$$q^{\frac{n_1^2 + n_2^2 + n_3^2}{8}} \exp(2\pi i (v, \mathfrak{z}_{A_2}))$$

where

$$v := \frac{1}{6} [n_3(\varepsilon_1 + 2\varepsilon_2 - \varepsilon_3) + n_1(2\varepsilon_1 + \varepsilon_2 + \varepsilon_3) + n_2(\varepsilon_1 - \varepsilon_2 + 2\varepsilon_3)] .$$

The hyperbolic norm of the last expression equals

$$\begin{aligned} 2 \frac{n_1^2 + n_2^2 + n_3^2}{8} - (v, v) \\ &= \frac{n_1^2 + n_2^2 + n_3^2}{4} - \frac{n_1^2 + n_2^2 + n_3^2}{6} - \frac{n_1 n_2 + n_1 n_3 - n_2 n_3}{6} \\ &= \frac{(-n_1 + n_2 + n_3)^2}{12} \geq \frac{1}{12} \end{aligned}$$

since the parity of n_1, n_2, n_3 is odd. This shows that the hyperbolic norm of ϑ_{1, A_2} is greater or equal to zero since

$$\eta(\tau)^{-1} = q^{-1/24}(1 + q(\dots)) .$$

Hence the hyperbolic norm of ψ_{9, A_2} is bounded by $\frac{4}{3}$.

- (b) From the transformation behaviour of $\phi_{0,1}$ we deduce immediately that the image of ${}_{A_1} \prod^{A_2(2)}(\cdot, \phi_{0,1})$ is a weak Jacobi form of the correct weight and index for $A_2(2)$. We consider the Fourier expansion of φ which is

$$\varphi(\tau, z) = \sum_{\substack{n \geq 0 \\ l \in A_1^\vee \\ 2n - (l, l) \geq 0}} f(n, l) q^n \exp(2\pi i(l, z\varepsilon_1)_{A_1}) .$$

We have to analyze the Fourier expansion of the product

$$\begin{aligned} \sum_{\substack{n \geq 0 \\ l \in \mathbb{Z} \\ 4n - l^2 \geq M}} f(n, l) q^n \exp(2\pi i l z_1) &\times \sum_{\substack{m \geq 0 \\ h \in \mathbb{Z} \\ 4m - h^2 \geq -1}} g(m, h) q^m \exp(2\pi i h z_2) \\ &\times \sum_{\substack{k \geq 0 \\ r \in \mathbb{Z} \\ 4k - r^2 \geq -1}} a(k, r) q^k \exp(2\pi i r(z_1 - z_2)) . \end{aligned} \quad (2.8)$$

If we set

$$v := \frac{1}{3} [r(\varepsilon_1 + 2\varepsilon_2 - \varepsilon_3) + l(2\varepsilon_1 + \varepsilon_2 + \varepsilon_3) + h(\varepsilon_1 - \varepsilon_2 + 2\varepsilon_3)]$$

each term of the Fourier expansion of the product construction looks like

$$q^{n+m+k} \exp(2\pi i(v, \mathfrak{z}_{A_2}))$$

and the estimation

$$2(n + m + k) - (v, v) \geq 2(n + m + k) - \frac{1}{9}(l^2 + h^2 + r^2) \geq M/2 - 1$$

follows from (2.8) and the facts that $-2lh \geq -(l^2 + h^2)$ and $2hr \geq -h^2 - r^2$. One easily calculates the lower bound M for φ_{k,A_1} to be 3 for $k = 10, 12$. This proves (b).

- (c) The statement can be proven with arguments similar to the proof of Theorem 2.4.5. In particular the lower bound for all non-vanishing Fourier coefficients of $\varphi_{12,D_3(2)}$ is $\frac{1}{2}$.
- (d) The application of $_{D_3(2)}\prod^{D_4(2)}(\cdot, \phi_{0,1})$ changes the lower bound of the hyperbolic norm for the Fourier coefficients by adding $-1/2$. This shows that $\varphi_{k,D_4(2)}$ are still holomorphic Jacobi forms. $\square$

2.5 Eisenstein-like Jacobi forms for the root lattice $4A_1$

In section 2.1 we have already observed that the space of Jacobi forms $J_{k,L;t}$ of integral weight and index is contained in the $\mathcal{O}(\mathbb{H})$ -module $\langle \theta_{\mu,t}^L; \mu \in L^\vee/tL \rangle_{\mathcal{O}(\mathbb{H})}$ which is free of dimension $t^{\text{rank}(L)} d(L)$. If the lattice is unimodular then the space of Jacobi forms of weight k and index one is isomorphic to the space of elliptic modular forms of weight $k - \text{rank}(L)/2$ as a complex vector space. In particular for $k = \text{rank}(L)/2 = N/2$ we know that $J_{N/2,L;1}$ is one-dimensional and spanned by the singular Jacobi form

$$\theta_{0,1}^L = \vartheta_L \quad , \quad \vartheta_L(\tau, \mathfrak{z}) = \sum_{l \in L} \exp(\pi i(l, l)\tau + 2\pi i(l, \mathfrak{z})) .$$

For the rest of this section we shall fix a chain of embeddings

$$A_1 \hookrightarrow 2A_1 \hookrightarrow 3A_1 \hookrightarrow 4A_1 \hookrightarrow E_8 .$$

From the unimodular lattice E_8 we obtain the unique singular Jacobi form ϑ_{E_8} of weight 4. We will investigate the pullbacks

$$\epsilon_{4,NA_1} = \vartheta_{E_8} \downarrow_{NA_1} \in J_{4,NA_1,1} .$$

The next Proposition will be important later in view of the maximal group of the modular forms induced by ϵ_{4,NA_1} .

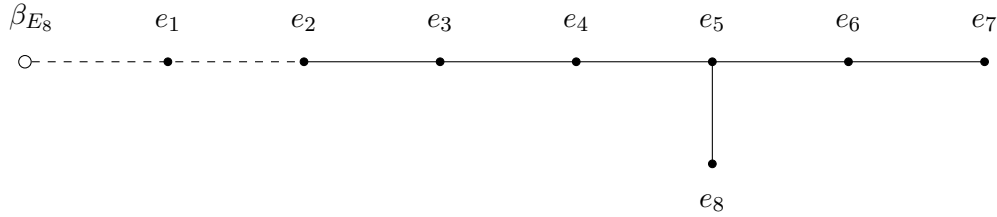
Proposition 2.5.1 *Let $N \in \{1, 2, 3, 4\}$ and let $\sigma \in \text{O}(NA_1)$. Then σ can be extended to $\text{O}(E_8)$ and any sublattice $L \leq E_8$ where $L \cong NA_1$ is equivalent to NA_1 with respect to the action of $\text{O}(E_8)$. Moreover ϵ_{4,NA_1} is invariant under the transformation induced by σ such that for all $\tau, \mathfrak{z}$ one has $\epsilon_{4,NA_1}(\tau, \sigma \cdot \mathfrak{z}) = \epsilon_{4,NA_1}(\tau, \mathfrak{z})$.*

Proof

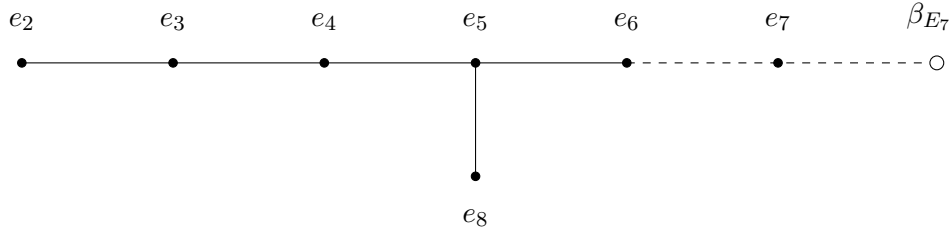
For the proof we will make use of Proposition 1.4.11. For this purpose we first have to calculate $NA_1^\perp_{E_8}$. This can be done by using the extended Coxeter diagrams from figure 1.4. Let us consider the extended Coxeter diagram for E_8 in the following way:

We identify A_1 with the highest root in E_8 . From the extended Coxeter diagram we can read off all basis vectors of E_8 perpendicular to the highest root. The subset of these roots is again a root lattice where we again have a highest root for every connected component. Hence the procedure is to identify NA_1 with N additional roots which are pairwise perpendicular.

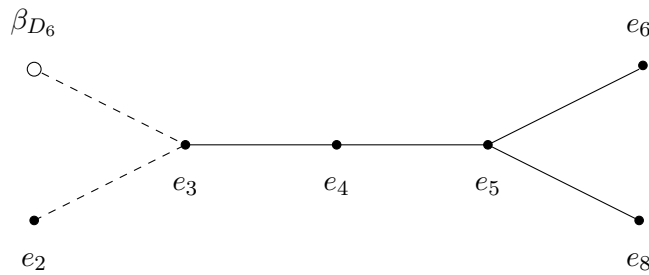
We start with β_{E_8} which is of type A_1 . We see from the diagram that a copy of E_7 is contained in $A_1^\perp_{E_8}$. Since the discriminant $d(A_1 \oplus E_7) = 4$ we conclude from the formula $[E_8 : A_1 \oplus E_7] \geq 2$ that the orthogonal complement does actually coincide with E_7 .



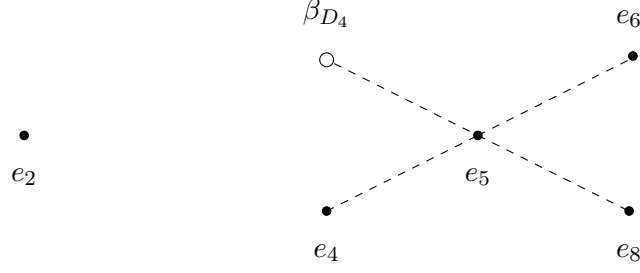
We compute the orthogonal complement of $2A_1$ in E_8 by considering



We know that β_{E_8} is perpendicular to β_{E_7} confer e.g. [E], p. 33. Hence the last figure shows that (a copy) of D_6 is contained in the orthogonal complement and by Proposition 1.4.11 the discriminant group of the orthogonal complement is isomorphic to $D(2A_1) \cong \mathcal{C}_2 \times \mathcal{C}_2$. Hence $2A_1^\perp = D_6$. Now the same procedure can be used for $3A_1$ and $4A_1$, respectively. In the first case we consider the extended diagram



and again observe that β_{D_6} is perpendicular to each β_{E_7}, β_{E_8} . By cutting off the root e_3 and the same argument as before considering the isomorphy of the discriminant group $D(3A_1)$ and the discriminant group of the orthogonal complement we obtain $3A_1^\perp = A_1 \oplus D_4$. Finally we find another additional root β_{D_4} perpendicular to $\beta_{E_8}, \beta_{E_7}, \beta_{D_6}$ as the next figure shows



The picture shows that β_{D_4} is perpendicular to the sublattice spanned by e_2, e_4, e_6, e_8 which is isomorphic to $4A_1$. The same argument as before yields $4A_1^\perp = 4A_1$.

Let $M, M' \leq E_8$ be isomorphic sublattices and denote by K, K' its orthogonal complements in E_8 . By Proposition 1.4.11 we know that there exist isomorphisms $\gamma : D(M) \rightarrow D(K)$ and $\gamma' : D(M') \rightarrow D(K')$. For the second part of the assertion we have to investigate according to Corollary 1.4.10 whether for every $\phi : M \rightarrow M'$ there exists an isomorphism $\psi : K \rightarrow K'$ such that

$$\overline{\psi} = \gamma' \circ \overline{\phi} \circ \gamma^{-1}.$$

Since $K \cong K'$ this amounts to saying that the natural homomorphism $O(K) \rightarrow O(D(K))$ is surjective. In Section 1.9 we have verified that this is the case for all of the corresponding orthogonal complements of the lattices given in the assertion. Now the invariance property follows immediately. $\square$

In the next step we construct a differential operator. This operator is well-known and a treatment can be found in [Cho] for the general case or in [EZ] for classical Jacobi forms. Denote by S the Gram matrix of L . The heat operator is given by

$$H = 4\pi i \det(S) \frac{\partial}{\partial \tau} - \det(S) S^{-1} \left[\frac{\partial}{\partial \mathfrak{z}} \right].$$

In [Cho] it was shown by a direct calculation that for every $\varphi \in J_{k,1}(L)$ we have the identities

$$(H\varphi) |_{k+2,1} M = H \left(\varphi |_{k,1} M \right) + 4\pi i \det(S) \left(k - \frac{N}{2} \right) \left(\frac{c}{c\tau + d} \right) \left(\varphi |_{k,1} M \right)$$

for all $M = \begin{pmatrix} a & b \\ c & d \end{pmatrix} \in \mathrm{SL}(2, \mathbb{Z})$ and

$$(H\varphi) |_{k+2,1} [x, y : (x, y)/2] = H \left(\varphi |_{k,1} [x, y : (x, y)/2] \right) \quad \text{for all } x, y \in L.$$

This operator can be seen as a generalization of the heat operator for Jacobi forms in [EZ], p. 32. Here we make a different automorphic correction by considering the quasi-modular Eisenstein series of weight 2, i.e.

$$G_2(\tau) = -\frac{1}{24} + \sum_{n \geq 1} \sigma_1(n) e^{2\pi i n \tau}$$

which transforms under $\mathrm{SL}(2, \mathbb{Z})$ as

$$G_2\left(\frac{a\tau + b}{c\tau + d}\right) = (c\tau + d)^2 G_2(\tau) - \frac{c(c\tau + d)}{4\pi i}.$$

and denote by $G_2 \bullet$ the operator which multiplies a function by G_2 . By virtue of the transformation property of G_2 we obtain a quasi-modular operator.

Lemma 2.5.2 *For every $k \in \mathbb{N}$ there is a quasi-modular differential operator $H_k : J_{k,L;1} \rightarrow J_{k+2,L;1}$ defined by the formula*

$$H_k = H + (4\pi i)^2 \det(S) \left(k - \frac{N}{2}\right) G_2 \bullet.$$

The operator H acts on $q^n r^l$, $n \in \mathbb{N}, l \in L^\vee$ by multiplication with $(2\pi i)^2 \det(S) (2n - (l, l))$.

Proof

It remains to proof the second part. We will proof the assertion by a suitable transformation of the abelian variable $\mathfrak{z}$. Since S is positive definite there exists some $W \in \mathrm{GL}(N, \mathbb{R})$ such that $S = W^{\mathrm{tr}} W$. Now we introduce new coordinates $\hat{\mathfrak{z}} = W\mathfrak{z}$ and observe that the differentials are connected by the identity

$$\frac{\partial}{\partial \hat{\mathfrak{z}}} = (W^{\mathrm{tr}})^{-1} \frac{\partial}{\partial \mathfrak{z}}$$

(note that we use a column convention for $\frac{\partial}{\partial \mathfrak{z}}$, here.) In the new coordinates the quadratic form is represented by the identity matrix and we obtain

$$\begin{aligned} H(q^n r^l) &= H[\exp(2\pi i n \tau) \exp(2\pi i (\hat{l}, \hat{\mathfrak{z}}))] \\ &= \det(S) (2\pi i)^2 (2n - (\hat{l}, \hat{l})) \exp(2\pi i n \tau) \exp(2\pi i (\hat{l}, \hat{\mathfrak{z}})) \\ &= \det(S) (2\pi i)^2 (2n - (l, l)) q^n r^l. \end{aligned} \quad \square$$

Using this operator we define

$$\epsilon_{6,NA_1} = H_4(\epsilon_{4,NA_1}) \in J_{6,NA_1;1} \quad N \in \{1, 2, 3, 4\}.$$

This functions inherit the invariance under coordinate permutations from ϵ_{4,NA_1} .

Corollary 2.5.3 *Let $N \in \{1, 2, 3, 4\}$. For all $\sigma \in \mathrm{O}(NA_1)$ we have $\epsilon_{6,NA_1}(\tau, \sigma \cdot \mathfrak{z}) = \epsilon_{6,NA_1}(\tau, \mathfrak{z})$. In particular ϵ_{6,NA_1} is invariant with respect to the action of $\mathrm{O}(D(NA_1)) \cong \mathcal{S}_N$.*

Proof

This follows from Proposition 2.5.1 and Lemma 2.5.2 since H acts on the monomials $q^n r^l$ by multiplication of the hyperbolic norm which is invariant under $\mathrm{O}(NA_1)$. Note that due to Section 1.9 each automorphism of the discriminant group can be realized as a preimage in $\mathrm{O}(NA_1)$. $\square$

The next lemma is about cusp forms and follows immediately from Lemma 2.5.2.

Lemma 2.5.4 *Let $\varphi \in J_{k,L;1}$. Then φ is a cusp form if and only if $H_k(\varphi)$ is a cusp form.*

Now let $N \in \{1, 2, 3\}$. We consider the Fourier expansions of $\epsilon_{4,NA_1}, \epsilon_{4,(N+1)A_1}$

$$\begin{aligned} \epsilon_{4,(N+1)A_1}(\tau, \mathfrak{z}_{(N+1)A_1}) &= \sum_{\substack{n \geq 0 \\ l \in (N+1)A_1^\vee \\ 2n - (l, l) \geq 0}} f(n, l) \exp(2\pi i(n\tau + (l, \mathfrak{z}))), \\ \epsilon_{4,(N+1)A_1} \downarrow_{NA_1}(\tau, \mathfrak{z}_{NA_1}) &= \sum_{\substack{n \geq 0 \\ l \in NA_1^\vee \\ 2n - (l, l) \geq 0}} \left(\sum_{\substack{h \in A_1^\vee \\ (h, h) \leq 2n - (l, l)}} f(n, l + h) \right) \exp(2\pi i(n\tau + (l, \mathfrak{z}))). \end{aligned}$$

If we consider the Fourier coefficients of $H(\epsilon_{4,NA_1})$ and $H(\epsilon_{4,(N+1)A_1}) \downarrow_{NA_1}$ we obtain for $n \geq 0$ and $l \in NA_1^\vee$ the first function

$$(2\pi i)^2 \det(NA_1)(2n - (l, l)) \left(\sum_{\substack{h \in A_1^\vee \\ (h, h) \leq 2n - (l, l)}} f(n, l + h) \right)$$

and for the second

$$(2\pi i)^2 \det((N+1)A_1) \left(\sum_{\substack{h \in A_1^\vee \\ (h, h) \leq 2n - (l, l)}} (2n - (l + h, l + h)) f(n, l + h) \right).$$

In order to decide whether the difference $H_4(\epsilon_{4,NA_1}) - H_4(\epsilon_{4,(N+1)A_1}) \downarrow_{NA_1}$ is zero it would be necessary to have explicit formulae for the Fourier coefficients of the pullbacks ϵ_{4,NA_1} .

2.6 The arithmetic lifting of Jacobi forms

In [G] Gritsenko gave a lifting construction for Jacobi forms. In this section we give a brief survey of this lifting operator. We denote by $\mathfrak{M}^+$ the group

$$\mathfrak{M}^+ = \{A \in \mathrm{GL}(2, \mathbb{Z}) , \det(A) > 0\}.$$

According to [G], Lemma 2.7 we have an embedding of the Hecke ring for $\mathrm{SL}(2, \mathbb{Z})$ into the Hecke ring for the Jacobi group given by

$$\mathrm{Im}_- : \mathrm{SL}(2, \mathbb{Z}) A \mathrm{SL}(2, \mathbb{Z}) \rightarrow \Gamma^J \mathrm{diag}(\det(A)A^*, I_N, \det(A)^{-1}A)\Gamma^J \quad \text{where } A \in \mathfrak{M}^+$$

and $A^* = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} (A^{-1})^{\text{tr}} \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$. It is well-known that the Hecke Algebra for the group $\text{SL}(2, \mathbb{Z})$ over $\mathbb{Q}$ is generated by elements of the shape

$$T(n) = \sum_{\substack{ab=n \\ a|b}} \text{SL}(2, \mathbb{Z}) \begin{pmatrix} a & 0 \\ 0 & b \end{pmatrix} \text{SL}(2, \mathbb{Z}) \quad , n \in \mathbb{N}.$$

If we set $T_-(n) := \text{Im}_-(T(n))$ then we obtain an operator

$$J_{k,L;t} \rightarrow J_{k,L;nt} \quad , \quad \varphi \mapsto \varphi|_{k,t} T_-(n) \quad , \text{ for } t \in \mathbb{Z}$$

and the action can be explicitly described as

$$\varphi|_{k,t} T_-(n)(\tau, \mathfrak{z}) = \sum_{\substack{ad=n \\ b \bmod d}} a^k n^{-1} \varphi\left(\frac{a\tau + b}{d}, a\mathfrak{z}\right)$$

Theorem 2.6.1 (Arithmetic Lift) *Let $\varphi \in J_{k,L;1}$ with Fourier expansion*

$$\varphi(\tau, \mathfrak{z}) = \sum_{\substack{n \geq 0 \\ l \in L^\vee \\ 2n - (l, l) \geq 0}} f(n, l) \exp(2\pi n\tau + 2\pi i(l, \mathfrak{z})).$$

If the zeroth Fourier coefficient $f(0, 0)$ is not 0 we also assume $k \geq 4$. Then there exists a modular form $\text{A-Lift}(\varphi)$ of weight k defined on $\mathcal{H}(L_1)$ with respect to $\tilde{\text{O}}(L_2)^+$. The operator A-Lift is a linear operator from the space of $J_{k,L;1}$ to the space $\mathcal{M}_k(\tilde{\text{O}}(L_2)^+)$

$$\text{A-Lift} : J_{k,L;1} \rightarrow \mathcal{M}_k(\tilde{\text{O}}(L_2)^+).$$

More explicitly we have

$$\text{A-Lift}(\varphi)(\omega, \mathfrak{z}, \tau) = f(0, 0)e_k(\tau) + \sum_{m=1}^{\infty} (m^{-1} \varphi|_{k,1} T_-(m))(\tau, \mathfrak{z}) \exp(2\pi i m \omega).$$

For a maximal even lattice L the function $\text{A-Lift}(\varphi)$ is a cusp form if φ is a Jacobi cusp form.

Chapter 3

Modular Groups

3.1 Embeddings of modular groups

In this chapter we consider again a positive definite even lattice L in $\mathbb{R}^N$ and the associated lattices L_1 and L_2 of signatures $(1, N+1)$ and $(2, N+2)$, respectively. We remind that there is a morphism

$$\pi : O(L_2) \rightarrow O(D(L)) \quad , \quad \ker(\pi) = \tilde{O}(L_2)$$

The following Proposition is a consequence of a theorem from Nikulin [N], Theorem 1.14.2 and Remark 1.14.3.

Proposition 3.1.1 *Let $l(D(L_2))$ denote the minimal number of generators of the discriminant group $D(L_2) \cong D(L)$. Assume that $N+1 \geq l(D(L_2))$. Then the genus of L_2 contains only one class and π is surjective.*

We want to analyze embeddings of modular groups. We have already seen that there is a natural embedding of groups $O(L) \hookrightarrow O(L_2)$. Since $\tilde{O}(L_2)$ is a normal subgroup of $O(L_2)$ the group $O(L)$ acts on $\tilde{O}(L_2)$ by conjugation. Thus we can consider the exterior semi-direct product

$$O(L) \ltimes \tilde{O}(L_2) .$$

Lemma 3.1.2 *Assume that $O(L) \rightarrow O(D(L))$ is surjective. Then*

$$O(L_2) \cong O(L) \ltimes \tilde{O}(L_2) / \{(g, g^{-1}) \mid g \in O(L) \cap \tilde{O}(L_2)\} .$$

Proof

Consider the homomorphism

$$\Psi : O(L) \ltimes \tilde{O}(L_2) \rightarrow O(L_2) \quad , \quad (g, n) \mapsto g \cdot n$$

which is well-defined since $O(L)$ acts on $\tilde{O}(L_2)$ by conjugation. Let $\gamma \in O(L_2)$. Since $O(L) \rightarrow O(D(L))$ is surjective there exists a $g \in O(L)$ such that $\pi(g) = \pi(\gamma)$. This shows $\Psi((g, n)) = gn = \gamma$ for some $n \in \tilde{O}(L_2)$ and yields that Ψ is surjective. Since

$$\ker(\Psi) = \{(g, g^{-1}) \mid g \in O(L) \cap \tilde{O}(L_2)\}$$

the statement follows from the homomorphism theorem. $\square$

We can formulate a criterion which gives us embeddings of the full orthogonal groups.

Corollary 3.1.3 *Let $L' \leq L$ be a sublattice of the positive definite lattice L and assume that $O(M) \rightarrow O(D(M))$ is surjective for $M = L, L'$. We assume that there exists an embedding $O(L') \hookrightarrow O(L)$. Then there is an embedding $O(L_2(L')) \hookrightarrow O(L_2(L))$.*

Proof

From the preceding Lemma we know that there is a homomorphism of groups

$$O(L') \ltimes \tilde{O}(L_2(L')) \hookrightarrow O(L) \ltimes \tilde{O}(L_2(L)) \rightarrow O(L_2(L))$$

with kernel $\{(h, h^{-1}) \mid h \in O(L') \cap \tilde{O}(L_2(L'))\}$ and the assertion follows from Lemma 3.1.2. $\square$

Corollary 3.1.4 *Assume that $O(L) \rightarrow O(D(L))$ is surjective and $\tilde{SO}(L_2) = E(L_2)$ where the last group is the group of unimodular Eichler transvections defined in section 1.4. We additionally assume that $O(L) \cap \tilde{O}(L_2)$ is contained in the commutator subgroup of $O(L_2)$. Then every character of the Jacobi group $\Gamma^J(L)$ can be extended to $O(L_2)$.*

Proof

Let χ be a character of $O(L) \ltimes \tilde{O}(L_2)$. The assumptions imply $\tilde{O}(L_2) = \langle \Gamma^J(L), \sigma_1 \rangle$ where $\sigma_1 := \sigma_{e-f}$, confer Theorem 3.2.3 in the next Section. Note that by Lemma 3.1.2 we have $O(L_2) \cong O(L) \ltimes \tilde{O}(L_2) / \{(g, g^{-1}) \mid g \in O(L) \cap \tilde{O}(L_2)\}$. The character χ is also defined modulo $\{(g, g^{-1}) \mid g \in O(L) \cap \tilde{O}(L_2)\}$ since every element of $O(L) \cap \tilde{O}(L_2)$ is a commutator and we have

$$\chi((g, g^{-1})) = \chi((g, 1)) \chi((1, g^{-1})).$$

Hence it remains to show that every character of $\Gamma^J(L)$ can be extended to $O(L) \ltimes \tilde{O}(L_2)$. Write $n \in \tilde{O}(L_2)$ as a product of σ_1 and elements of $\Gamma^J(L)$. Since $\Gamma^J(L) \subseteq \tilde{SO}(L_2)$ and $\det \sigma_1 = -1$ we know that the parity of the number of σ_1 in any representation of n must be the same. Since σ_1 is an involution this shows that $\chi(n)$ is uniquely determined by $\chi|_{\Gamma^J(L)}$ and $\chi(\sigma_1)$. Thus χ is uniquely determined by $\chi|_{\Gamma^J(L)}$, $\chi|_{O(L)}$ and $\chi(\sigma_1)$. $\square$

3.2 Computation of the commutator subgroup of $O(L_2)$

Before we start with the computation of the commutator subgroup for the orthogonal group $O(L_2)$ we formulate certain conditions on the positive lattice L which impose that the orthogonal group has a nice set of generators. In [GHS] the authors treated a family of lattices whose orthogonal groups have this property.

Definition 3.2.1 Let M be a lattice and $p \in \mathbb{P}$ be a prime. The p -rank of M denoted by $\text{rank}_p(M)$ is the maximal rank of sublattices $K \leq M$ such that $d(K)$ is coprime to p . We say that the lattice L_2 satisfies the *Kneser condition* if $\text{rank}_2(L_2) \geq 6$ and $\text{rank}_3(L_2) \geq 5$.

Remark 3.2.2 Our Kneser condition is more restrictive than the formulation given in [GHS]. The condition given there only needs the assumptions on the 2- and 3-rank of M being an even integral lattice of Witt index ≥ 2 over $\mathbb{R}$ and represents -2. The lattice L_2 satisfies the Kneser condition if L contains a sublattice isomorphic to A_2 .

A proof of the next Theorem can be found in [GHS], Proposition 3.4.

Theorem 3.2.3 Assume that the lattice L_2 satisfies the Kneser condition. Then we have

$$\widetilde{SO}(L_2)^+ = E_U(L_1)$$

and

$$\widetilde{O}(L_2)^+ = \langle \Gamma^J(L), \sigma_1 \rangle$$

where σ_1 is the reflection $\sigma_{e_1 - f_1}$.

Proof

We repeat the proof for the statement on the generators of the Jacobi group since this will be important for the analysis of the commutator subgroup. The Jacobi group was defined in Section 2.1. An immediate calculation gives us representations of the generators of the Jacobi group as Eichler transvections, namely

$$t(e, y) = [0, y : 0], \quad t(e_1, x) = [x, 0 : 0], \quad t(e, e_1) = [0, 0 : 1] \quad \text{for } x, y \in L$$

and

$$t(f, e_1) = \left\{ \begin{pmatrix} 1 & 0 \\ -1 & 1 \end{pmatrix} \right\}, \quad t(e, f_1) = \left\{ \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix} \right\}.$$

Hence it only remains to show that the elements above generate the stable orthogonal group. Define $S := t(f, e_1)t(e, f_1)t(f, e_1) = \left\{ \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix} \right\}$. One has

$$S(e) = -e_1, \quad S(f) = f_1, \quad \sigma_1(e_1) = f_1, \quad \sigma_1(f_1) = e_1.$$

This implies

$$\begin{aligned} \sigma_1 t(f, e_1) \sigma_1 &= t(\sigma_1(f), \sigma_1(e_1)) = t(f, f_1) \\ (S \sigma_1 S \sigma_1) t(e, -y) (S \sigma_1 S \sigma_1)^{-1} &= t(f, -y) \quad \text{for all } y \in L \end{aligned}$$

since $S\sigma_1 S\sigma_1(e) = f$ and $S\sigma_1 S\sigma_1(-y) = -y$. Thus $\langle \Gamma^J(L), \sigma_1 \rangle$ contains all the generators of $E_U(L_1)$ and the reflection σ_1 has determinant -1. $\square$

This Theorem can be used to determine generators for the full orthogonal group. The next statement follows immediately from Lemma 3.1.2.

Corollary 3.2.4 *Assume that the lattice L_2 satisfies the Kneser condition and $O(L) \rightarrow O(D(L))$ is surjective. Then we have*

$$O(L_2)^+ = \langle O(L), \Gamma^J(L), \sigma_1 \rangle.$$

We put $G := \langle E_U(L_1), \sigma_1 \rangle$. We remind to the definition of the scale of the lattice L which is defined as the minimal $s(L) \in \mathbb{N}$ such that $s(L)\mathbb{Z} = ((x, y), x, y \in L)$. We denote by $[G, G]$ the commutator subgroup of G given by

$$[G, G] = \langle [g, h] \mid g, h \in G \rangle \quad \text{where} \quad [g, h] := g^{-1}h^{-1}gh.$$

Lemma 3.2.5 *Assume that the lattice L_2 satisfies the Kneser condition or $s(L) = 2$. Then the set*

$$\{t(e, \mu e_1) \mid \mu = 0, \dots, s(L) - 1\}$$

contains a system of representatives of $E_U(L_1)$ modulo $[G, G]$.

Proof

The group $E_U(L_1)$ is generated by the following elements

$$\begin{aligned} t(e, e_1), \quad t(f, e_1) &= Jt(e, f_1)J \\ t(e, f_1), \quad t(f, f_1) &= Jt(e, e_1)J \\ t(e, y), \quad t(f, y) &= Jt(e, -y)J \quad \text{where } y \in L \end{aligned}$$

and

$$J = t(e, e_1)t(f, f_1)t(e, f_1)t(f, e_1)t(e, f_1)t(e, e_1) \quad , \quad J = \begin{pmatrix} 0 & 0 & 0 & 0 & -1 \\ 0 & 0 & 0 & -1 & 0 \\ 0 & 0 & I_N & 0 & 0 \\ 0 & -1 & 0 & 0 & 0 \\ -1 & 0 & 0 & 0 & 0 \end{pmatrix}$$

with respect to the standard basis of L_2 . From Lemma 2.1.5 we deduce that

$$t(e, s(L)e_1) = [0, 0 : s(L)] \in [G, G].$$

We compute a few commutators. Using Lemma 1.5.6 and (1.10) we obtain for $x \in L$

$$\begin{aligned}
[t(e_1, x), t(e, f_1)] &= t(e_1, -x) t(e_1, x)^{t(e, f_1)} = t(e_1, -x) t(e + e_1, x) \\
[S, t(e_1, x)] &= t(e, -x) t(e_1, x) \\
S &= \left\{ \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix} \right\} \\
t(e, -x) t(e + e_1, x) &= t(e, -x) t(e_1, x) t(e_1, -x) t(e + e_1, x) \\
[t(e, e_1), \sigma_1] &= t(e, -e_1) t(e, f_1) \\
T &:= t(e, s(L)e_1) t(e, -e_1) t(e, f_1)
\end{aligned}$$

where

$$T = \begin{pmatrix} 1 & -1 & 0 & 1 - s(L) & 1 - s(L) \\ 0 & 1 & 0 & 0 & s(L) - 1 \\ 0 & 0 & I_N & 0 & 0 \\ 0 & 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 0 & 1 \end{pmatrix}.$$

Now if the lattice L_2 satisfies the Kneser condition we deduce from [GHS], Theorem 1.7 that $[E_U(L_1), E_U(L_1)] = E_U(L_1)$. Since $[G, G] \leq E_U(L_1)$ we obtain the assertion in this case. If $s(L) = 2$ one computes $(TJ)^3 = I_{N+4}$. Hence $J^3 = J$ is a commutator. It remains to show that $t(e, f_1), t(e, x)$ are equivalent to one of the elements $t(e, 0), t(e, e_1)$ modulo $[G, G]$. For $x \in L$ we have

$$t(e, 2e_1) t(e, x) t(e + e_1, -x) t(e, x) = t(e, (s(L) - (x, x)/2)e_1) \quad (3.1)$$

Another computation shows

$$T^{-1} t(e, f_1) = t(e, -e_1).$$

From (3.1) we deduce that the forementioned set contains indeed a system of representatives for $E_U(L_1)$. $\square$

For a group G we define the abelianisation of G as

$$G^{\text{ab}} := G/[G, G]$$

which is isomorphic to the group of characters. This leads us to the main theorem of this section.

Theorem 3.2.6 *Let L be a lattice with nonempty root system which has one of the following properties:*

- (i) L_2 satisfies the Kneser condition and $O(L) \rightarrow O(D(L))$ is surjective.
- (ii) $s(L) \leq 2$ and $O(L_2)^+ = \langle O(L), E_U(L_1) \rangle$.

Then we have

$$\#(\mathrm{O}(L_2)^+)^{\mathrm{ab}} \leq \delta(L) \cdot \#\mathrm{O}(L)^{\mathrm{ab}}$$

where $\delta(L) = 1$ if L satisfies the Kneser condition and $\delta(L) = s(L)$ else.

Proof

From Lemma 3.1.2, Corollary 3.2.4 and the fact that the Weyl group of L is nontrivial we obtain

$$\mathrm{O}(L_2)^+ = \langle \mathrm{O}(L), E_U(L_1) \rangle$$

in both cases. We define $G := [\mathrm{O}(L_2^+), \mathrm{O}(L_2)^+]$. For each $g \in E_U(L_1)$ we have

$$g \cdot G \in \bigcup_{m=1}^{\delta(L)} t(e, \mu e_1)G$$

by the last Lemma. This yields $[M : G] \leq \delta(L)$ where $M := \langle E_U(L_1), G \rangle$. Now Lemma 3.1.2 tells us that every element $g \in \mathrm{O}(L_2)^+$ can be written as $g = a\tau$ where $a \in \mathrm{O}(L)$ and $\tau \in E_U(L_1)$. We observe that $H := \langle E_U(L_1), [\mathrm{O}(L), \mathrm{O}(L)] \rangle$ is a subgroup of M and the construction of H yields that g can be written in the form $g = ah$ where $a \in \mathrm{O}(L)/[\mathrm{O}(L), \mathrm{O}(L)]$ and $h \in H$. This shows

$$[\mathrm{O}(L_2)^+ : G] = [\mathrm{O}(L_2)^+ : M] \cdot [M : G] \leq \delta(L) \cdot \#\mathrm{O}(L)^{\mathrm{ab}}. \quad \square$$

The crucial point in the proof of the last theorem was the existence of a nice set of generators. This question also arised in [FH] and [Kl]. In [FH], Proposition 4.8, the orthogonal group $\mathrm{O}(L_2)^+$ is called *nicely generated* if

$$\mathrm{O}(L_2)^+ = \langle E_U(L_1), \mathrm{O}(L) \rangle.$$

In Lemma 4.6 the authors present a list of nine lattices L where this property does apply, namely

$$D_4, A_3, 3A_1, 2A_1, A_1 \oplus A_1(2), A_2, A_1, A_1(2), A_1(3).$$

Theorem 3.2.6 gives us one criterion for the full orthogonal group being nicely generated. In the following we present another criterion to decide this question.

Definition 3.2.7 The positive definite lattice L is called *Euclidean* if there exists $c(L) \in (0, 1)$ such that

$$\forall v \in L \otimes \mathbb{R} \exists l \in L : \frac{1}{2}(v + l, v + l) \leq c(L).$$

The next Lemma can be found in [Kl], Proposition 1.22 where some different notation was used. However, we rewrite the proof by using our terminology since we will need it later. In the next Lemma we consider L_1 as a sublattice of L_2 .

Lemma 3.2.8 (i) Let L be a Euclidean lattice and let $l_1 \in L_1^\vee$ be an isotropic vector, i.e. $(l_1, l_1) = 0$. Then there exists some $g \in \langle J, t(e_1, x) | x \in L \rangle$ such that $g(l_1) = \alpha e_1$ where $\alpha \in \mathbb{Z}$.

(ii) If L is Euclidean then $O(L_2)^+$ is nicely generated.

Proof

- (i) Write $l_1 = ae_1 + l + bf_1$ for some $l \in L^\vee, a, b \in \mathbb{Z}$. By assumption we have $ab = \frac{1}{2}(l, l)$. Without loss of generality we can assume $|b| \leq |a|$ since otherwise we can interchange a and b by multiplication with J . If $l = 0$ we have $b = 0$. So assume $l \neq 0$. This yields $b \neq 0$ since L is positive definite. By assumption there exists some $x \in L$ such that $\frac{1}{2}(\frac{l}{b} + x, \frac{l}{b} + x) \leq c(L) < 1$. This yields

$$a'e_1 + (l + bx) + bf_1 = t(e_1, x)(l_1) \quad \text{where} \quad (l + bx, l + bx) < (l, l), \quad a' \in \mathbb{Z}.$$

Iteratively we can transform l_1 to the desired shape.

- (ii) From Proposition 1.5.8 we know that $O(L_2)^+$ can be generated by Eichler transvections and $O(L_1)^+$. Hence it is sufficient to show $O(L_1)^+ \subseteq \langle E_U(L_1), O(L) \rangle$. So let $g \in O(L_1)^+$. With respect to our standard basis of L_2 we have

$$g = \begin{pmatrix} 1 & 0 & 0 \\ 0 & M & 0 \\ 0 & 0 & 1 \end{pmatrix}.$$

Let l_1 be the first column of M . Then we have

$$(l_1, l_1) = (g(e_1), g(e_1)) = 0.$$

According to part (i) there exists an $h \in \langle J, t(e_1, x) | x \in L \rangle \subseteq E_U(L_1)$ such that

$$hg = \begin{pmatrix} 1 & 0 & 0 & 0 & 0 \\ 0 & \alpha & * & * & 0 \\ 0 & 0 & A & * & 0 \\ 0 & 0 & d & \delta & 0 \\ 0 & 0 & 0 & 0 & 1 \end{pmatrix}. \quad (3.2)$$

Since hg is orthogonal we obtain $d = 0$ and $A \in O(L)$. After multiplication with A^{-1} we can assume that hg is of the shape

$$hg = \begin{pmatrix} 1 & 0 & 0 & 0 & 0 \\ 0 & \alpha & * & * & 0 \\ 0 & 0 & I_N & x & 0 \\ 0 & 0 & 0 & \delta & 0 \\ 0 & 0 & 0 & 0 & 1 \end{pmatrix}.$$

Moreover we have $\alpha, \delta \in \mathbb{Z}$ and $\alpha\delta = 1$. Since hg has to fix the connected component we obtain $\alpha = \delta = 1$. Now by multiplying hg by $t(e_1, x)$ we obtain the identity matrix. The proof is complete. $\square$

All the lattices listed above are Euclidean and hence possess nicely generated orthogonal groups. There is one exceptional case which we will treat in the sequel. The lattice $4A_1$ is not Euclidean. However the group $O(L_2(4A_1))^+$ is nicely generated, too, as we will see in a while. For the proof we will have to consider the Hurwitz order. Let $Q := \mathbb{Q}(-1, -1)$ be the rational quaternion algebra of signature $(-1, -1)$. As a vector space over $\mathbb{Q}$ we have

$$Q = \mathbb{Q} + \mathbb{Q}i + \mathbb{Q}j + \mathbb{Q}ij \quad , \quad i^2 = j^2 = -1.$$

The norm of a quaternion is defined as $N_Q(z) = z_0^2 + z_1^2 + z_2^2 + z_3^2$ where $z = z_0 + z_1i + z_2j + z_3ij$. A maximal order in Q is given by $\mathfrak{o} = \mathbb{Z} + \mathbb{Z}i + \mathbb{Z}j + \mathbb{Z}\omega$ where $\omega = \frac{1}{2}(1 + i + j + ij)$. The order $\mathfrak{o}$ is a positive definite even lattice and isomorphic to D_4 , if we equip $\mathfrak{o}$ with the bilinear form $2N_Q$. The sublattice $\mathfrak{o}_0 = \mathbb{Z} + \mathbb{Z}i + \mathbb{Z}j + \mathbb{Z}ij$ is isomorphic to $4A_1$. Using this identification it is possible to modify the proof of Lemma 3.2.8 for this case. The proof of this fact was communicated to the author by Till Dieckmann and can be found in [D]. We consider the two-sided ideal of even quaternions

$$\wp = \{a \in \mathfrak{o} ; N_Q(a) \in 2\mathbb{Z}\} = (1 + i)\mathfrak{o}.$$

Then $\wp$ is also an ideal of $\mathfrak{o}_0$ and according to [H], p. 35, we have

$$\mathfrak{o} = \wp \cup (1 + \wp) \cup (\omega + \wp) \cup (\bar{\omega} + \wp)$$

where $\bar{\omega}$ denotes the element $\frac{1}{2}(1 - i - j - ij)$. Moreover one has

$$\mathfrak{o}_0 = \wp \cup (1 + \wp).$$

Lemma 3.2.9 *For $a, b \in \mathfrak{o}_0$ where $b \neq 0$ there exists some $q \in \mathfrak{o}_0$ such that*

$$N_Q(a - qb) < N_Q(b) \quad \text{or} \quad a - qb = \omega b.$$

In the second case we always have $a, b \in \wp$.

Proof

It is clear that there exists some $q \in \mathfrak{o}_0$ such that $N_Q(\frac{a}{b} - q) \leq 1$. Moreover we choose q such that in the basis of Q for any component one has $(\frac{a}{b} - q)_k \leq \frac{1}{2}$, $k \in \{0, 1, 2, 3\}$. If we have equality one can add ± 1 to any component of q such that $\frac{a}{b} - q = \omega$. The first statement then follows by the multiplicativity of N_Q . Now assume $a - qb = \omega b$. By multiplying this equation with $\bar{b}$ we obtain $a\bar{b} - qN_Q(b) = \omega N_Q(b) \in \mathfrak{o}_0$. This yields $N_Q(b) \in 2\mathbb{Z}$ such that $b \in \wp$ and hence $a \in \wp$. $\square$

We need another Lemma.

Lemma 3.2.10 *Let $b \in \mathfrak{o}_0$ and let $\alpha, \delta \in \mathbb{Z}$ such that $N_Q(b) = \alpha\delta$. Then there exist $u, v \in \mathfrak{o}_0$ such that*

$$b = u\bar{v}, N_Q(u) = \alpha, N_Q(v) = \delta.$$

Proof

Since $\mathfrak{o}_0 = \wp \cup 1 + \wp$ and $\wp$ is a prime ideal it is sufficient to consider the case $b \in 1 + \wp$, the other one works similar. From [K1], I.1.13 we know there exist $u', v' \in \mathfrak{o}$ such that $b = u'\overline{v'}$ and $N_Q(u') = \alpha$, $N_Q(v') = \delta$. One has

$$\mathfrak{o} = \mathfrak{o}_0 \cup \omega \mathfrak{o}_0 \cup \overline{\omega} \mathfrak{o}_0. \quad (3.3)$$

From $b \in 1 + \wp$ we conclude $u' \equiv v' \pmod{\wp}$ and (3.3) yields $u := u'\varepsilon, v := v'\varepsilon \in \mathfrak{o}_0$ for some $\varepsilon \in \{1, \omega, \overline{\omega}\}$. Since ω is a unit one has $N_Q(\varepsilon) = 1$ and $b = u\overline{v}$. $\square$

This enables us to adapt the proof of Lemma 3.2.8 to the case $4A_1$.

Lemma 3.2.11 *The orthogonal group $O^+(L_2(4A_1))$ is nicely generated.*

Proof

We make some refinements in part (i) of Lemma 3.2.8. We identify $4A_1$ with $\mathfrak{o}_0$. Let $g \in O^+(L_1)$ such that

$$g = \begin{pmatrix} 1 & 0 & 0 \\ 0 & M & 0 \\ 0 & 0 & 1 \end{pmatrix}, \quad M = \begin{pmatrix} \alpha & * & \beta \\ b & A & c \\ \gamma & * & \delta \end{pmatrix}.$$

Since g orthogonal we obtain

$$N_Q(b) = \alpha\gamma \quad \alpha\delta + \beta\gamma - (b, c) = 1.$$

From Lemma 3.2.10 we obtain $u, v \in \mathfrak{o}_0$ such that

$$b = u\overline{v}, N_Q(u) = \alpha, N_Q(v) = \gamma.$$

Since the scale $s(4A_1) = 2$ we know that exactly one of the numbers α, γ is odd. Hence exactly one of the elements u, v is in $\wp$. Assume $b \neq 0$ and $|\gamma| \leq |\alpha|$. From Lemma 3.2.9 we obtain some $q \in \mathfrak{o}_0$ such that

$$N_Q(b - q\gamma) = N_Q(u\overline{v} - qv\overline{v}) = N_Q(\overline{v})N_Q(u - qv) < N_Q(v)N_Q(\overline{v}) = \gamma^2 \leq \alpha\gamma = N_Q(b)$$

where we have used the assumption $b \neq 0$ as in Lemma 3.2.8. This shows that we can again reduce the first column of M as in (3.2) and the rest of the proof is done as in the proof of Lemma 3.2.8. $\square$

3.3 Correspondences between some classical groups

In this section we are dealing with some examples of lattices M . For the sake of a better readability we will therefore write $L_j(M), j = 1, 2$ for the lattices of signature $(j, N + j)$ defined before. In [FH], Proposition 2.5 and Proposition 2.6 the authors describe the general procedure of how to construct the exceptional isogenies of signature $(2, 3), (2, 4)$ and $(2, 6)$ for the real orthogonal groups. Since our special interest is devoted to integral orthogonal groups, here, we recall the constructions for some special integral cases.

Signature (2, 3)

We start with the lattice A_1 . We consider Siegel's upper half-plane of genus 2

$$\mathbb{H}_2 = \left\{ \begin{pmatrix} \tau & z \\ z & \omega \end{pmatrix} \in \text{Mat}(2, \mathbb{C}) ; \text{Im}(\tau), \text{Im}(\omega) > 0 \right\}.$$

This half-space is biholomorphically equivalent to $\mathcal{H}(L_1(A_1))$ via

$$\sigma : \mathcal{H}(L_1(A_1)) \rightarrow \mathbb{H}_2 \quad , \quad (\tau, \mathfrak{z}, \omega) \mapsto \begin{pmatrix} \tau & z \\ z & \omega \end{pmatrix}.$$

In [GH] we can find an integral realization of the isogeny for $\text{Sp}(4, \mathbb{Z})$ which consists of all matrices $M \in \text{Mat}(4, \mathbb{Z})$ satisfying

$$M^{\text{tr}} \begin{pmatrix} 0 & -I_2 \\ I_2 & 0 \end{pmatrix} M = \begin{pmatrix} 0 & -I_2 \\ I_2 & 0 \end{pmatrix}.$$

The group $\text{Sp}(4, \mathbb{Z})$ acts on $\mathbb{H}_2$ by fractional linear transformations as a group of biholomorphic automorphisms. The following Lemma is proven in [GH].

Proposition 3.3.1 *There exists an isomorphism*

$$\Psi : \widetilde{\text{SO}}(L_2(A_1))^+ \rightarrow \text{Sp}(4, \mathbb{Z}) / \{\pm I_4\}$$

such that the following diagram commutes for every $g \in \widetilde{\text{SO}}(L_2(A_1))^+$

$$\begin{array}{ccc} \mathcal{H}(L_1(A_1)) & \xrightarrow{g} & \mathcal{H}(L_1(A_1)) \\ \sigma \downarrow & & \downarrow \sigma \\ \mathbb{H}_2 & \xrightarrow{\Psi(g)} & \mathbb{H}_2 \end{array}$$

Signature (2, 4)

To treat the case of signature (2, 4) the following Lemma is useful. For this purpose we recall the definition of the Hermitian upper half-plane of degree two

$$\mathbb{H}_2(\mathbb{C}) = \left\{ Z = \begin{pmatrix} \omega & z_1 \\ z_2 & \tau \end{pmatrix} \in \text{Mat}(2, \mathbb{C}) ; \frac{1}{2i}(Z - \bar{Z}^{\text{tr}}) > 0 \right\},$$

confer [DK]. Now let M be a lattice of rank two in $\mathbb{R}^2$. Any such lattice can be realized as $\mathbb{Z}\omega_1 + \mathbb{Z}\omega_2 \subseteq \mathbb{C}$ such that $\omega_2/\omega_1 \notin \mathbb{R}$. The quadratic form is given by $q(m) := |m|^2$ where $|\cdot|$ denotes the usual complex norm and the bilinear form is defined as $b(m_1, m_2) := 2 \text{Re}(\overline{m_1} m_2)$. We use the notation $(M, |\cdot|^2)$ for this kind of realization and call it a *lattice in $\mathbb{C}$* . For elements in $\mathfrak{z} \in M \otimes_{\mathbb{R}} \mathbb{C}$ we have a decomposition as

$$\mathfrak{z} = \mathfrak{z}_r + i\mathfrak{z}_i.$$

Lemma 3.3.2 *Let $(M, |\cdot|^2)$ be a lattice in $\mathbb{C}$. Then we have a biholomorphic equivalence between $\mathbb{H}_2(\mathbb{C})$ and $\mathcal{H}(L_1(M))$ given by*

$$\sigma : \mathcal{H}(L_1(M)) \rightarrow \mathbb{H}_2(\mathbb{C}) \quad , \quad (\omega, \mathfrak{z}, \tau) \mapsto \begin{pmatrix} \omega & \mathfrak{z}_r + i\mathfrak{z}_i \\ \overline{\mathfrak{z}_r + i\mathfrak{z}_i} & \tau \end{pmatrix}$$

and the identity

$$\tau\omega - q(\mathfrak{z}) = \det(\sigma(\omega, \mathfrak{z}, \tau)) \quad (3.4)$$

is valid for all $(\omega, \mathfrak{z}, \tau) \in \mathcal{H}(L_1(M))$.

Proof

The first part is clear and the second follows from the definition of the complexified quadratic form as

$$q(\mathfrak{z}) = q(\mathfrak{z}_r) - q(\mathfrak{z}_i) + ib(\mathfrak{z}_r, \mathfrak{z}_i). \quad \square$$

Now let $m \in \mathbb{N}_{>0}$ be square-free. We consider the imaginary quadratic number field $K = \mathbb{Q}(\sqrt{-m})$ and its ring of integers $\mathfrak{o}_K = \mathbb{Z} + \mathbb{Z}\omega$ where

$$\omega = \begin{cases} \sqrt{-m} & \text{if } m \not\equiv -1 \pmod{4} \\ \frac{1+\sqrt{-m}}{2} & \text{if } m \equiv -1 \pmod{4} \end{cases}$$

Then $\mathfrak{o}_K$ is a positive definite even root lattice of rank two. The ring of the Gaussian integers $\mathfrak{o}_{\mathbb{Q}(\sqrt{-1})}$ is of type $2A_1$ and the ring of the Eisenstein integers $\mathfrak{o}_{\mathbb{Q}(\sqrt{-3})}$ is of type A_2 . Obviously $\mathfrak{o}_K$ is a lattice in $\mathbb{C}$. The group $O(L_2(\mathfrak{o}_K))^+$ acts on $\mathbb{H}_2(\mathbb{C})$ via

$$\Psi : O(L_2(\mathfrak{o}_K))^+ \times \mathbb{H}_2(\mathbb{C}) \rightarrow \mathbb{H}_2(\mathbb{C}) \quad , \quad \Psi(g) = (Z \mapsto \sigma(g < \sigma^{-1}Z >))$$

as a group of biholomorphic automorphisms. Let us consider the matrices

$$R_M = \begin{pmatrix} I_2 & 0 & 0 \\ 0 & M & 0 \\ 0 & 0 & I_2 \end{pmatrix} \quad , \quad M \in O(\mathfrak{o}_K)$$

$$T_x = \begin{pmatrix} 1 & -x^{\text{tr}}S_1 & -(x, x)_1 \\ 0 & I_{N+2} & x \\ 0 & 0 & 1 \end{pmatrix} \quad , \quad x = me_1 + y + nf_1 \in L_1 \quad (3.5)$$

$$J = \begin{pmatrix} 0 & 0 & 0 & 0 & -1 \\ 0 & 0 & 0 & -1 & 0 \\ 0 & 0 & I_N & 0 & 0 \\ 0 & -1 & 0 & 0 & 0 \\ -1 & 0 & 0 & 0 & 0 \end{pmatrix}$$

where $y \in \mathfrak{o}_K$. The first and the last type of matrices certainly belong to $O(L_2(\mathfrak{o}_K))^+$ as we have observed earlier. From the identity

$$T_x = t(e, y)t(e, e_1)^m t(e, f_1)^n$$

we conclude that all elements listed in (3.5) belong to $O(L_2(\mathfrak{o}_K))^+$. The action on the tube domain $\mathcal{H}(L_1)$ is described by

$$R_M < Z > = (\omega, M\mathfrak{z}, \tau)^{\text{tr}} \quad , \quad T_x < Z > = Z + x \quad , \quad J < Z > = -\frac{1}{2}(Z, Z)^{-1}(\tau, -\mathfrak{z}, \omega)^{\text{tr}}$$

where $Z = (\omega, \mathfrak{z}, \tau)^{\text{tr}} \in \mathcal{H}(L_1)$. The *unitary group* $SU(\mathfrak{o}_K)$ consists of all matrices $M = \begin{pmatrix} A & B \\ C & D \end{pmatrix} \in \text{SL}(4, \mathfrak{o}_K)$ such that

$$\overline{M}^{\text{tr}} \begin{pmatrix} 0 & -I_2 \\ I_2 & 0 \end{pmatrix} M = \begin{pmatrix} 0 & -I_2 \\ I_2 & 0 \end{pmatrix} .$$

The group $SU(\mathfrak{o}_K)$ acts on $\mathbb{H}_2(\mathbb{C})$ as a group of biholomorphic transformations via

$$(M, Z) \mapsto (AZ + B)(CZ + D)^{-1}$$

and M acts trivially if and only if $M = \varepsilon I_4$ where $\varepsilon \in \mathbb{C}$ such that $|\varepsilon| = 1$. There is an isomorphism between orthogonal and unitary groups.

Proposition 3.3.3 *Let K be an imaginary quadratic number field and assume that $O(L_2(\mathfrak{o}_K))^+$ is nicely generated. If the lattice $L_2(\mathfrak{o}_K)$ does not satisfy the Kneser condition we additionally assume*

$$O(\mathfrak{o}_K) \cap \widetilde{SO}(L_2(\mathfrak{o}_K))^+ = \langle R_{-I_2} \rangle .$$

Then we have an isomorphism

$$\widetilde{SO}(L_2(\mathfrak{o}_K))^+ / \{\pm I_6\} \cong SU(\mathfrak{o}_K) / \{\pm I_4\}$$

and for every $g \in \widetilde{SO}(L_2(\mathfrak{o}_K))^+$ the diagram

$$\begin{array}{ccc} \mathcal{H}(L_1(\mathfrak{o}_K)) & \xrightarrow{g} & \mathcal{H}(L_1(\mathfrak{o}_K)) \\ \sigma \downarrow & & \downarrow \sigma \\ \mathbb{H}_2(\mathbb{C}) & \xrightarrow{\Psi(g)} & \mathbb{H}_2(\mathbb{C}) \end{array}$$

commutes.

Proof

We first note that the additional assumption in the case where L_2 does not satisfy the Kneser condition just guarantees that $-I_6$ is contained in the group generated by

the elements listed above. In particular R_{-I_2} induces the same transformation as J . Hence the group $\widetilde{\mathrm{SO}}(L_2(\mathfrak{o}_K))^+$ is generated by the elements listed in (3.5) where $M \in \mathrm{O}(\mathfrak{o}_K) \cap \widetilde{\mathrm{SO}}(L_2(\mathfrak{o}_K))^+$. In [De] it is shown that $\mathrm{SU}(\mathfrak{o}_K)$ is generated by

$$\begin{pmatrix} 0 & -I_2 \\ I_2 & 0 \end{pmatrix} \text{ and } \begin{pmatrix} I_2 & H \\ 0 & I_2 \end{pmatrix} \quad \text{where } H \in \mathrm{Her}(2, \mathfrak{o}_K) := \{M \in \mathrm{Mat}(2, \mathfrak{o}_K) | \overline{M}^{\mathrm{tr}} = M\}.$$

Now Lemma 3.3.2 implies

$$\Psi(J)(Z) = -\det(Z)^{-1} \begin{pmatrix} \tau & -\mathfrak{z}_r - i\mathfrak{z}_i \\ -\overline{\mathfrak{z}_r} - i\overline{\mathfrak{z}_i} & \omega \end{pmatrix} = -Z^{-1}$$

and for $x = me_1 + l + nf_1 \in L_1(\mathfrak{o}_K)$ we have

$$\Psi(T_x)(Z) = \sigma(T_x < (\omega, \mathfrak{z}, \tau) >) = \sigma((\omega, \mathfrak{z}, \tau) + x) = Z + H$$

where $Z = (\omega, \mathfrak{z}, \tau)^{\mathrm{tr}} \in \mathcal{H}(L_1)$ and

$$H = \begin{pmatrix} m & l \\ \overline{l} & n \end{pmatrix} \in \mathrm{Her}(2, \mathfrak{o}_K).$$

Since these elements generate the group of special unitary transformations we are done. $\square$

Remark 3.3.4 For the sake of a short proof we have required that the group is nicely generated. In general this is too restrictive. In a more general situation one also has to consider other types of generators and has to determine the correct image in the above diagram. We like to mention that $L_2(\mathfrak{o}_K)$ for both $K = \mathbb{Q}(\sqrt{-1})$ and $K = \mathbb{Q}(\sqrt{-3})$ satisfies the conditions of the Proposition.

Signature (2, 6)

At the end of this section we finally treat the case of signature (2, 6). We consider again the Hurwitz order

$$\mathfrak{o} = \mathbb{Z} + \mathbb{Z}i + \mathbb{Z}j + \mathbb{Z}\omega$$

where i, j, ω were defined in the last section. This case has already been treated in [FH], so we just cite the results and omit most of the details, here. A careful treatment of Hamilton quaternions their modular groups and the associated half-spaces can be found in [K1]. The case of the Hurwitz order can be found in [K2]. The skew field of *Hamilton quaternions* is

$$Q_H := \mathbb{R} + \mathbb{R}i + \mathbb{R}j + \mathbb{R}ij.$$

The quaternionic half-plane of degree two is defined as

$$\mathbb{H}_2(Q_H) := \{Z = X + iY \in \mathrm{Mat}(2, Q_H) \otimes_{\mathbb{R}} \mathbb{C}; Z = \overline{Z}^{\mathrm{tr}}, Y > 0\}.$$

In the same manner as before we can define a biholomorphic equivalence between $\mathbb{H}_2(Q_H)$ and $\mathcal{H}(L_1(D_4))$. The group $\mathrm{Sp}(4, \mathfrak{o})$ consists of all matrices $M \in \mathrm{GL}(4, \mathfrak{o})$ such that

$$\overline{M}^{\mathrm{tr}} \begin{pmatrix} 0 & -I_2 \\ I_2 & 0 \end{pmatrix} M = \begin{pmatrix} 0 & -I_2 \\ I_2 & 0 \end{pmatrix}.$$

For a proof of the next statement confer [FH], Remark 7.2.

Proposition 3.3.5 $\mathrm{Sp}(4, \mathfrak{o})/\{\pm I_4\}$ is a normal subgroup in $\mathrm{O}(L_2(\mathfrak{o}))^+/\{\pm I_8\}$ of index 4 and there is a commutative diagram

$$\begin{array}{ccc} \mathcal{H}(L_1(\mathfrak{o})) & \longrightarrow & \mathcal{H}(L_1(\mathfrak{o})) \\ \downarrow & & \downarrow \\ \mathbb{H}_2(Q_H) & \longrightarrow & \mathbb{H}_2(Q_H) \end{array}$$

which is compatible with the action of the corresponding groups on the domains. The quotient is isomorphic to $(\mathbb{Z}/2\mathbb{Z}) \times (\mathbb{Z}/2\mathbb{Z})$ and it is generated by the cosets of the orthogonal transformations

$$\mathbb{H}_2(Q_H) \rightarrow \mathbb{H}_2(Q_H) \quad , \quad Z \mapsto Z^{\mathrm{tr}} , Z \mapsto \alpha Z \alpha \quad \text{where} \quad \alpha = \frac{1+i}{\sqrt{2}} .$$

Chapter 4

Vector valued modular forms

In this chapter we introduce vector valued modular forms. Our main focus is on Jacobi-Eisenstein series. The arithmetic lifting of Jacobi-Eisenstein series appears to be a quite natural construction which already appeared in [EZ], Section 6. By using the well-known theta-correspondence Jacobi-Eisenstein series and the vector valued Eisenstein series treated in [Br] coincide. We show that the Jacobi-Eisenstein series form a basis of the complementary space $J_{k,L;1}^{(cusp)}$ and show that the dimension is associated to the number of (non-equivalent) zero-dimensional cusps. Moreover we will see that these functions are a normalized basis with respect to a variant of Siegel's Φ -operator adapted to Jacobi forms. At the end of the second section we derive explicit formulae for the Fourier expansion of Jacobi-Eisenstein series for every cusp. The explicit formula makes them important in view of generators of graded rings of modular forms. Most of the facts in the first section are well-known and can be found in the first chapter of [Br]. The second chapter is motivated by [BrK] where the authors gave an explicit formula for the Fourier coefficients of the simplest vector valued Eisenstein series. Throughout this chapter let L be a positive definite even lattice. We start with

Definition 4.0.6 Let G be a group and V be a finite-dimensional vector space over $\mathbb{C}$. A *representation* of G on V is a homomorphism of groups

$$G \rightarrow \mathrm{GL}(V).$$

The *degree* of the representation is $\dim_{\mathbb{C}} V$. We call a representation *finite* if the image of G in $\mathrm{GL}(V)$ is finite.

4.1 Weil representation and vector valued modular forms

We start by regarding the elliptic factor of automorphy $j(A, \tau) = c\tau + d$ where $A = \begin{pmatrix} a & b \\ c & d \end{pmatrix} \in \mathrm{SL}(2, \mathbb{Z})$. In order to allow half-integral weights we have to consider branches of the square root. For $z \in \mathbb{C}$ we put $\sqrt{z} = z^{\frac{1}{2}}$ such that $\arg(\sqrt{z}) \in (-\pi/2, \pi/2]$. For $A \in \mathrm{SL}(2, \mathbb{Z})$ let ϕ be a holomorphic function on $\mathbb{H}$ satisfying $\phi(\tau)^2 = j(A, \tau)$, $\tau \in \mathbb{H}$.

The action of $\mathrm{SL}(2, \mathbb{Z})$ on the space of holomorphic functions defined on $\mathbb{H}$ given by the *Petersson slash operator* gives rise to the cocycle relation

$$j(AB, \tau) = j(A, B\tau) \cdot j(B, \tau) \text{ where } A, B \in \mathrm{SL}(2, \mathbb{Z}).$$

This relation leads us to the *metaplectic group* $\mathrm{Mp}(2, \mathbb{Z})$ whose elements are pairs

$$(A, \phi(\tau))$$

and the multiplication is given by

$$(A, \phi_A(\tau))(B, \phi_B(\tau)) = (AB, \phi_A(B\tau)\phi_B(\tau)).$$

The map

$$\begin{pmatrix} a & b \\ c & d \end{pmatrix} \mapsto \widetilde{\begin{pmatrix} a & b \\ c & d \end{pmatrix}} = \left(\begin{pmatrix} a & b \\ c & d \end{pmatrix}, \sqrt{c\tau + d} \right)$$

defines a locally isomorphic embedding of $\mathrm{SL}(2, \mathbb{R})$ into $\mathrm{Mp}(2, \mathbb{R})$. The group $\mathrm{Mp}(2, \mathbb{Z})$ is the inverse image of $\mathrm{SL}(2, \mathbb{Z})$ under the covering map $\mathrm{Mp}(2, \mathbb{R}) \rightarrow \mathrm{SL}(2, \mathbb{R})$. We denote the standard generators of $\mathrm{SL}(2, \mathbb{Z})$ by

$$T = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}, \quad J = \begin{pmatrix} 0 & -1 \\ -1 & 0 \end{pmatrix}.$$

Lemma 4.1.1 *The group $\mathrm{Mp}(2, \mathbb{Z})$ is generated by $\tilde{T}$ and $\tilde{J}$. The centre is generated by*

$$C = \left(\begin{pmatrix} -1 & 0 \\ 0 & -1 \end{pmatrix}, i \right).$$

Proof

Every element of the centre has to commute with $\tilde{T}$ and $\tilde{J}$. A simple computation yields that the centre consists of the elements

$$(I_2, 1), (I_2, -1), (-I_2, i), (-I_2, -i)$$

which form a cyclic group of order 4 generated by C . The group $\mathrm{Mp}(2, \mathbb{Z})$ can be generated by $\tilde{T}, \tilde{J}$ and C^2 since T, J generate $\mathrm{SL}(2, \mathbb{Z})$ and multiplication by C^2 changes between the two possible branches of the square root. The proof is completed by virtue of the relation

$$\tilde{J}^2 = (\tilde{J}\tilde{T})^3 = C. \quad \square$$

There is a unitary representation ϱ_L of $\mathrm{Mp}(2, \mathbb{Z})$ on the group algebra $\mathbb{C}[L^\vee/L]$. Let $(\mathbf{e}_\mu)_{\mu \in L^\vee/L}$ denote the standard basis of $\mathbb{C}[L^\vee/L]$. Then ϱ_L can be defined by the action on the generators of $\mathrm{Mp}(2, \mathbb{Z})$ as follows:

$$\begin{aligned} \varrho_L(\tilde{T})\mathbf{e}_\mu &= \exp(\pi i(\mu, \mu))\mathbf{e}_\mu \\ \varrho_L(\tilde{J})\mathbf{e}_\mu &= \frac{\sqrt{i}^N}{\sqrt{|d(L)|}} \sum_{\nu \in L^\vee/L} \exp(-2\pi i(\mu, \nu))\mathbf{e}_\nu. \end{aligned} \quad (4.1)$$

This representation is called the *Weil representation* of the lattice L . We note that this implies

$$\varrho_L(C)\mathfrak{e}_\mu = i^N \mathfrak{e}_{-\mu}.$$

If we consider the representation matrix of $\varrho_L(A, \phi)$ with respect to the basis $(\mathfrak{e}_\mu)$ and denote it by $(\varrho_{L, \nu}(A, \phi))$ there is an explicit formula for these coefficients, confer for example [Br], Proposition 1.1.

Proposition 4.1.2 *Let $A = \begin{pmatrix} a & b \\ c & d \end{pmatrix} \in \mathrm{SL}(2, \mathbb{Z})$. The coefficient $\varrho_{L, \nu}(\tilde{A})$ is given by*

$$\varrho_{L, \nu}(\tilde{A}) = \begin{cases} \sqrt{i}^{N(1-\mathrm{sgn}(d))} \delta_{\mu, a\nu} \exp(\pi i a b(\mu, \mu)) & , c = 0 \\ \varepsilon(\tilde{A}) \sum_{l \in L/cL} \exp\left(\frac{\pi i}{c}(a(\mu + l, \mu + l) - 2(\nu, \mu + l) + d(\nu, \nu))\right) & , c \neq 0 \end{cases}$$

where

$$\varepsilon(\tilde{A}) = \frac{|c|^{\frac{-N}{2}} \sqrt{i}^{N \mathrm{sgn}(c)}}{\sqrt{|L^\vee/L|}}$$

Here δ denotes the Kronecker delta.

We note that ϱ_L is a finite representation. Let V be a finite-dimensional vector space over $\mathbb{C}$ and let $f : \mathbb{H} \rightarrow V$ be a holomorphic function. For $k \in \mathbb{Z}/2$ and $(A, \phi) \in \mathrm{Mp}(2, \mathbb{Z})$ we define the Petersson slash operator as usual by

$$f|_k(A, \phi)(\tau) = \phi(\tau)^{-2k} f(A\tau).$$

This leads us to the definition of a vector valued modular form.

Definition 4.1.3 Suppose that ϱ is a finite representation of the group $\mathrm{Mp}(2, \mathbb{Z})$ on the finite-dimensional vector space V and let $k \in \mathbb{Z}/2$. A *holomorphic modular form with respect to $\mathrm{Mp}(2, \mathbb{Z})$* is a holomorphic function $f : \mathbb{H} \rightarrow V$ satisfying

- (i) $f|_k(A, \phi) = \varrho(A, \phi)f$ for all $(A, \phi) \in \mathrm{Mp}(2, \mathbb{Z})$
- (ii) f is holomorphic at the cusp ∞ .

If additionally f has a zero at the cusp ∞ then we call f a *cusp form*.

We would like to make the behaviour of a modular form at the cusp ∞ more precise. The invariance under $\mathrm{Mp}(2, \mathbb{Z})$ implies $f(\tau + a) = \varrho(\tilde{T})^a f(\tau)$ for all $a \in \mathbb{N}$. Since ϱ is a finite representation there exists some $\kappa \in \mathbb{N}$ such that $f(\tau + \kappa) = f(\tau)$. This implies that f possesses a Fourier expansion of the shape

$$f(\tau) = \sum_{\gamma} \sum_{n \in \mathbb{Z}} c(n, \gamma) \exp\left(\frac{2\pi i n \tau}{\kappa}\right) \mathfrak{e}_\gamma$$

where γ runs through some basis of V . Then condition (ii) is equivalent to

$$f(\tau) = \sum_{\gamma} \sum_{n \geq 0} c(n, \gamma) \exp\left(\frac{2\pi i n \tau}{\kappa}\right) \mathbf{e}_{\gamma}$$

and being a cusp form is equivalent to saying $c(n, \gamma) \neq 0$ implies $n > 0$. The space of holomorphic modular forms of weight k is denoted by $\mathcal{M}_k^V(\varrho)$ and the space of cusp forms is denoted by $\mathcal{S}_k^V(\varrho)$. If we consider $\varrho_L(A, \phi)$ as a matrix the image of the dual representation ϱ_L^* is given by the complex conjugate of ϱ_L . Note that $\varrho_L^* = \varrho_{-L}$. We like to give some examples.

Example 4.1.4 (a) The multiplier system of Dedekind's eta function becomes a finite character for the metaplectic group defined by

$$v_{\eta}(\tilde{J}) = \exp(3\pi i/4) \quad v_{\eta}(\tilde{T}) = \exp(\pi i/12).$$

If χ is any (finite) character for $\mathrm{Mp}(2, \mathbb{Z})$ then χ^2 becomes an abelian character for $\mathrm{SL}(2, \mathbb{Z})$ defined by $\chi^2(A) := \chi^2(\tilde{A})$ for $A \in \mathrm{SL}(2, \mathbb{Z})$. Note that we only have to check that the order of $\chi^2(J)$ divides 4. This shows that every finite character of $\mathrm{Mp}(2, \mathbb{Z})$ is a power of v_{η} and hence $\mathrm{Mp}(2, \mathbb{Z})^{\mathrm{ab}} \cong \mathcal{C}_{24}$. So we have $\eta \in \mathcal{S}_{\frac{1}{2}}^V(v_{\eta})$.

(b) Let $f \in \mathcal{M}_k^V(\varrho_L^*)$. From (4.1) we derive the shape of the Fourier expansion as

$$f(\tau) = \sum_{\gamma} \sum_{n \in -\frac{(\gamma, \gamma)}{2} + \mathbb{Z}_{\geq 0}} c(n, \gamma) \exp(2\pi i n \tau) \mathbf{e}_{\gamma}.$$

Following this example we will also consider vector valued modular forms with respect to a character χ for $\mathrm{Mp}(2, \mathbb{Z})$ which is a power of v_{η} . There is a well-known correspondence between Jacobi forms for the lattice L and vector valued modular forms with respect to the Weil representation.

Proposition 4.1.5 *Let $\varphi \in J_{k, L; 1}(v_{\eta}^a \times \nu_{H(L)})$ where $k \in \mathbb{Z}$ and consider the decomposition from Lemma 2.1.10*

$$\varphi(\tau, \mathfrak{z}) = \sum_{\mu \in L^{\vee}/L} \varphi_{\mu}(\tau) \theta_{\mu, 1}^L(\tau, \mathfrak{z}).$$

The map

$$J_{k, L; 1}(v_{\eta}^a \times \nu_{H(L)}) \rightarrow \mathcal{M}_{k-N/2}(\varrho_L^*, v_{\eta}^a) \quad , \quad \varphi \mapsto (\varphi_{\mu})_{\mu}$$

defines an isomorphism between the linear spaces.

This follows from Proposition 4.1.2 and the behaviour of $\theta_{\mu, 1}^L$ under the action of the slash operator, confer for the latter [Sh], Proposition 1.6. Let $\beta \in L^{\vee}/L$. Following [EZ] we define the linear operator

$$\Phi_{\beta} : J_{k, L, t}(1) \rightarrow \mathbb{C} \quad , \quad \varphi \mapsto \lim_{s \rightarrow \infty} [\varphi|_{k, t} [\beta, 0 : 0] (si, 0)].$$

Note that for $t \in \mathbb{Z}$ we have

$$\varphi|_{k,t}[\beta, 0 : 0](\tau, \mathfrak{z}) = \exp(\pi i t((\beta, \beta)\tau + 2(\beta, \mathfrak{z}))) \sum_{\substack{n \geq 0 \\ l \in \bar{L}^\vee}} f(n, l) e^{2\pi i(n\tau + (l, \mathfrak{z} + \tau\beta))}. \quad (4.2)$$

That this map is in fact an operator is the contents of the next lemma which gives us another description of cuspidal Jacobi forms.

Lemma 4.1.6 (a) Let $\varphi \in J_{k,L;t}^{(weak)}(1)$ and suppose $t \in \mathbb{Z}$. Then we have

$$\varphi \in J_{k,L;t}(1) \iff \forall \beta \in L^\vee/L : \Phi_\beta(\varphi) \text{ does exist.}$$

(b) The function $\varphi \in J_{k,L;t}(1)$ is a Jacobi cusp form if and only if φ is in the kernel of Φ_β for all zero-dimensional cusps of $\mathcal{D}(L_2)$, i.e.

$$\Phi_\beta(\varphi) = 0 \text{ for all isotropic } \beta \in L^\vee/L.$$

Proof

Since $J_{k,L;t}(1) = J_{k,L(t);1}(1)$ it suffices to consider the index one case.

(a) Let $\varphi \in J_{k,L;1}(1)$ be a Jacobi form and let $\beta \in L^\vee/L$. For n, l satisfying $f(n, l) \neq 0$ we have by definition

$$[2n + (\beta, \beta) + 2(l, \beta)] - (l + \beta, l + \beta) = 2n - (l, l) \geq 0$$

and $(l + \beta, l + \beta) \geq 0$. Hence the limit

$$\Phi_\beta(\varphi) = \lim_{s \rightarrow \infty} \sum_{\substack{n \geq 0 \\ l \in \bar{L}^\vee}} f(n, l) e^{-\pi(2n + (\beta, \beta) + 2(l, \beta))s} \quad (4.3)$$

does exist. If the last expression does exist for all $\beta \in L^\vee/L$ we consider n, l such that $f(n, l) \neq 0$. Due to Lemma 2.1.9 we can choose l such that there exists a $\beta \in L^\vee/L$ satisfying $\beta \equiv l \pmod{L}$. From (4.3) we deduce $2n + (\beta, \beta) + 2(l, \beta) \geq 0$ for all $\beta \in L^\vee/L$. This implies

$$2n - (l, l) = 2n + (\beta, \beta) + 2(l, \beta) - (l + \beta, l + \beta) \geq -(l + \beta, l + \beta).$$

If we choose $\beta = -l$ the last inequality yields $2n - (l, l) \geq 0$ which was to be shown.

(b) Suppose φ is a cusp form. The same argument as in part (a) considering the strict inequality yields $\Phi_\beta(\varphi) = 0$ for all cusps. Now suppose the last statement is true for all cusps β . Let n, l satisfy $2n - (l, l) = 0$. By choosing $\beta = -l$ we conclude $2n + (\beta, \beta) + 2(l, \beta) = 0$ (note that l is isotropic). Then (4.3) and $\Phi_{-l}(\varphi) = 0$ imply $f(n, l) = 0$. $\square$

Remark 4.1.7 If β is isotropic then $\Phi_\beta(\varphi)$ is the value of the Jacobi form φ at the zero-dimensional boundary component associated to c_β .

4.2 Jacobi-Eisenstein series

Let $k, t \in \mathbb{N}$. The finite-dimensional vector space of Jacobi forms of fixed index and fixed weight admits a decomposition as

$$J_{k,L;t} = J_{k,L;t}^{(cusp)} \oplus J_{k,L;t}^{(eis)}$$

where *cusp* denotes the subspace of Jacobi cusp forms and the second component is the complementary space. Our next aim is to investigate the codimension of $J_{k,L;t}^{(cusp)}$. For the classical setting where $L = A_1$ this was already done in [EZ]. The answer is the following. Write $m = ab^2$ where a is square-free. If k is even the dimension of $J_{k,L;m}^{(eis)}$ is exactly $\lfloor \frac{b}{2} \rfloor + 1$. This can be interpreted as follows. Each Jacobi form (for an arbitrary lattice) can be considered as a Jacobi form with lattice-valued index (depending on t and L) see for example [Z]. In the classical setting this is reflected by the index being a lattice of type $A_1(t)$ namely the even lattice $\mathbb{Z}$ of rank one with quadratic form $2tx^2$. The discriminant group of this lattice contains no nontrivial isotropic elements if and only if t is square-free. In this case $\dim J_{k,L;t}^{(eis)} = 1$. If t is not square-free the formula above tells us that

$$\dim J_{k,L,t}^{(eis)} = \# \mathcal{D}(L_2(A_1))_{\Gamma}^0, \quad \Gamma = \widetilde{\mathrm{SO}}(L_2(A_1))^+,$$

confer Lemma 1.6.7. We will show that this result is also true if L_2 is arbitrary. We define the subgroup

$$\tilde{\Gamma}_{\infty} = \langle \tilde{T} \rangle = \left\{ \left(\begin{pmatrix} 1 & n \\ 0 & 1 \end{pmatrix}, 1 \right), n \in \mathbb{Z} \right\} \leq \mathrm{Mp}(2, \mathbb{Z})$$

Following [Br] we define vector valued Eisenstein series.

Definition 4.2.1 Let $\beta \in L^{\vee}/L$ be isotropic and $k \in \mathbb{Z}/2$ such that $k \geq \frac{5}{2}$. The *vector valued Eisenstein series* of weight k is defined as

$$\vec{e}_{L,k,\beta} := \frac{1}{2} \sum_{(A,\phi) \in \tilde{\Gamma}_{\infty} \backslash \mathrm{Mp}(2,\mathbb{Z})} \varrho_L^*(A, \phi)^{-1} \mathbf{e}_{\beta}|_k(A, \phi).$$

For a proof of the next Theorem we refer to [Br], Theorem 1.6.

Theorem 4.2.2 Let $k \geq 5/2$ and let $\beta \in L^{\vee}/L$ be isotropic. Then we have $\vec{e}_{L,k,\beta} \in \mathcal{M}_k^V(\varrho_L^*)$. The Fourier expansion is given by

$$\vec{e}_{L,k,\beta}(\tau) = \sum_{\gamma \in L^{\vee}/L} \sum_{\substack{n \geq 0 \\ n \in \mathbb{Z} - q(\gamma)}} c_{\beta}(n, \gamma) \exp(2\pi i n \tau) \mathbf{e}_{\gamma}$$

where

$$c_{\beta}(n, \gamma) = \begin{cases} \delta_{\beta,\gamma} + \delta_{-\beta,\gamma} & \text{if } n = 0 \\ \frac{(2\pi)^k n^{k-1}}{\Gamma(k)} \sum_{c \in \mathbb{Z} \setminus \{0\}} |c|^{1-k} H_c^*(\beta, 0, \gamma, n) & \text{if } n > 0. \end{cases}$$

Here $H_c^*(\beta, 0, \gamma, n)$ denotes a generalized Kloosterman sum. If σ is any permutation of the classes of $L^\vee/L$ then we have $H_c^*(\sigma(\beta), 0, \gamma, n) = H_c^*(\beta, 0, \sigma^{-1}(\gamma), n)$.

In the next step we are going to define Jacobi-Eisenstein series for every 0-dimensional cusp of $\mathcal{D}(L_2)_\Gamma^0$ where $\Gamma = \widetilde{\mathrm{SO}}(L_2)^+$. For this purpose we give a modified description of the Jacobi group $\Gamma^J(L)$. The *real Heisenberg group* is defined as

$$H(L \otimes \mathbb{R}) := \langle [x, y : r] ; x, y \in L \otimes \mathbb{R}, r \in \mathbb{R} \rangle .$$

The group law is

$$[x, y : r] \cdot [x', y' : r'] = [x + x', y + y', r + r' + \frac{1}{2}((x, y') - (x', y))] .$$

We observe that the projection

$$H(L \otimes \mathbb{R}) \rightarrow (L \otimes \mathbb{R}) \times (L \otimes \mathbb{R}) \quad , \quad [x, y : r] \mapsto (x, y)$$

has kernel $\{[0, 0 : r] ; r \in \mathbb{R}\}$ which equals the centre of $H(L \otimes \mathbb{R})$. Therefore the Heisenberg group is the central extension of $\mathbb{R}$ with $(L \otimes \mathbb{R}) \times (L \otimes \mathbb{R})$

$$0 \rightarrow \mathbb{R} \rightarrow H(L \otimes \mathbb{R}) \rightarrow (L \otimes \mathbb{R}) \times (L \otimes \mathbb{R}) \rightarrow 0 .$$

We have an action of $\mathrm{SL}(2, \mathbb{R})$ on the Heisenberg group which is given by

$$\mathrm{SL}(2, \mathbb{R}) \curvearrowright H(L \otimes \mathbb{R}) \quad , \quad A.[x, y : r] := \{A\}[x, y : r]\{A^{-1}\} = [dx - cy, -bx + ay : r] .$$

As usual we denote the 1-sphere by

$$S^1 = \{\zeta \in \mathbb{C}; |\zeta| = 1\} .$$

Following [EZ] we consider the group

$$C_{\mathbb{Z}} = \{[0, 0 : r] ; r \in \mathbb{Z}\}$$

which is a subgroup of the centre of $H(L \otimes \mathbb{R})$. This gives us the following description of the factor group

Lemma 4.2.3 *There is a central extension of $H(L \otimes \mathbb{R})/C_{\mathbb{Z}}$ given by*

$$1 \rightarrow S^1 \rightarrow H(L \otimes \mathbb{R})/C_{\mathbb{Z}} \rightarrow (L \otimes \mathbb{R}) \times (L \otimes \mathbb{R}) \rightarrow 0 .$$

Proof

We define an embedding $\iota : S^1 \rightarrow H(L \otimes \mathbb{R})/C_{\mathbb{Z}}$, $e^{2\pi i \kappa} \mapsto [0, 0 : \kappa]$. This is obviously well-defined and injective and since $\iota(S^1)$ is contained in the centre of the Heisenberg group we also have $\iota(S^1) \subseteq C(H(L \otimes \mathbb{R})/C_{\mathbb{Z}})$. If we define the projection as above we obtain that the sequence is exact. $\square$

The factor group inherits the $\mathrm{SL}(2, \mathbb{R})$ -action from $H(L \otimes \mathbb{R})$ so that we can define

$$G^J := \mathrm{SL}(2, \mathbb{R}) \ltimes H(L \otimes \mathbb{R})/C_{\mathbb{Z}}.$$

According to Lemma 4.2.3 we write $\{A, [x, y], \zeta\}$ for the elements of G^J where $A \in \mathrm{SL}(2, \mathbb{R})$, $x, y \in L \otimes \mathbb{R}$ and $\zeta \in S^1$ and the group law reads

$$\begin{aligned} & \{A, [x, y], \zeta\} \cdot \{A', [x', y'], \zeta'\} \\ &= \left\{ AA', [d'x - c'y, -b'x + a'y] + [x', y'], \zeta\zeta' \cdot e^{\pi i((d'x - c'y, y') - (x', -b'x + a'y))} \right\} \end{aligned}$$

We now consider the discrete subgroup

$$\Gamma_s^J(L) := \mathrm{SL}(2, \mathbb{Z}) \ltimes (L \times L) = \langle \{A, [x, y], 1\} \mid A \in \mathrm{SL}(2, \mathbb{Z}), x, y \in L \rangle \leq G^J.$$

This group corresponds to the classical Jacobi group in [EZ]. We introduce the subgroup

$$\Gamma_{\infty}^J(L) = \left\{ g \in \Gamma_s^J ; 1 \mid_{k,1} g = 1 \right\}$$

We now assume that k is even. Since every element in Γ_s^J can be written in the form $\{I, [x, y]\} \times \{A, [0, 0]\}$ where $x, y \in L$ and $A \in \mathrm{SL}(2, \mathbb{Z})$ we obtain

$$\begin{aligned} \left(1 \mid_{k,1} \{I, [x, y]\} \right) \Big|_{k,1} \{A, [0, 0]\} &= e^{\pi i m((x, x)\tau + 2(x, \mathfrak{z}))} \Big|_{k,m} A \\ &= (c\tau + d)^{-k} \cdot e^{-\pi i m \frac{c(\mathfrak{z}, \mathfrak{z})}{c\tau + d}} \cdot e^{\pi i m((x, x)\frac{a\tau + b}{c\tau + d} + 2(x, \frac{\mathfrak{z}}{c\tau + d}))} \end{aligned}$$

If we put $(\tau, \mathfrak{z}) = (2i, 0)$ and $\varphi = \arg(2ci + d)$ the last expression equals

$$\begin{aligned} & (2ci + d)^{-k} \cdot e^{-\pi i m \frac{c(0,0)}{2ci+d}} \cdot e^{\pi i m((x, x)\frac{2ai+b}{2ci+d} + 2(x, \frac{0}{2ci+d}))} \\ &= (4c^2 + d^2)^{-k/2} \cdot e^{-ik\varphi} \cdot e^{i\pi m \frac{(2i+4ac+bd)(x, x)}{4c^2+d^2}} \\ &= (4c^2 + d^2)^{-k/2} e^{-\frac{\pi m(x, x)}{4c^2+d^2}} \cdot e^{-ik\varphi} \cdot e^{i\pi m \frac{(4ac+bd)(x, x)}{4c^2+d^2}}. \end{aligned}$$

So one has

$$\{I, [x, y]\} \{A, [0, 0]\} \in \Gamma_{\infty}^J \iff 1 = (4c^2 + d^2)^{-k/2} e^{-\frac{\pi m(x, x)}{4c^2+d^2}} \cdot e^{-ik\varphi} \cdot e^{i\pi m \frac{(4ac+bd)(x, x)}{4c^2+d^2}}$$

This immediately implies $x = 0$ and $c = 0$. This gives us the alternative description

$$\Gamma_{\infty}^J = \left\{ \left\{ \pm \begin{pmatrix} 1 & n \\ 0 & 1 \end{pmatrix}, [0, y] \right\} ; y \in L, n \in \mathbb{Z} \right\}.$$

One has the following decomposition into cosets

$$\begin{aligned} \Gamma_s^J &= \bigcup_{\substack{c, d \in \mathbb{Z}, c \geq 0 \\ (c, d) \neq (0, -1) \\ \gcd(c, d) = 1}} \bigcup_{x \in L} \Gamma_{\infty}^J \left\{ \begin{pmatrix} a & b \\ c & d \end{pmatrix}, [x, 0] \right\} \\ &= \bigcup_{\substack{c, d \in \mathbb{Z}, c \geq 0 \\ (c, d) \neq (0, -1) \\ \gcd(c, d) = 1}} \bigcup_{x \in L} \Gamma_{\infty}^J \left\{ \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}, [x, 0] \right\} \cdot \left\{ \begin{pmatrix} a & b \\ c & d \end{pmatrix}, [0, 0] \right\} \end{aligned} \tag{4.4}$$

where a, b are chosen such that $\begin{pmatrix} a & b \\ c & d \end{pmatrix} \in \mathrm{SL}(2, \mathbb{Z})$.

Remark 4.2.4 Note that the centre of $H(L)$ acts trivially on Jacobi forms of integral index. We have

$$\Gamma_s^J \cong \mathrm{SL}(2, \mathbb{Z}) \ltimes H_s(L).$$

In this sense g runs through a system of representatives of $\Gamma_\infty^J \backslash \Gamma_s^J$ if and only if $g[x, y : r]$ does, where $[x, y : r] \in H(L)$.

We define a subgroup of index two

$$\tilde{\Gamma}_\infty^J = \left\{ \left\{ \begin{pmatrix} 1 & n \\ 0 & 1 \end{pmatrix}, [0, y] \right\} ; y \in L, n \in \mathbb{Z} \right\} \leq \Gamma_\infty^J.$$

Definition 4.2.5 Let $k \in \mathbb{N}$ where $k \geq 2 + \frac{N+1}{2}$. Suppose $\beta \in L^\vee/L$ is isotropic, i.e., $(\beta, \beta) \in 2\mathbb{Z}$. The *Jacobi-Eisenstein series* of weight k at the cusp associated to c_β is given as

$$e_{k,\beta}^L := \frac{1}{2} \sum_{g \in \tilde{\Gamma}_\infty^J \backslash \Gamma_s^J} 1|_{k,1} [\beta, 0 : 0] g.$$

Lemma 4.2.6 (a) The Jacobi-Eisenstein series $e_{k,\beta}^L$ is a well-defined holomorphic function on $\mathbb{H} \times (L \otimes \mathbb{C})$.

(b) Let $\beta, \gamma \in L^\vee/L$ be isotropic and define $\sigma = [\gamma - \beta, \beta - \gamma : \varepsilon] \in H(L \otimes \mathbb{Q})$ as in Remark 1.6.14. Then we have

$$e_{k,\beta}^L|_{k,1} \sigma = e_{k,\gamma}^L.$$

Remark 4.2.7 Note that the element σ acts by the same formula as the integral Heisenberg group $H(L)$ since $\beta - \gamma$ is isotropic. The last statement is true since β, γ are elements in some overlattice M of L showing that the difference $\beta - \gamma$ also belongs to M and thus is isotropic, too.

Proof

Considering the holomorphicity we just mention at this moment that by $e_{k,\beta}^L$ we indeed obtain a uniformly convergent series on all vertical strips

$$V(\delta) := \{(\tau, \mathfrak{z}) \in \mathbb{H} \times (L \otimes \mathbb{C}) ; \tau = x + iy, y \geq \delta, x^2 \leq \delta^{-1}, \bar{\mathfrak{z}}^{\mathrm{tr}} \mathfrak{z} \leq \delta^{-1}\}.$$

The proof will follow from the identification in the next Proposition. We show that the definition does not depend on the choice of the representative in $\tilde{\Gamma}_\infty^J \backslash \Gamma_s^J$. Let $g \in \tilde{\Gamma}_\infty^J$. From the above description we obtain

$$1|_{k,1} [\beta, 0] g = 1|_{k,1} [\beta, m\beta] \quad , \quad m \in \mathbb{Z}.$$

We note that

$$1|_{k,1} [\beta, r] = 1|_{k,1} [\beta, 0] \quad , \quad \text{for any } r \in L \otimes \mathbb{Q}.$$

This shows that $e_{k,\beta}^L$ is well-defined. Now if γ is chosen accordingly we let $g \in \tilde{\Gamma}_\infty^J \backslash \Gamma_s^J$. According to (4.4) we can assume g to be of the shape

$$g = \left\{ \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}, [x, 0] \right\} \cdot \left\{ \begin{pmatrix} a & b \\ c & d \end{pmatrix}, [0, 0] \right\}.$$

Then there is some $r \in L \otimes \mathbb{Q}$ such that

$$g\sigma = \left\{ \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}, [0, r] \right\} \left\{ \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}, [\gamma - \beta, 0] \right\} g.$$

If we apply the slash operator this yields

$$1|_{k,1} [\beta, 0] g\sigma = 1|_{k,1} [\gamma, 0] g$$

showing the second assertion. □

In the next Proposition we establish the theta correspondence for Eisenstein series.

Proposition 4.2.8 *Let $k \geq 2 + \frac{N+1}{2}$ and $\beta \in L^\vee/L$ be isotropic. Then we have*

$$e_{k,\beta}^L = \frac{1}{2} \sum_{\mu \in L^\vee/L} (\vec{e}_{L,k-N/2,\beta}, \mathfrak{e}_\mu) \cdot \theta_{\mu,1}^L$$

where $(\cdot, \cdot)$ denotes the standard scalar product on $\mathbb{C}[L^\vee/L]$.

Proof

Let $N = \text{rank}(L)$. By Lemma 2.1.13 we know that

$$\Theta_1^L \left(A\tau, \frac{\mathfrak{z}}{c\tau + d} \right) = \exp \left(\pi i \left(\frac{c(\mathfrak{z}, \mathfrak{z})}{c\tau + d} \right) \right) \phi(\tau)^N \varrho_L^*(A, \phi)^{-1} \Theta_1^L(\tau, \mathfrak{z}), \quad (A, \phi) \in \text{Mp}(2, \mathbb{Z})$$

where

$$\Theta_1^L = \sum_{\mu \in L^\vee/L} \theta_{\mu,1}^L \mathfrak{e}_\mu.$$

A computation using (4.4) yields

$$\begin{aligned} e_{k,\beta}^L(\tau, \mathfrak{z}) &= \frac{1}{2} \sum_{\substack{c,d \in \mathbb{Z} \\ (c,d)=1}} (c\tau + d)^{-k} \exp \left(-\pi i \left(\frac{c(\mathfrak{z}, \mathfrak{z})}{c\tau + d} \right) \right) \\ &\quad \times \theta_{\beta,1}^L \left(A\tau, \frac{\mathfrak{z}}{c\tau + d} \right) \\ &= \frac{1}{2} \sum_{\substack{c,d \in \mathbb{Z} \\ (c,d)=1}} \phi(\tau)^{N-2k} \left(\varrho_L^*(\tilde{A})^{-1} \Theta_1^L(\tau, \mathfrak{z}), \mathfrak{e}_\beta \right) \end{aligned}$$

where ϕ is defined by $\tilde{A} = (A, \phi)$. We have

$$\varrho_L(C)\mathfrak{e}_\mu = i^N \mathfrak{e}_{-\mu} \quad \text{hence} \quad \varrho_L^*(C^2)^{-1}\mathfrak{e}_\mu = (-1)^N \mathfrak{e}_\mu.$$

Moreover one has

$$\phi(\tau)^{N-2k} \left(\varrho_L^*(\tilde{A})^{-1} \Theta_1^L(\tau, \mathfrak{z}), \mathfrak{e}_\beta \right) = \sum_{\mu \in L^\vee/L} \left(\varrho_L^*(\tilde{A})^{-1} \mathfrak{e}_\beta \Big|_{k-N/2} \tilde{A}, \mathfrak{e}_\mu \right) \cdot \theta_{\mu,1}^L.$$

Now the cosets $\langle \tilde{\Gamma}_\infty, C^2 \rangle \setminus \text{Mp}(2, \mathbb{Z})$ are exactly parametrized by $c, d \in \mathbb{Z}$ such that c, d are coprime. We observe that

$$\varrho_L^*(C^2)^{-1} \mathfrak{e}_\beta \Big|_{k-N/2} C^2 = (-1)^{N-2k} (-1)^N \mathfrak{e}_\beta.$$

Since k is integral C^2 acts trivially what explains the appearance of the factor $\frac{1}{2}$ in the above formula. Now the assertion follows from the definition of $\vec{e}_{L,k-N/2,\beta}$ and the fact that $\tilde{A}C^{2\epsilon}$ where $\epsilon \in \{0, 1\}$ runs through a complete set of representatives of $\tilde{\Gamma}_\infty \setminus \text{Mp}(2, \mathbb{Z})$ if A runs through a complete set of representatives of $\langle T^n | n \in \mathbb{Z} \rangle \setminus \text{SL}(2, \mathbb{Z})$. $\square$

In the next Theorem we can use Jacobi-Eisenstein series to construct an explicit basis for the space $J_{k,L;1}^{(eis)}$.

Theorem 4.2.9 *Let $k \in \mathbb{N}$ such that $k \geq 2 + \frac{N+1}{2}$ where N is the rank of the positive definite even lattice L .*

(a) *For every isotropic $\beta \in L^\vee/L$ one has $e_{k,\beta}^L \in J_{k,L;1}$.*

(b) *Let $\beta, \gamma \in L^\vee/L$ be isotropic. The operator $\Phi = \bigoplus_\beta \Phi_\beta$ is a normalized projector*

$$\Phi : J_{k,L;1} \rightarrow J_{k,L;1}^{(eis)} \quad , \quad \varphi \mapsto \sum_{\substack{\beta \in L^\vee/L \\ \beta \text{ isotropic}}} \Phi_\beta(\varphi)$$

in the following sense:

$$\Phi_\beta(e_{k,\gamma}^L) = \begin{cases} 1 & \text{if } \beta = \pm\gamma \text{ mod } L \\ 0 & \text{else.} \end{cases}$$

(c) *The Jacobi-Eisenstein series $e_{k,\beta}^L$ where β runs through representatives of isotropic vectors in $L^\vee/L$ span $J_{k,L;1}^{(eis)}$. We have $e_{k,-\beta}^L = e_{k,\beta}^L$ and*

$$\dim J_{k,L;1}^{(eis)} = \sharp \mathcal{D}(L_2)_\Gamma^0 \quad , \quad \Gamma = \widetilde{\text{SO}}(L_2)^+.$$

(d) If M is an overlattice of L then we have

$$e_{k,0}^M = \sum_{\beta \in L \setminus M} e_{k,\beta}^L.$$

Proof

(a) This follows from Proposition 4.2.8 and Proposition 4.1.5.

(b) We calculate using Proposition 4.2.8 and Theorem 4.2.2

$$\begin{aligned} \Phi_\beta(e_{k,\gamma}^L) &= \lim_{t \rightarrow \infty} (e_{k,\gamma}^L|_{k,1}[\beta, 0 : 0](ti, 0)) \\ &= \lim_{t \rightarrow \infty} (e_{k,\beta+\gamma}^L(ti, 0)) \\ &= \frac{1}{2} \lim_{t \rightarrow \infty} \sum_{\mu \in L^\vee/L} \sum_{\substack{n \in \mathbb{Z}-q(\mu) \\ n \geq 0}} c_{\beta+\gamma}(n, \mu) \exp(-2\pi t n) \theta_{\mu,1}^L(ti, 0) \end{aligned}$$

Note that

$$\lim_{t \rightarrow \infty} \theta_{\mu,1}^L(it, 0) = \begin{cases} 0 & \text{if } \mu \in L^\vee \setminus L \\ 1 & \text{if } \mu \in L. \end{cases}$$

This yields

$$\Phi_\beta(e_{k,\gamma}^L) = \frac{1}{2} c_{\beta+\gamma}(0, 0) = \frac{\delta_{\beta+\gamma,0} + \delta_{-\beta-\gamma,0}}{2}.$$

Now the claim follows from $e_{k,-\beta}^L = e_{k,\beta}^L$.

(c) From part (b) follows that $e_{k,\beta}^L$ where $\beta \in L^\vee/L$ runs through the isotropic classes in $D(L)$ and $\beta + L, -\beta + L$ are identified are linearly independent. Let us consider the Fourier expansion of a holomorphic Jacobi form at the simplest cusp. By virtue of Lemma 2.1.9 the Fourier coefficients depend only on the hyperbolic norm $2n - (l, l)$ and the class of $l \in L^\vee$. If $f(n, l)$ denotes the corresponding Fourier coefficient the same Lemma tells us that $f(n, -l) = f(n, l)$. Hence the codimension of $J_{k,L;1}^{(cusp)}$ is bounded by the number of isotropic elements in $D(L)$ where for each such $l \in D(L)$ satisfying $l \neq -l$ we count either l or $-l$. Now Proposition 1.6.7 shows that the forementioned formula for the dimension is true.

(d) Suppose $[M : L] = m$ and consider a set of representatives $l_1, \dots, l_m \in L^\vee$ for $L \setminus M$. We first note that $[H(M) : H(L)] = m^2$ and $[l_\alpha, l_{\alpha'} : 0]$ where $1 \leq \alpha, \alpha' \leq m$ is a complete set of representatives for $H(L) \setminus H(M)$. With some obvious identifications this yields

$$\Gamma_\infty^J(M) \setminus \Gamma_s^J(M) = \bigcup_{\alpha=1}^m \Gamma_\infty^J(L) \setminus (\Gamma_s^J(L) \cdot [l_\alpha, 0 : 0]).$$

This completes the proof. □

We note another useful property of Jacobi-Eisenstein series.

Lemma 4.2.10 *For any $\sigma \in \mathrm{O}(L)$ and k, β chosen as before we have*

$$e_{k,\beta}^L|_k\sigma = e_{k,\sigma\beta}^L$$

where the action of $\mathrm{O}(L)$ is defined as before by

$$\varphi(\tau, \mathfrak{z})|_k\sigma = \varphi(\tau, \sigma\mathfrak{z}).$$

This corresponds to the usual action of the modular group. The factor of automorphy for $\mathrm{O}(L)$ is trivial.

Proof

We make again use of the theta correspondence for Jacobi-Eisenstein series given in Proposition 4.2.8. For any $\mu \in L^\vee/L$ one observes

$$\theta_{\mu,1}^L|_{N/2}\sigma = \theta_{\sigma^{-1}\mu,1}^L.$$

Theorem 4.2.2 implies

$$c_\beta(n, \sigma^{-1}\gamma) = c_{\sigma\beta}(n, \gamma).$$

Hence the Fourier expansion of $\vec{e}_{L,k-N/2,\beta}$ yields

$$e_{k,\beta}^L|_k\sigma = \frac{1}{2} \sum_{\mu \in L^\vee/L} (\vec{e}_{L,k-N/2,\beta}, \mathfrak{e}_\mu) \cdot \theta_{\sigma^{-1}\mu,1}^L = \frac{1}{2} \sum_{\mu \in L^\vee/L} (\vec{e}_{L,k-N/2,\sigma\beta}, \mathfrak{e}_\mu) \cdot \theta_{\mu,1}^L = e_{k,\sigma\beta}^L.$$

□

The next corollary tells us how to compute the Fourier coefficients of $e_{k,0}^L$ from $\vec{e}_{L,k-N/2,0}$.

Corollary 4.2.11 *Let $k \in \mathbb{N}$ such that $k \geq 2 + \frac{N+1}{2}$. We denote the Fourier Jacobi coefficients of $e_{k,0}^L$ with respect to the simplest cusp by $f_0(n, l)$ where $n \in \mathbb{N}$ and $l \in L^\vee$ such that $\Delta(n, l) := 2n - (l, l) \geq 0$. Then $f_0(n, l)$ equals*

$$\begin{cases} \delta_{0,l} & \text{if } \Delta(n, l) = 0 \\ \frac{(2\pi)^{k-N/2}(\Delta(n, l)/2)^{k-1-N/2}}{2\Gamma(k-N/2)} \sum_{c \in \mathbb{Z} \setminus \{0\}} |c|^{1-k+N/2} H_c^*(0, 0, l, \Delta(n, l)/2) & \text{if } \Delta(n, l) > 0. \end{cases}$$

By abuse of notation l is identified with $\mu \in L^\vee/L$ such that $l \cong \mu \bmod L$ in the above formula.

Proof

Theorem 4.2.2 and Proposition 4.2.8 imply

$$\begin{aligned} e_{k,0}^L(\tau, \mathfrak{z}) &= \sum_{\substack{n \in \mathbb{N} \\ l \in L^\vee \\ 2n - (l, l) \geq 0}} f_0(n, l) \exp(2\pi i n \tau) \exp(2\pi i (l, \mathfrak{z})) \\ &= \frac{1}{2} \sum_{\mu \in L^\vee/L} \sum_{\substack{n \in \mathbb{Z} - q(\mu) \\ n \geq 0}} \sum_{l \in L} c_0(n, \mu) \exp(\pi i [2n + (l + \mu, l + \mu)]\tau) \exp(2\pi i (l + \mu, \mathfrak{z})). \end{aligned}$$

Hence we obtain the formula

$$f_0(n + (l, l)/2, l) = \frac{1}{2}c_0(n, l)$$

where $n \in \mathbb{Z} - q(l)$, $n \geq 0$. □

What about the Fourier expansion of $e_{k,\beta}^L$? We like to develop a formula for $e_{k,\beta}^L$ out of the one for $e_{k,0}^L$.

Lemma 4.2.12 *Let $k \in \mathbb{N}$ such that $k \geq 2 + \frac{N+1}{2}$ and $\beta \in L^\vee/L$ be isotropic. We consider the Fourier expansion of $\varphi \in J_{k,L;1}$ with respect to the simplest cusp*

$$\sum_{\substack{n \in \mathbb{N} \\ l \in L^\vee \\ 2n - (l, l) \geq 0}} f(n, l) \exp(2\pi i n \tau) \exp(2\pi i (l, \mathfrak{z})).$$

(a) *The value of φ at the boundary component associated to c_β is given by*

$$\Phi_\beta(\varphi) = f(n_\beta, -\beta) \quad , \quad n_\beta = \frac{(\beta, \beta)}{2}.$$

(b) *One has*

$$\varphi|_{k,1}[\beta, -\beta, 0] = \sum_{\substack{n \in 2\mathbb{Z}/q(L) \\ n \geq 0 \\ l \in L^\vee \\ 2n - (l, l) \geq 0}} f_\beta(n, l) \exp(2\pi i n \tau) \exp(2\pi i (l, \mathfrak{z}))$$

where

$$f_\beta(n, l) = \exp(-2\pi i (\mu, \beta)) \cdot f(n - q(\beta + \mu) + q(\mu), l - \beta)$$

for $n \in \mathbb{Z} + q(\beta + \mu) - q(\mu)$, $l \in L^\vee$, $n \geq 0$. Here $q(L)$ denotes the level of L .

Proof

(a) We have

$$\Phi_\beta(\varphi) = \lim_{s \rightarrow \infty} \sum_{\substack{n \geq 0 \\ l \in L^\vee}} f(n, l) e^{-\pi(2n + (\beta, \beta) + 2(l, \beta))s}.$$

Note that

$$2n + (\beta, \beta) + 2(l, \beta) = 2n - (l, l) + (l + \beta, l + \beta) \geq 0.$$

Since φ is a Jacobi form and L is positive definite this inequality shows that

$$2n + (\beta, \beta) + 2(l, \beta) = 0 \implies l = -\beta \bmod L \text{ and } n = \frac{(l, l)}{2}.$$

Hence we conclude using the same inequality that $2n + (\beta, \beta) + 2(l, \beta)$ is minimal if and only if $l = -\beta$ and $n = \frac{(\beta, \beta)}{2}$. In this case $2n + (\beta, \beta) + 2(l, \beta) = 0$ and the assertion follows.

(b) One easily verifies that

$$\exp(\pi i[(\beta, \beta)\tau + 2(\beta, \mathfrak{z})]) \theta_{\alpha,1}^L(\tau, \mathfrak{z} + \beta\tau + \gamma) = \exp(2\pi i(\mu, \gamma)) \theta_{\alpha+\beta,1}^L(\tau, \mathfrak{z})$$

where $\alpha, \beta, \gamma \in L^\vee$. Hence in view of Lemma 2.1.10 we obtain

$$\varphi|_{k,1}[\beta, -\beta : 0](\tau, \mathfrak{z}) = \sum_{\mu \in L^\vee/L} \varepsilon_{\mu,\beta}(\tau) \sum_{n \in \mathbb{Z}-q(\mu+\beta)} f(n+q(\mu+\beta), \mu) e^{2\pi i n \tau} \theta_{\mu+\beta,1}(\tau, \mathfrak{z})$$

where

$$\varepsilon_{\mu,\beta}(\tau) = \exp(-2\pi i(\mu, \beta)) \cdot \exp(\pi i[(\beta, \beta) + 2(\mu, \beta)]\tau).$$

The condition $f(n+q(\mu+\beta), \mu) \neq 0$ implies $2n + (\beta, \beta) + 2(\mu, \beta) \geq 0$. The Fourier coefficient of

$$\exp(\pi i[2n + (\beta, \beta) + 2(\mu, \beta) + (x + \mu + \beta, x + \mu + \beta)]\tau + 2\pi i(x + \mu + \beta, \mathfrak{z}))$$

where $n \in \mathbb{Z} - q(\mu + \beta)$, $x \in L$ equals $\exp(-2\pi i(\mu, \beta))f(n+q(\mu+\beta), \mu)$. Hence

$$f_\beta(n, l) = \exp(-2\pi i(\mu, \beta)) \cdot f(n - q(\beta + \mu) + q(\mu), l - \beta)$$

for $n \in \mathbb{Z} + q(\beta + \mu) - q(\mu)$, $l \in L^\vee$, $n \geq 0$ and the hyperbolic norm satisfies

$$2n - (l, l) \geq 0.$$

□

The above formula can be considerably improved. In [BrK] the authors found an explicit formula showing that $f_0(n, l)$ is proportional to a fraction of Riemann's ζ -function and a generalized Dirichlet L -series $L(s, \chi_D)$. In the next Theorem we show that a similar result also holds for $f_\beta(n, l)$. In this case the Fourier coefficients belong to a cyclotomic field.

Corollary 4.2.13 *Let k, β be chosen such that $e_{k,\beta}^L$ is well-defined. Consider the Fourier expansion of $e_{k,\beta}^L$ which is given as*

$$e_{k,\beta}^L(\tau, \mathfrak{z}) = \sum_{\mu \in L^\vee/L} \sum_{\substack{n \in \mathbb{Z}-q(\mu) \\ n \geq 0}} f_\beta(n+q(\mu+\beta), \mu) \exp(2\pi i n \tau) \theta_{\mu+\beta,1}(\tau, \mathfrak{z}).$$

Define $\Delta_\beta(n, l) := 2n + (l, l) - (l - \beta, l - \beta)$. Then $f_\beta(n, l)$ equals

$$\begin{cases} \delta_{\beta,l} & , \Delta_\beta(n, l) = 0 \\ \frac{(2\pi)^{k-N/2} (\Delta_\beta(n, l)/2)^{k-1-N/2}}{\exp(2\pi i(l, \beta)) 2\Gamma(k-N/2)} \sum_{c \in \mathbb{Z} \setminus \{0\}} \frac{H_c^*(0, 0, l - \beta, \Delta_\beta(n, l)/2)}{|c|^{k-1-N/2}} & , \Delta_\beta(n, l) > 0 \end{cases}$$

for $n \in \mathbb{Z} - q(l)$ and $l \in L^\vee$. The Fourier coefficients $f_\beta(n, l)$ belong to the cyclotomic field $\mathbb{Q}(\zeta)$ where $\zeta := \exp(4\pi i/q(L))$ and $q(L)$ denotes the level of L .

Proof

This follows from Lemma 4.2.12, Corollary 4.2.2 and Corollary 4.7 in [BrK] where the authors showed that the Fourier coefficients of $\vec{e}_{L,k,0}$ are rational. □

Chapter 5

Borcherds Automorphic Products

In [Bo] Borcherds developed his theory of automorphic products for the orthogonal group. His paper has had an enormous impact on the theory of automorphic forms and provided a machinery to construct automorphic forms with a known divisor. In [G2] Gritsenko gives a modification of Borcherds result, who uses the language of vector valued modular forms. The first section repeats this variant of Borcherds theory. In the second section we apply the results from the first part to the modular forms for the A_1 -tower and D_N -tower. Some parts already appear in [G1] whereas the construction of the anti-symmetric modular forms for the lattices $3A_1, 4A_1$ is new. The last two sections are essentially a summary of Grandpierre's thesis [Gr] where the author uses the technique of quasi-pullbacks to construct reflective modular forms for several primitive embeddings $L \hookrightarrow II_{2,26}$. We again only recall the parts concerned with the towers for A_1, D_N and A_2 due to the two facts that these are the only cases we are interested in this context and secondly the procedure is particularly easy to perform for this examples.

5.1 Multiplicative Liftings

In this section we follow [G2] in the presentation of the material. In order to construct multiplicative liftings in the next section we are mainly interested in the computational aspects of Borcherds' theory. The purpose of this section is to understand how the Fourier coefficients of a suitable weak Jacobi form are used to construct a holomorphic modular form whose weight and divisor are completely determined by the input function. Moreover we will discuss the method used in [GN] which gives us an algorithm for the construction of the input function under the hypothesis that a modular form has a presentation as an arithmetic lifting which contains a "good divisor". In this section let L be a positive definite even lattice.

Definition 5.1.1 Let $X \subseteq L^\vee$ be finite. Let $m : X \rightarrow \mathbb{Z}$ be a function such that there exists some $C \in \mathbb{R}$ satisfying

$$\forall \mathfrak{z} \in L \otimes \mathbb{C} : \sum_{l \in X} m(l)(l, \mathfrak{z})^2 = 2C(\mathfrak{z}, \mathfrak{z}).$$

The pair (X, m) is called a *vector system*.

Let $V = L \otimes \mathbb{R}$ and (X, m) be a vector system. A connected component of

$$V \setminus \left(\bigcup_{x \in X \setminus \{0\}} \{v \in V; (x, v) = 0\} \right)$$

is called a *Weyl chamber*. Let W be a fixed Weyl chamber. Then for $l \in L^\vee$ we define an ordering on $L^\vee$ by

$$l > 0 : \iff \exists w \in W : (l, w) > 0.$$

We first want to show the existence of a vector system. For this sake we need to introduce weakly holomorphic Jacobi forms.

Definition 5.1.2 Let $k \in \frac{1}{2}\mathbb{Z}$ and $t \in \mathbb{Q}$. A holomorphic function $\varphi : \mathbb{H} \times (L \otimes \mathbb{C}) \rightarrow \mathbb{C}$ is called a *weakly holomorphic Jacobi form* of weight k and index t with character χ if it satisfies

$$\forall g \in \Gamma^J(L) : \varphi|_{k,t} g = \chi(g) \varphi$$

and if additionally there is some $K \in \mathbb{N}$ such that

$$\varphi(\tau, \mathfrak{z}) = \sum_{\substack{n \in \mathbb{Q}, n \equiv \frac{D}{24} \pmod{\mathbb{Z}} \\ n \geq -K \\ l \in \frac{1}{2}L^\vee}} f(n, l) e^{2\pi i(n\tau + (l, \mathfrak{z}))}.$$

The space of weakly holomorphic Jacobi forms is denoted by $J_{k,L;t}^{(wh)}(\chi)$.

For a proof of the next statement confer [G2], Proposition 2.2.

Proposition 5.1.3 Let $\varphi \in J_{0,L;1}^{(wh)}(1)$ with Fourier expansion

$$\varphi(\tau, \mathfrak{z}) = \sum_{n \in \mathbb{Z}, l \in L^\vee} f(n, l) q^n \exp(2\pi i(l, \mathfrak{z})).$$

Then there exists a $C \in \mathbb{C} \setminus \{0\}$ such that

$$\forall \mathfrak{z} \in L \otimes \mathbb{C} : \sum_{l \in L^\vee} f(0, l) (l, \mathfrak{z})^2 = 2C(\mathfrak{z}, \mathfrak{z}) \quad (5.1)$$

More precisely we have

$$C = \frac{1}{24} \sum_{l \in L^\vee} f(0, l) - \sum_{n > 0, l \in L^\vee} f(-n, l) \sigma_1(n) = \frac{1}{2 \operatorname{rank}(L)} \sum_{l \in L^\vee} f(0, l) (l, l)$$

and the pair $(\mathcal{V}_\varphi, f(0, \cdot))$ is a vector system where $\mathcal{V}_\varphi \subseteq L^\vee$ such that $f(0, l) \neq 0$ for all $l \in \mathcal{V}_\varphi$

The fact that $\mathcal{V}_\varphi$ is indeed a finite set will follow from the proof of 5.1.6.

Corollary 5.1.4 *Let $\varphi \in J_{0,L;1}^{(wh)}$. Assume that all Fourier coefficients $f(0,l)$ of φ are integral. We define*

$$\psi_\varphi(\tau, \mathfrak{z}) = \prod_{l \in L^\vee, l > 0} \left(\frac{\vartheta(\tau, (l, \mathfrak{z}))}{\eta(\tau)} \right)^{f(0,l)}.$$

Then ψ_φ transforms like a Jacobi form of weight 0 and index C where

$$C = \frac{1}{2 \operatorname{rank}(L)} \sum_{l \in L^\vee} f(0,l)(l, l)$$

with a character of the shape $v_\eta^{2A} \times \nu$ where $A = \sum_{l \in L^\vee, l > 0} f(0,l)$ and ν is a character for the Heisenberg group. Let $B = \frac{1}{2} \sum_{l \in L^\vee, l > 0} f(0,l)l \in \frac{1}{2}L^\vee$. If $C \cdot n(L) \in 2\mathbb{Z}$ then we have $C \cdot s(L) \in \mathbb{Z}$ and $B \in L^\vee$.

Proof

By the transformation behaviour of φ we obtain $\varphi(\tau, -\mathfrak{z}) = \varphi(\tau, \mathfrak{z})$. In particular this yields $f(0, -l) = f(0, l)$ for all $l \in \mathcal{V}_\varphi$ and from the last Proposition we obtain the formula

$$\forall \mathfrak{z} \in L \otimes \mathbb{C} : \sum_{l \in L^\vee, l > 0} f(0,l)(l, \mathfrak{z})^2 = C(\mathfrak{z}, \mathfrak{z}).$$

The transformation properties of ϑ and η under the group $\operatorname{SL}(2, \mathbb{Z})$ imply

$$\begin{aligned} \psi_\varphi \left(\frac{a\tau + b}{c\tau + d}, \frac{\mathfrak{z}}{c\tau + d} \right) &= \prod_{l \in L^\vee, l > 0} (v_\eta(M)^2)^{f(0,l)} \\ &\times \prod_{l \in L^\vee, l > 0} \exp \left(\frac{\pi i c(l, \mathfrak{z})^2 f(0,l)}{c\tau + d} \right) \psi_\varphi(\tau, \mathfrak{z}) \\ &= v_\eta(M)^{\sum_{l \in L^\vee, l > 0} 2f(0,l)} \cdot \exp \left(\frac{\pi i c C(\mathfrak{z}, \mathfrak{z})}{c\tau + d} \right) \psi_\varphi(\tau, \mathfrak{z}) \end{aligned}$$

Thus ψ_φ transforms like a Jacobi form of weight 0 and index C under the group $\operatorname{SL}(2, \mathbb{Z})$. To check the transformation behaviour under the Heisenberg group we note that (5.1) can be written in the form

$$\sum_{l \in L^\vee, l > 0} f(0,l)(l, \mathfrak{z}_1)(l, \mathfrak{z}_2) = C(\mathfrak{z}_1, \mathfrak{z}_2)$$

for $\mathfrak{z}_1, \mathfrak{z}_2 \in L \otimes \mathbb{C}$. Hence we obtain for $x, y \in H(L)$

$$\begin{aligned} \psi_\varphi(\tau, \mathfrak{z} + x\tau + y) &= \prod_{l \in L^\vee, l > 0} (-1)^{f(0,l)[(l,x)+(l,y)]} \\ &\times \exp(-\pi i f(0,l)[(l,x)^2\tau + 2(l,x)(l,\mathfrak{z})]) \psi_\varphi(\tau, \mathfrak{z}) \\ &= \exp(-\pi i C[(x,x)\tau + 2(x,\mathfrak{z})]) (-1)^{\sum_{l \in L^\vee, l > 0} f(0,l)(l,x+y)} \\ &\times \psi_\varphi(\tau, \mathfrak{z}) \end{aligned}$$

From Lemma 2.1.5 we deduce that $C \cdot s(L) \in \mathbb{Z}$ if $C \cdot n(L) \in 2\mathbb{Z}$ which is equivalent to the fact that the Heisenberg part of the character of ψ_φ is trivial. In this case $\sum_{l \in L^\vee, l > 0} f(0, l)l \in 2L^\vee$ and $B \in L^\vee$ and the proof is completed by the last Proposition. $\square$

Lemma 5.1.5 *Let $\varphi \in J_{0, L; 1}^{(wh)}$ be a weakly holomorphic Jacobi form with Fourier expansion*

$$\varphi(\tau, \mathfrak{z}) = \sum_{n \in \mathbb{Z}, l \in L^\vee} f(n, l) q^n r^l$$

where we have put $q = \exp(2\pi i \tau)$, $r^l := \exp(2\pi i(l, \mathfrak{z}))$, $s := \exp(2\pi i \omega)$. Assume that $f(n, l) \in \mathbb{Z}$ if $2n - (l, l) \leq 0$. Then the identity

$$q^A r^B s^C \prod_{(n, l, 0) > 0} (1 - q^n r^l)^{f(0, l)} = \eta(\tau)^{f(0, 0)} \prod_{l > 0} \left(\frac{\vartheta(\tau, (l, \mathfrak{z})) e^{\frac{2\pi i}{\text{rank } L}(l, l)\omega}}{\eta(\tau)} \right)^{f(0, l)}$$

is valid where

$$A = \frac{1}{24} \sum_{l \in L^\vee} f(0, l), \quad B = \frac{1}{2} \sum_{l > 0} f(0, l)l \in \frac{1}{2}L^\vee, \quad C = \frac{1}{2 \text{rank}(L)} \sum_{l \in L^\vee} f(0, l)(l, l)$$

and $(n, l, 0) > 0$ if either $(n > 0 \text{ and } l \text{ arbitrary})$ or $(n = 0 \text{ and } l < 0)$.

Proof

We have a bijection between positive and negative vectors which is given by $l \mapsto -l$. From the product expansions of $\eta(\tau)$ and $\vartheta(\tau, z)$ we obtain

$$\begin{aligned} & \eta(\tau)^{f(0, 0)} \prod_{l > 0} \left(\frac{\vartheta(\tau, (l, \mathfrak{z})) e^{\frac{2\pi i}{\text{rank } L}(l, l)\omega}}{\eta(\tau)} \right)^{f(0, l)} \\ &= q^{\frac{1}{24}f(0, 0)} \prod_{l > 0} (-q^{\frac{1}{12}})^{f(0, l)} \cdot \prod_{l > 0} r^{-\frac{1}{2}f(0, l)l} \cdot \prod_{l > 0} s^{\frac{f(0, l)(l, l)}{\text{rank } L}} \\ & \times \prod_{n=1}^{\infty} (1 - q^n)^{f(0, 0)} \prod_{l > 0} \left(\prod_{n=1}^{\infty} (1 - q^{n-1} r^l)^{f(0, l)} (1 - q^n r^{-l})^{f(0, l)} \right) \\ &= q^{\frac{1}{24}f(0, 0)} \prod_{l > 0} (-q^{\frac{1}{12}})^{f(0, l)} \cdot \prod_{l > 0} r^{-\frac{1}{2}f(0, l)l} \cdot \prod_{l > 0} s^{\frac{1}{\text{rank } L}f(0, l)(l, l)} \\ & \times \prod_{l > 0} r^{f(0, l)l} \cdot \prod_{l < 0} (-1)^{f(0, l)} (1 - r^l)^{f(0, l)} \prod_{n=1}^{\infty} \prod_{l \in L^\vee} (1 - q^n r^l)^{f(0, l)} \\ &= \prod_{l \in L^\vee} q^{\frac{1}{24}f(0, l)} \cdot \prod_{l > 0} r^{\frac{1}{2}f(0, l)l} \cdot \prod_{l \in L^\vee} s^{\frac{f(0, l)(l, l)}{2 \text{rank } L}} \\ & \times \prod_{l < 0} (1 - r^l)^{f(0, l)} \prod_{n=1}^{\infty} \prod_{l \in L^\vee} (1 - q^n r^l)^{f(0, l)} \end{aligned}$$

as stated in the assertion. $\square$

We recall the definition of the rational quadratic divisor. For $h \in L_2 \otimes \mathbb{Q}$ such that $(h, h) < 0$ we have

$$[\text{pr}(Z)] \in \mathcal{D}_{[h]}(L_2) \iff (\text{pr}(Z), h) = 0$$

If we consider the special vectors $h_{n,l} = (0, 1, l, n, 0)$ with respect to our standard basis of L_2 where $l \in L^\vee$ and $n \in \mathbb{Z}$ such that $(h_{n,l}, h_{n,l}) < 0$ we see that $[\text{pr}(Z)] \in \mathcal{D}_{[h_{n,l}]}(L_2)$ if and only if $n\tau - (l, \mathfrak{z}) + \omega = 0$. For the purpose of the next theorem we define the affine divisor with respect to $h_{n,l}$ associated to $\mathcal{D}_{[h_{n,l}]}(L_2)$ as

$$\mathcal{H}_{n,l}(L_1) := \{(\omega, \mathfrak{z}, \tau)^{\text{tr}} \in \mathcal{H}(L_1); n\tau - (l, \mathfrak{z}) + \omega = 0\}.$$

Obviously

$$[\text{pr}(Z)] \in \mathcal{D}_{[h_{n,l}]}(L_2) \iff Z \in \mathcal{H}_{n,l}(L_1).$$

The choice of $h_{n,l}$ and the definition of $\mathcal{H}_{n,l}(L_1)$ are justified by the Eichler criterion in Proposition 1.5.8. If $h \in L_2^\vee$ is vector with $(h, h) < 0$ we can always find a transvection $\tau \in E(L_2)$ such that $h = \tau h_{n,l}$ for some $n \in \mathbb{N}$ and $l \in L^\vee$ where h and $h_{n,l}$ belong to the same class in $D(L_2)$. Hence

$$\tau \mathcal{D}_{[h]}(L_2) = \mathcal{D}_{[h_{n,l}]}(L_2) = [\text{pr}(\mathcal{H}_{n,l}(L_1))].$$

The next Theorem is a special case of Borchers automorphic products [Bo]. The formulation and proof of the next Theorem can be found in [G2].

Theorem 5.1.6 (Borchers, Gritsenko) *Let $\varphi \in J_{0,L;1}^{(wh)}(1)$ be a weakly holomorphic Jacobi form with Fourier expansion*

$$\varphi(\tau, \mathfrak{z}) = \sum_{n \in \mathbb{Z}, l \in L^\vee} f(n, l) q^n r^l$$

where we have put $q = \exp(2\pi i \tau)$, $r^l := \exp(2\pi i (l, \mathfrak{z}))$, $s := \exp(2\pi i \omega)$. Assume that $f(n, l) \in \mathbb{Z}$ if $2n - (l, l) \leq 0$. Then the product

$$\text{M-Lift}(\varphi)(Z) = q^A r^B s^C \prod_{\substack{n, m \in \mathbb{Z}, l \in L^\vee \\ (n, l, m) > 0}} \left(1 - q^n r^l s^m\right)^{f(nm, l)},$$

where $Z = (\tau, \mathfrak{z}, \omega) \in \mathcal{H}(L_1)$ and

$$A = \frac{1}{24} \sum_{l \in L^\vee} f(0, l), \quad B = \frac{1}{2} \sum_{l > 0} f(0, l) l \in \frac{1}{2} L^\vee, \quad C = \frac{1}{2 \text{rank}(L)} \sum_{l \in L^\vee} f(0, l) (l, l)$$

defines a meromorphic modular form of weight $k = \frac{1}{2} f(0, 0)$ with respect to $\tilde{\mathcal{O}}^+(L_2)$ with a character χ . Here we make the convention

$$(n, l, m) > 0 : \iff m > 0 \text{ or } (m = 0 \text{ and } n > 0) \text{ or } (m = n = 0 \text{ and } l < 0).$$

The zeroes and poles of M-Lift (φ) lie on the affinization of the rational quadratic divisors $\mathcal{D}_{[h]}$ where $h \in L_2^\vee$ is a primitive vector such that $(h, h) < 0$ represented by $\mathcal{H}_{n,l}(L_1)$. The multiplicity of the divisor is given by

$$\text{mult } D_{[h]} = \sum_{\substack{d>0 \\ (h,h)=2n-(l,l) \\ h \equiv l \pmod{L_2}}} f(d^2 n, dl).$$

If this number is in $\mathbb{N}$ then M-Lift (φ) has a zero and if it is negative then it has a pole along $\mathcal{H}_{n,l}(L_1)$.

Proof

We only give a sketch of the proof here. The assertion is a special case of [Bo], Theorem 13.3. We like to remark that Borcherds uses vector valued modular forms to formulate this theorem which correspond to weakly holomorphic Jacobi forms by the following observation: The correspondence between vector valued modular forms and Jacobi forms was given in Proposition 4.1.5 and Proposition 2.1.10. A holomorphic vector valued modular form with respect to the Weil representation which is meromorphic at ∞ has a Fourier expansion of the shape

$$\phi(\tau) = \sum_{\gamma \in L^\vee/L} \sum_{r \geq -K} c(r, \gamma) \exp\left(\frac{2\pi i r \tau}{\kappa}\right) \mathbf{e}_\gamma$$

for some $K \in \mathbb{N}$. This type of automorphic forms are used in Borcherds' paper. If we use the variable $r \in \mathbb{Z}$ to parametrize the hyperbolic norm of a weakly holomorphic Jacobi form and the variables $x \in L$ and $\gamma \in L^\vee/L$ we obtain as in the proof of Corollary 2.1.12 some $\kappa \in \mathbb{N}$ such that

$$(f(n, x + \gamma) \neq 0 \implies n \geq -K + \kappa \min_{v \in \gamma + L} (v, v)) \iff (c(r, \gamma) \neq 0 \implies r \geq -K)$$

by identifying ϕ and φ according to the natural generalization of Proposition 4.1.5 to meromorphic vector valued modular forms. To observe the modularity we decompose the function as

$$\text{M-Lift}(\varphi) = \left(\prod_{(n,l,0)>0} \dots \right) \times \left(\prod_{m>0} \dots \right).$$

By Lemma 5.1.5 the first factor equals

$$\tilde{\psi}_{L,C}(Z) = \eta(\tau)^{f(0,0)} \prod_{l>0} \left(\frac{\vartheta(\tau, (l, \mathfrak{z}))}{\eta(\tau)} \right)^{f(0,l)} \exp(2\pi i C\omega)$$

and is according to Corollary 5.1.4 a modular form for the Jacobi group $\Gamma^J(L)$ of weight $f(0,0)/2$ with character $v_\eta^{2A+f(0,0)} \times \nu$. The second factor can be rewritten as

$$\exp \left(- \sum_{m \geq 1} m^{-1} \tilde{\varphi} | T_-(m)(Z) \right).$$

This formula is explained in the proof of Theorem 2.1 in [GN]. Hence we have the identity

$$\text{M-Lift}(\varphi)(Z) = \tilde{\psi}_{L,C}(Z) \cdot \exp \left(- \sum_{m \geq 1} m^{-1} \tilde{\varphi} | T_{-}(m)(Z) \right)$$

where $\tilde{\varphi}$ is the associated modular form from (2.3). This shows that $\text{M-Lift}(\varphi)$ is a modular form of weight $f(0,0)/2$ for the Jacobi group with the character defined above. One can work out the behaviour of this function under the additional generators of $\tilde{\text{O}}(L_2)^+$ to determine the character as it was done in [G2]. The singularities of $\text{M-Lift}(\varphi)$ lie on the rational quadratic divisors $\mathcal{D}_{[h]}$ defined by the primitive vectors $h \in L_2^\vee$. From the above Remark we conclude that this divisor is completely determined by a vector $h_{n,l}$ such that $2n - (l, l) = (h, h) < 0$ and $l \equiv h \pmod{L_2}$. The multiplicity of the divisor is given by the finite quantity

$$\sum_{d > 0} f(d^2 n, dl)$$

where n, l are chosen as before. If this quantity is positive the function has a zero. It has a pole if it is a negative integer. $\square$

The representation of $\text{M-Lift}(\varphi)$ in the proof of the theorem yields the formula

$$\begin{aligned} \text{M-Lift}(\varphi)(Z) &= \psi(\tau, \mathfrak{z}) \exp(2\pi i C\omega) \\ &\times (1 - \varphi(\tau, \mathfrak{z}) \exp(2\pi i \omega) + \frac{1}{2}(\varphi(\tau, \mathfrak{z})^2 - \varphi(\tau, \mathfrak{z}) | T_{-}(2)) \exp(4\pi i \omega) \\ &- \frac{1}{6}(\varphi(\tau, \mathfrak{z})^3 - 3\varphi(\tau, \mathfrak{z}) \cdot \varphi(\tau, \mathfrak{z}) | T_{-}(2) + 2\varphi(\tau, \mathfrak{z}) | T_{-}(3)) \exp(6\pi i \omega) \\ &+ \dots) \end{aligned}$$

where we have made use of the power-series representation of $\exp(z)$. Assume that $\text{M-Lift}(\varphi) = \text{A-Lift}(\phi)$ where $\phi \in J_{k,L;1}(1)$ and $k = f(0,0)/2 \in \mathbb{N}$. We assume that there is some $m_0 \geq 0$ such that

$$\text{A-Lift}(\phi)(Z) = \sum_{m \geq m_0} \phi_m(\tau, \mathfrak{z}) \exp(2\pi i m \omega) \quad \text{where } \phi_{m_0} \neq 0.$$

A comparison of the Fourier expansions yields the next Corollary.

Corollary 5.1.7 *Let $F(Z) = \sum_{m \geq m_0} \phi_m(\tau, \mathfrak{z}) \exp(2\pi i m \omega)$ be a modular form of weight k for the group $\tilde{\text{O}}(L_2)^+$ such that $\phi_{m_0} \neq 0$ and $\phi_m \in J_{k,L;m}(1)$, $m \in \mathbb{N}$. If there is some $\varphi \in J_{0,L;1}^{(wh)}(1)$ such that $F = \text{M-Lift}(\varphi)$ then we have*

$$\varphi = -\frac{\phi_{m_0+1}}{\phi_{m_0}}.$$

5.2 Construction of reflective modular forms

We start with the consideration of the Jacobi forms $\psi_{12-2N,NA_1} \in J_{12-2N,NA_1;1}(1)$ from Proposition 2.3.5 which were defined as

$$\psi_{12-2N,NA_1}(\tau, \mathfrak{z}_{NA_1}) = \eta(\tau)^{24-6N} \vartheta_{NA_1}^2(\tau, \mathfrak{z}_{NA_1}) \quad \text{where } N = 1, \dots, 4.$$

Now these functions can be used to create modular forms by taking the arithmetic lifting from Theorem 2.6.1 which yields functions $G_{12-2N}^{NA_1} \in \mathcal{M}_{12-2N}(\tilde{\mathcal{O}}(L_2(NA_1)))^+$

$$G_{12-2N}^{NA_1} := \text{A-Lift}(\psi_{12-2N,NA_1}).$$

From Example 2.1.17 we obtain that $2\mathcal{H}_{0,\varepsilon_j/2}(L_1(4A_1))$ where $j = 1, \dots, N$ and all $\tilde{\mathcal{O}}(L_2(NA_1))^+$ -translates are contained in the divisor of $G_{12-2N}^{NA_1}$. If this functions have a representation as a multiplicative lifting then $G_{12-2N}^{NA_1} = \text{M-Lift}(\varphi_{0,NA_1})$ where $\varphi_{0,NA_1} \in J_{0,NA_1;1}^{(wh)}(1)$ according to Corollary 5.1.7. From Example 2.1.17 we deduce that the coefficient $f(0,0)$ in the Fourier expansion of ψ_{12-2N,NA_1} is always zero. Hence the definition of the arithmetic lifting yields

$$\varphi_{0,NA_1} = - \frac{2^{-1} \psi_{12-2N,NA_1}|T_-(2)}{\psi_{12-2N,NA_1}} \quad (5.2)$$

and we have

$$\begin{aligned} & (\psi_{12-2N,NA_1}|T_-(2))(\tau, \mathfrak{z}) \\ &= 2^{12-2N} \psi_{12-2N,NA_1}(2\tau, 2\mathfrak{z}) + \psi_{12-2N,NA_1}\left(\frac{\tau}{2}, \mathfrak{z}\right) + \psi_{12-2N,NA_1}\left(\frac{\tau+1}{2}, \mathfrak{z}\right). \end{aligned} \quad (5.3)$$

We consider the product expansion of $\eta(\tau)$ and $\vartheta(\tau, z)$

$$\vartheta(\tau, z) = -q^{1/8} r^{-1/2} \prod_{n \geq 1} (1 - q^{n-1} r)(1 - q^n r^{-1})(1 - q^n) \quad , \quad \eta(\tau) = q^{1/24} \prod_{n \geq 1} (1 - q^n).$$

We introduce the variables

$$r_j = \exp(2\pi i z_j) \quad \text{for } j = 1, \dots, N$$

with respect to our standard model of NA_1 and define the rational functions

$$p_N(r_1, \dots, r_N) := r_1^{-1} \cdots r_N^{-1} \cdot (1 - r_1)^2 \cdots (1 - r_N)^2.$$

This leads to the following sum expansion of $\psi_{12-2N,NA_1}(\tau, \mathfrak{z}_N)$

$$q p_N(r_1, \dots, r_N) - q^2 \left[2 \sum_{j=1}^N \sum_{\epsilon_j \in \{0, \pm 1\}} r_j^{\epsilon_j} p_N(r_1, \dots, r_N) + \sum_{j=1}^{24-6N} p_N(r_1, \dots, r_N) \right] + \dots \quad (5.4)$$

Here the first contribution of the q^2 -term is due to the product expansion of $\vartheta_{NA_1}^2$ and the second is due to the product formula of $\eta(\tau)^{24-6N}$. If we combine formula (5.2), (5.3) and (5.4) we end up with

$$\begin{aligned} & \varphi_{0,NA_1}(\tau, \mathfrak{z}_{NA_1}) \\ &= -2^{-1} \cdot 2(-2N - (24 - 6N) - 2r_1 - \cdots - 2r_N - 2r_1^{-1} - \cdots - 2r_N^{-1}) + q(\cdots) \\ &= (24 - 4N) + 2r_1 + \cdots + 2r_N + 2r_1^{-1} + \cdots + 2r_N^{-1} + q(\cdots) \end{aligned} \quad (5.5)$$

In order to make the formulation of the next theorem more compatible we will denote the divisor of the functions in the projective terminology although the modular form to be considered has been realized on the affine cone $\mathcal{H}(L_1)$. The formulation is a variant of Theorem 5.1 in [G1].

Theorem 5.2.1 *Let $N \in \{1, 2, 3, 4\}$. We have the identity*

$$G_{12-2N}^{NA_1} = \text{M-Lift}(\varphi_{0,NA_1})$$

and $G_{12-2N}^{NA_1} \in \mathcal{M}_{12-2N}(\text{O}(L_2(NA_1))^+, 1)$. The modular form $G_{12-2N}^{NA_1}$ is reflective and its divisor is equal to

$$\bigcup_{\substack{\pm r \in L_2(NA_1) \\ (r,r)=-2 \\ \text{div}(r)=2}} \mathcal{D}_{[r]}(L_2(NA_1)) = \text{O}(L_2(NA_1))^+ \cdot \mathcal{D}_{[\varepsilon_1/2]}(L_2(NA_1))$$

where $G_{12-2N}^{NA_1}$ vanishes of order two along this set. If $N \neq 4$ then $G_{12-2N}^{NA_1}$ is a cusp form. The different orbits under the group $\tilde{\text{O}}(L_2(NA_1))^+$ are represented by

$$\mathcal{D}_{[\varepsilon_1/2]}(L_2(NA_1)), \dots, \mathcal{D}_{[\varepsilon_N/2]}(L_2(NA_1)).$$

A convenient notation for the divisor is

$$\text{div}(G_{12-2N}^{NA_1}) = 2\mathcal{D}_{[\varepsilon_1/2]}(L_2) + \cdots + 2\mathcal{D}_{[\varepsilon_N/2]}(L_2)$$

if one considers $G_{12-2N}^{NA_1}$ as a modular form with respect to the stable orthogonal group $\tilde{\text{O}}(L_2(NA_1))^+$. For the proof of the Theorem we need another elementary Lemma.

Lemma 5.2.2 *Let $\Gamma < \text{O}(L_2)^+$ be a subgroup of finite index containing $\tilde{\text{O}}(L_2)^+$. Then we have $\mathcal{M}_0(\Gamma, 1) = \mathbb{C}$.*

Proof

According to our standard notation the lattice L_2 contains the hyperbolic plane $\mathbb{Z}e + \mathbb{Z}f$. Let $\mathcal{H}(L_1)$ be the affine model of the domain $\mathcal{D}(L_2)$. We can consider modular forms as functions given in affine coordinates due to Lemma 1.6.4. According to (1.14) a modular form F has a Fourier expansion of the shape

$$F(Z) = \sum_{\substack{(m,l,n) \in L_1^\vee \\ m,n \geq 0 \\ 2mn - (l,l) \geq 0}} f(m, l, n) \exp(2\pi i[n\tau + m\omega + (l, \mathfrak{z})]) = \sum_{m \geq 0} \varphi_m(\tau, \mathfrak{z}) \exp(2\pi i m \omega).$$

Since $\Gamma^J(L) \leq \Gamma$ we deduce that F is invariant with respect to the action of $\Gamma^J(L)$. In view of Lemma 2.1.6 we deduce that $\varphi_m \in J_{0,L;m}$. From the dimension estimation (2.5) for Jacobi forms we deduce that $J_{0,L;m} = \{0\}$ whenever $m > 0$ and we have $J_{0,L;0}(1) = \mathbb{C}$. This proves the assertion. $\square$

Now we are able to prove the Theorem.

Proof

We consider the Fourier expansion of the Jacobi form φ_{0,NA_1} in (5.5) and deduce from the formula that it is in fact a weak Jacobi form. Hence Corollary 2.1.12 implies that the hyperbolic norm in the Fourier expansion is bounded from below by the quantity

$$-\max_{l \in L^\vee/L} \left(\min_{v \in l+L} (v, v) \right) = -\frac{N}{2}, \quad L = NA_1.$$

Now assume there is a pair $(n, l) \in \mathbb{N} \times L^\vee$ such that $2n - (l, l) < 0$. Since $(l, l) \in \frac{1}{2}\mathbb{N}$ this implies

$$2n - (l, l) \in [-N/2, -1/2].$$

One observes that

$$2n - (l, l) = -\frac{1}{2} \iff l \equiv \varepsilon_j/2 \pmod{NA_1} \text{ for some } j = 1, \dots, N.$$

Since the q^0 -terms with hyperbolic norm equal to $-1/2$ of φ_{0,NA_1} are all equal to two and all other terms having a negative norm vanish we conclude that all the information about the divisor is completely contained in the q^0 part of φ_{0,NA_1} where we have taken into account $r_j = \exp(2\pi i(\varepsilon_j/2, \mathfrak{z}))$ and Lemma 2.1.9. Now Theorem 5.1.6 yields that the divisor is of the shape stated above with multiplicity two and the weight of the multiplicative lifting equals $12 - 2N$. The reflectiveness of the lifted form follows from the Eichler criterion 1.5.8, Proposition 1.6.2 and Proposition 1.5.2. The statement on cusp forms follows from Theorem 2.6.1. It remains to show that $G_{12-2N}^{NA_1}$ coincides with $\text{M-Lift}(\varphi_{0,NA_1})$. Since $G_{12-2N}^{NA_1}$ contains the divisor of the second function we can divide to obtain

$$\frac{G_{12-2N}^{NA_1}}{\text{M-Lift}(\varphi_{0,NA_1})} \in \mathcal{M}_0(\tilde{\mathcal{O}}(L_2))^+$$

due to Koecher's principle 1.6.5. By the last Lemma we deduce that there exists some $\alpha \in \mathbb{C}^*$ such that $G_{12-2N}^{NA_1} = \alpha \cdot \text{M-Lift}(\varphi_{0,NA_1})$. Moreover we can compute the Fourier expansion of the Borcherds lifting using the formula from the last section which yields

$$\text{M-Lift}(\varphi_{0,NA_1}) = \psi_{12-2N,NA_1} \exp(2\pi i\omega)(1 + \dots).$$

Since $G_{12-2N}^{NA_1}$ has the same Fourier coefficient for $\exp(2\pi i\omega)$ we conclude $\alpha = 1$. From the shape of ψ_{12-2N,NA_1} and the formula for the arithmetic lifting in Theorem 2.6.1 we deduce that the maximal modular group is indeed $\mathcal{O}(L_2)^+$. $\square$

We now consider the D_8 -tower and remind to the definition of ψ_{12-2N, D_N}

$$\psi_{12-N, D_N}(\tau, \mathfrak{z}_{D_N}) = \eta(\tau)^{24-3N} \vartheta_{D_N}(\tau, \mathfrak{z}_{D_N}).$$

As before we will investigate the arithmetic lifting defined by

$$G_{12-N}^{D_N} := \text{A-Lift}(\psi_{12-N, D_N}).$$

We define the variables s_j with respect to our standard model for D_N as

$$s_j = \exp(2\pi i(\varepsilon_N, \mathfrak{z}_{D_N})).$$

The Fourier expansion of ψ_{12-N, D_N} can be computed analogously to (5.4)

$$q p_N(s_1, \dots, s_N) - q^2 \left[\sum_{j=1}^N \sum_{\epsilon_j \in \{0, \pm 1\}} s_j^{\epsilon_j} p_N(s_1, \dots, s_N) + \sum_{j=1}^{24-3N} p_N(s_1, \dots, s_N) \right] + \dots \quad (5.6)$$

As before the necessary condition for $G_{12-N}^{D_N}$ being an arithmetic and mutiplicative lifting yields the input function φ_{0, D_N} which has exactly the same shape as $\varphi_{0, N A_1}$ in (5.2). Hence we can repeat the computation of the q^0 -term for this function and obtain

$$\begin{aligned} \varphi_{0, D_N}(\tau, \mathfrak{z}_{D_N}) &= -2^{-1} \cdot (-2N - 2(24 - 3N) - 2s_1 - \dots - 2s_N - 2s_1^{-1} - \dots - 2s_N^{-1}) + q(\dots) \\ &= (24 - 2N) + s_1 + \dots + s_N + s_1^{-1} + \dots + s_N^{-1} + q(\dots) \end{aligned} \quad (5.7)$$

This leads us to the next Theorem, see [G1].

Theorem 5.2.3 (Gritsenko '10) *Let $N \in \{1, \dots, 8\}$. We have the identity*

$$G_{12-N}^{D_N} = \text{M-Lift}(\varphi_{0, D_N})$$

and $G_{12-N}^{D_N} \in \mathcal{M}_{12-N}(\Gamma, \chi)$ where

$$\Gamma = \begin{cases} \text{O}(L_2(D_N))^+ & \text{if } N \neq 4 \\ \langle \tilde{\text{O}}(L_2(D_4))^+, \sigma_{\varepsilon_1} \rangle & \text{if } N = 4 \end{cases}.$$

In the last case $\Gamma \leq \text{O}(L_2(D_4))^+$ is a subgroup of index three and the character is defined as

$$\chi : \Gamma \rightarrow \Gamma / \tilde{\text{O}}(L_2(D_N))^+ \xrightarrow{\sim} \text{O}(D_N) \xrightarrow{\text{sign}} \{\pm 1\}.$$

The modular form $G_{12-N}^{D_N}$ is strongly reflective and its divisor is equal to

$$\bigcup_{\substack{\pm r \in L_2(D_N) \\ (r, r) = -4 \\ \text{div}(r) = 2}} \mathcal{D}_{[r]}(L_2(D_N)) = \tilde{\text{O}}(L_2(D_N))^+ \cdot \mathcal{D}_{[\varepsilon_1]}(L_2(D_N))$$

if $N \neq 4$ and for D_4 it is given by

$$\tilde{O}(L_2(D_N))^+ \cdot \mathcal{D}_{[\varepsilon_1]}(L_2(D_N)).$$

If $N \neq 8$ then $G_{12-N}^{D_N}$ is a cusp form.

Proof

As in the proof of Theorem 5.2.1 we conclude from (5.7) that φ_{0,D_N} is a weak Jacobi form. Hence Corollary 2.1.12 implies that the hyperbolic norm in the Fourier expansion is bounded from below by the quantity

$$-\max_{l \in L^\vee/L} \left(\min_{v \in l+L} (v, v) \right) = -\frac{N}{4}, \quad L = D_N.$$

Now assume there is a pair $(n, l) \in \mathbb{N} \times L^\vee$ such that $2n - (l, l) < 0$. Then one observes

$$2n - (l, l) \in \{-N/4, -1\}.$$

Since the lattice $D_N^\vee$ represents all such values under $-(\cdot, \cdot)_{D_N}$ we know that all the information about the divisor is completely contained in the q^0 part of φ_{0,D_N} due to Lemma 2.1.9. Now Theorem 5.1.6 yields that the divisor is of the shape stated above with multiplicity one and the weight of the multiplicative lifting equals $12 - 2N$ where we have to consider the special case D_4 where all non-trivial classes in $D(D_4)$ have representatives of minimal norm -1 . Note that the order of $\varepsilon_1 \in D_N$ is always $2 = \text{div}(2\varepsilon_1)$. From Section 1.9 we know that $\tilde{O}(L_2(D_N))^+$ has index two in the full modular group if $N \neq 4$ and index six if $N = 4$. Note that we have the following representation of ψ_{12-N,D_N}

$$\psi_{12-N,D_N}(\tau, \mathfrak{z}_{D_N}) = \eta(\tau)^{24-3N} \vartheta(\tau, (\varepsilon_1, \mathfrak{z}_{D_N})) \cdots \vartheta(\tau, (\varepsilon_N, \mathfrak{z}_{D_N})).$$

If $N \neq 4$ then $O(L_2(D_N))^+$ is generated by the stable orthogonal group and σ_{ε_1} according to the considerations in Section 1.9. We conclude that $G_{12-N}^{D_N}$ transforms with the character χ defined above and the maximal modular group is the full modular group if $N \neq 4$. In the case $N = 4$ the maximal modular group is the subgroup of index three generated by $\tilde{O}(L_2(D_4))^+$ and σ_{ε_1} . In the last case we obtain a reflective modular form because the divisor equals

$$\bigcup_{g \in \tilde{O}(L_2(D_4))^+} \mathcal{D}_{[g(\varepsilon_1)]}$$

and due to the identity

$$\sigma_{g(\varepsilon)} = g\sigma_{\varepsilon_1}g^{-1} \in \Gamma \text{ for } g \in \tilde{O}(L_2(D_4))^+$$

and if $N \neq 4$ every reflection σ_r where $(r, r) = -4$ and $\text{div}(r) = 2$ certainly belongs to $\Gamma = O(L_2(D_N))^+$. The remaining parts of the proof follow with the same arguments as in the proof of Theorem 5.2.1. $\square$

The case $L = D_4$ is special for the D_N -series. In this case we know from Lemma 2.1.19 that there exist two additional Jacobi forms of singular weight. The three singular Jacobi forms induce Jacobi forms for $4A_1$ as we have seen in Proposition 2.3.8

$$\begin{aligned}\vartheta_{4A_1}^{(1)}(\tau, \mathfrak{z}_{\mathcal{B}_{4A_1}}) &= \vartheta(\tau, z_1 - z_2)\vartheta(\tau, z_1 + z_2)\vartheta(\tau, z_3 - z_4)\vartheta(\tau, z_3 + z_4) \\ \vartheta_{4A_1}^{(2)}(\tau, \mathfrak{z}_{\mathcal{B}_{4A_1}}) &= \vartheta(\tau, z_3 - z_2)\vartheta(\tau, z_3 + z_2)\vartheta(\tau, z_1 - z_4)\vartheta(\tau, z_1 + z_4) \\ \vartheta_{4A_1}^{(3)}(\tau, \mathfrak{z}_{\mathcal{B}_{4A_1}}) &= \vartheta(\tau, z_1 - z_3)\vartheta(\tau, z_1 + z_3)\vartheta(\tau, z_2 - z_4)\vartheta(\tau, z_2 + z_4).\end{aligned}$$

where $\mathcal{B}_{NA_1}$ denotes another model of $4A_1$ namely

$$(\varepsilon_2 - \varepsilon_1, \varepsilon_2 + \varepsilon_1, \varepsilon_4 - \varepsilon_3, \varepsilon_4 + \varepsilon_3, (\cdot, \cdot)_N).$$

For abbreviation we call this basis $u_1, \dots, u_4$. In this model the construction can be interpreted by means of the identity

$$\vartheta(\tau, z_j \pm z_k) = \vartheta(\tau, (\mathfrak{z}_{\mathcal{B}_{NA_1}}, (u_j \pm u_k)/2)). \quad (5.8)$$

We define $\psi_{8,4A_1}^{(j)}(\tau, \mathfrak{z}_{\mathcal{B}_{NA_1}}) = \eta(\tau)^{12}\vartheta_{4A_1}^{(j)}$ for $j = 1, 2, 3$ which yields a series of holomorphic Jacobi form of weight 8 and index 1. If we define $\varphi_{0,4A_1}^{(j)}$ as before then one obtains a Fourier expansion as in (5.7) where one has to modify the coordinates $s_1, \dots, s_4$ according to the choice of j . This yields three new functions with a good divisor.

Proposition 5.2.4 *There are three modular forms $\Delta_{8,4A_1}^{(j)} \in \mathcal{M}_8(\widetilde{\mathrm{O}}(L_2(4A_1))^+, 1)$ where $j = 1, 2, 3$ whose divisor is given by all $\widetilde{\mathrm{O}}(L_2(4A_1))^+$ -translates of*

$$\begin{aligned}\mathcal{D}_{[u_1+u_2]}(L_2(4A_1)) + \mathcal{D}_{[u_3+u_4]}(L_2(4A_1)) &\quad \text{if } j = 1, \\ \mathcal{D}_{[u_1+u_4]}(L_2(4A_1)) + \mathcal{D}_{[u_2+u_3]}(L_2(4A_1)) &\quad \text{if } j = 2, \\ \mathcal{D}_{[u_1+u_3]}(L_2(4A_1)) + \mathcal{D}_{[u_2+u_4]}(L_2(4A_1)) &\quad \text{if } j = 3\end{aligned}$$

and we have

$$\Delta_{8,4A_1}^{(j)} = \mathrm{A-Lift}(\psi_{8,4A_1}^{(j)}).$$

Each of them is a cusp form.

Proof

The proof is essentially the same as for the Theorem before. The structure of the divisor can be derived from Theorem 5.1.6, (5.8) and the obvious modifications of (5.7). The statement on cusp forms follows from [GHS1], Theorem 4.2, compare Remark 1.7.10 $\square$

As an immediate consequence we construct a function which is anti-symmetric. The character v_π of this function is explained by

$$v_\pi : \mathrm{O}^+(L_2(NA_1)) \twoheadrightarrow \mathrm{O}(NA_1^\vee/NA_1) \xrightarrow{\sim} \mathcal{S}_N \xrightarrow{\mathrm{sign}} \{\pm 1\}. \quad (5.9)$$

Corollary 5.2.5 *There is a cusp form*

$$\Delta_{24}^{4A_1} = \text{M-Lift}(\varphi_{0,4A_1}^{(1)} \varphi_{0,4A_1}^{(2)} \varphi_{0,4A_1}^{(3)}) \in \mathcal{M}_{24}(\text{O}(L_2(4A_1))^+, v_\pi)$$

satisfying

$$\Delta_{24}^{4A_1} = \Delta_{8,4A_1}^{(1)} \Delta_{8,4A_1}^{(2)} \Delta_{8,4A_1}^{(3)}$$

which is strongly reflective. The divisor equals

$$\bigcup_{\substack{\pm r \in L_2(4A_1) \\ (r,r)=-4 \\ \text{div}(r)=2}} \mathcal{D}_{[r]}(L_2(4A_1)) = \text{O}(L_2(4A_1))^+ \cdot \mathcal{D}_{[u_1+u_2]}(L_2(4A_1)).$$

Proof

The statement on the existence follows from Proposition 2.3.8 and 5.2.4. The divisor is the sum of the rational quadratic divisors listed in Proposition 5.2.4 and all of them are equivalent under the action of the full modular group because the symmetric group $\mathcal{S}_4$ acts 4-fold transitive on the set $\{1, 2, 3, 4\}$. $\square$

If we consider the pullbacks $\psi_{8,4A_1}^{(j)}|_{3A_1}$ we immediately obtain another anti-symmetric function for $3A_1$.

Corollary 5.2.6 *There is a cusp form $\Delta_{24}^{3A_1} \in \mathcal{M}_{24}(\text{O}(L_2(3A_1))^+, v_\pi)$ which is reflective. The divisor equals*

$$\bigcup_{\substack{\pm r \in L_2(3A_1) \\ (r,r)=-2 \\ \text{div}(r)=2}} 2\mathcal{D}_{[r]}(L_2(3A_1)) \cup \bigcup_{\substack{\pm r \in L_2(3A_1) \\ (r,r)=-4 \\ \text{div}(r)=2}} \mathcal{D}_{[r]}(L_2(3A_1))$$

and the different orbits with respect to the full modular group are represented by

$$\mathcal{D}_{[u_1+u_2]}(L_2(3A_1)) \quad , 2\mathcal{D}_{[u_1]}(L_2(3A_1)).$$

We can even construct a strongly reflective modular form

$$\Delta_{18}^{3A_1} := \frac{\Delta_{24}^{3A_1}}{G_6^{3A_1}}$$

Corollary 5.2.7 *There is a cusp form $\Delta_{18}^{3A_1} \in \mathcal{M}_{18}(\text{O}(L_2(3A_1))^+, v_\pi)$ which is strongly reflective. The divisor equals*

$$\bigcup_{\substack{\pm r \in L_2(3A_1) \\ (r,r)=-4 \\ \text{div}(r)=2}} \mathcal{D}_{[r]}(L_2(3A_1))$$

and the different orbits with respect to the full modular group are represented by

$$\mathcal{D}_{[u_1+u_2]}(L_2(3A_1)).$$

5.3 The Quasi-Pullback of a modular form

In this section we consider $II_{p,q}$, the unique (up to isomorphism) even unimodular lattice of signature (p, q) where $p, q > 0$. The following statement and its application to $\Delta_{12}(\tau)^{-1}$ are essential for the construction of the quasi-pullback and can be found in [Bo2], Theorem 10.1 and Example 2 of the same section.

Theorem 5.3.1 *Suppose that f is a nearly holomorphic modular form of weight $-\frac{s}{2}$ for $\mathrm{SL}(2, \mathbb{Z})$, i.e. a function on the upper half-plane transforming like a $\mathrm{SL}(2, \mathbb{Z})$ modular form with a Fourier expansion of the shape*

$$f(\tau) = \sum_{n \geq -K} c(n)q^n \quad , \quad K \in \mathbb{N}$$

such that $c(n) \in \mathbb{Z}$ and the property $24|c(0)$ if $s = 0$. There is a unique vector $\varrho \in II_{1,1+s}$ such that

$$\Phi(Z) = e^{2\pi i(\varrho, Z)} \prod_{\substack{r \in II_{1,1+s} \\ r > 0}} (1 - e^{2\pi i(r, Z)})^{c((r, r)/2)} \quad , \quad Z \in \mathcal{H}(II_{1,1+s})$$

is a meromorphic automorphic form of weight $c(0)/2$ for the group $\mathrm{O}(II_{2,2+s})^+$. All the zeroes and poles of Φ lie on rational quadratic divisors and the multiplicity of Φ at the rational quadratic divisor of the primitive negative norm vector $r \in II_{2,2+s}$ is

$$\sum_{n > 0} c(n^2(r, r)/2) .$$

In particular if this number is always nonnegative then Φ is holomorphic.

Note that Borcherds considers the lattice $II_{2+s,2}$, instead. For the purpose of the next Corollary we define

$$R_{-2}^{II_{2,26}} := \{r \in II_{2,26} \mid (r, r) = -2\} .$$

Corollary 5.3.2 *There is a holomorphic modular form $\Phi_{12} \in \mathcal{M}_{12}(\mathrm{O}(II_{2,26})^+, \det)$ and its divisor is given by*

$$\mathrm{div}(\Phi_{12}) = \bigcup_{r \in R_{-2}^{II_{2,26}}} \mathcal{D}_{[r]}(II_{2,26})$$

where the vanishing order is exactly one on each irreducible component.

Proof

This is exactly the contents of Remark 2 in Section 10 of [Bo2]. It is well-known that $1/\Delta_{12}$ is a nearly holomorphic modular form of weight $-12 = -s/2$ for the group $\mathrm{SL}(2, \mathbb{Z})$ and its Fourier expansion equals

$$1/\Delta_{12}(\tau) = \sum_{n \geq 0} p_{24}(n)q^{n-1} = q^{-1} + 24 + 324q + 3200q^2 + 25650q^3 + \dots .$$

In view of the last Theorem the function $Z \mapsto \Phi_{12}(Z)$ has weight $p_{24}(1)/2 = 12$ and vanishes if and only if $(Z, r) = 0$ for some $r \in R_{-2}^{II_{2,26}}$. The multiplicity of the divisor is

$$\sum_{n>0} p_{24}(1 + n^2(r, r)/2) = p_{24}(0) = 1.$$

The determination of the character was done in section 10 of [Bo2]. $\square$

The function Φ_{12} has singular weight and Borcherds determined its Fourier expansion as

$$\Phi_{12}(Z) = e^{2\pi i(\varrho, Z)} \prod_{\substack{r \in II_{1,25} \\ r > 0}} (1 - e^{2\pi i(r, Z)})^{p_{24}(1+(r, r)/2)} = \sum_{w \in W} \det(w) \Delta_{12}((w(\varrho), Z))$$

where W is the reflection group of the lattice $II_{1,25}$. A comparison of the Fourier coefficients yields the Weyl denominator formula for the fake monster Lie algebra. Now let M be a lattice of signature $(2, 2 + N)$. The rational quadratic divisor for $r \in M \otimes \mathbb{Q}$, $(r, r) < 0$ was defined as

$$\mathcal{D}_{[r]}(M) = \{[\mathcal{Z}] \in \mathcal{D}(M) \mid (\mathcal{Z}, r) = 0\}.$$

We consider the following situation. Let $L \leq M$ be a primitive sublattice of signature $(2, 2 + N')$ where $N' < N$. The lattice $L_M^\perp$ is negative definite. Hence each vector $r \in M$ can be written as $r = \alpha + \beta$ where $\alpha \in L^\vee$ and $\beta \in (L^\perp)^\vee$. The following Lemma is fundamental for the determination of a function obtained as a quasi-pullback.

Lemma 5.3.3 *Let L, M as above and $r \in M$ such that $(r, r) < 0$. We have*

$$\mathcal{D}(L) \cap \mathcal{D}_{[r]}(M) = \begin{cases} \mathcal{D}_{[\alpha]}(L) & \text{if } (\alpha, \alpha) < 0 \\ \emptyset & \text{if } (\alpha, \alpha) \geq 0, \alpha \neq 0 \\ \mathcal{D}(L) & \text{if } \alpha = 0, \text{ i.e. } r \in L^\perp. \end{cases}$$

Proof

Since $L \leq M$ we have $\mathcal{D}(L) \subseteq \mathcal{D}(M)$. Let $\mathcal{Z} \in \mathcal{D}(L)^\bullet$ such that $(\mathcal{Z}, r) = 0$. We have already observed that $\mathcal{Z} = \mathcal{X} + i\mathcal{Y} \in \mathcal{D}(L)^\bullet$ is equivalent to

$$(\mathcal{X}, \mathcal{Y}) = 0 \quad , \quad (\mathcal{X}, \mathcal{X}) = (\mathcal{Y}, \mathcal{Y}) > 0$$

where $\mathcal{X}, \mathcal{Y} \in L \otimes \mathbb{R}$. Moreover $(\mathcal{Z}, r) = 0$ if and only if $(\mathcal{Z}, \alpha) = 0$. Hence the vector space $\langle \mathcal{X}, \mathcal{Y}, \alpha \rangle_{\mathbb{R}} \leq L \otimes \mathbb{R}$ is of real dimension at most three and $\alpha \in \langle \mathcal{X}, \mathcal{Y} \rangle^\perp$. Assume that $\alpha \neq 0$. If $(\alpha, \alpha) > 0$ this yields a subspace of signature $(3, 0)$ of $L \otimes \mathbb{R}$. Hence the intersection is empty if $(\alpha, \alpha) > 0$. Now assume that α is isotropic, i.e. $(\alpha, \alpha) = 0$. Since the signature of L is $(2, N' + 2)$ there exists an isotropic vector $w \in L \otimes \mathbb{R}$ such that $(w, \alpha) = 1$. By assumption $w \perp \langle \mathcal{X}, \mathcal{Y} \rangle$. Then $\langle \mathcal{X}, \mathcal{Y} \rangle \oplus \langle w, \alpha \rangle$ is a subspace of signature $(3, 1)$ which yields again a contradiction. Hence no such $\mathcal{Z}$ can exist if $\alpha \neq 0$ and $(\alpha, \alpha) \geq 0$. Hence if such a $\mathcal{Z}$ exists and $\alpha \neq 0$ we necessarily have $(\alpha, \alpha) < 0$. The additional statement in the case $\alpha = 0$ follows from Proposition 1.4.9 which states that the class of β in $D(L^\perp)$ is uniquely determined by the class of α in $D(L)$. $\square$

In the above situation we consider the set of -2 -roots

$$R_{-2}(L^\perp) = \{r \in M \mid (r, r) = -2, r \perp L\}$$

which is contained in the negative definite lattice $L^\perp$ and hence finite. Hence we can define $N(L^\perp) = \frac{\#R_{-2}(L^\perp)}{2} \in \mathbb{N}$. The next Theorem describes the construction of a quasi-pullback from Borcherds function Φ_{12} .

Theorem 5.3.4 *Let $L \hookrightarrow II_{2,26}$ be a primitive embedding of an even lattice of signature $(2, N+2)$ where $1 \leq N \leq 24$. The function $\Phi|_L^{(QP)}$ belongs to $\mathcal{M}_{12+N(L^\perp)}(\tilde{O}(L)^+, \det)$ where*

$$\Phi|_L^{(QP)}(\mathcal{Z}) = \frac{\Phi_{12}(\mathcal{Z})}{\prod_{r \in R_{-2}(L^\perp)/\{\pm 1\}} (\mathcal{Z}, r)} \Big|_{\mathcal{D}(L)}$$

where in the product one fixes a set of representatives for $R_{-2}(L^\perp)/\{\pm 1\}$. The function vanishes exactly on all rational quadratic divisors of the type $\mathcal{D}_{[\alpha(r)]}(L)$ where r runs through the set $R_{-2}^{II_{2,26}}$ and $\alpha \in L^\vee$ is chosen according to the decomposition in Lemma 5.3.3. If $N(L^\perp) > 0$ we say that $\Phi|_L^{(QP)}$ is a quasi-pullback of Φ_{12} .

Proof

The function Φ_{12} is an element of $\mathcal{M}_{12}(\mathcal{O}(II_{2,26})^+, \det)$ and its divisor is given by

$$\text{div}(\Phi_{12}) = \bigcup_{r \in R_{-2}^{II_{2,26}}} \mathcal{D}_{[r]}(II_{2,26})$$

where the vanishing order is exactly one on each irreducible component according to Corollary 5.3.2. Hence for each $r \in R_{-2}^{II_{2,26}}$ we have

$$\Phi_{12}|_{\mathcal{D}_{[r]}(II_{2,26})}(\mathcal{Z}) = (\mathcal{Z}, r)\Psi(\mathcal{Z})$$

for a holomorphic function $\Psi : \mathcal{D}_{[r]}(II_{2,26}) \rightarrow \mathbb{C}^*$ which is a holomorphic modular form of weight $12+1=13$. The roots $r, -r$ determine the same divisor. This explains why we only expand the product over representatives mod ± 1 . Therefore $\Phi|_L(\mathcal{Z}) = 0$ if and only if $\mathcal{Z} \in \mathcal{D}_{[r]}(II_{2,26}) \cap \mathcal{D}(L)$ for some $r \in R_{-2}^{II_{2,26}}$. Hence $\Phi|_L$ vanishes identically on $\bigcup_{r \in R_{-2}(L^\perp)} \mathcal{D}_{[r]}(II_{2,26}) \cap \mathcal{D}(L)$ by Lemma 5.3.3. This shows that $\Phi|_L^{(QP)}$ has the divisor as stated above and the function is certainly holomorphic. Now we investigate the modularity. By Lemma 1.3.1 we have an embedding $\tilde{O}(L)^+ \hookrightarrow \tilde{O}(II_{2,26})^+ = \mathcal{O}(II_{2,26})^+$. Let $\mathcal{Z} \in \mathcal{D}(II_{2,26})$ and $g \in \tilde{O}(L)^+$. We write $\mathcal{Z} = \mathcal{Z}_L \oplus \mathcal{Z}_{L^\perp}$ and $\hat{g}$ for the extension to $\tilde{O}(II_{2,26})^+$ according to Lemma 1.3.1. This yields

$$\Phi|_L^{(QP)}(g\mathcal{Z}_L) = \frac{\Phi_{12}(\hat{g}\mathcal{Z})}{\prod_{r \in R_{-2}(L^\perp)/\{\pm 1\}} (\hat{g}\mathcal{Z}, r)} \Big|_{\mathcal{D}(L)} = \frac{\det(\hat{g})\Phi_{12}(\mathcal{Z})}{\prod_{r \in R_{-2}(L^\perp)/\{\pm 1\}} (\mathcal{Z}, \hat{g}^{-1}r)} \Big|_{\mathcal{D}(L)}.$$

Since the extension $\hat{g}$ acts trivially on $L^\perp$ we obtain $\det(\hat{g}) = \det(g)$ and $\hat{g}^{-1}r = r$ for all $r \in R_{-2}(L^\perp)$ such that $\Phi|_L^{(QP)}(g\mathcal{Z}_L) = \det(g) \Phi|_L^{(QP)}(\mathcal{Z}_L)$. Now let $t \in \mathbb{C}^*$. Since $\Phi_{12}(t\mathcal{Z}_L) = t^{-12}\Phi_{12}(\mathcal{Z}_L)$ the quasi-pullback transforms like a modular form for $\tilde{O}(L)^+$ of weight $12 + N(L^\perp)$. If the root system $R_{-2}(L^\perp)$ is empty then we obtain a usual pullback of weight 12 with character $\det$. $\square$

The next theorem can be found in [GHS2], Corollary 8.12. It states that the quasi-pullback usually produces cusp forms.

Theorem 5.3.5 *Under the assumptions of Theorem 5.3.4 we have*

$$\Phi|_L^{(QP)} \in \mathcal{S}_{12+N(L^\perp)}(\tilde{O}(L)^+, \det)$$

if $N(L^\perp) > 0$.

In the sequel our task is to investigate the quasi-pullback for several root lattices. Our special interest is focussed on the D_N - and NA_1 -series. The simplest cases occur for those root lattices which split as an orthogonal summand of the root system of a Niemeier lattice $\mathcal{N}$. We first consider the tower

$$L_2(D_N) \quad , \quad 3 \leq N \leq 8.$$

As $D_3 \cong A_3$ we consider the *Niemeier lattices*

$$\begin{aligned} &\mathcal{N}(3D_8), \quad \mathcal{N}(E_6 \oplus D_7 \oplus A_{11}), \quad \mathcal{N}(4D_6), \quad \mathcal{N}(2A_9 \oplus D_6), \\ &\mathcal{N}(2A_7 \oplus 2D_5), \quad \mathcal{N}(6D_4), \quad \mathcal{N}(4A_5 \oplus D_4), \quad \mathcal{N}(8A_3) \cong \mathcal{N}(8D_3) \end{aligned}$$

In the following a Niemeier lattice is always considered as a negative definite lattice. In each of the cases let us fix an embedding

$$\begin{aligned} L_2(D_N) &\hookrightarrow L_2(D_N \oplus K(N)) \text{ if } N \neq 4, 6 \\ L_2(D_4) &\hookrightarrow L_2(D_4 \oplus K^{(j)}(4)), \quad j = 1, 2 \\ L_2(D_6) &\hookrightarrow L_2(D_6 \oplus K^{(j)}(6)), \quad j = 1, 2 \end{aligned}$$

where $K^{(1)}(4) = 4A_5, K^{(2)}(4) = 5D_4, K^{(1)}(6) = 2A_9, K^{(2)}(6) = 3D_6$. In each case the procedure is the following:

Algorithm 5.3.6 (a) Check if the following condition is satisfied:

For all $l \in L_2(D_N)^\vee$ such that $-2 \leq (l, l) < 0$ there exists a primitive vector $x \in D_N(-1)^\vee$ such that $l \equiv x \pmod{L_2(D_N)}$ and $-2 \leq (x, x) < 0$. If this is the case continue with the next step.

(From Section 1.9 we know that this is true for $D_N, N \leq 8$.)

- (b) (This step is automatically satisfied in our special situation and becomes interesting if we drop the condition of D_N being an orthogonal component of a root system for some Niemeier lattice.)

For each embedding $L_2(D_N) \hookrightarrow L_2(\mathcal{N}(D_N \oplus K(N)))$ into the root system of one of the Niemeier lattices listed above we have $II_{2,26} \cong L_2(\mathcal{N}(D_N \oplus K(N)))$ since the lattices have isomorphic discriminant forms, confer [N] Theorem 1.3.1 and Proposition 3.1.1. Now for $r = \alpha + \beta \in II_{2,26}$ where $0 > (\alpha, \alpha) \geq -2$ and $(r, r) = -2$ we know that there is some $x \in D_N^\vee$ satisfying (a). Moreover it follows that

$$r' = x + \beta \in \mathcal{N}(-1) \cap (D_N(-1)^\vee \oplus K(N)^\vee)$$

is in the same class as $r \bmod D_N \oplus K(N)$ and $-2 = (r', r')$. Hence r' is a -2-root in $\mathcal{N}(-1)$. We have to decide whether it is possible to express r' as a sum of two non-roots in $D_N(-1)^\vee \oplus K(N)^\vee$. Every root of $\mathcal{N}(-1)$ is contained in the root sublattice $\mathcal{N}_R$. Hence any sum of non-roots obtained in this way does certainly not belong to $\mathcal{N}(-1)$. This shows that either x is a root in $D_N(-1)$ or β is a root in $K(N)_R$. Note that we have used that every element $\mathcal{N}(-1)$ has a unique expression as a direct sum in $D_N(-1)^\vee \oplus K(N)^\vee$. Since $x \neq 0$ we have $(x, x) = -2$ and $\beta = 0$. This implies that $\alpha \in L_2(D_N)$ is a -2-root.

- (c) We need to determine representatives for all orbits under the action of $\tilde{O}(L_2(D_N))^+$ on $\bigcup_\alpha \mathcal{D}_{[\alpha]}(L_2(D_N))$. For this purpose we use the Eichler criterion 1.5.8.
- (d) The multiplicity of each divisor is equal to the number of -2-roots $r \in II_{2,26}$ such that the projection α to $L_2(D_N)^\vee$ is a -2-root in $L_2(D_N)$. From the correspondence in Proposition 1.4.9 it follows that this quantity is equal to the number of vectors $m \in L_2(D_N)^\perp$ such that $(m, m) = -2 - (\alpha, \alpha) = 0$. Since this lattice is negative definite only the zero element satisfies this condition. This shows that each divisor has multiplicity one.

In order to classify the different orbits by using the Eichler criterion we state a useful Lemma.

Lemma 5.3.7

Let $\alpha \in L_2(D_N)$ be a -2-root where $3 \leq N \leq 8$. Then α is primitive in $L_2(D_N)$ and in $L_2(D_N)^\vee$. In particular we have $\text{div}(\alpha) = 1$.

Proof

The first part follows from the fact that the lattice $L_2(D_N)$ is even. Since $(\alpha, \alpha) = -2$ we have $\text{div}(\alpha) \in \{1, 2\}$. Assume that $\text{div}(\alpha) = 2$. In this case according to Lemma 1.4.1 there exists a primitive $l \in L_2(D_N)^\vee$ such that $\alpha = \text{div}(\alpha)l$ and the order of l in the discriminant group is $2 = \text{div}(\alpha)$. Assume that $(l, l) \equiv -1 \pmod{2\mathbb{Z}}$. In this case $(\alpha, \alpha) \equiv 0 \pmod{4\mathbb{Z}}$ which is a contradiction. If $(l, l) \equiv -N/4 \pmod{2\mathbb{Z}}$ then we obtain $(\alpha, \alpha) \equiv -N \pmod{8\mathbb{Z}}$. Since $3 \leq N \leq 8$ this yields a contradiction, too. Since $l \in L_2(D_N)$ also leads to a contradiction we have shown that the assumption $\text{div}(\alpha) = 2$ was false which completes the proof. $\square$

In accordance with Eichler's criterion we introduce the following convention for rational quadratic divisors and $h \in L_2$:

$$\mathcal{D}_{[n,l]}(L_2) := \mathcal{D}_{[h]}(L_2) \quad \text{if} \quad (h, h) = n \text{ and } h^* \equiv l \pmod{L_2}.$$

Theorem 5.3.8 *There are the following 8 modular forms which are obtained as quasi-pullbacks from Φ_{12} for the D_N -tower where the subscripts indicate the weight*

$$F_{54}^{D_3}, F_{72}^{D_4}, F_{88}^{D_5}, F_{102}^{D_6}, F_{114}^{D_7}, F_{124}^{D_8}$$

and the extra functions for D_4, D_6 coincide with $F_{72}^{D_4}$ and $F_{102}^{D_6}$ up to multiplication in $\mathbb{C}^*$. The divisor is given in each case by the union

$$\bigcup_{r \in R_{-2}(L_2(D_N))} \mathcal{D}_{[r]}(L_2(D_N)) = \tilde{\mathcal{O}}(L_2(D_N))^+ \cdot \mathcal{D}_{[-2,0]}(L_2(D_N))$$

and all the modular forms stated above are reflective.

Proof

As mentioned above we can apply algorithm 5.3.6 since there are always primitive representatives of norm smaller than 2 in the dual lattice. To determine the different orbits of the divisor we use Lemma 5.3.7. From the Eichler criterion 1.5.8 we deduce that all rational quadratic divisors belong to the same orbit under the action of $\tilde{\mathcal{O}}(L_2(D_N))^+$. Moreover by Proposition 1.6.2 $\sigma_r \in \tilde{\mathcal{O}}(L_2(D_N))^+$. In Section 1.8 we have listed the number of roots for all irreducible root lattices except for E_6, E_7 . However the number of roots for those lattices is well-known, confer for example [E], p. 29. One has $\#R_2(E_7) = 126$ and $\#R_2(E_6) = 72$. This completes the proof. $\square$

We now apply the algorithm to the series $A_1, 2A_1, 3A_1, 4A_1$. In our standard model we have

$$NA_1^\vee = (\langle \frac{\varepsilon_1}{2}, \dots, \frac{\varepsilon_N}{2} \rangle, 2(\cdot, \cdot)_N).$$

Since $(\varepsilon_j/2, \varepsilon_j/2) = \frac{1}{2}$ for all j we deduce that assumption (a) of algorithm 5.3.6 is satisfied for NA_1 if and only if $N \leq 4$. Considering the Niemeier lattice

$$\mathcal{N}(24A_1)$$

we fix embeddings

$$L_2(NA_1) \hookrightarrow L_2(NA_1 \oplus K(N)) \text{ for } N = 1, \dots, 4.$$

We again calculate the possible divisors of the -2-roots in this case. We obtain 4 reflective modular forms.

Theorem 5.3.9 *Let $N \in \{1, 2, 3, 4\}$. There are the following 4 modular forms which are obtained as quasi-pullbacks from Φ_{12} for the A_1 tower where the subscripts indicate the weight*

$$F_{35}^{A_1}, F_{34}^{2A_1}, F_{33}^{3A_1}, F_{32}^{4A_1}.$$

The divisor is given in each case by the union

$$\bigcup_{r \in R_{-2}(L_2(NA_1))} \mathcal{D}_{[r]}(L_2(NA_1))$$

and the different orbits under the action of the group $\tilde{O}(L_2(NA_1))^+$ are represented by

$$\mathcal{D}_{[-2,0]}(L_2(NA_1)), \mathcal{D}_{[-2,\varepsilon_1/2]}(L_2(NA_1)), \dots, \mathcal{D}_{[-2,\varepsilon_N/2]}(L_2(NA_1))$$

and all the modular forms stated above are reflective.

We will write

$$\mathcal{D}_{[-2,0]}(L_2(NA_1)) + \mathcal{D}_{[-2,\varepsilon_1/2]}(L_2(NA_1)) + \dots + \mathcal{D}_{[-2,\varepsilon_N/2]}(L_2(NA_1))$$

for the divisor which is a more convenient notation in algebraic geometry.

Proof

Let $r \in R_{-2}(L_2(NA_1))$ be a primitive -2-root. From the considerations in Proposition 1.6.2 and (1.11) we know that $\text{div}(r)|(r, r) = -2$. If r belongs to NA_1 then we have $\text{div}(r) = 2$ since the scale of NA_1 is 2. But we also have the vector $h_{-1,0} \in L_2(NA_1)$ which yields a -2-vector with divisor equal to 1. Any element $g \in \tilde{O}(L_2(NA_1))^+$ satisfies $\text{div}(gr) = \text{div}(r)$ and $gr^* \equiv r^* \pmod{L_2(NA_1)}$. Up to the sign there are N different roots in NA_1 which yield N different classes in $D(NA_1)$ namely $\varepsilon_1/2, \dots, \varepsilon_N/2$. These vectors are exactly the primitive representatives of the minimal norm classes. If on the other hand α is another -2-root in $L_2(NA_1)$ where $\text{div}(\alpha) = 2$ then the class of $\alpha^* = \alpha/2$ belongs to one of the N classes of minimal norm. Hence there are exactly $N + 1$ different orbits of $R_{-2}(L_2(NA_1))$ under the action of $\tilde{O}(L_2(NA_1))^+$ represented by $h_{-1,0}, \varepsilon_1/2, \dots, \varepsilon_N/2$. Since $N(kA_1^\perp) = 24 - k$ we obtain the assertion. $\square$

We consider a second variant of algorithm 5.3.6. We first apply this variant to the D_N -tower and repeat the construction for other lattices afterwards. We now consider the Niemeier lattices

$$\begin{aligned} &\mathcal{N}(A_{15} \oplus D_9), \quad \mathcal{N}(3D_8), \quad \mathcal{N}(E_6 \oplus D_7 \oplus A_{11}), \quad \mathcal{N}(4D_6), \\ &\mathcal{N}(2A_9 \oplus D_6), \quad \mathcal{N}(2A_7 \oplus 2D_5), \quad \mathcal{N}(6D_4), \quad \mathcal{N}(4A_5 \oplus D_4) \end{aligned}$$

and fix embeddings

$$\begin{aligned} L_2(D_N) &\hookrightarrow L_2(D_{N+1} \oplus K(N)) \text{ if } N \neq 3, 5 \\ L_2(D_3) &\hookrightarrow L_2(D_4 \oplus K^{(j)}(4)), \quad j = 1, 2 \\ L_2(D_5) &\hookrightarrow L_2(D_6 \oplus K^{(j)}(6)), \quad j = 1, 2 \end{aligned}$$

where we identify $D_N \subseteq D_{N+1}$ as the sublattice generated by $b_1, \dots, b_N \in \mathbb{Z}^{N+1}$, compare Figure 2.1.

Algorithm 5.3.10 (a) Check if the following condition is satisfied:

For all $l \in L_2(D_N)^\vee$ such that $-2 \leq (l, l) < 0$ there exists a primitive vector $x \in D_N(-1)^\vee$ such that $l \equiv x \pmod{L_2(D_N)}$ and $-2 \leq (x, x) < 0$. If this is the case continue with the next step.

(From Section 1.9 we know that this is true for D_N , $N \leq 8$.)

- (b) For each embedding $L_2(D_N) \hookrightarrow L_2(\mathcal{N}(D_{N+1} \oplus K(N)))$ into the root system of one of the Niemeier lattices listed above we have $II_{2,26} \cong L_2(\mathcal{N}(D_{N+1} \oplus K(N)))$. Now for $r = \alpha + \beta \in L_2(D_N)^\vee \oplus (L_2(D_N))_{L_2(\mathcal{N})}^\perp$ where $0 > (\alpha, \alpha) \geq -2$ and $(r, r) = -2$ we know that there is some $x \in D_N^\vee$ satisfying (a). We again set $r' = x + \beta$ and can conclude as in part (b) of 5.3.6 that r' belongs to the root system $\mathcal{N}_R(-1)$ which is $D_{N+1} \oplus K(N)_R$. From Lemma 2.4.2 we know that $(D_N)_{D_{N+1}}^\perp = \langle 2\varepsilon_{N+1} \rangle$. Since $x \neq 0$ this implies that either x is a -2-root in D_N or $x \in D_N^\vee$ and there exists an element $x' \in (D_N^\perp)^\vee$ such that $x + x'$ is a -2-root in D_{N+1} . However, the only nontrivial class in $D_N^\vee$ which allows such a continuation is the class $\varepsilon_1 + D_N$ and a primitive representative is ε_1 . This shows that α is either a -2-root of $L_2(D_N)$ or $\alpha \equiv \varepsilon_1 \pmod{L_2(D_N)}$ and $(\alpha, \alpha) = -1$.
- (c) Determine the different orbits of the divisor by using the Eichler criterion. Note that for any $n \in \mathbb{N}$ we have $\mathcal{D}_{[r]}(L_2(D_N)) = \mathcal{D}_{[nr]}(L_2(D_N))$.
- (d) If α is a -2-root in $L_2(D_N)$ the associated divisor has of course multiplicity one. If α is a -1-root then the multiplicity of the corresponding divisor is equal to the number of elements r in the class $\varepsilon_N + (D_N)_{D_{N+1}}^\perp$ such that $(r, r) = -1$ and this number is exactly two. Hence the divisor which belongs to the -1-roots has multiplicity 2.

Lemma 5.3.11 *Let $\alpha \in L_2(D_N)$ be a -4-root which means that $(\alpha, \alpha) = -4$ and assume that $\alpha/2 \equiv \varepsilon_1 \pmod{L_2(D_N)}$ for $3 \leq N \leq 8$. Then α is primitive in $L_2(D_N)$ and all such α belong to the same orbit under the action of $\tilde{O}(L_2(D_N))^+$*

Proof

If α was not primitive in $L_2(D_N)$ then the absolute value of the quadratic form would be greater than 4 which is a contradiction. Now α is certainly not primitive in $L_2(D_N)^\vee$ because $\alpha/2$ belongs to $L_2(D_N)^\vee$. Hence $\text{div}(\alpha) \neq 1$. The minimal norm of the vectors in the dual lattice is $\frac{N}{4}$ if $N = 3$ and 1 else. This shows that $\text{div}(\alpha) \neq 4$ and we always have $\text{div}(\alpha) = 2$. Now the second part follows from the Eichler criterion. $\square$

Theorem 5.3.12 *We have the following modular forms for the D_N -tower*

$$F_{72}^{D_3}, F_{88}^{D_4}, F_{102}^{D_5}, F_{114}^{D_6}, F_{124}^{D_7}, F_{132}^{D_8}$$

obtained as quasi-pullbacks where the divisor is given in each case by

$$\mathcal{D}_{[-2,0]}(L_2(D_N)) + 2\mathcal{D}_{[-4,\varepsilon_1]}(L_2(D_N)).$$

Proof

From algorithm 5.3.10 we know that the divisor is determined by -2-roots in $L_2(D_N)$ and -4-roots α such that $\alpha/2 \equiv \varepsilon_1 \pmod{L_2(D_N)}$. The Eichler criterion and Lemma 5.3.11 tell us that these forms vanish with multiplicity one on the orbits of $\mathcal{D}_{[-2,0]}(L_2(D_N))$ and with multiplicity two on the orbits of $\mathcal{D}_{[-4,\varepsilon_1]}(L_2(D_N))$. Since $(D_N)_{D_{N+1}}^\perp$ contains no -2-roots we can read off the quantity $N(L_2(D_N)^\perp)$ directly. $\square$

We want to apply the modified algorithm to the $4A_1$ -tower. For this purpose we consider the natural embedding $A_1 \hookrightarrow A_2$. The next Lemma tells us how we have to modify the calculation of the multiplicity of the divisor for the lattices NA_1 and A_2 .

Lemma 5.3.13 *We assume that $1 \leq N \leq 4$.*

- (a) *We have $(A_N)_{A_{N+1}}^\perp = \langle (N+1)\varepsilon_{N+2} - \varepsilon_{N+1} - \cdots - \varepsilon_1 \rangle_{\mathbb{Z}}$ where we consider A_N as a subset of $\mathbb{Z}^{N+1}$.*
- (b) *Let $\alpha \in A_1^\vee$ such that $-2 < (\alpha, \alpha) < 0$ and such that there exists a continuation to a -2-root in A_2 . Then $\alpha \equiv \frac{1}{2}(\varepsilon_2 - \varepsilon_1) \pmod{A_1}$ and there are exactly two possibilities to continue α which correspond to the roots $\pm(\varepsilon_3 - \varepsilon_1), \pm(\varepsilon_2 - \varepsilon_3)$.*
- (c) *Let $\alpha \in NA_1^\vee$ such that $-2 < (\alpha, \alpha) < 0$ and such that there exists a continuation to a -2-root in NA_2 . We write $\varepsilon_2^j - \varepsilon_1^j$ for the basis of the j -th copy of A_1 in NA_1 . Then $\alpha \equiv \frac{1}{2}(\varepsilon_2^j - \varepsilon_1^j) \pmod{NA_1}$ for some j and there are exactly two possibilities to continue α .*
- (d) *Let $\alpha \in A_2^\vee$ such that $-2 < (\alpha, \alpha) < 0$ and such that there exists a continuation to a -2-root in A_3 . Then $\alpha \equiv \frac{1}{3}(2\varepsilon_1 - \varepsilon_2 - \varepsilon_3) \pmod{A_2}$ or $\alpha \equiv -\frac{1}{3}(2\varepsilon_1 - \varepsilon_2 - \varepsilon_3) \pmod{A_2}$ and there is exactly one possibility to continue α which corresponds to the roots $(\varepsilon_1 - \varepsilon_4)$ and $-(\varepsilon_1 - \varepsilon_4)$, respectively.*

Proof

- (a) The lattice A_N is the subset of $\mathbb{Z}^{N+1}$ defined as

$$\{(x_1, \dots, x_{N+1}) \in \mathbb{Z}^{N+1} \mid x_1 + \cdots + x_{N+1} = 0\}.$$

This immediately shows that the element $(-1, -1, \dots, -1, N+1)$ is in the orthogonal complement. Since $A_N^\perp$ has rank 1 the lattice is generated by the last vector.

- (b) The discriminant group $D(A_N)$ is a cyclic group of order $N+1$ and is generated by $\frac{1}{N+1}(N\varepsilon_1 - \varepsilon_2 - \cdots - \varepsilon_{N+1}) + A_N$. Now let $x \in A_{N+1}$ be a -2-root and write

$$x = \frac{a}{N+1}(N\varepsilon_1 - \varepsilon_2 - \cdots - \varepsilon_{N+1}) + b \frac{(N+1)\varepsilon_{N+2} - \varepsilon_{N+1} - \cdots - \varepsilon_1}{(N+1)(N+2)} \in A_N^\vee + (A_N^\perp)^\vee.$$

We would like to analyze which values for a and b are possible. We thus have to solve the equation

$$2 = a^2 \frac{N(N+1)}{(N+1)^2} + b^2 \frac{(N+1)(N+2)}{(N+1)^2(N+2)^2} = a^2 \frac{N}{N+1} + b^2 \frac{1}{(N+1)(N+2)}.$$

We will do this only in the cases where $N = 1, 2$ because we only need these two results later. If $N = 1$ we obtain the integral solutions

$$\frac{a^2}{4} + \frac{b^2}{6} = 2 \iff (a, b) = (\pm 2, 0) \text{ or } (a, b) = (\pm 1, \pm 3).$$

Hence there are exactly two ways to extend a vector $\alpha \in \frac{\varepsilon_2 - \varepsilon_1}{2} + A_1$ where $(\alpha, \alpha) = -1/2$ to a -2-root in A_2 . One verifies indeed

$$x = \pm(\varepsilon_2 - \varepsilon_1) \text{ if } a = \pm 2, \quad x = \pm(\varepsilon_3 - \varepsilon_1), \pm(\varepsilon_2 - \varepsilon_3) \text{ if } a = \pm 1.$$

(c) This follows immediately from the last part.

(d) Now we consider the case $N = 2$ where we obtain the following integral solutions of

$$\frac{2a^2}{3} + \frac{b^2}{12} = 2 \iff (a, b) = (1, \pm 4), (-1, \pm 4).$$

In fact we only obtain a -2-root in A_3 if ab is negative which yields the root $\pm(\varepsilon_1 - \varepsilon_4)$. Hence in this case there is exactly one way to extend a vector $\alpha \in N\varepsilon_1 - \varepsilon_2 - \cdots - \varepsilon_{N+1} + A_2$ of minimal norm to a -2-root in A_3 and this class is the only class in $D(A_2)$ which allows such a continuation. $\square$

For the next Theorem we use a different model of the lattice NA_1 compatible with the last Lemma. The Niemeier lattice

$$\mathcal{N}(12A_2)$$

yields four modular forms for the $4A_1$ -tower.

Theorem 5.3.14 *We have four modular forms for $NA_1, N = 1, \dots, 4$ which are obtained as a quasi-pullback from Φ_{12} given by*

$$F_{45}^{A_1}, F_{42}^{2A_1}, F_{39}^{3A_1}, F_{36}^{4A_1}.$$

The divisor is given by

$$\mathcal{D}_{[-2,0]}(L_2(NA_1)) + 3\mathcal{D}_{[-2, \frac{1}{2}(\varepsilon_2^1 - \varepsilon_1^1)]}(L_2(NA_1)) + \cdots + 3\mathcal{D}_{[-2, \frac{1}{2}(\varepsilon_2^N - \varepsilon_1^N)]}(L_2(NA_1)).$$

Proof

We apply algorithm 5.3.10 to the lattice $L_2(NA_1)$. The first component of the divisor is determined by the -2-roots in $L_2(NA_1)$ and this yields

$$\mathcal{D}_{[-2,0]}(L_2(NA_1)) + \mathcal{D}_{[-2, \frac{1}{2}(\varepsilon_2^1 - \varepsilon_1^1)]}(L_2(NA_1)) + \cdots + \mathcal{D}_{[-2, \frac{1}{2}(\varepsilon_2^N - \varepsilon_1^N)]}(L_2(NA_1)).$$

The second component is determined according to Lemma 5.3.13 by all -2-vectors $\alpha \in L_2(NA_1)$ such that $\alpha/2 \equiv \frac{1}{2}(\varepsilon_2^j - \varepsilon_1^j) \pmod{L_2(NA_1)}$. Then it is clear that α cannot be primitive in $L_2(NA_1)^\vee$ which yields $\text{div}(\alpha) = 2$. From Lemma 5.3.13 we know that the divisor for the second type of such α always has multiplicity 2 so that the whole divisor

has multiplicity 3. By using the Eichler criterion we obtain the above description. The last Lemma shows that we do not get additional roots in the orthogonal complement coming from the embedding $NA_1 \hookrightarrow NA_2$ such that

$$N(L_2(NA_1)^\perp) = \frac{1}{2} \sharp R_{-2}((12 - N)A_2) = 3(12 - N)$$

which completes the proof. $\square$

Finally we consider the Niemeier lattices

$$\mathcal{N}(12A_2), \mathcal{N}(8A_3)$$

to obtain quasi-pullbacks for the lattice $L_2(A_2)$. We note that this lattice also possesses a primitive representative $l \in A_2^\vee$ for each class in $D(L_2(A_2))$ such that $-2 \leq (l, l) < 0$. We can either take -2-roots or

$$\pm \varrho = \pm \frac{1}{3}(2\varepsilon_1 - \varepsilon_2 - \varepsilon_3).$$

Theorem 5.3.15 (a) *There is a reflective modular form $F_{45}^{A_2}$ obtained by the method of quasi-pullback for the lattice A_2 with divisor*

$$\mathcal{D}_{[-2,0]}(L_2(A_2)).$$

(b) *We have a modular form $F_{54}^{A_2}$ for the lattice A_2 with divisor*

$$\mathcal{D}_{[-2,0]}(L_2(A_2)) + \mathcal{D}_{[-6,\varrho]}(L_2(A_2)) + \mathcal{D}_{[-6,-\varrho]}(L_2(A_2)).$$

Proof

- (a) Let us first consider the embedding $A_2 \hookrightarrow 12A_2$. Let $\alpha \in L_2(A_2)$ be a -2-root. Then α is certainly primitive in $L_2(A_2)$ since the lattice is even. Assume that α is not primitive in the dual lattice. Then $\alpha/2 \in L_2(A_2)^\vee$ and we have $(\alpha, \alpha) = -1/2$. Since the discriminant form only takes values in the set $\{2\mathbb{Z}, \frac{2}{3} + 2\mathbb{Z}\}$ this is a contradiction. Hence α must be primitive in the dual lattice and we obtain $\text{div}(\alpha) = 1$. Then α yields the divisor $\mathcal{D}_{[-2,0]}(L_2(A_2))$ and the form is reflective of weight $12 + \frac{1}{2} \sharp R_{-2}(11A_2) = 45$.
- (b) Now we consider $A_2 \hookrightarrow 8A_3$. In this case we obtain a composite divisor. The first part is determined by the -2-roots in $L_2(A_2)$. The second part according to Lemma 5.3.13 is determined by a -6-root α where $\alpha/3$ belongs to one of the two non-trivial classes and both cases occur. Then we can again conclude that α is primitive in $L_2(A_2)$ and α is not primitive in the dual lattice which yields $\text{div}(\alpha) \neq 1$. Since the divisor has to divide the order of the discriminant group we have $\text{div}(\alpha) = 3$. Now the Eichler criterion tells us that there are exactly three different orbits for the divisor with each representative having multiplicity one. The -2-roots correspond to reflective divisors. The weight is $12 + \frac{1}{2} \sharp R_2(7A_3) = 54$. $\square$

5.4 Maximal modular groups and the character $\det$

In the last section we have found modular forms for the character $\det$. In this context the natural question arises if the maximal modular group is larger than $\widetilde{\mathrm{O}}(L_2)^+$. If this is the case we would like to know the character which corresponds to this extension. The first question can be answered using the technique of discriminant forms and Proposition 1.4.11 as well as Corollary 1.4.10. Now let $L \leq L_2(\mathcal{N})$ be a primitive embedding and let ϕ be an automorphism of L . According to Proposition 1.4.11 and Corollary 1.4.10 ϕ extends to an element of $\mathrm{O}(L_2(\mathcal{N}))^+ \cong \widetilde{\mathrm{O}}(L_2(\mathcal{N}))^+$ if and only if there exists a $\psi \in \mathrm{O}(K)$ such that

$$\overline{\psi} = \gamma \circ \overline{\phi} \circ \gamma^{-1} \text{ where } K = L_{L_2(\mathcal{N})}^\perp.$$

The next Lemma tells us how one can determine the character if a continuation exists.

Lemma 5.4.1 *Let $L, K, \mathcal{N}$ be chosen as above and assume that $g \in \mathrm{O}(L)^+$ extends to $\hat{g} \in \mathrm{O}(L_2(\mathcal{N}))^+$. Then for every $\mathcal{Z} \in \mathcal{D}(L)^\bullet$ we have*

$$\Phi|_L^{(QP)}(g\mathcal{Z}_L) = \det(\hat{g}) \Phi|_L^{(QP)}(\mathcal{Z}_L)$$

and any such $\hat{g}$ satisfies $\hat{g}|_K \in \mathrm{O}(K)$.

We consider the lattices NA_1 where $N = 2, 3, 4$ and D_N where $N = 3, \dots, 8$. From Theorem 3.2.6 we can compute an upper bound for the order of the group of finite characters. More precisely we have

$$\sharp(\mathrm{O}(L_2(L))^+)^{\mathrm{ab}} \leq \begin{cases} 4 & \text{if } L = D_N \\ 8 & \text{if } L = NA_1 \end{cases}$$

and in the case $L = A_1$ the bound is 4. In fact we even have

Proposition 5.4.2 *We have*

(a) $(\mathrm{O}(L_2(D_N))^+)^{\mathrm{ab}} = \langle \det, v_\pi \rangle \cong \mathcal{C}_2 \times \mathcal{C}_2$ where $N = 3, \dots, 8$

(b) $(\mathrm{O}(L_2(NA_1))^+)^{\mathrm{ab}} = \langle \det, v_\pi, v_2 \rangle \cong \mathcal{C}_2 \times \mathcal{C}_2 \times \mathcal{C}_2$ where v_2 is the character induced by $v_\eta^{12} \times \nu_{H(NA_1)}$ for $N = 2, 3, 4$. If $N = 1$ then v_π is trivial and the group of the characters is isomorphic to $\mathcal{C}_2 \times \mathcal{C}_2$. We have

$$\ker(v_2) \cap \widetilde{\mathrm{SO}}(L_2(NA_1))^+ \subsetneq \widetilde{\mathrm{SO}}(L_2(NA_1))$$

and

$$\mathrm{O}(NA_1) \leq \ker(v_2).$$

Proof

Since the orthogonal groups for a definite lattice L are finite we can compute its commutator subgroup. The character v_π has been constructed before. A construction for the character v_2 is given in [Kl], section 1.6.3 as

$$v_2 : \mathrm{O}(L_2(NA_1))^+ \rightarrow \mathrm{Sp}(4, \mathbb{F}_2) \rightarrow \mathcal{S}_6 \xrightarrow{\text{sign}} \{\pm 1\}$$

and this construction satisfies $v_2|_{\Gamma^J(NA_1)} = v_\eta^{12} \times \nu_{H(NA_1)}$. □

In [Gr] the author investigated whether there is a continuation for a given $g \in \mathrm{O}(L)^+$ to $\mathrm{O}(L_2(\mathcal{N}))^+$ or not by studying the automorphisms of the Niemeier lattices out of explicit constructions. Since we do not want to go into details in this context we only state the results for some of the forms found in the last section.

Proposition 5.4.3 *We put for abbreviation $\Gamma = \Gamma(L_2)$ for the full modular group of the lattice L_2 .*

- (a) *The modular forms in Theorem 5.3.9 and Theorem 5.3.8 are all modular with respect to the full modular group except of $F_{124}^{D_8}$. The character is of the shape $\det \cdot v_\pi^m$ where $m \in \{0, 1\}$ and $F_{124}^{D_8}$ is only modular with respect to the stable orthogonal group.*
- (b) *We have $F_{35}^{A_1} \in \mathcal{M}_{35}(\Gamma, \det)$, $F_{34}^{2A_1} \in \mathcal{M}_{34}(\Gamma, \det)$ and $F_{33}^{3A_1} \in \mathcal{M}_{33}(\Gamma, \det \cdot v_\pi)$.*
- (c) *The character of the modular forms $F_{k_N}^{D_N}$ in Theorem 5.3.8 is $\det \cdot v_\pi^N$ for $N \in \{3, \dots, 7\}$.*

Proof

- (a) Compare [Gr], chapter 4 and use Proposition 5.4.2.
- (b) We note that the character v_π is trivial for A_1 . In the case of the lattice $2A_1$ we can use Proposition 3.3.3 to identify $F_{34}^{2A_1}$ with the product $\phi_4\phi_{30}$ in [DK], Corollary 4 where the authors constructed ϕ_k as Borcherds products for the unitary group $\mathrm{SU}(2, 2)_{\mathfrak{o}(\sqrt{-1})}$. The character of $F_{33}^{3A_1}$ was determined in [Kl], Theorem 5.12 where this form appears as the the product $\psi_3\psi_{30}$.
- (c) The characters of the forms appearing in this part have been determined in [Gr], Section 4.3. □

The following list contains all modular forms which appear in this chapter. For the modular forms with respect to A_2 we do not write down the character because we are mainly interested in the other two towers. The two blank spaces mean that the maximal modular groups are smaller in these cases.

modular form	construction	character	divisor
$F_{35}^{A_1}$	$\Phi _{A_1}^{(QP)}$	det	$\mathcal{D}_{[-2,0]}(L_2) + \mathcal{D}_{[-2,\varepsilon_1/2]}(L_2)$
$F_{34}^{2A_1}$	$\Phi _{2A_1}^{(QP)}$	det	$\mathcal{D}_{[-2,0]}(L_2) + \mathcal{D}_{[-2,\varepsilon_1/2]}(L_2)$
$F_{33}^{3A_1}$	$\Phi _{3A_1}^{(QP)}$	det v_π	$\mathcal{D}_{[-2,0]}(L_2) + \mathcal{D}_{[-2,\varepsilon_1/2]}(L_2)$
$F_{32}^{4A_1}$	$\Phi _{4A_1}^{(QP)}$	det v_π^ε	$\mathcal{D}_{[-2,0]}(L_2) + \mathcal{D}_{[-2,\varepsilon_1/2]}(L_2)$
$G_{10}^{A_1} = \frac{F_{45}^{A_1}}{F_{35}^{A_1}}$	A-Lift(ψ_{10,A_1})	1	$2\mathcal{D}_{[-2,\varepsilon_1/2]}(L_2)$
$G_8^{2A_1} = \frac{F_{42}^{2A_1}}{F_{34}^{2A_1}}$	A-Lift($\psi_{8,2A_1}$)	1	$2\mathcal{D}_{[-2,\varepsilon_1/2]}(L_2)$
$G_6^{3A_1} = \frac{F_{39}^{3A_1}}{F_{33}^{3A_1}}$	A-Lift($\psi_{6,3A_1}$)	1	$2\mathcal{D}_{[-2,\varepsilon_1/2]}(L_2)$
$G_4^{4A_1} = \frac{F_{36}^{4A_1}}{F_{32}^{4A_1}}$	A-Lift($\psi_{4,4A_1}$)	1	$2\mathcal{D}_{[-2,\varepsilon_1/2]}(L_2)$
$\Delta_{10}^{2A_1}$	A-Lift(ψ_{10,D_2})	v_π	$\mathcal{D}_{[-4,\frac{\varepsilon_1-\varepsilon_2}{2}]}(L_2)$
$\Delta_{18}^{3A_1}$	Corollary 5.2.6	v_π	$\mathcal{D}_{[-4,\frac{\varepsilon_1-\varepsilon_2}{2}]}(L_2)$
$\Delta_{24}^{4A_1}$	Corollary 5.2.5	v_π	$\mathcal{D}_{[-4,\frac{\varepsilon_1-\varepsilon_2}{2}]}(L_2)$
$F_{45}^{A_2}$	$\Phi _{A_2}^{(QP)}$	see [Gr]	$\mathcal{D}_{[-2,0]}(L_2)$
$\frac{F_{54}^{A_2}}{F_{45}^{A_2}}$	A-Lift(ψ_{9,A_2})	see [Gr]	$\mathcal{D}_{[-6,\varrho]}(L_2) + \mathcal{D}_{[-6,-\varrho]}(L_2)$

modular form	construction	character	divisor
$F_{54}^{D_3}$	$\Phi _{D_3}^{(QP)}$	$\det v_\pi$	$\mathcal{D}_{[-2,0]}(L_2)$
$F_{72}^{D_4}$	$\Phi _{D_4}^{(QP)}$	$\det$	$\mathcal{D}_{[-2,0]}(L_2)$
$F_{88}^{D_5}$	$\Phi _{D_5}^{(QP)}$	$\det v_\pi$	$\mathcal{D}_{[-2,0]}(L_2)$
$F_{102}^{D_6}$	$\Phi _{D_6}^{(QP)}$	$\det$	$\mathcal{D}_{[-2,0]}(L_2)$
$F_{114}^{D_7}$	$\Phi _{D_7}^{(QP)}$	$\det v_\pi$	$\mathcal{D}_{[-2,0]}(L_2)$
$F_{124}^{D_8}$	$\Phi _{D_8}^{(QP)}$	$\det$	$\mathcal{D}_{[-2,0]}(L_2)$ (stable orthogonal group)
$(G_9^{D_3})^2 = \frac{F_{72}^{D_3}}{F_{54}^{D_3}}$	$A\text{-Lift}(\psi_{9,D_3})^2$	1	$2\mathcal{D}_{[-4,\varepsilon_1]}(L_2)$
$(G_8^{D_4})^2 = \frac{F_{88}^{D_4}}{F_{72}^{D_4}}$	$A\text{-Lift}(\psi_{8,D_4})^2$		$2\mathcal{D}_{[-4,\varepsilon_1]}(L_2)$
$(G_7^{D_5})^2 = \frac{F_{102}^{D_5}}{F_{88}^{D_5}}$	$A\text{-Lift}(\psi_{7,D_5})^2$	1	$2\mathcal{D}_{[-4,\varepsilon_1]}(L_2)$
$(G_6^{D_6})^2 = \frac{F_{114}^{D_6}}{F_{102}^{D_6}}$	$A\text{-Lift}(\psi_{6,D_6})^2$	1	$2\mathcal{D}_{[-4,\varepsilon_1]}(L_2)$
$(G_5^{D_7})^2 = \frac{F_{124}^{D_7}}{F_{114}^{D_7}}$	$A\text{-Lift}(\psi_{5,D_7})^2$	1	$2\mathcal{D}_{[-4,\varepsilon_1]}(L_2)$
$(G_4^{D_8})^2 = \frac{F_{132}^{D_8}}{F_{124}^{D_8}}$	$A\text{-Lift}(\psi_{4,D_8})^2$		$2\mathcal{D}_{[-4,\varepsilon_1]}(L_2)$

Chapter 6

Applications

6.1 Modular forms for the A_1 -tower

We start with some basic properties of modular forms for this tower. The first observation is that there are no non-trivial modular forms of negative weight.

Lemma 6.1.1 *Let $\Gamma < \mathrm{O}(L_2)^+$ be a subgroup of finite index containing $\tilde{\mathrm{O}}(L_2)^+$ and let χ be a character for Γ of finite order $a \in \mathbb{N}$. Then we have*

$$\mathcal{M}_k(\Gamma, \chi) = \{0\} \quad \text{if } k < 0$$

and if $k \neq 0$ then $\mathcal{M}_k(\Gamma, \chi) \neq \{0\}$ implies $k \geq \frac{N}{2a}$. Moreover we have $\mathcal{M}_0(\Gamma, 1) = \mathbb{C}$.

Proof

According to our standard notation the lattice L_2 contains the hyperbolic plane $\mathbb{Z}e + \mathbb{Z}f$. Let $G(Z) := (F^a)_e(Z) = F^a(\mathrm{pr})(Z)$, where $Z \in \mathcal{H}(L_1)$, be the affine realization of F^a with respect to the cusp e . As we have already observed in Section 1.6 G has a Fourier expansion of the shape

$$G(Z) = \sum_{\substack{(m,l,n) \in L_1^\vee \\ m,n \geq 0 \\ 2mn - (l,l) \geq 0}} g(m,l,n) \exp(2\pi i[n\tau + m\omega + (l, \mathfrak{z})]) = \sum_{m \geq 0} \varphi_m(\tau, \mathfrak{z}) \exp(2\pi i m \omega).$$

Since the Jacobi group $\Gamma^J(L) \leq \Gamma$ we deduce that G is invariant with respect to the action of $\Gamma^J(L)$. In view of Lemma 2.1.6 we deduce that $\varphi_m \in J_{ak,L;m}$. Since there are no nontrivial holomorphic Jacobi forms of negative weight we deduce that $\dim \mathcal{M}_k(\Gamma, \chi) = 0$ whenever $k < 0$. Moreover if $k \neq 0$ and $\varphi_m \neq 0$ for some m then $ak \geq N/2$ by Corollary 2.1.15 where $N = \mathrm{rank}(L)$. The last statement was already proven in Lemma 5.2.2. $\square$

The next Lemma is elementary, too.

Lemma 6.1.2 *Let $\Gamma < \mathrm{O}(L_2)^+$ be a subgroup of finite index containing $\tilde{\mathrm{O}}(L_2)^+$. We have $\mathcal{M}_k(\Gamma, 1) = \{0\}$ if k is odd.*

Proof

Since $-I_{N+4} \in \tilde{O}^+(L_2(NA_1))$ the claim follows from

$$F(-I_{N+4} \cdot \mathcal{Z}) = F(-\mathcal{Z}) = (-1)^{-k} F(\mathcal{Z}). \quad \square$$

We now consider modular forms for the lattice $L = NA_1$.

Lemma 6.1.3 *Let $F \in \mathcal{M}_k(\tilde{O}^+(L_2(NA_1)), 1)$. If F has a zero in $z_N = 0$ then the order of F in $z_N = 0$ is at least 2.*

Proof

Let $\mathcal{Z} \in \mathcal{D}(L_2(NA_1))^\bullet$ and write $\mathcal{Z} = (\dots, z_1, \dots, z_N, \dots)$ such that $(z_1, \dots, z_N) \in NA_1(-1) \otimes \mathbb{C}$. The transformation

$$\mathcal{Z} \mapsto \mathcal{Z}' = (\dots, z_1, \dots, -z_N, \dots)$$

belongs to $\tilde{O}^+(L_2(NA_1))$. Since F is modular we obtain $F(\mathcal{Z}) = F(\mathcal{Z}')$. By considering the Taylor expansion of F in z_N we obtain that the order of zero must be even. This yields the assertion. $\square$

We are now able to prove our first result on the structure of the graded ring of modular forms for the NA_1 -tower. Here we will use the Jacobi forms for the NA_1 -tower defined in Section 2.3 and Section 2.5. We set

$$\mathcal{E}_4^{NA_1} := \text{A-Lift}(\epsilon_{4,NA_1}) \quad , \quad \mathcal{E}_6^{NA_1} := \text{A-Lift}(\epsilon_{6,NA_1}) \quad \text{for } N = 1, \dots, 4.$$

According to Proposition 2.5.1 and Corollary 2.5.3 we have

$$\mathcal{E}_k^{NA_1} \in \mathcal{M}_k(\text{O}(L_2(NA_1))^+, 1) \quad \text{for } k = 4, 6; N = 1, \dots, 4$$

and we have the following diagram

$$\mathcal{E}_4^{4A_1} \xrightarrow{z_4=0} \mathcal{E}_4^{3A_1} \xrightarrow{z_3=0} \mathcal{E}_4^{2A_1} \xrightarrow{z_2=0} \mathcal{E}_4^{A_1}. \quad (6.1)$$

From Theorem 5.2.1 we obtain the reflective modular forms $G_{10}^{A_1}, G_8^{2A_1}, G_6^{3A_1}, G_4^{4A_1}$ which are symmetric with respect to the action of the discriminant group. In a first step we want to determine the graded rings of modular forms with trivial character. The starting point is the following classical result of Igusa which can be found in [Ig], Theorem 1.

Theorem 6.1.4 (Igusa) *The graded ring of Siegel modular forms of even weight is a polynomial ring in four homogeneous generators*

$$E_4, E_6, \chi_5^2, \psi_{12}$$

where E_k denote the Eisenstein series of Siegel type and weight k and χ_5^2, ψ_{12} are cusp forms of weight 10 and 12, respectively.

According to Theorem 3.3.1 we have an isomorphism

$$\mathrm{Sp}(4, \mathbb{Z}) / \{\pm I_4\} \xrightarrow{\sim} \widetilde{\mathrm{SO}}(L_2(A_1))^+.$$

We note that $\widetilde{\mathrm{O}}(L_2(A_1))^+ = \mathrm{O}(L_2(A_1))^+$. From this we deduce

$$\mathcal{M}_k(\mathrm{Sp}(4, \mathbb{Z}), 1) = \mathcal{M}_k(\widetilde{\mathrm{SO}}(L_2(A_1))^+, 1).$$

Since the lattice A_1 is Euclidean we know from Lemma 3.2.8 that the orthogonal group is nicely generated. Hence we can apply Theorem 3.2.6 and conclude that every character of $\Gamma^J(A_1)$ can be extended to a character of $\mathrm{O}(L_2(A_1))^+$ and the order of the group of characters is bounded by 4. We know that the group of (finite) characters contains the group generated by the determinant character. Hence for a modular form $F \in \mathcal{M}_k(\widetilde{\mathrm{O}}(L_2(A_1))^+, \det^\kappa)$ we have

$$(-1)^\kappa F(\mathcal{Z}) = F((-I_5 \mathcal{Z})) = F(-\mathcal{Z}) = (-1)^k F(\mathcal{Z}).$$

This shows that

$$\bigoplus_{k \in 2\mathbb{Z}} \mathcal{M}_k(\mathrm{Sp}(4, \mathbb{Z}), 1) \cong \bigoplus_{k \in \mathbb{Z}} \mathcal{M}_k(\mathrm{O}(L_2(A_1))^+, 1)$$

and

$$\bigoplus_{k \in \mathbb{Z}} \mathcal{M}_k(\mathrm{Sp}(4, \mathbb{Z}), 1) \cong \bigoplus_{k \in \mathbb{Z}} \mathcal{M}_k(\mathrm{O}(L_2(A_1))^+, 1) \oplus \bigoplus_{k \in \mathbb{Z}} \mathcal{M}_k(\mathrm{O}(L_2(A_1))^+, \det).$$

The last algebra equals

$$\mathbb{C}[E_4, E_6, \chi_5^2, \psi_{12}, \chi_{35}]$$

and χ_{35} corresponds to the modular form $F_{35}^{A_1}$ from Theorem 5.3.9 obtained as a quasi-pullback. The second character for the full orthogonal group arises from the fact that $\mathrm{Sp}(4, \mathbb{Z})$ has exactly one nontrivial character of order two and there exists an interpretation of this character for the group $\mathrm{O}(L_2(A_1))^+$ which can be found in [GN], Theorem 1.12 and in a more general variant in [CG].

Theorem 6.1.5 *Let $\varphi \in J_{k,t;L}(v_\eta^D \times \nu)$, $k \in \mathbb{Z}, t \in \mathbb{Q}$ and let D be an even divisor of 24. If $Q = \frac{24}{D}$ is odd we assume the ν is trivial. Fix $\mu \in (\mathbb{Z}/Q\mathbb{Z})^*$. Then there exists a modular form $\mathrm{A}\text{-Lift}_\mu(\varphi)$ of weight k with respect to $\widetilde{\mathrm{O}}(2U + L(-Qt))^+$ with a character of order Q which is induced by the binary character v of $H_s(L(Qt))$ and the character $e^{2\pi i \mu \frac{h}{Q}}$ of the center of $H(L(Qt))$.*

As an immediate application we find a modular form $\chi_5^{A_1} \in \mathcal{M}_5(\mathrm{O}(L_2(A_1))^+, v_2)$ which is strongly reflective and whose square equals $G_{10}^{A_1}$, confer [G1], Theorem 5.1. Using this form we get a new formulation of Igusa's result in [Ig] for the structure of

$$\mathcal{A}(\Gamma) := \bigoplus_{k \in \mathbb{Z}} \mathcal{M}_k(\Gamma) \quad , \quad \Gamma = [\mathrm{O}(L_2(A_1))^+, \mathrm{O}(L_2(A_1))^+].$$

We define

$$H_{12}^{A_1} := \text{A-Lift}(\varphi_{12,A_1}) .$$

It is classically known that $H_{12}^{A_1}, \mathcal{E}_4^{A_1}, \mathcal{E}_6^{A_1}$ are algebraically independent. This follows easily from the algebraic independency of the elliptic Eisenstein series e_4, e_6 combined with the facts

$$\mathcal{E}_4^{A_1} = \text{A-Lift}(e_{4,0}^{A_1}) \quad , \quad \mathcal{E}_6^{A_1} = \text{A-Lift}(e_{6,0}^{A_1}) .$$

This yields

Corollary 6.1.6 *The full graded ring of modular forms for the lattice A_1 equals*

$$\mathcal{A}(\Gamma) = \mathbb{C}[\mathcal{E}_4^{A_1}, \mathcal{E}_6^{A_1}, \chi_5^{A_1}, H_{12}^{A_1}, F_{35}^{A_1}]$$

and the only relation among the generators is given by

$$\left(F_{35}^{A_1}\right)^2 \in \mathbb{C}[\mathcal{E}_4^{A_1}, \mathcal{E}_6^{A_1}, \chi_5^{A_1}, H_{12}^{A_1}] .$$

In the formulation of the next Theorem there appears the function

$$\Delta_{12,12} : \mathbb{H} \times \mathbb{H} \rightarrow \mathbb{C} \quad , \quad \Delta_{12,12}(\tau, \omega) = \Delta_{12}(\tau) \Delta_{12}(\omega)$$

which is an automorphic cusp form with respect to the group $\text{SL}(2, \mathbb{Z}) \times \text{SL}(2, \mathbb{Z})$. From the diagram in Proposition 2.3.5 we immediately obtain a pullback diagram of symmetric modular forms.

Theorem 6.1.7 *There is the following diagram of modular forms for $\text{O}(L_2(NA_1))^+$ and trivial character where $N = 1, \dots, 4$. The function at the top of the arrow equals up to multiplication in $\mathbb{C}^*$ the pullback of the lower function. The weights are indicated by the subscript and except for the first column all forms are cuspidal. The construction is explained in the proof.*

$$\begin{array}{ccccccc}
 G_4^{4A_1} & \xrightarrow{z_4=0} & G_6^{3A_1} & \xrightarrow{z_3=0} & G_8^{2A_1} & \xrightarrow{z_2=0} & G_{10}^{A_1} & \xrightarrow{z_1=0} & \Delta_{12,12} \\
 H_6^{4A_1} & \xrightarrow{z_4=0} & H_8^{3A_1} & \xrightarrow{z_3=0} & H_{10}^{2A_1} & \xrightarrow{z_2=0} & H_{12}^{A_1} & & \\
 H_8^{4A_1} & \xrightarrow{z_4=0} & H_{10}^{3A_1} & \xrightarrow{z_3=0} & H_{12}^{2A_1} & & & & \\
 H_{10}^{4A_1} & \xrightarrow{z_4=0} & H_{12}^{3A_1} & & & & & & \\
 H_{12}^{4A_1} & & & & & & & &
 \end{array}$$

Proof

We use the symmetrization operator from Section 2.3 to turn the Jacobi forms in Proposition 2.3.5 into functions of $J_{k,NA_1;1}^+$. Since the lattice NA_1 is maximal if $N \neq 4$ we can use Theorem 2.6.1 to produce cusp forms for the group $O(L_2(NA_1))^+$ whenever $N < 4$ and modular forms otherwise where the weights are inherited from the Jacobi forms in Proposition 2.3.5. The modularity with respect to the full group is achieved by application of the symmetrization operator. Finally we note that A-Lift is compatible with pullbacks such that for a Jacobi form $\varphi \in J_{k,NA_1;1}$ the following diagram commutes

$$\begin{array}{ccc} \varphi & \xrightarrow{\quad} & \varphi|_{(N-1)A_1} \\ \downarrow & & \downarrow \\ \text{A-Lift}(\varphi) & \xrightarrow{z_N=0} & \text{A-Lift}(\varphi|_{(N-1)A_1}) \end{array}$$

which completes the proof. $\square$

Theorem 6.1.8 *Let $N \in \{1, \dots, 4\}$. The graded ring $\mathcal{A}(O(L_2(NA_1))^+)$ is a polynomial ring in $N + 3$ variables over $\mathbb{C}$ generated by*

$$\mathcal{E}_4^{NA_1}, \quad \mathcal{E}_6^{NA_1}, \quad G_{12-2N}^{NA_1}, \quad H_{14-2N}^{NA_1}, \dots, \quad H_{12}^{NA_1}.$$

Proof

We prove the statement by a recursion process where it suffices to consider homogeneous elements in $\mathcal{A}$. The case $N = 1$ was treated in Theorem 6.1.4. For abbreviation write $\Gamma = \Gamma_N$ for the full modular group and $L = NA_1$. Note that there are no modular forms of negative weight and that $\mathcal{M}_0(\Gamma, 1) = \mathbb{C}$ due to Lemma 6.1.1. Note that Corollary 3.1.3 applies here and yields an embedding

$$O(L_2(NA_1))^+ \hookrightarrow O(L_2((N+1)A_1))^+.$$

The case $N = 2$. Let $F \in \mathcal{M}_k(\Gamma, 1)$. We know that the pullback $\mathcal{E}_6^{2A_1}|_{\mathcal{D}_{[-2, \varepsilon_2/2]}(L_2)} \in \mathbb{C}^\times \mathcal{E}_6^{A_1}$ from the initial step $N = 1$. Hence by virtue of Theorem 6.1.7 and diagram (6.1) there exists a homogeneous polynomial $p \in \mathbb{C}[X_1, \dots, X_4]$ such that

$$\tilde{F}|_{\mathcal{D}_{[-2, \varepsilon_2/2]}(L_2)} \equiv 0 \quad \text{where} \quad \tilde{F} := F - p(\mathcal{E}_4^{2A_1}, \mathcal{E}_6^{2A_1}, H_{10}^{2A_1}, H_{12}^{2A_1}).$$

By Lemma 6.1.3 we even know $2\mathcal{D}_{[-2, \varepsilon_2/2]}(L_2) \subseteq \text{div}(\tilde{F})$. Since Γ contains the permutation of the variables $z_1 \mapsto z_2, z_2 \mapsto z_1$ we conclude from Theorem 5.2.1 that $\text{div}(G_8^{2A_1}) \subseteq \text{div}(\tilde{F})$. Hence we can divide by $G_8^{2A_1}$ and from Koecher's principle 1.6.5 we deduce

$$\frac{\tilde{F}}{G_8^{2A_1}} \in \mathcal{M}_{k-8}(\Gamma, 1).$$

By induction on the weight we can conclude

$$F \in G_8^{2A_1} \mathbb{C}[\mathcal{E}_4^{2A_1}, \mathcal{E}_6^{2A_1}, G_8^{2A_1}, H_{10}^{2A_1}, H_{12}^{2A_1}] + \mathbb{C}[\mathcal{E}_4^{2A_1}, \mathcal{E}_6^{2A_1}, H_{10}^{2A_1}, H_{12}^{2A_1}]$$

which completes the proof in this case.

The case $N = 3$. Since $\mathcal{M}_6(\Gamma_2, 1) = \mathbb{C}\mathcal{E}_6^{2A_1}$ the proof is completely analogue to the last case.

The case $N = 4$. In this case we have to take into account $\mathcal{M}_6(\Gamma_3, 1) = \mathbb{C}\mathcal{E}_6^{3A_1} + \mathbb{C}G_6^{3A_1}$. Since

$$\mathcal{E}_6^{4A_1}|_{\mathcal{D}_{[-2, \varepsilon_4/2]}(L_2)} \in \mathbb{C}\mathcal{E}_6^{3A_1} + \mathbb{C}^\times G_6^{3A_1}$$

we can find a polynomial $p \in \mathbb{C}[X_1, \dots, X_6]$ with the property

$$\begin{aligned} & p(\mathcal{E}_4^{4A_1}, \mathcal{E}_6^{4A_1}, H_6^{4A_1}, H_8^{4A_1}, H_{10}^{4A_1}, H_{12}^{4A_1})|_{\mathcal{D}_{[-2, \varepsilon_4/2]}(L_2)} \\ &= p(\mathcal{G}_4^{3A_1}, \mathcal{G}_6^{3A_1}, G_6^{3A_1}, H_8^{3A_1}, H_{10}^{3A_1}, H_{12}^{3A_1}). \end{aligned}$$

Now we can apply the same reduction process as before to prove the claim in this case.

The algebraic independency of the generators in each case is a direct consequence of Theorem 5.2.1 and follows by the reduction process described before. $\square$

From Proposition 5.4.2 we know that $(O(L_2(NA_1)))^{+ab} = \langle \det, v_\pi, v_2 \rangle$ where v_π is trivial if $N = 1$. If Γ denotes the full orthogonal group and $\Gamma' := [\Gamma, \Gamma]$ its commutator subgroup we have the decomposition

$$\mathcal{M}_k(\Gamma') = \bigoplus_{\lambda \in \Gamma^{ab}} \mathcal{M}_k(\Gamma, \lambda).$$

The next Lemma is useful in order to determine the graded ring $\mathcal{A}(\Gamma')$.

Lemma 6.1.9 *Let $\Gamma = \Gamma_N$ denote the full modular group for the lattice NA_1 . Let $F \in \mathcal{M}_k(\Gamma, \lambda)$ for some $\lambda \in \Gamma^{ab}$.*

- (i) *If $\lambda = \det v_2^m v_\pi^n$ then $\mathcal{D}_{[-2, \varepsilon_1/2]}(L_2) \subseteq \text{div}(F)$.*
- (ii) *If $\lambda = v_2^m v_\pi$ where $m \in \{0, 1\}$ then $\mathcal{D}_{[-4, \frac{\varepsilon_1 - \varepsilon_2}{2}]}(L_2) \subseteq \text{div}(F)$.*
- (iii) *If $\lambda = v_2$ then $\mathcal{D}_{[-2, 0]}(L_2) \subseteq \text{div}(F)$.*

Proof

- (i) From Proposition 5.4.2 we deduce that all reflections of the shape σ_{ε_j} where j runs through $\{1, \dots, N\}$ satisfy $\lambda(\sigma_{\varepsilon_j}) = -1$ and hence the fixed locus $\mathcal{D}_{[-2, \varepsilon_j/2]}(L_2)$ is contained in $\text{div}(F)$.

- (ii) We have $\lambda(\sigma_{\varepsilon_j - \varepsilon_l}) = -1$ where j, l are distinct numbers in $\{1, \dots, N\}$ and this reflection induces the permutation $z_j \mapsto z_k, z_k \mapsto z_j$ with respect to our standard basis of NA_1 which shows that the fixed locus of this reflection is part of $\text{div}(F)$.
- (iii) We consider the reflection $\sigma_1 = \sigma_{e_1 - f_1}$ as in Theorem 3.2.3. From [Kl], Section 1.6.3 and the formula for P in the proof of Corollary 1.23 we deduce that $v_2(\sigma_1) = -1$. As $\mathcal{D}_{[-2,0]}(L_2)$ is exactly the fixed locus of this reflection we are done. $\square$

Now we are able to determine the structure of the graded rings $\mathcal{A}(\Gamma')$. The following list gives a summary of the forms constructed so far. The maximal modular group is the full orthogonal group Γ for each form.

modular form	construction	character	divisor
$\chi_5^{A_1}$	A-Lift ₁ (ϱ_{5,A_1})	$\det v_2$	$\mathcal{D}_{[-2,\varepsilon_1/2]}(L_2(A_1))$
$\chi_4^{2A_1}$	A-Lift ₁ ($\varrho_{4,2A_1}$)	$\det v_2$	$\mathcal{D}_{[-2,\varepsilon_2/2]}(L_2(2A_1))$
$\chi_3^{3A_1}$	A-Lift ₁ ($\varrho_{3,3A_1}$)	$\det v_2$	$\mathcal{D}_{[-2,\varepsilon_3/2]}(L_2(3A_1))$
$\chi_2^{4A_1}$	A-Lift ₁ ($\varrho_{2,4A_1}$)	$\det v_2$	$\mathcal{D}_{[-2,\varepsilon_4/2]}(L_2(4A_1))$
$F_{30}^{A_1}$	$F_{35}^{A_1} / \chi_5^{A_1}$	v_2	$\mathcal{D}_{[-2,0]}(L_2(A_1))$
$F_{30}^{2A_1}$	$F_{34}^{2A_1} / \chi_4^{2A_1}$	v_2	$\mathcal{D}_{[-2,0]}(L_2(2A_1))$
$F_{30}^{3A_1}$	$F_{33}^{3A_1} / \chi_3^{3A_1}$	$v_2 v_\pi$	$\mathcal{D}_{[-2,0]}(L_2(3A_1))$
$F_{30}^{4A_1}$	$F_{32}^{4A_1} / \chi_2^{4A_1}$	$v_2 v_\pi^\epsilon$	$\mathcal{D}_{[-2,0]}(L_2(4A_1))$
$\Delta_{10}^{2A_1}$	A-Lift(ψ_{10,D_2})	v_π	$\mathcal{D}_{[-4,\frac{\varepsilon_1 - \varepsilon_2}{2}]}(L_2(2A_1))$
$\Delta_{18}^{3A_1}$	Corollary 5.2.6	v_π	$\mathcal{D}_{[-4,\frac{\varepsilon_1 - \varepsilon_2}{2}]}(L_2(3A_1))$
$\Delta_{24}^{4A_1}$	Corollary 5.2.5	v_π	$\mathcal{D}_{[-4,\frac{\varepsilon_1 - \varepsilon_2}{2}]}(L_2(4A_1))$

Theorem 6.1.10 (a) Let $N \in \{2, 3, 4\}$. The graded ring

$$\mathcal{A}([O(L_2(NA_1))^+, O(L_2(NA_1))^+])$$

is generated by

$$\mathcal{E}_4^{NA_1}, \quad \mathcal{E}_6^{NA_1}, \quad \chi_{6-N}^{NA_1}, \quad H_{14-2N}^{NA_1}, \dots, \quad H_{12}^{NA_1}, \quad \Delta_{k_N}^{NA_1}, \quad F_{30}^{NA_1}$$

where k_N can be read off the table preceding the Theorem.

(b) Let $N \in \{2, 3, 4\}$. The modular forms

$$\mathcal{E}_4^{NA_1}, \quad \mathcal{E}_6^{NA_1}, \quad \chi_{6-N}^{NA_1}, \quad H_{14-2N}^{NA_1}, \dots, \quad H_{12}^{NA_1}$$

are algebraically independent. The only relations among the generators of $\mathcal{A}$ are given by

$$(\Delta_{k_N}^{NA_1})^2 = P_{NA_1}(\mathcal{E}_4^{NA_1}, \mathcal{E}_6^{NA_1}, \chi_{6-N}^{NA_1}, H_{14-2N}^{NA_1}, \dots, H_{12}^{NA_1})$$

and

$$(F_{30}^{NA_1})^2 = Q_{NA_1}(\mathcal{E}_4^{NA_1}, \mathcal{E}_6^{NA_1}, \chi_{6-N}^{NA_1}, H_{14-2N}^{NA_1}, \dots, H_{12}^{NA_1})$$

where $P_{NA_1}, Q_{NA_1} \in \mathbb{C}[X_1, \dots, X_{N+3}]$ are uniquely determined polynomials.

Proof

Part (a) is proved by induction on the weight and follows from the preceding table, Theorem 6.1.8 and Lemma 6.1.9. We will only illustrate the proof for the case $N = 2$. The other cases are done analogously. Let $P \in \mathcal{R} = \mathbb{C}[X_1, \dots, X_5]$ be homogeneous such that

$$P(\mathcal{E}_4^{2A_1}, \mathcal{E}_6^{2A_1}, H_{10}^{2A_1}, H_{12}^{2A_1}, \chi_4^{2A_1}) = 0.$$

We write $P = Q + R$ for $Q \in \mathbb{C}[X_1, \dots, X_4]$ and $R \in (X_5)_{\mathcal{R}}$. Since

$$\chi_4^{2A_1}|_{\mathcal{D}_{[-2, \varepsilon_1/2]}}(L_2(2A_1)) = 0$$

we conclude $Q = 0$ from the pullback diagram in Theorem 6.1.7 and Theorem 6.1.4. This yields

$$R(\mathcal{E}_4^{2A_1}, \mathcal{E}_6^{2A_1}, H_{10}^{2A_1}, H_{12}^{2A_1}, \chi_4^{2A_1}) = 0.$$

Since $\text{div}(\chi_4^{2A_1})$ is given by the modular translations of $\mathcal{D}_{[-2, \varepsilon_1/2]}(L_2(2A_1))$ we conclude

$$(R/X_5)(\mathcal{E}_4^{2A_1}, \mathcal{E}_6^{2A_1}, H_{10}^{2A_1}, H_{12}^{2A_1}, \chi_4^{2A_1}) = 0$$

by the identity theorem on holomorphic functions in many variables. Now an induction on the degree of R yields $R = 0$ and thus $P = 0$ showing the algebraic independency of the five generators. Obviously there are no linear relations among the generators due to the characters of $\chi_{6-N}^{2A_1}, \Delta_{10}^{2A_1}$ and $F_{30}^{2A_1}$. Hence the ideal of relations is generated by $X_6^2 - P_{2A_1}$ and $X_7^2 - Q_{2A_1}$. $\square$

The Maaß-Lift of symmetric Jacobi forms defines an embedding

$$J_{k, NA_1; 1}^+ \xrightarrow{\text{A-Lift}} \mathcal{M}_k(\text{O}(L_2(NA_1))^+)$$

The injectivity follows from the fact that the input function appears as the first Fourier-Jacobi coefficient. The well-definedness follows from the definition of A-Lift in Theorem 2.6.1 and the considerations on discriminant forms in Section 1.9. From the structure of the graded rings we derive a couple of conclusions and identities.

Corollary 6.1.11 (a) For every $N \in \{1, 2, 3, 4\}$, suitable $k \in \mathbb{N}$ and isotropic $\beta \in (NA_1)^\vee/NA_1$ one has $\text{A-Lift}(e_{k, \beta}^{NA_1}) \in \mathcal{M}_k(\text{O}(L_2(NA_1))^+)$.

(b) Let $N \in \{1, 2, 3\}$. We have an isomorphism of the linear spaces

$$J_{4,NA_1;1}^+ \cong \mathcal{M}_k(\mathrm{O}(L_2(NA_1))^+).$$

Moreover we have

$$e_{4,0}^{NA_1} = \varepsilon_{4,NA_1}.$$

Proof

(a) The group $\mathrm{O}(L_2(NA_1))^+$ is generated by $\widetilde{\mathrm{SO}}(L_2(NA_1))^+$ and the natural embedding of $\mathrm{O}(NA_1)$ according to Lemma 3.1.2 and the discussion of the root lattice NA_1 in Section 1.9. There we have also seen that $\mathrm{O}(NA_1)$ acts on the discriminant group by permutation of the coordinates $\varepsilon_1, \dots, \varepsilon_N$ of representatives in $L^\vee$. The lattices NA_1 are maximal if $N \neq 4$ and $4A_1$ has two distinct isotropic classes represented by 0 and $\frac{\varepsilon_1 + \varepsilon_2 + \varepsilon_3 + \varepsilon_4}{2}$. Hence $\mathrm{O}(NA_1)$ acts trivially on the isotropic classes of $D(NA_1)$ and the claim follows from Lemma 4.2.10.

(b) From the structure of the graded ring in Theorem 6.1.8 we know that $\dim J_{4,NA_1;1}^+ = 1$ for $N \in \{1, 2, 3\}$. In each case the estimation $4 \geq 2 + \frac{N+1}{2}$ holds such that $e_{4,0}^{NA_1}$ is well-defined and the arithmetic lifting yields modular forms with respect to $\mathrm{O}(L_2(NA_1))^+$ according to part (a). Hence the first part is proved and we know there is a constant $\alpha \in \mathbb{C}^*$ such that

$$e_{4,0}^{NA_1} = \alpha \varepsilon_{4,NA_1}.$$

The Fourier coefficient $f(0, 0)$ is 1 for both functions. Hence $\alpha = 1$. $\square$

Now we are able to derive a number theoretical application. We define

$$s_N(m, l) := \sharp\{x \in ((NA_1)_{E_8}^\perp)^\vee \mid l + x \in E_8, (l + x, l + x)/2 = m\} \in \mathbb{Z}.$$

Since the lattice E_8 is maximal there are no nontrivial isotropic subgroups of $D(E_8)$. Hence we can rewrite this quantity for $N \in \{1, 2, 3\}$ in the simplified shape

$$s_N(m, l) = \sharp\{x \in ((NA_1)_{E_8}^\perp)^\vee \mid (l + x, l + x)/2 = m\}, \quad m \in \mathbb{N}, l \in (NA_1)^\vee. \quad (6.2)$$

Proposition 6.1.12 *Using the notation from [BrK] the following identities are valid:*

$$\begin{aligned} s_1(m, y) &= \frac{(2\pi)^{7/2} \Delta(m, y)^{5/2}}{i 8 \Gamma(7/2)} \cdot \frac{L(3, \chi_D)}{\zeta(6)} \cdot \prod_{p|2\tilde{\Delta}(m, y)} \frac{1 - \chi_D(p)p^{-3}}{1 - p^{-6}} L_{y, \Delta(m, y)/2}(7/2, p), \\ s_2(m, y) &= \frac{(2\pi)^3 \Delta(m, y)^2}{(-1) 8 \Gamma(3)} \cdot \frac{1}{L(3, \chi_D)} \cdot \prod_{p|2\tilde{\Delta}(m, y)} \frac{L_{y, \Delta(m, y)/2}(3, p)}{1 - \chi_D(p)p^{-3}}, \\ s_3(m, y) &= \frac{(2\pi)^{5/2} \Delta(m, y)^{3/2}}{(-i) 8 \Gamma(5/2)} \cdot \frac{L(2, \chi_D)}{\zeta(4)} \cdot \prod_{p|2\tilde{\Delta}(m, y)} \frac{1 - \chi_D(p)p^{-2}}{1 - p^{-4}} L_{y, \Delta(m, y)/2}(5/2, p) \end{aligned}$$

where $\Delta(m, y) = 2m - (y, y) > 0$. If the last quantity is zero then we have $s_N(m, y) = 1$ in each case.

Remark 6.1.13 The last Proposition tells us that it is possible to determine the number of lattice points in $((NA_1)_{E_8}^\perp)^\vee$ lying on a sphere of radius $2m$ by evaluating a number theoretical function. On the other hand we know that the right hand side of the identities above is integral. In general the expressions of this type are only rational, confer Corollary 4.7 in [BrK].

Proof

The Theorem follows immediately from the Fourier expansion of $e_{4,0}^{NA_1}$ given in Corollary 4.2.13 and the explicit formula deduced in [BrK], Theorem 4.6 and the fact

$$\varepsilon_{4,NA_1}(\tau, \mathfrak{z}) = \sum_{\substack{m \in \mathbb{N} \\ y \in NA_1^\vee \\ 2m - (y, y) \geq 0}} s_N(m, y) \exp(2\pi i m \tau) \exp(2\pi i (l, \mathfrak{z})).$$

□

Now we consider the generators of $\mathcal{A}(\Gamma)$ of weight six for the lattices NA_1 , $N \in \{1, 2, 3\}$.

Corollary 6.1.14 *Let $N \in \{1, 2\}$. We have isomorphisms of the linear spaces*

$$J_{6,NA_1;1}^+ \cong \mathcal{M}_6(\mathrm{O}(L_2(NA_1))^+).$$

Moreover we have

$$\frac{\pi^2}{3}(2^{N+3} - N2^N) \cdot e_{6,0}^{NA_1} = \varepsilon_{6,NA_1}.$$

Proof

The first part follows by definition of the operator H_4 in Lemma 2.5.2 and comparison of the Fourier coefficient $f(0, 0)$. □

6.2 Applications to other towers of modular forms

In the last Section we were able to provide a very simple reduction process. This was essentially the contents of Theorem 6.1.7. In the situation of the A_1 -tower the following two conditions are satisfied simultaneously:

$$s(NA_1) = 2 \quad \text{and} \quad NA_1 \text{ is a reducible root lattice.}$$

As we have seen in Proposition 2.3.5 and Theorem 2.4.5 it is the first condition which allowed us to define operators

$$\otimes_{NA_1} \phi_{0,1} : J_{k,NA_1;1}^{(weak)}(1) \rightarrow J_{k,(N+1)A_1;1}^{(weak)}(1)$$

$$_{D_N(2)} \prod^{D_{N+1}(2)}(\cdot, \phi_{0,1}) : J_{k,D_N(2);1}^\Pi \rightarrow J_{k,D_{N+1}(2);1}^\Pi.$$

For the D_N -tower a similar construction does not exist since the scale of D_N is 1. Hence in the general situation of a root lattice we should use different coordinates to

describe the graded ring of modular forms. A possible strategy is to consider Jacobi-Eisenstein series. As we have seen before it is possible to compute the Fourier coefficients of the Eisenstein series $e_{k,\beta}^L$ using a computer. In order to determine the graded rings of modular forms one could consider the pullbacks of the Jacobi-Eisenstein series $e_{k,\beta}^L|_{L'}$ for suitable primitive embeddings $L' \hookrightarrow L$ using the explicit formulae. The construction of Eisenstein-like Jacobi forms in Section 2.5 does still work here. Instead of considering pullbacks of ϑ_{E_8} we might possibly have to replace E_8 by another even unimodular lattice in the general situation. Note that the genus of this lattice does usually contain more than one class. Hence one has to compare the restrictions of the different Jacobi theta series additionally. In this situation there are different one-dimensional boundary components. We have already treated this situation before when we considered quasi-pullbacks of Φ_{12} . Here we have 24 realizations of $II_{2,26}$ since there are exactly 24 classes of positive definite even unimodular lattices in dimension 24 represented by the 23 Niemeier lattices and the Leech lattice. Finally we indicate how we could apply our methods to the D_N -tower. The presentation of the reflective modular forms in Theorem 5.2.3 points to a little obstacle in the case of the lattice D_4 . In this case the lifting is only modular for a subgroup of index 3 in $O(L_2(D_4))^+$. As we have already mentioned at the end of Section 2.1 this is due to the additional symmetries of the Coxeter diagram for D_4 . If we replace this function by the product of the liftings of $\vartheta_{D_4}^{(j)}$ for $j = 1, 2, 3$ we obtain a function with a larger divisor.

Theorem 6.2.1 *There is a cusp form $\Delta_{24}^{D_4} \in \mathcal{S}_{24}(O(L_2(D_4))^+, \chi)$ with a binary character*

$$\chi : \Gamma \rightarrow \Gamma/\tilde{O}(L_2(D_4))^+ \xrightarrow{\sim} O(D_4) \xrightarrow{\text{sign}} \{\pm 1\} \quad \text{where } \Gamma = O(L_2(D_4))^+.$$

The modular form $\Delta_{24}^{D_4}$ is strongly reflective and its divisor is equal to

$$\bigcup_{\substack{\pm r \in L_2(D_4) \\ (r,r)=-4 \\ \text{div}(r)=2}} \mathcal{D}_{[r]}(L_2(D_4)).$$

This function appeared in [FH] first and later in [K2] where the author determined the graded ring of modular forms. The considerations in Section 1.9 and Corollary 3.1.3 yield embeddings

$$O(L_2(D_N))^+ \hookrightarrow O(L_2(D_{N+1}))^+.$$

Moreover we know the group of characters for the modular groups by virtue of Proposition 5.4.2. The corresponding modular forms appear in the list following Proposition 5.4.2. Finally we like to mention that it is possible to find an analogue of Theorem 2.4.5 for the lattices $A_N(2)$. This can be done by using the representation

$$A_N = e_{\mathbb{Z}^{N+1}}^\perp$$

and considering the basis from Section 1.8 where $e := \varepsilon_1 + \cdots + \varepsilon_{N+1} \in \mathbb{Z}^{N+1}$. Here we can again find a series of singular Jacobi forms of index 1. With respect to the basis

given above this is

$$\vartheta_{A_N}(\tau, \mathfrak{z}) = \eta(\tau)^{-1} \vartheta(\tau, (\varepsilon_1, \mathfrak{z})) \cdots \vartheta(\tau, (\varepsilon_{N+1}, \mathfrak{z})) .$$

If we write this down in coordinates we obtain

$$\vartheta_{A_N}^{[1]}(\tau, \mathfrak{z}) = \eta(\tau)^{-1} \vartheta(\tau, -z_1) \vartheta(\tau, z_1 - z_2) \cdots \vartheta(\tau, z_{N-1} - z_N) \vartheta(\tau, z_N) .$$

Here the superscript [1] indicates that we are dealing with an index 1 Jacobi form. In particular we have

$$\vartheta_{A_1}^{[1]} = -\eta(\tau)^{-1} \vartheta_{A_1}^2 \quad , \quad \vartheta_{A_2}^{[1]} = -\vartheta_{1, A_2} .$$

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$J_{k,L;t}^{(wh)}(\chi)$	Φ_{12}	137
$J_{k,L;t}^{(cusp)}(\chi)$	Φ_F	37
$J_{k,L;t}^{(weak)}(\chi)$	Φ_β	110
$J_{k,D_N(2);1}^{\text{II}}$	$\mathbb{Q}(-1, -1)$	100
$J_{k,L;t}$	$\mathbb{R}$	1
$J_{k,L;t}(\chi)$	$\mathbb{R}^{r,s}$	2
$J_{k,L;t}^{(eis)}$	$\mathbb{Z}$	1
$J_{k,NA_1;t}(1)^+$	ϵ_{4,NA_1}	86
$J_{k,NA_1;t}(1)^-$	ϵ_{6,NA_1}	89
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	$\text{GL}(V)$	2

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$\mathcal{B}_{NA_1}$	75	$\theta_{\mu,t}^L$	56
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$\mathcal{D}(M)^{\bullet}$	14	φ_{12,A_1}	70
$\mathcal{D}(M)_c$	14	ϱ_L^*	110
$\mathcal{D}_{[n,l]}(L_2)$	142	ϱ_L	108
$\mathcal{D}_{[r]}(L_2)$	23	$\vartheta(\tau, z)$	60
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$\mathcal{H}_{n,l}(L_1)$	127	$\vartheta_{4A_1}^{(2)}$	75
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$\mathcal{M}_k^V(\varrho)$	110	$\vartheta_{D_4}^{(3)}$	63
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